

Tanami Operations

Northern Territory, Australia

NI 43-101 Technical Report



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December 31, 2018

Prepared For
Newmont Mining Corporation

Report Author
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CERTIFICATE OF QUALIFIED PERSON

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I am a Registered Member of the Society for Mining, Metallurgy and Exploration, 4044636 and a Professional Engineer in Alberta, 44399. I graduated from Mining Engineering at the University of Alberta in 1986 (B.Sc.) and in 1991 (M.Sc.).

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As a result of my experience and qualifications, I am a Qualified Person for the content in the technical report, as the term Qualified Person is defined in National Instrument 43-101 *Standards of Disclosure for Mineral Projects* (NI 43-101).

I visited the Tanami site from 13 April to 17 April 2018.

I am responsible for all sections of the technical report.

I am not independent of Newmont as independence is described by Section 1.5 of NI 43-101.

I have been involved with the Tanami operation in a reserves governance role since 2014.

I have read NI 43-101 and all sections of the technical report have been prepared in compliance with that Instrument.

As of the effective date of the technical report, to the best of my knowledge, information and belief, all sections of the technical report contain all scientific and technical information that is required to be disclosed to make the technical report not misleading.

Dated: 4 March 2019

"Signed and sealed"

Donald Charles Doe, SME Registered Member 4044636

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1.0 SUMMARY

1.1 Introduction

Newmont Mining Corporation (Newmont) has prepared a report entitled “Technical Report, Tanami Operations, Northern Territory, Australia” (the Report) for the Newmont Tanami Operations (referred to as Tanami Operations or the Project), located in the Northern Territory (NT), Australia.

Mr. Donald Doe, Registered Member of the Society for Mining, Metallurgy & Exploration (RM-SME), Group Executive Reserves, Technical Services at Newmont, is the Qualified Person (QP) for the Report, as defined in National Instrument (NI) 43-101 Standards of Disclosure for Mineral Projects.

This Report supports declared Mineral Resources and Mineral Reserves and summarizes the Project development and current operations. Newmont will use this Report in support of disclosure and filing requirements with the Canadian securities regulators as specified in Section 4.2 (1) (c) of NI 43-101. This Report will be filed under Newmont’s System for Electronic Document Analysis and Retrieval (SEDAR) profile.

Newmont uses a wholly-owned Australian subsidiary, Newmont Australia Holdings Pty Ltd as the holding company of its company group in Australia, and Newmont Tanami Pty Ltd for the Tanami holdings. In this Report, the name Newmont is used interchangeably for the parent and subsidiary companies. The operations are known as “Newmont Tanami Operations” (NTO).

1.2 Project Setting

The Project is situated in the Western Tanami Desert Region of the NT, and currently comprises four distinct areas:

- **The Granites area:** Mined out Bullakitchie, Shoe, Quorn and Bunkers Hill deposits; site of The Granites processing plant, airstrip and Twin Hills accommodation village and associated infrastructure;
- **The Dead Bullock Soak (DBS) mining operations:** Callie underground mine, including the Callie, Auron, Federation and Liberator deposits and depleted open pits;
- **The Windy Hill area:** Depleted open pit at the Minotaur deposit;
- **The Oberon–Titania area:** Open pit Mineral Resource for the Oberon deposit.

The processing plant, camp and airstrip are located at The Granites, located on the Tanami Track approximately 550 kilometers (km) northwest of Alice Springs. The current mining operations are located at the DBS area, approximately 42 km to the west of The Granites. The nearest settlement is the Aboriginal settlement of Yuendumu, some 270 km southeast of The Granites. The region is sparsely populated.

Road access to the site is via the mostly unsealed Tanami Track. Personnel access to site is primarily via air with regularly scheduled flights from Darwin, Alice Springs, Brisbane and Perth.

1.3 Development History

Gold was discovered within the NT Tanami Desert region in 1900 by A.A Davidson, and over the next 30 years small quantities were extracted from alluvial deposits and quartz veins. In the 1930s, new gold deposits were discovered within the sedimentary sequence at The Granites, leading to the mining of approximately 10,000 ounces (oz) between 1932 and 1952. These operations provided the basis for prospecting in the region.

The Granites area was historically exploited for gold from narrow quartz veins and from limited alluvial and eluvial zones. A total of 420 kilograms (kg) of gold was produced from the area prior to 1961.

Normandy NFM Limited (NNFM) (as Newmont Tanami Pty Ltd was then known) has been involved in The Granites since 1975. In 1983, following agreement with the Traditional Owners (TOs) and the granting of mining titles by the NT Government, NNFM drilled a total of 40 drill holes at the Bullakitchie and Shoe deposits. NNFM went on to develop The Granites Gold Mine (TGGM), commencing mining at The Granites in 1986. In 1988, NNFM and the Central Land Council (CLC) concluded a deed providing for exploration and mining, over an area covering 80 square kilometers (km²) and including The Granites deposits.

Open pit and underground mines were established in The Granites area, including East and West Bullakitchie, Shoe, West Shoe, Quorn and Bunker's Hill, operating from 1984 to 1995, with a short resumption of open pit mining during 2002–2003. A total of approximately 1.1 million ounces (Moz) of gold was produced from the underground and open pit sources.

Commencing in 1988, regional exploration activities at DBS resulted in the discovery of the Triumph Hill, Colliewobble and Dead Bullock Ridge deposits, with open pit mining commencing in 1990. Further exploration success at DBS followed, with the discovery and subsequent development of the Villa and Callie deposits in 1992. Underground mining of the Callie deposit commenced in 1995 and is ongoing, open pit mining at DBS ceased in 2003. Extensional exploration led to the discovery of the Federation and Auron deposits. Total production from DBS is currently approximately 50 million tonnes (Mt) at 5.5 grams per tonne (g/t) Au for 8.8 Moz from all open pit and underground sources.

Concurrently the Minotaur deposit was identified in 1994 at Windy Hill. Mining at Minotaur commenced in June 2003 with ore being trucked to The Granites plant for processing. Mining activity ceased in February 2004, with a total of 53,987 oz of gold produced.

The Oberon deposit was discovered in June 1994. It has not been mined to date. An updated mining study and Mineral Resource estimate for Oberon was completed in 2018 to assess the viability of developing Oberon as an open pit, which remains at an early evaluation stage.

In Q3 2017, the Tanami Expansion Project (TEP) was commissioned. The TEP comprised construction of a second decline in the underground mine and incremental increase in the capacity in the processing plant.

The second decline was completed in mid-2016, enabling a step change in mining rates up to 2.6 Mtpa. The processing plant expansion included adding a ball mill, thickener and gravity circuit to improve recoveries and expanding mill capacity from 2.3 to 2.6 Mtpa. The mill reached commercial production by achieving sustained throughput, mill availability and grind size results in late August 2017.

With the underground mine currently reaching a depth of approximately 1.4 km below the surface, a project named the Tanami Expansion 2 (TE2) is being evaluated.

TE2 is focussed primarily on a 1,460 m production shaft and associated infrastructure. A shaft is considered to present the best long-term value for the Project to reduce operating costs and to access ore at depth. The shaft option requires fewer trucks underground, thus reducing labor costs, and reducing energy requirements both for fuel and refrigeration. The TE2 project allows for debottlenecking of the mine and is not considered a material expansion. This project is currently at feasibility study level.

TE2 will facilitate future growth opportunities through optimized production, lower operating costs and improved access to exploration platforms.

1.4 Mineral Tenure and Surface Rights

1.4.1 Tenure History

NNFM applied for and was granted the original mining tenements used for the Project in 1975 (MLS 134 to MLS 144 inclusive). In 1983, NNFM was granted MLS 8 and in 1990 MLS 154, which is now the location of the Callie underground mine. ML 23283 was granted in 2003. NNFM was part of the Normandy Mining Limited group and in 2002, Newmont Mining Corporation acquired this group, which included the Project assets.

1.4.2 Mineral Tenure

The Project has an interest in 46 tenements covering an area of approximately 3,500 km². The tenements are spread across a noncontiguous area that is approximately 180 km by 90 km in size. Newmont Tanami Pty Ltd. is the registered owner of 45 of the tenements comprising 15 Mining Leases and 30 Exploration Licences, with Nova Minerals Pty Ltd (Nova) the registered holder of the Officer Hill tenement (EL 23150), which is subject to the Officer Hill Farm-in and Joint Venture Agreement (OHFJVA). Newmont holds a 70% interest in this tenement.

Mineral Resources and Mineral Reserves that are currently reported occur within MLS 154. Previously mined areas are located within MLS 8 and ML 23283.

1.4.3 Surface Rights

The Project is located on Aboriginal Freehold Land held by the Central Desert Aboriginal Lands Trust (CDALT) and administered by the CLC on behalf of Traditional Owners in accordance with the requirements of the Aboriginal Land Rights (Northern Territory) Act 1976.

The Mining Leases are subject to the Consolidated Mining Agreement (CMA) and the Windy Hill Agreement (WHA), executed by Normandy NFM Limited (now called Newmont Tanami Pty Ltd) and the CLC in 2003. This agreement provides for all current activities conducted on the Mining Leases including land access, land usage, rehabilitation, compensation payments, sacred site management as well as heritage issues. Other agreements are in place to cover haul roads and the Billabong and Schist Hills bore fields and the Exploration Licences. The CLC Agreements cover the leases in the Project for the life of the leases.

1.5 Royalties

Under the terms of the CLC Agreement, which allows access to the Project, a 2.5% royalty is payable based on ounces of gold and silver sold. A royalty of AU\$0.12/tonne is also imposed

for haulage, bringing the average royalty to 2.7%. This average royalty is considered in the cut-off grade calculation.

The Project is subject to the Northern Territory Government Minerals Royalty that is calculated as 20% of the Net Value of mine production, where Net Value equals the gross revenue less the operating costs for the year, a capital allowance on eligible capital assets expenditure, eligible exploration expenditure and additional deductions as approved by the NT Minister for Mines. This royalty is considered in the overall economic analysis.

1.6 Geology and Mineralization

The Project lies in the eastern part of the Early Proterozoic Granites-Tanami Inlier. The basement geology of the Project is predominantly comprised of the sedimentary rocks of the Tanami Group. The remaining basement is predominantly composed of granitic intrusions. The Tanami Group consists of two Formations; the Dead Bullock Formation and the overlying Killi Killi Formation.

The Killi Killi Formation is the most widespread unit across the Project tenements. It is dominated by a thick interval of turbidites, with a felsic volcanic component and a distinctive unit of jasper chert.

The Dead Bullock Formation has been informally divided by the geologists of the Project into two Members; the Callie Member and underlying Ferdies Member. These units host all the known gold mineralisation on the project tenements.

The Project and the Tanami Region overall have undergone multiple phases of structural deformation. Most areas display several generations of folding and later episodes of faulting. Due to the geographic dispersion of deposits across the Project area and their local stratigraphy, the appearance of these events is locally variable.

The gold mineralization at the DBS area is predominantly hosted by the Schist Hills Iron Member (SHIM), Orac Formation, Callie Laminated Beds (+/- Magpie Schist) and the Lower Auron Beds. The Granites, Oberon and Minotaur deposits are hosted in the Davidson Beds.

No geological or mineralization descriptions are provided in this section for The Granites and Minotaur deposits as these have been mined out.

At DBS there are two sub-units (Members) within the Dead Bullock Formation; the Callie Member (which includes SHIM, Orac and the Callie Laminated Beds) and the Ferdies Member (which includes the Lower Auron Beds). Post-mineralization faulting has repeated parts of the stratigraphy and mineralization.

The Callie mineralization is developed in the DBS anticlinorium which is an upright isoclinal fold with local complex parasitic folding. Mineralization is present in the Lantin and Callie Anticlines, the Challenger Syncline and their respective limbs. The Auron deposit sits stratigraphically below the Callie deposit. The Federation deposit represents an un-mined mineralization envelope hosted predominantly within the hinge and southern limb of the Federation Anticline which is parallel to the Callie and Lantin Anticlines.

The Callie deposit has a complex shape, which is a function of the difference in strike of the quartz vein corridors (070°) and the plunge of the folded sequence (100°). The gold mineralization occurs in steeply dipping sheeted veins within the folded sequence. The orientation of the veins is 65° towards 154°, hence they are locally referred to as "70/70" veins. Vein density in the ore zone averages five veins per meter; however, it can reach more than

12 veins per meter in “corridors” that range in thickness from 20 to 50 meters (m). The width of mineralization varies significantly due to the interaction of the fold, fault and quartz vein corridor geometries, at its thickest point it is ~200 m wide in the hinge of the Callie Anticline. Mineralization has a down-plunge extent of at least 2,800 m from the surface to the easternmost deepest drill intercept. The mineralization currently remains open at depth. Gold is characteristically nuggety and occurs exclusively within the veins as grains from sub-millimeter sizes through to >20 millimeters (mm).

Auron shares the same structural, metamorphism, alteration and vein generations as Callie, and the mineralized quartz vein corridors that control the Callie mineralization are continuous through Auron. The majority of the economic Auron mineralization is represented by the Callie-style 70/70 sheeted vein mineralization. Auron shows a similarly complex shape as the Callie deposit due to the intersection of the folded sequence and the vein corridors. This is further complicated by fault repetitions. The Auron mineralization has been demonstrated to have a plunge extent of at least 2,500 m and remains open to the east. Like Callie, the width of the Auron mineralization varies significantly due to the interaction of the fold, fault and quartz vein corridor geometries, at its thickest point it is ~150 m wide. It gradually thins out up-plunge and is attenuated on the fold limbs.

Gold at Federation is hosted by a similar sheeted 70/70 vein set. The Federation mineralization has been demonstrated to have a plunge extent of at least 1,000 m. Mineralization has reduced widths relative to intersections within Callie and Auron.

Exploration drilling conducted during 2015 confirmed a new discovery in the Federation Anticline. The new orebody, referred to as Liberator, was similar to Federation in its structure and geometry but hosted in a rock mass similar to that of the Auron orebody.

Callie, Auron, Federation and Liberator are referenced as the four key underground mining areas in this Report.

1.7 Exploration

Exploration activities completed in the modern era, post 1983, have included geochemical sampling, geological mapping of open pit and underground exposures, airborne and ground geophysical surveys, shallow percussion, Aircore (AC), Rotary Air Blast (RAB) and vacuum drilling, Reverse Circulation (RC) and diamond core drilling, and metallurgical testwork.

1.8 Drilling and Sampling

Approximately 19,240 core and RC drill holes are recorded as having been completed on the Project. A further 39,280 shallow percussion, AC, RAB and vacuum drill holes have been completed to support exploration activities.

Standard logging and sampling conventions were used to capture information from the drill core and, where applicable, RC chips. Data captured included lithology, mineralization, alteration (visual), structural and geotechnical data, with provision for geologists to add comments on the core logging if required.

Recoveries have been measured for most core. Core recovery is typically very good and generally exceeds 98%.

Core drill collars have been surveyed by professional survey staff. Instruments have included a theodolite and total station instruments. RC drill collars were laid out and surveyed post-drilling using a differential global positioning system (DGPS) instrument.

Most drill holes were down-hole surveyed using magnetic single shot tools with lesser number of drill holes surveyed using a non-magnetic gyroscope or strain gauge based down-hole tool.

Density data were collected by Newmont staff, using the water immersion method.

Underground drill core is sampled and logged at a nominal 1 m interval, with a 0.3 m minimum sample length and a 1.3 m maximum sample length. Whole core sampling has been routine for underground holes since 1998, although one hole is sawn, and half core sampled per 100 m spaced section to allow a permanent record of the targeted deposits to be retained. Occasionally drill holes are sawn and half core sampled when sample is required for metallurgical or other testwork. Surface core is sawn in half and half core sampled.

Prior to 1999, all exploration RC and diamond core samples were assayed by Analabs in Adelaide. ALS Minerals (ALS) has performed sample preparation and fire assay on all geology samples between 1999 and 2015 and Bureau Veritas Minerals laboratories in Adelaide performed the sample preparation and fire assay on geology samples in 2015 and 2016. The laboratory currently holds ISO17025 accreditation as well as ISO9001, ISO14001 and ISO18001 accreditations.

Gold analysis prior to 2010 was completed with a fire assay with an atomic absorption spectrometry (AAS) finish, unless visible gold was observed, in which case a screen fire assay was completed. Post 2010 the screen fire assay was replaced with five repeat fire assays which were averaged with the original, this process was also applied for any original assay over 2 g/t Au. In 2013, the AAS finish was replaced with an Inductively-Coupled Plasma (ICP) finish. In late 2014, the number of fire assays performed for samples over 2 g/t Au was reduced to four and from mid-2015 any samples above 20 g/t Au were finished using a gravimetric determination on four fire assayed repeats.

Samples from the Auron host units were separately analyzed for sulfur (S) and arsenic (As), utilizing a four-acid complete digest. The resulting solution was then analyzed using an ICP finish.

Quality Assurance and Quality Control (QA/QC) programs are documented to have been in place since 1998, with a comprehensive system since 2004 consisting of the insertion of blank, Standard Reference Material (SRM) and duplicate samples. Newmont's QA/QC submission rate meets industry-accepted standards of insertion rates.

1.9 Data Verification

The QP has conducted personal inspections of the Project as part of his data verification. Mr. Doe has visited the Project numerous times during his career at Newmont, most recently between 13 and 17 April 2018.

An extensive program of data verification has been part of the development of the Project, including audits and reviews by internal staff, and external consultants, primarily in support of Mineral Resource and Mineral Reserves estimation.

The data verification programs undertaken on the data collected provide sufficient confidence in the geological interpretations, the analytical and database quality, and therefore support the use of the data in Mineral Resource and Mineral Reserves estimation.

1.10 Metallurgical Testwork

Metallurgical testwork programs have been undertaken at both Newmont's Technical Services facility in Denver, Colorado, as well as ALS AMMTEC Laboratories in Perth, Western Australia (WA). These programs were supported by a seven-year period of continuous processing of the underground ores (without oxide blending) and the process plant performance has met or exceeded laboratory results. Recent plant performance has been slightly better than that achieved under laboratory conditions and provides confidence in the results achieved to date.

Comminution and gold recovery testwork confirmed that the three-stage crushing, grinding, gravity separation and Carbon-In-Pulp (CIP) gold recovery circuit could process up to 2.4 million tonnes per annum (Mtpa) of the underground ores without further modification (pre-TEP). To process higher throughput rates and increase the recovery of gold further, the TEP recommended an increase in ball milling capacity, installation of a pre-leach thickener and an upgrade to the gravity circuit. The gravity gold component of the ore is high with over 60% recoverable by gravity and overall gold recovery between 94% and 96% based on pre-TEP plant performance. The post-TEP flowsheet is now operational, with additional capacity included in the grinding and gravity sections. Recoveries since the upgrade have been between 95% and 99% from throughput rates of typically 360 tonnes per hour (tph) to 380 tph with final grind sizes of between 80% passing (P_{80}) 60 micrometer (μm) and 80 μm versus a design throughput rate of 333 tph and a target grind size P_{80} of 125 μm .

Metallurgical testing programs will continue as required to evaluate different ore lithologies from new mining areas that may require changes in processing to improve throughput and/or recoveries. Metallurgical testwork programs are active in the Auron, Callie Deeps and Federation areas of the Project as well as the initial testwork on the recently accessed Liberator deposit.

Feed to the processing plant consists of ore from several stopes at a time and contains predominantly Auron ores with occasional, minor quantities of Callie ore. To assist the geologists in tracking and monitoring their model performance, ore is kept in discrete, consecutive weekly parcels. The ore from one week is usually completely exhausted before the subsequent week's parcel is fed to the process plant.

Blending of the mined ore occurs at the run in mine (RIM) and run of mine (ROM) pads and during haulage between the two. Recovery models were not updated in 2018, therefore the 2017 recovery models were retained. These models included a 1.55% recovery increase to allow for the TEP processing plant improvements. These models have been in line with the reconciled monthly recoveries achieved.

The recovery models used in preparing the 2015 to 2018 budgets were based on the initial lab testwork on Auron Beds (AB) and Lower Auron Beds (LAB) material. Typically, process plant actual performance had outperformed the models by 1 to 2%. Therefore, development of new models was warranted and completed in 2015. These models were used for preparation of the 2016, 2017 and 2018 budgets. The 2018 update was prepared with the same recovery models as 2017 to maintain consistency around TEP expectations.

Gold recovery functions are summarized below:

- **Callie Gold Recovery** = $0.6166 \times (\text{Au g/t}) + 94.08\% + 1.55$ (actual plant data) – Capped at 98%;

- **Auron Beds Tails Gold Grade** = $((\text{Au g/t}) - (0.0656 \times (\text{Au g/t}) + 0.0771)) / (\text{Au g/t}) \times 100 + 1.55$ (based on laboratory testwork);
- **Lower Auron Beds Tails Gold Grade** = $((\text{Au g/t}) - (0.0064 \times (\text{Au g/t}) + 0.1978)) / (\text{Au g/t}) \times 100 + 1.55$ (based on laboratory testwork);
- **Federation Tails Gold Grade** = $((\text{Au g/t}) - (0.0074 \times (\text{Au g/t}) + 0.1861)) / (\text{Au g/t}) \times 100 + 1.55$ (based on laboratory testwork);
- **Liberator Tails Gold Grade** = $((\text{Au g/t}) - (0.0064 \times (\text{Au g/t}) + 0.1978 + 0.05)) / (\text{Au g/t}) \times 100 + 1.55$.

These recovery functions continue to be monitored and adjusted accordingly based on actual plant production data and further laboratory testing.

1.11 Mineral Resource Estimates

Surface and solid wireframes were constructed representing interpreted structure (faults) and stratigraphy units. Stratigraphic logging is the primary driver for the geological modeling; however, geological mapping, structural logging and Au and S assay data also guided the interpretation. Grade shell solids were constructed to enclose high-grade portions of the sheeted quartz vein corridor.

The data were flagged by fault bound domains and stratigraphy and analyzed statistically to determine domain selection for Mineral Resource estimation. Exploratory Data Analysis (EDA) was performed on the raw drilling data and the composited data.

A nominal compositing length of 2 m was applied for Callie and Auron and a 1 m compositing length for Federation.

The Auron, Federation and Kerril South South Mineral Resource models were estimated by Ordinary Kriging (OK). The Callie Deeps Mineral Resource model was estimating using an inverse distance (ID) method. The Callie and Auron grade control model areas are interpolated utilizing Multiple Indicator Kriging (MIK).

Grade caps were applied to drill composites at the time of grade estimation. For Auron Upper and Lower areas, high grades were controlled through the MIK estimation process. Density values are assigned to Mineral Resource models by stratigraphic unit, varying between 2.82 grams per cubic centimeter (g/cm³) and 2.91 g/cm³. Density determinations within units are very consistent and exhibit low variability.

Blocks were classified as Measured, Indicated, or Inferred Mineral Resource based on a combination of the drill spacing and number of drill holes used to interpolate each block. A grade control estimate is required for classification of Measured Mineral Resource.

The block model was validated using visual inspection, swath plots, and comparison with Nearest-Neighbor (NN) models. No significant areas of error or bias were identified.

Mineral Resources for the Callie, Auron and Federation and Liberator underground deposits at DBS have been evaluated and reported within designed stope and development solids, with modifying factors applied to allow for dilution and mining recovery losses. Reasonable prospects of eventual economic extraction was evaluated separately for each area, mine level and stope using a gold price of US\$1,400/oz or AU\$1,750/oz. Material within mineable/profitable shapes was reported as Measured Mineral Resource, Indicated Mineral Resource or Inferred Mineral Resource.

The Mineral Resource estimate for the Oberon open pit deposit was reported within a resource pit design based on a gold price of US\$1,400/oz or AU\$1,750/oz to satisfy reasonable prospects of eventual economic extraction as an open pit operation.

1.12 Mineral Resource Statement

Mineral Resources consider geological, mining, processing and economic constraints, and have been confined within preliminary stope designs or open pit resource designs (for Oberon), and therefore are classified in accordance with the 2014 CIM Definition Standards for Mineral Resources and Mineral Reserves (the CIM Definition Standards).

Underground Mineral Resources are exclusive of those Mineral Resources that have been converted to Mineral Reserves and are reported at gold price of US\$1,400/oz or AU\$1,750/oz on a 100% basis, with an effective date of 31 December 2018.

Mineral Resources for Oberon open pit are reported at a gold price of US\$1,400/oz or AU\$1,750/oz on a 100% basis, with an effective date of 31 December 2018.

Newmont cautions that Mineral Resources that are not Mineral Reserves do not have demonstrated economic viability.

The Mineral Resource estimates were prepared by Mr. Shaun Schneider, Fellow of the Australian Institute of Geoscientists (FAIG), Geology Manager and Newmont employee, under the supervision of the QP.

Table 1-1 presents the economic parameters used in the 2018 underground Mineral Resource stope evaluation.

Table 1-1: Economic Parameters used in Underground Mineral Resource Stope Evaluation

Description	Unit	Rate	\$/ore tonne
Variable Mining Costs			
Lateral Ore Development	per metre advanced	\$3,320	\$42.36
Slot Rise Development	per metre	\$1,066	\$0.55
Diamond Drilling	per drill metre	\$146	\$6.89
Production Drilling	per stope drilling m	\$29.0	\$1.94
Stope Blasting	per stope tonne	\$2.01	\$2.01
Loading	per tonne	\$3.78	\$3.78
Haulage	per tkm	\$1.29	\$3.71
Hoisting	per tonne	\$4.87	\$4.87
Surface Haulage	per ore tonne	\$4.96	\$4.96
Paste Filling	per fill tonne	\$21.35	\$14.21
Utilities	per tonne	\$0.76	\$0.76
UG Mine Support	per tonne	\$0.43	\$0.43
Other Variable Costs			
Process	per ore tonne	\$12.60	\$12.60
Support (Site & Region)	per ore tonne	\$1.36	\$1.63
General Sustaining Capital	per ore tonne	\$7.12	\$11.02
Fixed Costs			
Mining	per year	\$88,000,000	\$33.85
Sustaining Capital	per year	\$11,400,000	\$5.69
Milling	per year	\$26,600,000	\$10.23
Support (Site & Region)	per year	\$38,600,000	\$14.85

The sustaining capital includes an allowance for the Capital Recovery Factor (CRF). The cut-off grade varies with depth. All the costs listed in Table 1-1 were used in a cashflow evaluation for each stope and each level.

Table 1-2 presents the economic parameters used in Oberon open pit Mineral Resource evaluation.

Table 1-2: Economic Parameters used in Oberon Mineral Resource Evaluation

Description	Unit	Cover	Oxide	Transition	Fresh
Processing – Incremental Ore Haulage/D&B Costs	AU\$/tonne	0.03	0.12	0.08	0.17
Processing – Surface Ore Haulage to Granites	AU\$/tonne	9.50	9.50	9.50	9.50
Processing – Variable	AU\$/tonne	21.1	21.1	21.1	21.1
Processing – Fixed	AU\$/tonne	5.4	5.4	5.4	5.4
Dewatering/Aquifer Reinjection operating costs	AU\$/tonne	1.2	1.2	1.2	1.2
Closure Cost	AU\$/tonne	0.0	0.0	0.0	0.0
Tailings Capacity – Sustaining Capital	AU\$/tonne	1.2	1.2	1.2	1.2
CRF on Sustaining Capital (TSF)	AU\$/tonne	0.2	0.2	0.2	0.2
Total Process Costs	AU\$/tonne	38.56	38.66	38.61	38.71
Processing Sales – WGC	AU\$/oz	0.08	0.08	0.08	0.08
Processing – Recovery	%	95.2	95.2	95.2	95.2
Price	AU\$/oz	1,600	1,600	1,600	1,600
Cut-Off Grade	g/t Au	0.788	0.789	0.789	0.790

Table 1-3 presents the total gold Mineral Resource for the Project.

Table 1-3: Mineral Resource – Gold at the Effective Date of 31 December 2018

	Measured Resource			Indicated Resource			Measured + Indicated Resource			Inferred Resource		
	Tonnage (kt)	Au Grade (g/t)	Au Metal (koz)	Tonnage (kt)	Au Grade (g/t)	Au Metal (koz)	Tonnage (kt)	Au Grade (g/t)	Au Metal (koz)	Tonnage (kt)	Au Grade (g/t)	Au Metal (koz)
Tanami UG	400	3.15	40	3,800	4.73	570	4,200	4.57	610	6,300	5.67	1,140
Tanami OP	-	-	-	4,200	2.21	300	4,200	2.21	300	3,900	1.88	230
Total UG + OP	400	3.15	40	8,000	3.40	870	8,400	3.39	910	10,200	4.22	1,370

Notes to accompany the Mineral Resource table:

- UG = underground and OP = open pit;
- UG Mineral Resources presented are aggregations for the Auron, Callie, Federation and Liberator UG mine areas and comprise weighted average values;
- OP Mineral Resources presented represent Mineral Resources for the Oberon deposit;
- Mineral Resources have an effective date of 31 December 2018;
- Mineral Resources are reported exclusive of Mineral Reserves; and are reported on a 100% basis;
- Mineral Resources are reported using a gold price of US\$1,400/oz, equivalent to AU\$1,750/oz at an exchange rate of US\$0.80 = AU\$1.00;
- UG Mineral Resources (Auron, Callie, Federation and Liberator) are evaluated and reported within conceptual stope designs;
- Mining, processing, site and regional G&A costs, metallurgical recoveries, and royalties are considered in the cut-off determination and in individual stope economic evaluations. The lowest cut-off grade applied was 1.6 g/t Au (ore development). Table 1-1 presents the economic parameters used in the 2018 underground Mineral Resource stope evaluation;
- For UG operations, the Mineral Resource was evaluated following a similar process to that used for the Mineral Reserves; tonnages are based on technically and economically viable minable shapes applied to the Mineral Resource model and include modifying factors for stope dilution (7 to 20% dilution) and mining recovery losses (92.5% recovery);
- OP Mineral Resources (Oberon deposit) are reported within a Mineral Resource pit design based on a gold price of US\$1,400/oz, equivalent to AU\$1,750/oz. Table 1-2 presents the economic parameters used in Oberon open pit Mineral Resource evaluation;
- Tonnages are rounded to the nearest 100,000 tonnes;
- Ounces are estimates of gold metal contained in the Mineral Resource and do not include allowances for processing losses. Ounces are rounded to the nearest 10,000 ounces;
- Rounding as required by reporting guidelines may result in apparent summation differences between tonnes, grade and contained metal content;
- Tonnage and grade measurements are in metric units. Gold ounces are reported as troy ounces.

1.13 Mineral Reserves Estimates

Mineral Reserves were derived by applying modifying factors to the Measured or Indicated Mineral Resources in the 2018 Mineral Reserves Plan. Stope and development designs were evaluated considering economic, mine operating, processing and other relevant modifying factors.

Mineral Reserves estimation incorporated the interface of all development, production and backfilling activities in the mine. A cash-flow analysis was carried out for every stope and every level. Uneconomic stopes and their associated activities were removed from the 2018 Mineral Reserves Plan.

The Mineral Reserves estimate for the Project and specifically the DBS Underground is based on a long-term mine plan that is economically viable and technically feasible. The analysis completed exceeds PFS level and meets the conditions for Mineral Reserves reporting as outlined in the CIM Definition Standards. The planning process conforms to the pertinent Newmont Guidelines.

Stope and underground development design criteria have been based on site geotechnical guidance. The guidance has been developed through operational experience in combination with sound analysis and modelling. Applied cut-off grades were based on the site Business Plan 2019 (BP19) Financial Model in combination with the design pricing guidance of US\$1,200/oz or AU\$1,600/oz. Technical reviews of the proposed ventilation and backfilling systems have also been completed to validate the mine plan that underpins the Mineral Reserves estimate.

Detailed scheduling in combination with financial modelling was used to test the economic viability of the Mineral Reserves. Variable and fixed operating cost estimates were based on current operating experience. A capital expenditure model was also developed using the BP19 capital expenditure estimate.

Table 1-4 presents the economic parameters used in the 2018 underground Mineral Reserves stope evaluation.

Table 1-4: Economic Parameters used in Underground Mineral Reserves Stope Evaluation

Description	Unit	Rate	\$/ore tonne
Variable Mining Costs			
Lateral Ore Development	per metre advanced	\$3,320	\$42.36
Slot Rise Development	per metre	\$1,066	\$0.55
Diamond Drilling	per drill metre	\$146	\$6.89
Production Drilling	per stope drilling m	\$29.0	\$1.94
Stope Blasting	per stope tonne	\$2.01	\$2.01
Loading	per tonne	\$3.78	\$3.78
Haulage	per tkm	\$1.29	\$3.71
Hoisting	per tonne	\$4.87	\$4.87
Surface Haulage	per ore tonne	\$4.96	\$4.96
Paste Filling	per fill tonne	\$21.35	\$14.21
Utilities	per tonne	\$0.76	\$0.76
UG Mine Support	per tonne	\$0.43	\$0.43
Other Variable Costs			
Process	per ore tonne	\$12.60	\$12.60
Support (Site & Region)	per ore tonne	\$1.36	\$1.63
General Sustaining Capital	per ore tonne	\$7.12	\$11.02

Description	Unit	Rate	\$/ore tonne
Fixed Costs			
Mining	per year	\$88,000,000	\$33.85
Sustaining Capital	per year	\$11,400,000	\$5.69
Milling	per year	\$26,600,000	\$10.23
Support (Site & Region)	per year	\$38,600,000	\$14.85

The sustaining capital includes an allowance for the CRF. The cut-off grade varies with depth. All the costs listed in Table 1-4 were used in a cashflow evaluation for each stope and each level.

1.14 Mineral Reserves Statement

Proven and Probable Mineral Reserves were declared for the underground mineralization.

The Mineral Reserves estimate was prepared by Mr. Johannes Grobler, Technical Services Superintendent and Newmont employee, under the supervision of the QP.

Mineral Reserves have an effective date of 31 December 2018 and are reported to a gold price of US\$1,200/oz or AU\$1,600/oz and an assumed exchange rate of US\$0.75 = AU\$1.00.

At the effective date of the Report, the exchange rate was US\$1.00 = approx. AU\$1.42¹.

Table 1-5 presents the total gold Mineral Reserves for the Project.

¹ Source: Reserve Bank of Australia (RBA) website (<https://rba.gov.au/statistics/frequency/exchange-rates.html>)

Table 1-5: Mineral Reserves – Gold at the Effective Date of 31 December 2018

	Proven Reserves			Probable Reserves			Proven + Probable Reserves		
	Tonnage (kt)	Au Grade (g/t)	Au Metal (koz)	Tonnage (kt)	Au Grade (g/t)	Au Metal (koz)	Tonnage (kt)	Au Grade (g/t)	Au Metal (koz)
Tanami UG	10,200	5.46	1,780	16,400	5.54	2,910	26,600	5.51	4,690
Total UG	10,200	5.46	1,780	16,400	5.54	2,910	26,600	5.51	4,690

Notes to accompany the Mineral Reserves table:

- UG = underground;
- UG Mineral Reserves presented are aggregations for UG operations and comprise weighted average values for several distinct underground mine areas;
- Mineral Reserves are reported on a 100% basis;
- Mineral Reserves are reported to a gold price of US\$1,200/oz, equivalent to AU\$1,600/oz at an exchange rate of US\$0.75 = AU\$1.00;
- UG Mineral Reserves are evaluated and reported within stope and development designs;
- Mining, processing, site and regional G&A costs, metallurgical recoveries, and royalties are considered in the cut-off calculations and in individual stope economic evaluations. The lowest cut-off grade applied was 1.6 g/t Au (ore development). Table 1-4 presents the economic parameters used in the 2018 underground Mineral Reserves stope evaluation;
- For UG operations, the Mineral Reserves tonnages are based on technically and economically viable minable shapes applied to the Mineral Resource model and include modifying factors for stope dilution (7 to 20% dilution) and mining recovery losses (92.5% recovery);
- Tonnages are rounded to the nearest 100,000 tonnes;
- Ounces are estimates of metal contained in the Mineral Reserves and do not include allowances for processing losses. Ounces are rounded to the nearest 10,000 ounces;
- Rounding as required by reporting guidelines may result in apparent summation differences between tonnes, grade and contained metal content;
- Tonnage and grade measurements are in metric units. Gold ounces are reported as troy ounces.

1.15 Mine Plan

Access to the mine is via the Callie Decline located in the footwall of the Callie mine area. The decline has reached the elevation of M20mRL, where M represents levels located below 0mRL (surface is 1390mRL). Nominal level spacing is 40 m, with additional sub-levels in the upper part of the mine.

There are four main underground mining areas at DBS: Callie, Auron, Federation and Liberator. Liberator is yet to declare a Mineral Reserve.

Development of an independent decline system to access the Auron deposit commenced in 2013 and was completed second quarter of 2016.

The mining method used at the Tanami is long-hole open stoping with paste or rock backfill, using a 92.5% mine recovery factor and a stope dilution factor that varies according to geotechnical recommendations (7% to 20%).

Stope design is dependent on orebody shape, accessibility, mining method and dilution factors. A cut-off grade of 3.5 g/t Au was used for the Mineral Reserves, which yielded the best Net Present Value (NPV) over the current cost and revenue profile defined by BP19. Marginal stopes with overall grades between 3.0 and 3.5 g/t Au were also included in the Mineral Reserves, based on individual economic evaluation of each stope.

Automated stope shapes were adjusted manually to ensure that they were mineable. The ground support strategy uses the concept of rock reinforcement and surface control to construct a stable support arch for the specified excavation geometry. The Tanami Operations are considered dry mining operations, with annual pumping from operations being in the range of 20 L/min.

Development and face production activities are performed by a fleet of twin boom drilling jumbos. Blasting is carried out with mobile explosives vehicles utilizing ANFO. Ground support installation is performed using jumbos for installation, and a cable bolter machine is used for cable bolting. Vertical development is typically carried out using a raise-bore hole, followed by stripping with long holes.

The mine plan from 2019 onward is designed and scheduled for 2.7 Mtpa. The mine plan is developed on a month by month basis for 2019, 2020 and 2021. It is then reported on an annual basis for 2022 through to the end of mine life (2028).

Conventional underground mining equipment is used to support the underground mining activities. Material handling is performed with a fleet of loaders and 60t underground haul trucks. All ore is trucked out of the mine to the surface ROM stockpile. Waste is generally placed in previously mined-out secondary or stand-alone stopes.

The mining operation is ventilated using an exhausting system with a capacity of 1,080 m³/s. The underground air is cooled using a refrigeration plant with a capacity of 18 megawatts of refrigeration (MWR).

1.16 Recovery Plan

The processing facility has been in operation since 1986 and has undergone several expansions and upgrades since that time.

The Granites processing plant comprises a three-stage crushing circuit followed by two stage series ball milling to grind the ore to a cyclone overflow P_{80} of typically 60 to 100 μm ahead of gravity concentration, cyanide leaching, and gold recovery. All gold recovered is processed on site to produce gold doré bars.

Within the grinding circuit, gravity concentration is used to recover liberated gold, utilizing rougher and cleaner spirals and Wilfley tables as well as a Knelson centrifugal concentrator to produce a gold concentrate. The gravity recoverable gold concentrate is cyanide leached using an intensive leach reactor (Acacia reactor), producing an eluate solution suitable for electrowinning.

Gold which is present in the ore but not gravity recoverable (e.g. present in cyclone overflow) is further processed in a conventional CIP circuit. Lime is added to the ore to increase the pH to greater than 10, sodium cyanide solution and oxygen are added to leach the gold from the ore, and activated carbon is added in a counter current flow direction to recover the gold from solution. The activated carbon is then stripped of the adsorbed gold using an intensive cyanide solution and high temperature elution process, recovering an eluate solution suitable for electrowinning and barren carbon that can be re-activated and returned to the CIP circuit.

The electrowinning cell cathodes are cleaned, the concentrate dried, and direct smelted producing gold doré bars.

Tailings from the processing plant are treated with Caro's Acid to destroy residual cyanide in solution and then pumped to a nearby Tailings Storage Facility (TSF), where the water is decanted and returned to the processing plant for re-use.

Additional fresh water for the operation is sourced from several bore fields in the region and captured from rainfall run off during the wet season.

1.17 Infrastructure

Due to the remote geographic location of the Project, completely stand-alone infrastructure has been established to cater for situations of isolation and absence of local services. The site infrastructure is split between The Granites and DBS operational areas.

The Granites includes the processing plant, power station, tailing storage facilities, airstrip, accommodation village and associated support facilities. DBS includes facilities for the support of the underground mine including power station, fuel facilities, workshops, stores and offices.

1.18 Marketing

Newmont has an operative refining agreement for refining of gold and silver doré produced from the Project. Newmont's bullion is sold on the spot market, by marketing experts retained in-house by Newmont.

1.19 Environmental, Permitting and Social Considerations

The Project has been the subject of extensive baseline studies. Environmental monitoring is ongoing and addresses areas such as flora, fauna, acid rock drainage (ARD), weeds, air quality, surface and groundwater, process control and operational water monitoring, tailings characterization, meteorological, regional biodiversity monitoring and rehabilitation assessment.

The Project currently holds authorization (0086-02) to carry out mining activities under the NT *Mining Management Act (2016)*. As required by this legislation the Project operates under an active approved Mining Management Plan (MMP). The MMP outlines the mining activities of the project and the associated environmental management. The Project must ensure it meets the commitments outlined within the MMP or risk penalties from the Department of Primary Industry and Resources (DPIR). The rights of the project to extract groundwater are also managed through the MMP. The MMP includes details on the mine closure plan and sets out the security bond that is required to be lodged with the NT Government to cover the anticipated cost of the rehabilitation commitments associated with the Project.

Newmont holds all required permits to support Life of Mine (LOM) operations. Some permits must be updated on an annual basis.

Newmont has developed a Stakeholder Relationship Management Plan (SRMP) to identify and ensure an understanding of the needs of the surrounding communities and to determine appropriate programs for filling those needs. Newmont appropriately monitors socio-economic trends, community perceptions and mining impacts.

1.20 Capital Costs

The capital cost estimate is based on expected sustaining capital requirements. Development capital costs have been estimated based on previous experience, and equipment replacement and rebuilds have been included.

1.21 Operating Costs

The operating costs were derived from BP19. The operating costs used in the economic evaluation of the Mineral Reserves were based on an average of future costs.

In BP19, mining costs are estimated at US\$48.44/ore tonne mined, process costs at US\$19.82/ore tonne mined, and site and regional general and administrative (G&A) costs at US\$12.14/ore tonne mined (averaged over the mine life).

1.22 Economic Analysis

Newmont is using the provision for producing issuers, whereby producing issuers may exclude the information required under Item 22 for technical reports on properties currently in production and no material expansion of current production is included in the Report.

Mineral Reserves declaration is supported by performing an economic test on each stope independently on a stand-alone basis. For Mineral Reserves, a stope needs to be cash flow positive. All stopes in the Mineral Reserves Plan have satisfied the Mineral Reserves qualification and all levels return positive discounted cash flows at a 7% discount rate.

1.23 Interpretation and Conclusions

In the opinion of the QP:

- Information provided by Newmont's legal and tenure experts on the mining tenure held by Newmont in the Project area supports that the company has valid title that is sufficient to support declaration of Mineral Resources and Mineral Reserves;
- Information provided by Newmont's legal and tenure experts supports that the Operations hold sufficient surface rights to enable mining operations, and the declaration of Mineral

Resources and Mineral Reserves. Appropriate steps, where required, have been taken to lodge either extensions or renewals of tenements as such fall due;

- Knowledge of and understanding of the deposit settings, lithologies, mineralization style, and structural and alteration controls on mineralization is sufficient to support Mineral Resource and Mineral Reserves estimation;
- The exploration programs completed to date are appropriate to the style of the deposits and prospects within the Project. The Project retains significant brownfields exploration potential, and additional work may be undertaken;
- The quantity and quality of the lithological, geotechnical, collar and downhole survey data collected in the exploration and infill drill programs completed by Newmont and the predecessor operating companies are sufficient to support Mineral Resource estimation and conversion to Mineral Reserves;
- A reasonable level of data verification has been and that no material issues would have been left unidentified from the programs undertaken;
- Metallurgical testwork and associated analytical procedures were appropriate to the mineralization type, confirmed the optimal processing routes, and were performed using samples that are typical of the mineralization styles found within the Project;
- Samples selected for testing were representative of the various types and styles of mineralization. They were selected from a range of depths within the deposit and sufficient intersection selected so that tests were performed on a sufficient sample mass. As mining progresses deeper and/or new mining zones are identified, additional variability tests are planned to be undertaken as required;
- Testwork results have primarily been confirmed by production data in mining areas;
- Mill process recovery factors are based on production data for Callie and testwork results for Auron ore, as Auron has never been treated separately through the mill. Federation recovery is also based on testwork, but no Federation ore has yet been processed through the mill;
- Ore hardness, reagent consumptions and process conditions are based on both testwork and production data;
- Recovery factors vary on a day to day basis depending on grade and mineralization type being processed. These variations are expected to trend to the forecast average recovery value for monthly or longer reporting periods;
- The Mineral Resources and Mineral Reserves estimation for the Project have been performed to industry best practices and conform to the requirements of the CIM Definition Standards. There is an upside for Project if Inferred Mineral Resources can be upgraded to higher confidence Mineral Resource categories to support additional Mineral Reserves;
- The assumptions used in developing the mine plan are consistent with previous plant operating experience. Previous production throughputs and recoveries and the Project background history provide supporting data for the proposed production profile;
- The current processing facilities are appropriate for the mineralization types provided by the mine. The post-TEP flowsheet and equipment upgrade and existing infrastructure adequately support the current mine plan;
- Newmont is readily able to market the doré produced from the Project;

- Existing mining activities are supported under existing Project permits held by Newmont; however, additional permits would be required for Project development for any other areas other than the areas currently permitted for mining activities at DBS and processing activities at The Granites; permits may require annual approvals;
- Newmont has sufficiently addressed the environmental impact of the operation, and subsequent closure and remediation requirements that Mineral Resources and Mineral Reserves can be declared, and that the mine plan is appropriate and achievable. Closure provisions are appropriately considered. Monitoring programs are in place. The mine currently has the appropriate social license to operate;
- The capital and operating cost provisions for the mine plan that supports Mineral Reserves have been reviewed. The basis for the estimates that include mine budget data, vendor quotes, and operating experience, is appropriate to the known mineralization, mining and production schedules, marketing plans, and equipment replacement and maintenance requirements. Appropriate provision has been made in the estimates for the expected mine operating usages including labor, fuel and power and for closure and environmental considerations. Capital cost estimates include appropriate sustaining estimates;
- Under the assumptions in this Report, the Project has positive economics until the end of mine life, which supports Mineral Reserves estimate.

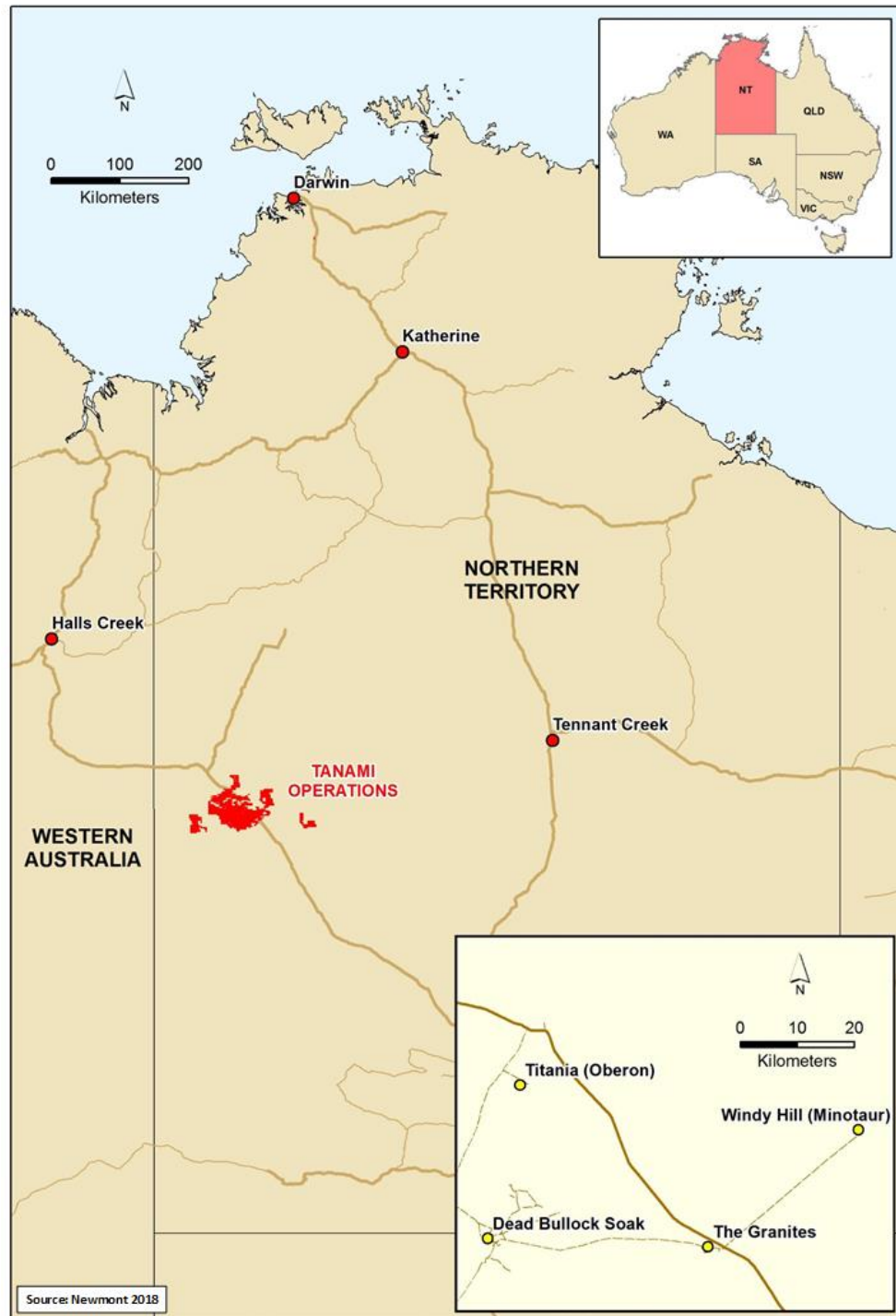
1.24 Recommendations

To facilitate future growth of the Mineral Resources and Mineral Reserves, the QP recommends the following:

- Continue to develop and implement plans to explore in-mine and more distant exploration targets on the Project tenements. The approximate expenditure over the next five years is expected to be AU\$44.8M for surface exploration drilling totalling approximately 170,000 m and AU\$72.58M for underground exploration drilling totalling approximately 407,000 m;
- Continue to infill drill areas of the existing Mineral Resources to ensure confidence is maintained at an appropriate level for ongoing planning, scheduling and ore-control. The approximate costs of underground ore control drilling over the next five years is expected to be AU\$61M for approximately 605,000 m;
- Continue with metallurgical testing programs to validate input factors and assumptions used for generating Mineral Reserves as new material is added to the Mineral Resource inventory. The approximate costs of these ongoing sampling and testing programs are expected to amount to AU\$850k over the next five years;
- Continue to review the potential for expanding mine production, particularly at depths below the current operations, and determine the requirements to convert Mineral Resource in these areas to Mineral Reserves.

2.0 INTRODUCTION

Newmont has prepared this Report for the Tanami Operations, located in the NT, Australia (refer to Figure 2-1).



Note: Figure aligned to True North

Figure 2-1: Project Location Plan

2.1 Terms of Reference

This Report supports declared Mineral Resources and Mineral Reserves and summarizes the Project development and current operations. Newmont will use this Report in support of disclosure and filing requirements with the Canadian securities regulators as specified in Section 4.2 (1) (c) of NI 43-101. This Report will be filed under Newmont's SEDAR profile.

Newmont uses an indirectly wholly-owned Australian subsidiary, Newmont Australia Holdings Pty Ltd, as the holding company of its company group in Australia, and Newmont Tanami Pty Ltd for the Tanami holdings. In this Report, the name Newmont is used interchangeably for the parent and subsidiary companies.

The Project is 100% owned by Newmont Australia Pty Ltd. The operations are known as NTO.

All measurement units used in this Report, except for troy ounces, are metric, and currency is expressed in AU\$ unless stated otherwise. The Report uses US English.

Mineral Reserves are reported to a gold price of US\$1,200/oz or AU\$1,600/oz. The assumed exchange rate for Mineral Reserves estimation was US\$0.75 = AU\$1.00. At the effective date of the Report, the exchange rate was US\$1 equal to approximately AU\$0.75. US\$1.00 = approx. AU\$1.42²

2.2 Qualified Person

Mr. Doe, RM-SME, Group Executive, Reserves at Newmont, is the QP for the Report, as QP is defined in NI 43-101 Standards of Disclosure for Mineral Projects.

Mr. Doe has been Newmont's Group Executive, Reserves since 2014 and is accountable for Newmont's governance system for Mineral Resources and Mineral Reserves. Newmont has a system of controls, standards and guidelines designed to support the estimation and reporting of Mineral Resources and Mineral Reserves. On an annual basis, Mr. Doe reviews and approves the estimates for Mineral Resources and Mineral Reserves provided by Newmont sites and projects, along with their compliance to the required controls, standards and guidelines. Mr. Doe also reviews the supporting documentation for Mineral Resources and Mineral Reserves provided by Newmont sites and projects.

2.3 Site Visits and Scope of Personal Inspection

Mr. Doe, the QP, visited the Project between 13 and 17 April 2018.

During his site visit to the Project, Mr. Doe inspected the underground operations, and viewed the process plant and associated general site infrastructure, including the current tailings storage facility (TSF) operations. While on site, he discussed aspects of the operation with site-based staff and assessed the knowledge and abilities of the site staff to carry out their duties as required. These site discussions included the overall approach to the mine plan, anticipated mining conditions, selection of the production target and potential options for improvement. Other areas of discussion included plant operation and recovery forecasts, capital and operating forecasts and results.

Mr. Doe receives and reviews monthly reconciliation reports from the mine. These reports include the industry standard reconciliation factors for tonnage, grade and metal; F1 (Mineral Reserve model compared to ore control model), F2 (mine delivered compared to mill received)

² Source: Reserve Bank of Australia (RBA) website (<https://rba.gov.au/statistics/frequency/exchange-rates.html>)

and F3 (F1 x F2) along with other measures such as compliance of actual production to mine plan and polygon mining accuracy. The reconciliation factors are recorded monthly and reported in a quarterly control document. Through the review of these reconciliation factors, the QP is able to ascertain the quality and accuracy of the data and its suitability for use in the assumptions underlying the Mineral Resource and Mineral Reserves estimates.

Mr. Doe also reviews Newmont's processes and internal controls at the mine site with operational staff on the work flow for determining Mineral Resource and Mineral Reserves estimates, mineral process performance, mining costs, and waste management.

The following Newmont employees contributed to various aspects of the Report under the supervision of the QP:

- Mr. Shaun Schmeider, Fellow of the Australian Institute of Geoscientists (FAIG), Geology Manager, Newmont – 22 years of experience in gold exploration, mining geology and geostatistical modelling and Mineral Resource estimation and 5.5 years' experience with the Project and Operation;
- Mr. Jorge Peres, Member and Chartered Professional (Geology) of the Australasian Institute of Mining and Metallurgy (MAusIMM), Senior Resource Geologist, Newmont – 20 years of experience in mining geology, exploration geology, geostatistical modelling and Mineral Resource estimation and 1.5 years' experience with the Project and Operation;
- Mr. Johannes Grobler, MAusIMM and Chartered Professional (Mining), Technical Services Superintendent – 22 years of experience in underground mining and 2.5 years' experience with the Project and Operation;
- Mr. Steven Hart, Fellow of the Australasian Institute of Mining and Metallurgy (FAusIMM), Principal Advisor Processing, Newmont – 30 years of experience in minerals processing operations and 6 years' experience with the Project and Operation;
- Mrs. Lucette Hugo, MAusIMM, Chartered Professional (Mining), Principal Mining Engineer, Newmont – 21 years of experience in open pit mining and six months supporting the Project and Operation;
- Ms. Nadine Wetzel, MAusIMM, Principal Mining Engineer, Newmont – 27 years of experience in underground mining and 10 years' experience supporting the Project and Operation.

2.4 Effective Dates

This Report has an effective date of 31 December 2018, which is the date of the Mineral Resource and Mineral Reserves estimates and the financial analysis.

There have been no material changes to the information on the Project between the effective date and the signature date of the Report.

2.5 Information Sources and References

Unless otherwise noted, all figures and tables were prepared by Newmont for the purposes of this Report.

As part of this Report, and in addition to day-to-day operations, supplementary background data were reviewed and new studies were completed as required, under the supervision of the QP.

Information that supports this Report has been obtained from Newmont or external consultants in the relevant field or has been prepared by or under the supervision of the QP.

Reference documents are cited in the text as appropriate and summarized in Section 27.0.

2.6 Previous Technical Reports

Newmont has not previously filed a Technical Report on the Project.

3.0 RELIANCE ON OTHER EXPERTS

The QP has relied upon Newmont experts for the information included in this Report on mineral tenure, surface rights, permitting, political, environmental and social considerations, taxation and markets.

4.0 PROPERTY DESCRIPTION AND LOCATION

4.1 Location

The Project is located in the Tanami Desert, approximately 550 km northwest of Alice Springs in the NT of Australia (refer to Figure 2-1). The Project centroid is situated at approximately 20° 32' S and 129° 56' E, approximately 300 km north of the Tropic of Capricorn.

The Project currently comprises four distinct areas:

- **The Granites area:** Mined out Bullakitchie, Shoe, Quorn and Bunkers Hill deposits; site of The Granites processing plant, airstrip and Twin Hills staff accommodation village and associated infrastructure;
- **The DBS mining operations:** Callie underground mine, including the Callie, Auron, Federation and Liberator deposits and depleted open pits;
- **The Windy Hill area:** Depleted open pit at the Minotaur deposit;
- **The Oberon–Titania area:** Popen pit Mineral Resource for the Oberon deposit.

4.2 Project Ownership

Newmont uses a wholly-owned Australian subsidiary, Newmont Australia Holdings Pty Ltd to hold its Australian company group, and the wholly indirectly-owned subsidiary Newmont Tanami Pty Ltd for the Tanami holdings.

4.3 Tenure History

NNFM applied for and was granted the original mining tenements used for the Project in 1975 (MLS 134 to MLS 144 inclusive). In 1983, NNFM was granted MLS 8 and in 1990 MLS 154, which is now the location of the Callie underground mine. ML 23283 was granted in 2003. NNFM was part of the Normandy Mining Limited group and in 2002, Newmont Mining Corporation acquired this group, which included the Tanami assets.

4.4 Current Mineral Tenure

The Project includes 46 mineral tenures covering an area of approximately 3,500 square kilometers (km²). The tenure is spread across a non-contiguous area that is approximately 180 km by 90 km in size.

Newmont Tanami Pty Ltd. is the registered holder of 45 of the tenements with Nova the registered holder of the Officer Hill tenement (EL 23150) that is subject to the OHJVA. Newmont holds a 70% interest in this tenement.

The current mineral tenure holdings are presented in Figure 4-1, and described in Table 4-1 (Mineral Leases) and Table 4-2 (Exploration Licences).

Estimated Mineral Resources and Mineral Reserves are covered by MLS154. Previously-mined areas are held under MLS 8 and ML 23283.

As at the date of reporting, the required annual exploration spending commitment over the Exploration Licences is AU\$2,810,988 (project commitment).

The mineral tenure is situated on Aboriginal Freehold Land within the Central Desert Aboriginal Land Trust (CDALT) and administered by the CLC.

4.5 Surface Rights and Property Agreements

The Project is located on Aboriginal Freehold Land which was granted as inalienable freehold title to the CDALT pursuant to the Aboriginal Land Rights Act 1976 (NT). The land is administered by the CLC on behalf of the TOs in accordance with the requirements of the Aboriginal Land Rights (Northern Territory) Act (1976).

ELs 10138, 23150, 8077 24864, 24865 and a small southern portion of EL 23662 are located on Tanami Downs pastoral operation, held by Mangkururpa Aboriginal Land Trust and administered by the CLC.

The Mineral Leases (Lease Areas) are subject to the CMA and the WHA, executed by Normandy NFM Limited (now Newmont Tanami Pty Ltd) and the CLC in 2003. These agreements provide for all current activities conducted on the mineral leases including land access, land usage, rehabilitation, compensation payments, sacred site management as well as heritage issues. Under Section 8 of these agreements, a Major Activity Notice (MAN) is required to be submitted to the CLC for activities which involve substantial disturbance to the Lease Areas or significant capital expenditure on the Lease Areas. The CMA is in force for the term of the mineral leases MLS8 (The Granites) and MLS 154 (DBS) and the WHA for ML 23283 (Windy Hill) remaining active.

The DBS haul road is managed under the Haul Road Agreement between Newmont Tanami Pty Ltd and the CLC and covers similar management requirements as the CMA.

The Billabong and Schist Hills water bore fields are managed separately under The Granites Mine Bore Field and Pipeline Memorandum of Lease and the Schist Hills Bore Field and Pipeline Memorandum of Lease agreements respectively.

The open pit Mineral Resource for the Oberon deposit is covered under EL 23662. As the registered holder of EL 23662, Newmont has exclusive right to apply for a Mineral Lease title within the EL title area. A Mineral Lease application to the NT Mineral Titles Office (MTO) is currently under development for a 4,110 ha land parcel encompassing the Oberon deposit and surrounding areas.

A Mineral Lease application (ML 27957) is progressing for an area directly south of MLS 8 and is currently subject to land access agreement discussions with the CLC. The ML 27957 area currently forms part of EL 4529.



Figure 4-1: Tanami Project Mineral Tenement Plan

Table 4-1: Mining Leases

Lease	Lease Name	Region	Status	Application Date	Grant Date	Expiry Date	Area	Commitment (AU\$)	Rent (AU\$)	Holders
MLA 27957*	Granites South	The Granites	Under Application	22/02/2010	-	-	737.95 ha	-	-	Newmont Tanami Pty Ltd
ML 23283	Minotaur-Windy Hill	The Granites	Granted	15/08/2001	10/06/2003	09/06/2023	708 ha	\$15,000	\$15,104	Newmont Tanami Pty Ltd
MLS 134	GML 60G	The Granites	Granted	14/10/1975	29/11/1983	28/11/2025	8 ha	\$0	\$404	Newmont Tanami Pty Ltd
MLS 135	GML 61G	The Granites	Granted	14/10/1975	29/11/1983	28/11/2025	8 ha	\$0	\$404	Newmont Tanami Pty Ltd
MLS 136	GML 62G	The Granites	Granted	14/10/1975	29/11/1983	28/11/2025	8 ha	\$0	\$404	Newmont Tanami Pty Ltd
MLS 137	GML 63G	The Granites	Granted	14/10/1975	29/11/1983	28/11/2025	8 ha	\$0	\$404	Newmont Tanami Pty Ltd
MLS 138	GML 64G	The Granites	Granted	14/10/1975	29/11/1983	28/11/2025	8 ha	\$0	\$404	Newmont Tanami Pty Ltd
MLS 139	GML 65G	The Granites	Granted	14/10/1975	29/11/1983	28/11/2025	7.95 ha	\$0	\$404	Newmont Tanami Pty Ltd
MLS 140	GML 66G	The Granites	Granted	14/10/1975	29/11/1983	28/11/2025	8 ha	\$0	\$404	Newmont Tanami Pty Ltd
MLS 141	GML 67G	The Granites	Granted	14/10/1975	29/11/1983	28/11/2025	7.5 ha	\$0	\$404	Newmont Tanami Pty Ltd
MLS 142	GML 68G	The Granites	Granted	14/10/1975	29/11/1983	28/11/2025	8 ha	\$0	\$404	Newmont Tanami Pty Ltd
MLS 143	GML 69G	The Granites	Granted	14/10/1975	29/11/1983	28/11/2025	8 ha	\$0	\$404	Newmont Tanami Pty Ltd
MLS 144	GML 70G	The Granites	Granted	14/10/1975	29/11/1983	28/11/2025	8 ha	\$0	\$404	Newmont Tanami Pty Ltd
MLS 154	DBS	The Granites	Granted	15/06/1990	12/02/1991	11/02/2036	1,477 ha	\$0	\$31,253	Newmont Tanami Pty Ltd
MLS 8	The Granites	The Granites	Granted	09/11/1983	15/05/1984	14/05/2034	2,186 ha	\$0	\$46,142	Newmont Tanami Pty Ltd

*MLA27957 is a Mineral Lease Application and has not yet been granted, therefore no Dates or financial commitments were available at the time of preparation of this Report. Newmont Tanami Pty Ltd is the Beneficial Holder for all tenements

Table 4-2: Exploration Licences

Lease	Region	Project	Status	Application Date	Grant Date	Expiry Date	Area	Rent (AU\$)	Holders	Beneficial Holders
EL 10138	Tanami	McFarlane	Granted	1/06/1998	8/06/2001	5/06/2019	54 Blocks	\$10,854.00	Newmont Tanami Pty Ltd	
EL 22170	Tanami	Central	Granted	1/09/1999	21/08/2003	19/08/2019	20 Blocks	\$1,755.00	Newmont Tanami Pty Ltd	
EL 22900	Tanami	Central	Granted	8/12/2000	25/05/2006	30/12/2020	13 Blocks	\$1,244.00	Newmont Tanami Pty Ltd	
EL 22933	Tanami	Central	Granted	21/12/2000	25/05/2006	30/12/2020	4 Blocks	\$587.00	Newmont Tanami Pty Ltd	
EL 23150	Tanami	Officer Hill JV	Granted	4/05/2001	29/07/2013	28/07/2019	64 Blocks	\$4,544.00	Nova Minerals Pty Ltd	Newmont Tanami Pty Ltd
EL 23308	Tanami	Windy Hill	Granted	15/08/2001	25/05/2006	30/12/2020	9 Blocks	\$952.00	Newmont Tanami Pty Ltd	
EL 23658	Tanami	Central	Granted	7/08/2002	3/04/2003	31/03/2019	55 Blocks	\$7,865.00	Newmont Tanami Pty Ltd	
EL 2366	Tanami	Central	Granted	21/11/1979	25/03/1988	31/12/2018	124 Blocks	\$25,839.00	Newmont Tanami Pty Ltd	
EL 23660	Tanami	Central	Granted	7/08/2002	3/04/2003	31/03/2019	67 Blocks	\$9,581.00	Newmont Tanami Pty Ltd	
EL 23662	Tanami	Central	Granted	7/08/2002	3/04/2003	31/03/2019	158 Blocks	\$22,594.00	Newmont Tanami Pty Ltd	
EL 2367	Tanami	Central	Granted	21/11/1979	25/03/1988	31/12/2018*	206 Blocks	\$42,731.00	Newmont Tanami Pty Ltd	
EL 23744	Tanami	Central	Granted	29/11/2002	25/05/2006	30/12/2020	3 Blocks	\$514.00	Newmont Tanami Pty Ltd	
EL 23833	Tanami	Central	Granted	10/03/2003	25/05/2006	30/12/2020	6 Blocks	\$733.00	Newmont Tanami Pty Ltd	
EL 24864	Tanami	Central	Granted	19/08/2005	12/04/2007	30/12/2020	7 Blocks	\$806.00	Newmont Tanami Pty Ltd	
EL 24865	Tanami	Central	Granted	19/08/2005	12/04/2007	30/12/2020	2 Blocks	\$441.00	Newmont Tanami Pty Ltd	
EL 24886	Tanami	Central	Granted	29/08/2005	12/04/2007	30/12/2020	3 Blocks	\$514.00	Newmont Tanami Pty Ltd	
EL 24888	Tanami	Central	Granted	29/08/2005	12/04/2007	30/12/2020	3 Blocks	\$514.00	Newmont Tanami Pty Ltd	
EL 24889	Tanami	Central	Granted	29/08/2005	12/04/2007	30/12/2020	2 Blocks	\$441.00	Newmont Tanami Pty Ltd	
EL 24890	Tanami	Central	Granted	29/08/2005	12/04/2007	30/12/2020	3 Blocks	\$514.00	Newmont Tanami Pty Ltd	
EL 24895	Tanami	Central	Granted	30/08/2005	12/04/2007	30/12/2020	25 Blocks	\$2,120.00	Newmont Tanami Pty Ltd	
EL 24896	Tanami	Central	Granted	30/08/2005	12/04/2007	30/12/2020	33 Blocks	\$2,704.00	Newmont Tanami Pty Ltd	
EL 24973	Tanami	Central	Granted	5/10/2005	12/04/2007	30/12/2020	32 Blocks	\$2,631.00	Newmont Tanami Pty Ltd	
EL 24974	Tanami	Central	Granted	5/10/2005	12/04/2007	30/12/2020	13 Blocks	\$1,244.00	Newmont Tanami Pty Ltd	
EL 25012	Tanami	Central	Granted	24/10/2005	12/04/2007	30/12/2020	2 Blocks	\$441.00	Newmont Tanami Pty Ltd	
EL 25013	Tanami	Central	Granted	24/10/2005	12/04/2007	30/12/2020	3 Blocks	\$514.00	Newmont Tanami Pty Ltd	
EL 30806	Tanami	Central	Granted	21/11/1979	25/03/1988	31/12/2018*	77 Blocks	\$16,157.00	Newmont Tanami Pty Ltd	
EL 4529	Tanami	Central	Granted	20/02/1984	9/05/1990	31/12/2018*	48 Blocks	\$10,183.00	Newmont Tanami Pty Ltd	
EL 8077	Tanami	Legend	Granted	4/01/1993	4/06/2001	2/01/2020	66 Blocks	\$13,891.00	Newmont Tanami Pty Ltd	
EL 8912	Tanami	Central	Granted	5/09/1994	9/09/1999	7/09/2019	105 Blocks	\$21,925.00	Newmont Tanami Pty Ltd	
EL 9737	Tanami	Central	Granted	20/06/1994	21/08/2003	19/08/2019	7 Blocks	\$1,324.00	Newmont Tanami Pty Ltd	
EL 9996	Tanami	Central	Granted	23/10/1997	21/08/2003	19/08/2019	5 Blocks	\$1,030.00	Newmont Tanami Pty Ltd	

4.5.1 The Tanami Exploration Agreement

In 2004, Newmont and the NT Government entered into the Tanami Exploration Agreement (TEA) to facilitate an exploration period during which Newmont had rights of exploration in the Tanami region, subject to the terms and conditions set out in the agreement.

Upon expiry of the TEA on 31 December 2014, residual terms were applied to the Licences which reverted to being administered in accordance with the provisions of the Mineral Titles Act 2015 (the Act).

4.5.2 Officer Hill Farm-in and Joint Venture Agreement

In 2018, Newmont earned a 70% interest in EL 23150 pursuant to the OHFJVA. Nova Minerals Pty Ltd (Nova) and Newmont are now in an exploration joint venture (JV) to further explore this tenement.

4.6 Royalties and Encumbrances

4.6.1 Central Land Council Agreement

Under the terms of the CLC Agreement, which covers the Project area, a 2.5% royalty (refer to Section 1.5) is payable based on the value of ounces of gold and silver sold. A further AU\$0.12/tonne is levied on ore transported along the DBS haul road from the mine to the processing plant, bringing the average royalty to 2.7%. This average royalty is considered in the cut-off grade determination. The royalties are paid twice annually, in July and January.

4.6.2 Northern Territory Government Minerals Royalty

The Project is subject to the NT Government Minerals Royalty (refer to Section 1.5). This royalty is calculated as 20% of the "Net Value" of mine production, where "Net Value" equals the gross revenue less the operating costs for the year, a capital allowance on eligible capital assets expenditure, eligible exploration expenditure, and additional deductions as approved by the NT Minister for Mines. This royalty is considered in the overall economic analysis.

4.7 Permits

Information on permitting for the Project is included in Section 20.0.

4.8 Environmental Considerations

Information on environmental considerations, including environmental liabilities for the Project is included in Section 20.0.

4.9 Social Considerations

Information on social considerations for the Project is included in Section 20.0.

4.10 Project Risks

The following significant factors or risks may affect access, title, or right or ability to perform work at the Project:

- Loss of access to land due to inability to meet Land Access Agreements (LAA) with Aboriginal TO Groups;
- Reputation damage with CLC/TOs due to poor stakeholder engagement (including staff and contractor behaviour);
- Reputational damage from publication of negative comments by employees or business partners;
- Dingo interaction with personnel and operations;
- Poor waste management resulting in long-term environmental degradation;
- Permitting failure for future projects i.e. TSF, shaft, airstrip, and exploration;
- Increased expectations and/or enforcement of environmental regulations by authorities;
- Unauthorized access/desecration/damage of culturally significant areas and/or species;
- Increase in royalty payments (NT Government and payments required under the Aboriginal Land Rights Act [ALRA]);
- Inadequate financial resource allocation for closure rehabilitation works;
- Extended disruption of operations due to fatality or major injury to Warlpiri (TO) personnel on site or at front gate;
- Failure to provide adequate assistance to community visitors needs/requests resulting in reputational damage;
- Depletion of groundwater resources from over extraction and mismanagement of resource, resulting in raw water supply restrictions;
- Newmont operational presence on Aboriginal owned land also implies social-cultural consequences for any environmental-water related risks.

4.11 Comments on Property Description and Location

In the opinion of the QP:

- Information from Newmont's legal and tenure experts supports that the mining tenure held is valid and is sufficient to support declaration of Mineral Resources and Mineral Reserves;
- Annual tenement fees have been paid to the relevant regulatory authorities and the required tenement reports submitted. Exploration work on the Exploration Licences to date has been sufficient to meet expenditure commitments;
- Licenses and Mineral Leases are located on Aboriginal Freehold Land held by the Central Desert Aboriginal Lands Trust and administered by the CLC on behalf of Traditional Owners;
- The Mining Leases are subject to the CMA and WHA with the CLC. These agreements provide for all current activities conducted on the mining leases including land access, land usage, rehabilitation, compensation payments, sacred site management as well as heritage related issues;
- Royalties are payable both to the NT Government and the CLC;
- Mining and exploration activities to date have been conducted within the regulatory framework required by the NT Government;

- Existing mining activities are supported under existing permits held by Newmont (refer to Section 20.0);
- Additional permits will be required for Project development for any other areas other than the areas currently permitted for mining activities at DBS, The Granites and Windy Hill.

5.0 ACCESSIBILITY, CLIMATE, LOCAL RESOURCES, INFRASTRUCTURE AND PHYSIOGRAPHY

5.1 Accessibility

The Project is located in the Western Tanami Desert Region of the NT. The processing plant, camp and airstrip are located at The Granites, which is located on the Tanami Track, approximately 550 km northwest of Alice Springs. The mining operations are located at the DBS area, approximately 42 km to the west of The Granites.

Project personnel primarily access the site via air with regularly-scheduled charter flights to The Granites airstrip, operated by Alliance Airlines from Darwin, Alice Springs, Brisbane and Perth. In addition to providing passenger transport, the charter flights provide a limited air-freight capacity. The Granites airstrip is sealed with bitumen, and flights can operate in most weather conditions.

Road access to the Project, which is some 550 km from Alice Springs, is via the Tanami Track. The nearest town is Yuendumu, approximately 270 km southeast of The Granites on the Tanami Track. The track is an unsealed gravel road between The Granites and Yuendumu and is susceptible to closure due to flooding, following heavy rain during the wet season. Freight access to the site is typically via three-trailer road-train trucks that operate along the Tanami Track.

The Granites is linked to the mining operations at DBS by a bituminized all-weather haul road. A basic network of pre-existing tracks link individual prospect areas to the main operating areas in MLS 8 and MLS 154. Once on site, buses and light vehicles are used to transport passengers between the accommodations camp and operational areas.

5.2 Climate

The Tanami Desert experiences a semi-arid and monsoonal climate, with approximately 90% of the annual rainfall occurring between November and April.

The nearest official meteorological station, Rabbit Flat, records a mean daily maximum temperature of 38.5 degrees Celsius (°C) for January and 25.7°C for July. Temperatures typically range from 47°C to -3.3°C.

Rabbit Flat receives a mean rainfall of 453 mm per annum, which is mostly derived from summer monsoonal and storm activity between November and April and records a mean evaporation rate of 2,788 mm (Class A Pan) per annum.

Mining operations are conducted year-round.

5.3 Local Resources and Infrastructure

The Tanami Desert is a remote, sparsely-populated region. Prior to development of Newmont's mining activities, infrastructure was almost completely absent in this part of the Tanami region.

The nearest settlements to the project are the Yuendumu and Lajamanu Aboriginal communities, located 270 km southeast and 370 km north respectively. Both are small remote communities and do not have any capacity to supply the operations. Alice Springs is the nearest large town and is the staging point for supplies to the project.

Surface rights and sufficiency of the rights to support current and planned mining operations is discussed in Section 4.5.

Project infrastructure is discussed in Section 18.0.

5.4 Physiography

The Project area has a desert red sand-plain terrain; transported and residual colluvial cover sediments and aeolian sand blanket a large portion of the Project area, with an estimated outcrop exposure of less than 10%. Mostly isolated low hills and ridges outcrop above the sand-plain; these are formed by the more resistant geological units that are mainly cherty iron-rich units.

The relief within the Project area is generally low, ranging from 330 m (meters above sea level (masl)) in drainage channels, and reaching a maximum of approximately 410 masl on ridge tops. Slope gradients are shallow.

The Tanami Desert is widely covered in aeolian sand with a vegetation cover dominated by spinifex with low bushes and scattered small trees. The area is mostly devoid of surface water except in small soaks and ephemeral streams after significant rainfall events.

5.5 Comments on Accessibility, Climate, Local Resources, Infrastructure and Physiography

In the opinion of the QP:

- The existing local infrastructure, availability of staff, methods whereby goods can be transported to the Project area are well-established and well understood by Newmont, and can support the declaration of Mineral Resources and Mineral Reserves;
- The Project covers operating mines, and all required infrastructure for the LOMP discussed in this Report are in place;
- Within Newmont's ground holdings, there is sufficient area to allow construction of any Project infrastructure that may be required in the future;
- There are no significant factors and risks that may affect access, title, or the right or ability to perform work on the property.

6.0 HISTORY

6.1 The Granites

Gold was first reported in the Project area at The Granites (MLS8 area) by Allan Davidson a government surveyor in 1900 (Davidson, 1905).

Prospecting between 1910 and 1932 led to minor production from narrow quartz veins and from limited alluvial and eluvial zones at Bunkers Hill (Ellis, 1927). The discovery of additional rich, but narrow, quartz veins at the Burdekin Duck Deposit on Chapmans Hill and the occurrence of a small patch of rich alluvial ground nearby produced a brief gold rush in 1932 (Baume, 1933; Madigan, 1944).

Charles Chapman took up leases at The Granites in 1932, and for the next 20 years, Chapmans Gold Mines NL (CGM) continued small-scale mining at several locations including Bullakitchie, Shoe and near Chapmans Hill. A total of 420 kg of gold was produced from the area prior to 1961.

The CGM leases were explored by Anglo Queensland Mining Pty Ltd between 1938 and 1948 outlining a Mineral Resource at Bullakitchie and Shoe (Hall, 1953).

Exploration by Northern Mines Development NL in 1954 and 1955 and Geopeko Ltd between 1965 and 1970, focused on testing prominent magnetic anomalies at Twin Hills and Ivy.

North Flinders Mines acquired tenure over The Granites area in 1975 but did not commence exploration until August 1983 due to prolonged land access negotiations and political and legal constraints (Dryver, 1984; Ireland and Mayer, 1984). Exploration activities from 1983 to 1984 consisted of surface geochemical surveys, geophysical mapping with limited geological mapping due to the Aeolian cover, 57 core drill holes, 75 RC drill holes and Mineral Resource and Mineral Reserves estimates at Bullakitchie and Shoe (Mayer, 1990). Development of the Bullakitchie Decline commenced in July 1984, with construction of the mill in December 1985. The first gold pour was in June 1986.

The first deposit to be mined was Bullakitchie, using both open pit and underground mining methods. Ore came primarily from the East Bullakitchie zone during 1984–1995. Some production also came from the Central Bullakitchie during development out to Shoe in 1988–1989. Open pit mining commenced at East Bullakitchie in 1986, and the West Bullakitchie extension began production in February 1992. Wall stability problems plagued the pits, and West Bullakitchie never reached completion due to failures on the southern wall. Upon cessation of mining activities in 1995, the Bullakitchie pit system was filled with tailings.

The Shoe open pit started in January 1987 and operated in parallel with the East Bullakitchie pit until a two-year hiatus starting from February 1988. In early 1990, the pit was extended out to the West Shoe area, and at the same time the original area of the pit was cut back and deepened by 30 m. The pit operations were abandoned after a major failure in the northern wall. In 1989, an underground decline to Shoe was initiated off the Bullakitchie Decline, eventually reaching the deposit in 1990. The Shoe Underground continued operation until the close of mining in 1995, eventually accounting for over 60% of annual underground production at The Granites.

The Quorn open pit operations commenced in February 1988, following the initial completion of mining at Shoe. The pit ceased production in 1995 as the focus of mining shifted to Callie. The pit was reopened in early 2002 as a source of low-grade oxide material, and production continued until depletion in February 2003.

The Bunkers Hill open pit commenced production in March 1990 as an exploration pit and continued operating until 1995. Production recommenced in early 2002 at the same time as Quorn, and the deposit was mined to depletion in April 2003.

During the combined 11 years of mining from 1984 to 2003, approximately 1.1 Moz of gold were produced from both the underground and open pit sources. A breakdown of production by deposit is presented in Table 6-1.

Table 6-1: The Granites Goldfield Production (1984 to 2003)

Deposit	Tonnage/Grade	Oz
Bullakitchie	2.8 Mt @ 6.5 g/t Au	589,000
Shoe	2.5 Mt @ 5.4 g/t Au	437,000
Quorn	1.1 Mt @ 2.8 g/t Au	91,000
Bunkers Hill	0.4 Mt @ 5.5 g/t Au	28,000
Total	6.8 Mt @ 5.5 g/t Au	1,114,500

In 1997, North Flinders Mines was renamed Normandy NFM following Normandy Mining taking a majority holding in the company. Newmont subsequently acquired Normandy Mining's stake in Normandy NFM and assumed control of the Project through its takeover of Normandy in 2002. Newmont subsequently acquired the remaining shares in the renamed Newmont NFM in early 2003 to obtain 100% control of the Project.

Following cessation of mining at The Granites, the processing plant at The Granites continued to process ore from remnant open pit stockpiles and ore sourced from the Windy Hill and DBS operations. Stockpiled ore derived from The Granites operations was exhausted in 2011, and The Granites processing plant now processes ore sourced entirely from the DBS underground operations.

Newmont has completed a limited quantity of exploration since acquisition of the Project, including a small RC drilling program to test the western extension of the Quorn mineralization in 2015 and evaluation of any Mineral Resource development opportunities around the historical mining areas at The Granites in 2018.

There are currently no Mineral Resources estimated and reported for The Granites.

6.2 Dead Bullock Soak

Commencing in 1988, regional exploration activities by North Flinders Exploration at DBS identified a geochemical surface anomaly which was followed up with infill sampling and then drilling activities. Geophysical surveys were also undertaken in the area. This resulted in the discovery of the Triumph Hill, Colliwobble, and Dead Bullock Ridge deposits, with open pit mining commencing in 1990.

The area rapidly became the focus of exploration with the mining lease (MLS 154) granted in early 1991. Further exploration success at DBS followed, with the discovery and subsequent development of the Villa and Callie deposits in 1992.

Open pit mining commenced in 1992 and continued for over 10 years to 2003. Production was sourced from four open pits: Callie, Villa, Triumph/Colliwobble and Dead Bullock Ridge.

Decline access development to the Callie deposit began in 1995, and the first underground stoping ore was extracted in May 1998. The extents of the Callie deposit were gradually

depth-extended down plunge with a combination of deep surface drilling and follow-up underground definition drilling between 1998 and 2008.

In 2003, the Federation mineralization was identified by Newmont to the south of the main Callie mineralization and an initial Mineral Resource for Federation was reported in 2006.

Following several years of exploration by Newmont, the Auron mineralization was identified in 2007 in units hosted stratigraphically below the Callie mineralization. Auron became a focus for exploration and numerous Mineral Resource updates were performed between 2008 and 2010.

A feasibility study (FS) was undertaken by Newmont between 2005 and 2011 to assess the viability of constructing a shaft to access deeper portions of the Callie and Auron deposits. This included very deep directional drilling being undertaken targeting the down plunge extents of the Callie and Auron mineralization. Drill holes were completed to a maximum downhole depth of 2,449 m. Approval for the construction of a shaft was given in 2011 and work commenced on underground development, surface site works, and pilot hole drilling.

Stoping production commenced in the Auron deposit in December 2012. Work on the shaft project was cancelled in late 2012, prior to shaft sinking work commencing due to a drop in the gold price. The underground mine plan was restructured to carry on hauling ore from the Callie and Auron deposits by truck.

Total production from DBS from 1990 to December 2018 is presented in Table 6-2.

Table 6-2: Total DBS Goldfield Production (1990 to 2018)

Deposit	Tonnage (Mt)	Au Grade (g/t)	Au Metal (Moz)
Callie Underground	36.0	5.9	6.99
Callie Open Cut	7.5	5.1	1.22
Villa Open Cut	2.2	3.1	0.21
Dead Bullock Ridge Open Cut	2.3	3.6	0.25
Triumph Hill, Colliwobble Open Cut	1.0	3.6	0.10
Total	49.0	5.5	8.77

Exploration drilling conducted during 2015 identified the Liberator orebody in the Federation Anticline. Liberator is similar in its structure and geometry to Federation but is hosted in a rock mass like that of the Auron orebody. A Maiden Mineral Resource estimate for Liberator was conducted during 2018.

Mineral Resources for the DBS Goldfield are reported in Section 14.0.

A summary of the major DBS Goldfield exploration, project and development milestones since Newmont ownership commenced in 2002 is presented in Table 6-3.

Table 6-3: Summary of DBS Milestones Since Commencement of Newmont Ownership

Year	Milestone
2006	First Inferred Resource declared in 2006 for the Federation Hinge Zone
2008	Initial Inferred Resource declared for Auron in 2008
2011-2012	Approval for the construction of a production shaft (2011), which was then cancelled (2012) due to a decrease in gold price
2012	First production from Auron commenced
2014	First Inferred Resource declared in 2014 for the southern limb of Federation
2016	Construction of a second underground decline down to the 380RL from near surface was constructed to support

Year	Milestone
	mining of Auron
2017	A mill expansion at The Granites and Tailings Filter Plant (TFP), both to support increased mining rates at the DBS operation commissioned
2018	Initial Inferred Resource for Liberator

6.3 Windy Hill

In 1989, the Tanami Joint Venture (TJV), formed by Zapopan NL (Zapopan) and Harlock Pty. Ltd. (Harlock), held and explored the ground now covered by ML 23283.

In 1990, an anomalous gold assay result was returned from Windy Hill, and drilling in 1993 returned significant gold results from four RAB drill holes. The Minotaur deposit was discovered in 1994 with an identified 400 m x 250 m mineralized area (Smith and Huntly, 1996).

The Windy Hill EL was purchased by North Flinders Mining from Pegasus Gold Ltd (Pegasus), (formerly Zapopan), as part of a purchase of the majority of Tanami Exploration Licences held by Pegasus. Tenement transfer took place in mid-June 1996. North Flinders Mining was subsequently acquired by Newmont.

Mining at Minotaur commenced in June 2003 with ore being trucked to The Granites plant for processing. Mining ceased in February 2004, after 53,987 oz of gold had been produced.

There are currently no Mineral Resources estimated and reported for the Minotaur deposit.

6.4 Oberon–Titania

The Oberon (formerly Titania) area was first tested in 1988 by North Flinders Mines during their Tanami regional exploration program. This survey consisted of vacuum drilling on a 2 km x 1 km grid, and soil, ferricrete and saprolite sampling. This yielded a single anomalous gold value in a vacuum drill hole. A 500 m x 500 m spaced vacuum program was initiated to follow up the anomaly. However, the presence of significant transported cover in a paleochannel led to the early abandonment of the program. The few results confirmed, but did not extend, the original anomaly. North Flinders Mines moved on to other targets, discouraged by the cover and the relatively low amplitude magnetic signature of the area, which did not match that of The Granites-style mineralization.

The area was left untested until 1993, when a single traverse of 100 m spaced RAB holes were completed ahead of possible relinquishment obligations. This phase of drilling returned several anomalous values from the original anomaly area, and from an adjacent site. It also revealed the bedrock consisted of coarse greywacke, previously considered to be a non-prospective lithology. Additional RAB drilling supplemented with AC drilling revealed similar anomalies 200 m further east and west, eventually identifying a system with the potential to host several deposits.

The Oberon deposit was discovered in June 1994. In late 1994, a RC program commenced to follow up the anomalous results over the Titania area. This RC program resulted in further extensions to the gold anomaly. In 1995, a core drill hole program was initiated, and the first Mineral Resource estimate completed the same year.

Between 1995 and 1999, work consisted of additional RC and core drilling, metallurgical, hydrogeological and geotechnical studies, and an initial shallow oxide open pit Mineral

Reserves was reported in 2000. However, due to low gold prices and a focus on the Dead Bullock Soak deposits, the Oberon deposit was not developed.

No further drilling occurred at Oberon from 2000 until Newmont commenced a core drill program in 2009 targeting postulated deeper Callie-style mineralization beneath the previously-identified Oberon mineralization. Initial results were encouraging, and a total of 26 core holes were completed between 2009 and 2012.

In late 2012, an updated Mineral Resource estimate was undertaken for the Oberon deposit, and a mining study was completed to evaluate the feasibility of potential open pit or underground mining. The results of these studies precluded consideration of Oberon as a Mineral Reserve with the deposit being evaluated as uneconomic at that time. Exploration of the Oberon deposit was suspended in 2013.

In 2018, an updated Newmont study, based on updated operating and capital costs supported the addition of Oberon (based on the 2013 Mineral Resource estimate) to the Mineral Resource declared in December 2018. Updated mining studies were also completed but to date no Mineral Reserves have been declared.

A summary of the major Oberon exploration, project and development milestones since Newmont ownership commenced in 2002 is presented in Table 6-4.

Table 6-4: Summary of Oberon Milestones Since Commencement of Newmont Ownership

Year	Milestone
2002	Completion of a pit optimization study at a gold price of \$600/oz. The study defined Probable Reserves
2005	FS was completed. The study deemed the deposit to be a potentially challenging open cut mine, due to the slim cost margin and several risks that could potentially render it uneconomic
2009-2012	Further exploration drilling of 26 core drill holes drilled targeting depth extensions well below historical drilling
2013	The deposit was re-estimated in 2013 to include the above drilling and a new open pit evaluation was completed. This study resulted in the removal of the 2002 Mineral Reserves
2018	Scoping study was completed, which demonstrated positive economics and justified declaration of Mineral Resources
2018	Infill drilling of the scoping study pit was commenced with three core drill holes completed

7.0 GEOLOGICAL SETTING AND MINERALIZATION

7.1 Regional Geology

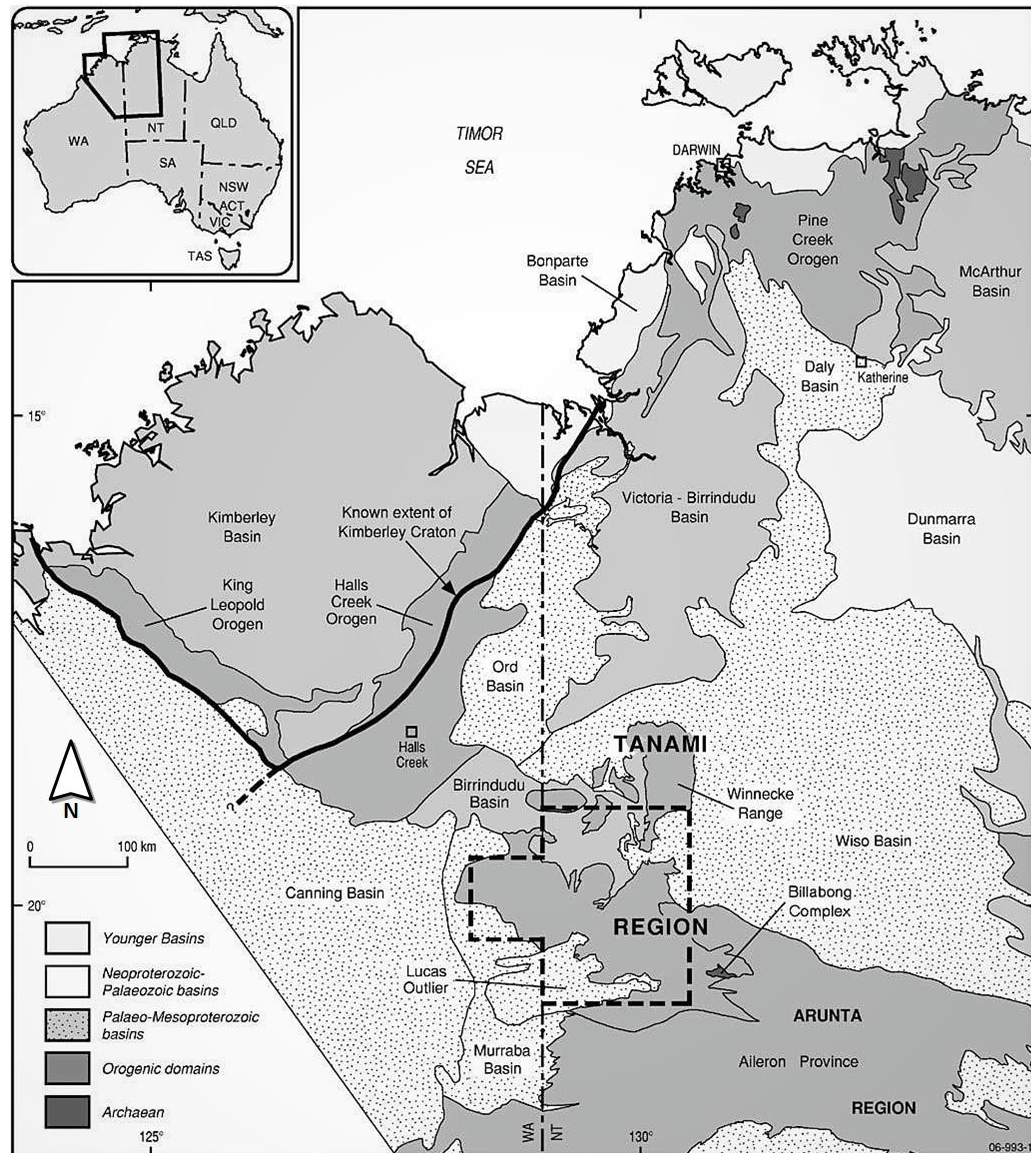
The Project is located in the eastern part of the Early Proterozoic Granites–Tanami Inlier that is part of the Northern Australian Orogenic Province (Plumb, 1990). The Inlier abuts the Arunta Complex to the south and east and is overlapped by younger cover sequences including the extensive Paleozoic Wiso Basin on its northeastern margin (refer to Figure 7-1). To the west, clastic sedimentary rocks of the Middle Proterozoic Birrindudu Basin overlie and separate the inlier from similar age rocks in the Halls Creek Province.

The oldest rocks of the Tanami region belong to the Billabong Complex, a suite of Archaean age gneiss and schist. This is unconformably overlain by the basal Proterozoic sequence known as the MacFarlanes Peak Group, which is dominated by mafic volcanic and volcanoclastic rocks suggestive of a rift setting. The MacFarlanes Peak Group has a maximum age of deposition of 1,880 million years (Ma). They are followed by a thick, possibly disconformable succession of clastic sediments making up the Tanami Group, which is considered to be representative of a passive margin sequence (Hendrickx et al, 2000). The Tanami Mine Group (refer to Figure 7-2) is subdivided into a thin basal meta-quartzite, the lower Tanami Group (Dead Bullock Formation) made up of carbonaceous siltstone, Banded Iron Formations (BIF) and calc-silicates, and an upper sequence of turbidites (Killi Killi Formation).

A suite of syn-to-post deformation dolerites and gabbros frequently intrude the carbonaceous rich components of the sequence. Large plutons of mostly undeformed late-to-post-orogenic adamellite and minor more mafic variants comprising The Granites granite suite are widespread throughout the area. Residual hills of gently folded Carpentarian Gardiner Sandstone unconformably overlie Early Proterozoic lithologies. Younger flat-lying Cambrian Antrim Plateau Basalts are also preserved as platform cover in areas protected from erosional stripping.

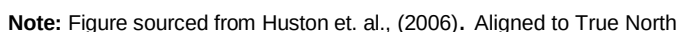
Complex, polyphase deformation during the Barramundi Orogeny (c. 1,880–1,890 Ma) has affected the entire Granites–Tanami Inlier. It appears to have been largely controlled by two sets of regional-scale, fundamental crustal fractures that trend north-northeast and west–northwest. This is evidenced by the orientation of successive phases of macroscopic folding, shearing and faulting in the region and the consistent sympathetic trends of later tectonic reactivated faults.

Tertiary drainage channels, now completely filled with alluvial sediment, lacustrine clays and calcrete, are a major feature of the region. Some drainage profiles exceed 10 km wide, and are more than 100 m in depth, presenting a significant barrier to mineral exploration.



Note: Figure sourced from Huston et. al., (2006). Aligned to True North

Figure 7-1: Major Crustal Elements of Northwest Australia Showing the Tanami Region



A desert terrain comprising transported and residual colluvial cover sediments and aeolian sand blanket a large portion of the Inlier, with an estimated outcrop exposure of less than 10% of the early Proterozoic lithological units. Owing to their more resistant nature, only the cherts, BIFs and associated interbedded graphitic schists tend to outcrop above the sand plain.

Gold mineralization within the Tanami is dominantly hosted by the Tanami Group and Mt Charles Formation, although mineralization has been recorded in all Proterozoic units older than the Birrindudu Group cover sequences.

7.2 Project Geology

The gold mineralization within the Project is split between four separate areas; The Granites, DBS, Windy Hill and Titania–Oberon. Due to their wide geographic dispersal, the geology and mineralization of each of these areas differs locally. The general project stratigraphy, intrusions, structural history, metamorphism and surficial geology are discussed in the following sub-sections.

7.2.1 Stratigraphy

The stratigraphy of the Project tenements is predominantly comprised of the sedimentary rocks of the Tanami Group. The Tanami Group consists of two Formations; the Dead Bullock Formation and the overlying Killi Killi Formation.

The Dead Bullock Formation is a fining-upward package comprising sandstone and siltstone grading up to siltstone, chert and BIF. The base of the Dead Bullock Formation has not been observed; the top is dominated by thin-bedded to laminated siltstone and iron-rich chert which, over a short interval, passes into interbedded greywacke and siltstone of the Killi Killi Formation. The contact is normally folded and sheared. The thickness of the Dead Bullock Formation remains unknown. In the DBS area it has an estimated minimum thickness of 1 km, but this is complicated by folding and faulting.

The Dead Bullock Formation has been informally divided by the geologists of the Tanami Operations into two Members; the Callie Member and underlying Ferdies Member. These units host all the known gold mineralization on the project tenements. At The Granites, Oberon and Minotaur, the deposits are hosted in the Davidson Beds.

The Killi Killi Formation is the most widespread unit across the Project tenements. It is dominated by a thick interval of turbidites, with a felsic volcanic component and a distinctive unit of jasper chert.

7.2.2 Intrusions

The Dead Bullock Formation is intruded by a suite of syn-to-post deformation dolerites and gabbros which frequently invade the carbonaceous-rich components of the sequence.

Large plutons of mostly undeformed late-to-post-orogenic adamellite and minor more mafic variants comprising The Granites granite suite are widespread throughout the area.

7.2.3 Structure

The Project and the Tanami Region overall has undergone multiple phases of structural deformation. Most areas display several generations of folding and later episodes of faulting. Due to the geographic dispersion of deposits across the Project area and their local stratigraphy, the appearance of these events varies locally. The timing, sequence and nomenclature of the regional structural events have evolved from between authors; the scheme published by Crispe et al. (2007) is outlined below.

- Sedimentation of the Tanami Group was terminated by regional deformation and greenschist to amphibolite facies metamorphism during the Tanami Event

(D1/M1), at around 1,830 Ma. Subsequent east–southeast–west–northwest to southeast–northwest directed shortening (D2), followed by northeast–southwest to east–west directed shortening (D3), has resulted in open northeast F2- and northwest F3-trending folds in both the Tanami Groups;

- Steep south to southeast dipping F4-fold structures of Tanami metasedimentary rocks, many spatially associated with granitoid intrusions, indicate a period of south–southeast-directed regional shortening (D4) syn-to-post the regional granitoid intrusive phase. A network of north- to northwest-striking faults, several of which are interpreted as oblique thrusts with a component of left lateral movement, indicates a period of D5 convergence during west– southwest–east–northeast to east–west directed shortening;
- The Paleoproterozoic basement and cover sequences have subsequently undergone several episodes of faulting, collectively termed D6+.

7.2.4 Metamorphism

Peak metamorphism during the Barramundi Orogeny reached amphibolite facies at The Granites, while more generally greenschist facies metamorphism is observed at DBS. Contact metamorphic aureoles, commonly identified in pelitic schist units by randomly orientated andalusite porphyroblasts, are well developed at the margins of the post-orogenic granite plutons.

7.2.5 Mineralization

Three types of gold mineralization have been recognized within the Project area:

- Callie-style mineralization is characterized by coarse and readily visible gold occurring in quartz veins hosted in carbonaceous siltstone. These contain very minor sulfides and are generally discordant to bedding. Mineralization is in subparallel, sheeted quartz veins within a structural corridor. Example deposits and prospects include Callie, Auron and Federation;
- Villa-style mineralization is associated with Au-sulfide (arsenopyrite, pyrite and pyrrhotite) $\pm$ quartz $\pm$ carbonate veins in BIF and chert. The veins follow a stratigraphic horizon of iron-rich, cherty metasedimentary rocks within the Dead Bullock Formation. While generally associated with veining, a vein is not necessary for gold deposition and gold may be disseminated throughout the host. Example deposits and prospects include Triumph, Colliwobble, Sleepy Hollow, Dead Bullock Ridge, and Villa–Fumerole;
- East Bullakatchie mineralization that developed in Au–carbonate–sulfide $\pm$ quartz veins. These are unique to the East Bullakatchie deposit and occur as concordant and discordant veins within metamorphosed sedimentary rock, which probably had a carbonate precursor.

7.2.6 Surficial Geology

The Project area is predominately covered with a desert terrain comprising transported and residual colluvial cover sediments and aeolian sand blanket. Outcrop above the sand plain is sparse and mostly confined to the granite intrusions. Outcrop of the early Proterozoic lithological units is rare and confined to the more resistant chert and iron-rich beds.

Tertiary drainage channels are now completely filled with alluvial and lacustrine clays and calcrete. A major paleo-channel with a depth to basement of up to 90 m transects the Project area to the north of The Granites and DBS.

7.3 Dead Bullock Soak Area

7.3.1 General Geology

The geology of the DBS area comprises a thick succession of fine to medium grained clastic sedimentary rocks, with minor chemical meta-sediments and intrusives. These rocks belong to the Dead Bullock Formation of the Paleoproterozoic Tanami Group.

At DBS there are two sub-units (members) within the Dead Bullock Formation; the Callie Member (which includes SHIM, Orac and the Callie Laminated Beds) and the Ferdies Member (which includes the Lower Auron Beds).

A table of the key lithological units (refer to Table 7-1) and a stratigraphic column (refer to Figure 7-3) have been presented for reference.

A semi-conformable dolerite (Coora Dolerite) intruded the hinge zone of the main regional anticline and separates the two mineralized portions of the Davidson Beds, the Orac Formation and Schist Hills Iron Member. Several generations of fine to medium-grained foliated dolerite sills occur throughout the Callie Boudin Chert–Magpie Schist sequence and are loosely grouped as the End It All Dolerite. Additional thin, cross-cutting "glimmerite" or lamprophyre dykes are observed locally.

7.3.2 Structure

The area has a complex deformation history (refer to Table 7-2). The dominant structural feature is the DBS anticlinorium which is an upright isoclinal fold with local complex parasitic folding which plunges at ~40° to the east–southeast, and has an amplitude and wavelength of greater than 3 km.

Table 7-1: General Stratigraphy of the DBS Area

Formation/Member	Abbreviation	Description
Killi Killi Formation		Grey quartz–feldspar–sericite, normally-graded wackes with minor siltstones and mudstones
Madigan Beds.		Graded turbiditic wackes commonly interbedded with thicker laminated carbonaceous siltstones and shales
Seldom Seen Schist		Massive, fine grained carbonaceous siltstone
Manganiferous Chert		Well bedded (mm–cm scale) fine grained, highly graphitic quartz–feldspar–chlorite–graphite cherty siltstone
Colgate Schist		Massive, fine grained feldspar–quartz–sericite–graphite unit
Schist Hills Iron Member	SHIM	Gruneritic cherts interbedded with fine grained grunerite–quartz–chlorite–pyrrhotite–pyrite–arsenopyrite schists and minor graphitic horizons
Dead Bullock Member		Sequence of massive to poorly bedded fine-grained pyritic quartz–feldspar–sericite coarse silts with minor graphitic silt units
Orac Formation: Lower Chert, Middle Schist, Upper Chert and Upper Schist Members		Chert units are typically well banded comprising large nodules of chert within a pyrrhotite–pyrite–arsenopyrite–chlorite–biotite–quartz ± grunerite schist. Schist units are typically moderately bedded feldspar–quartz–chlorite–biotite ± cordierite with interleaved graphitic horizons

Formation/Member	Abbreviation	Description
Upper Blake Beds	UBB	Alternate between fine graphitic beds and non-graphitic massive to banded quartz–sericite–chlorite–biotite schists; may have thin boudin chert horizons
Callie Boudin Chert	CBC	Banded to massive, graphitic schist
Magpie Schist	MS	Regularly bedded (1–5 cm beds), occasionally carbonaceous, chloritic siltstone
Callie Laminated Beds	CLB	Thin to medium-bedded quartz–biotite–chlorite–muscovite–albite schists with well-defined graded bedding and local ilmenite laminations
Lower Blake Beds/Lower Blake Laminations	LBB; LBL	Massive quartz–chlorite–muscovite–biotite–albite schists, with fine-bedded, occasionally laminated, turbidite horizons
Upper Auron Beds	UAB	Laminated quartz–magnetite–chlorite–hornblende schist
Auron Beds	AB	Laminated quartz–chlorite–biotite–hornblende schists, with significant iron-rich beds
Lower Auron Beds	LAB	Quartz–chlorite–muscovite–biotite–albite schists
Terran Beds	TB	Variably silt- to fine sand-dominated brown–grey quartz–chlorite–biotite unit
Zelda Beds	ZB	Grey–green chloritic, planar to cross-bedded fine-grained sandstone and siltstone

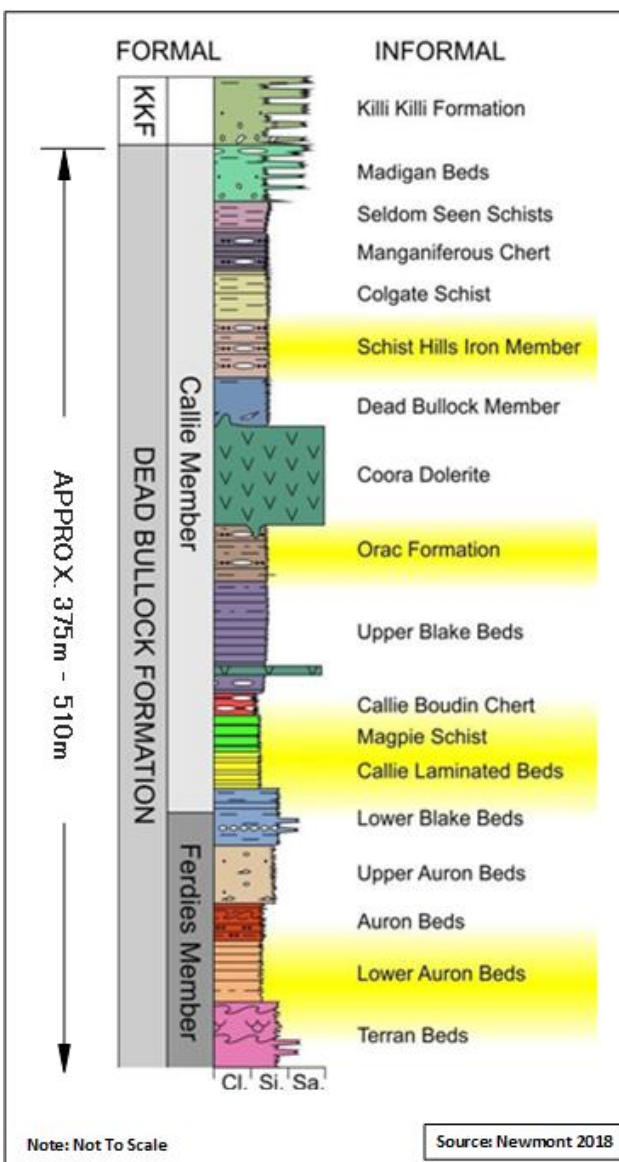


Figure 7-3: DBS Simplified Stratigraphic Column

Table 7-2: Major Deformation Events

Event	Description
D1	Upright tight folding formed the DBS anticlinorium and development of pervasive east–southeast-striking S1 axial plane cleavage.
Post D1	Intrusion of lamprophyre dykes. Dykes commonly strike sub-parallel S1 cleavage.
D2	Deformation event producing local east–northeast-striking S2 cleavage and minor folding.
Late or post D2	Formation of gold quartz veins sub-parallel to S2 cleavage in low strain environment.
D3	Westward directed reverse faulting along east dipping faults. Associated with buck-quartz veining and carbonate–sericite alteration. Local south-striking S3 fault cleavage.

Event	Description
D4	Brittle fault activation, dextral strike-slip with late dip-slip movement.
D5	Apparent fault sinistral strike slip movement.

Note: Table data modified after Vanderhor, 2003 and Miller 2010

The anticlinorium is crosscut by several generations of faults:

- West–northwest to north-trending, east-dipping cross faults;
- East–northeast-trending steeply south–southeast-dipping axial planar faults;
- West–northwest-trending, north–northeast dipping faults.

Deformation events resulted in the main vein styles presented in Table 7-3.

The 70/70 terminology used in Table 7-3 was derived as follows: On average, the sheeted quartz veins dip at 65° towards 154° hence they are locally referred to as 70/70 veins (i.e. dip 70° and strike 70°). These veins have a strong continuity over 5 to 10 m. The quartz veins are typically 2 to 30 mm wide and have a chlorite selvage.

Vein densities, for example at Callie, generally average five veins per meter; however, the vein density can reach more than 12 veins per meter. Where vein densities are elevated over several tens of meters, they are grouped into what are locally termed Quartz Vein Corridors (QVC). These corridors can range in thickness from 20 to 50 m, and their longitudinal extent is defined by the host lithology because this has a direct influence on gold mineralization.

Table 7-3: Vein Styles

Vein Type	Timing	Mineralization	Description
Bedding parallel quartz veins	D1	Non-mineralized	Thin (generally <1 cm) quartz ± biotite ± pyrite ± chalcopyrite
70/70 quartz vein set	Late t o Post D2	Mineralized veins	En-echelon sheeted quartz vein set with pinch and swell textures. Veins are mineralized in the right chemical environment
Buck quartz veins	D3	Non-mineralized	Quartz–carbonate ± sulfide veins associated with D3 deformation
Random	Various	Non-mineralized	Randomly oriented quartz ± carbonate veins

Note: Definition for 70/70 veins: On average, the sheeted quartz veins dip at 65° towards 154° hence they are locally referred to as 70/70 veins (i.e. dip 70° and strike 70°). These veins have a strong continuity over 5–10 m. The quartz veins are typically 2–30 mm wide and have a chlorite selvage.

7.3.3 Metamorphism

Regionally the sedimentary rocks have been metamorphosed to lower greenschist facies. However, locally at DBS the sequence has components of upper greenschist metamorphism, including the presence of biotite and amphibole (Lovett et al., 1993).

7.3.4 Alteration

Hydrothermal fluids responsible for the Callie mineralization have pervasively altered the sedimentary rocks, removing graphite and introducing chlorite, biotite, calcium carbonate and amphibole.

7.4 Dead Bullock Soak Deposits

The DBS sequence of rocks host several deposits which are described below. Geological plans showing the relative locations of the Callie, Auron, Federation and Liberator deposits are included as Figure 7-4 and Figure 7-5.

A geological cross-section through the deposits is included as Figure 7-6.

Figure 7-7 presents a long section of deposits accessed via underground mining.

7.4.1 Callie

Historically the terms “Lantin West”, “Wilson” and “Callie” shoots were used to refer to the mineralization hosted in the Lantin and Callie Anticlines respectively. The name “Callie Deeps” was applied to any mineralization down-plunge to the east of 60900mE. In more recent times these areas have been reported under the collective term “Callie” which has been subdivided into the following mining area names:

- **Lantin North Limb (LNH)** – Upper northern limb of the Lantin Anticline;
- **Lantin (LAN)** – Mineralization north of the Kerril Fault (including remnants hosted in the Callie Anticline in the upper part of the deposit);
- **Kerril South (KRS)** – Mineralization south of Kerril Fault, mostly hosted in the Callie Anticline and to a lesser extent the Lantin anticline;
- **Kerril South South (KSS)** – Mineralization on southern limb and parasitic folds of the Callie Anticline;
- **Crown (CRN)** – Remnant mineralization immediately below the mined-out Callie open pit on the southern limb of the Challenger syncline.

Mineralization at Callie is hosted within the Lantin and Callie Anticlines, Challenger Syncline and their respective limbs. The Magpie Schist and Callie Laminated Beds host most of the gold. The Lower Blake Laminations and to a much lesser extent, the Callie Boudin Chert contain minor amounts of economic mineralization. The deposit has been disrupted by at least three sets of post-mineralization faults.

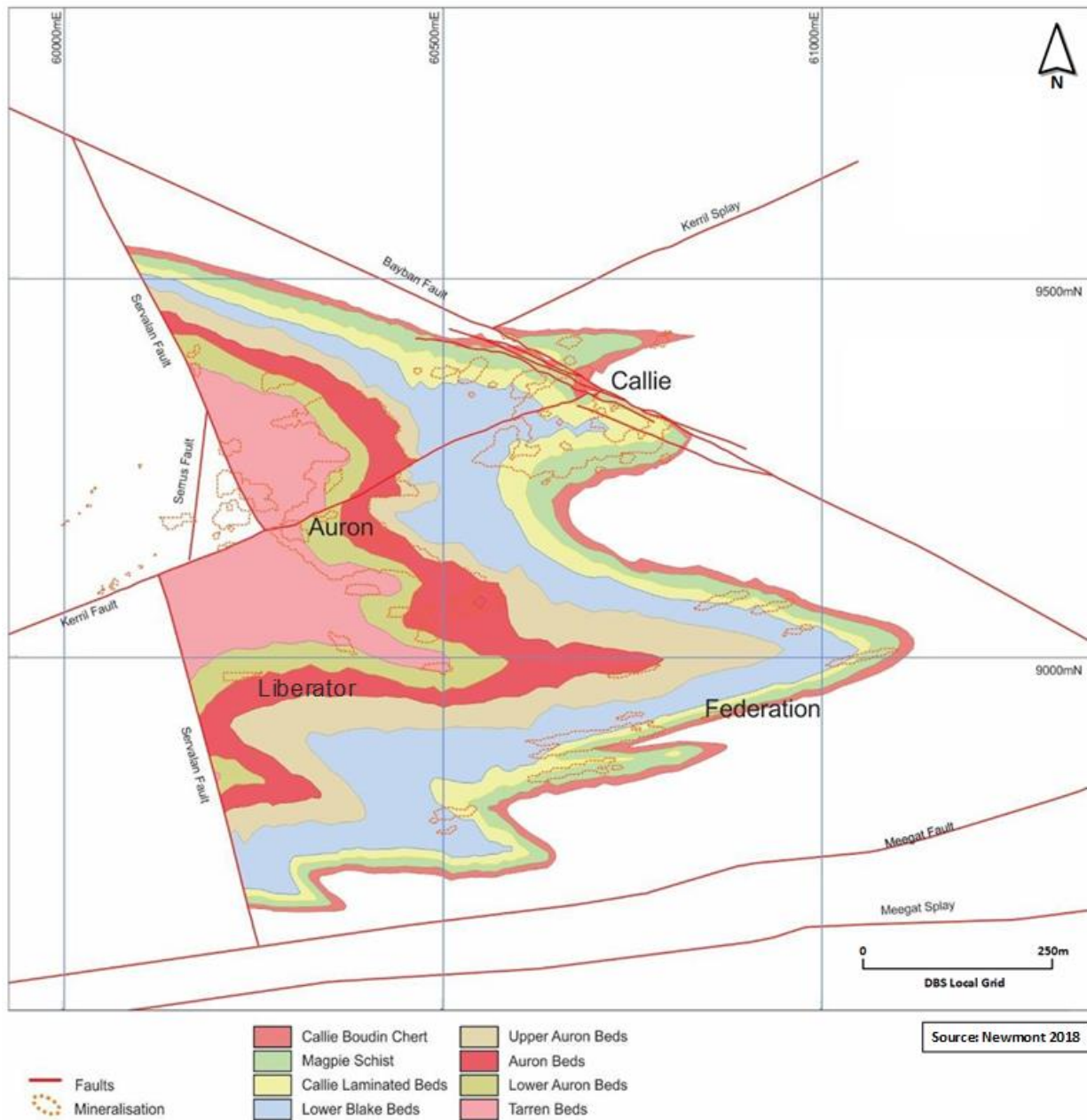
The deposit has a complex shape. This is a function of the difference in strike of the quartz vein corridors (070°) and the plunge of the folded sequence (100°), the acute intersection of which has allowed mineralization to develop over an extensive plunge.

Mineralization has been shown to have a down plunge extent of at least 2.8 km from the surface to the eastern-most deepest drill intercept. This equates to a depth below the surface of ~2 km and horizontal length of ~2.3 km. Mineralization remains open at depth. The width of mineralization varies significantly due to the interaction of fold, fault and quartz vein corridor geometries. At its thickest point it is ~200 m wide in the hinge of the Callie Anticline. At depth and on the fold limbs the width of mineralization is diminished and attenuated to less than 20 m. Similarly, the thickness of the deposit varies significantly depending on the fold geometry. The main host units are between 40 m and 50 m thick; however, they can locally be thicker in fold hinges or where duplicated by faulting.



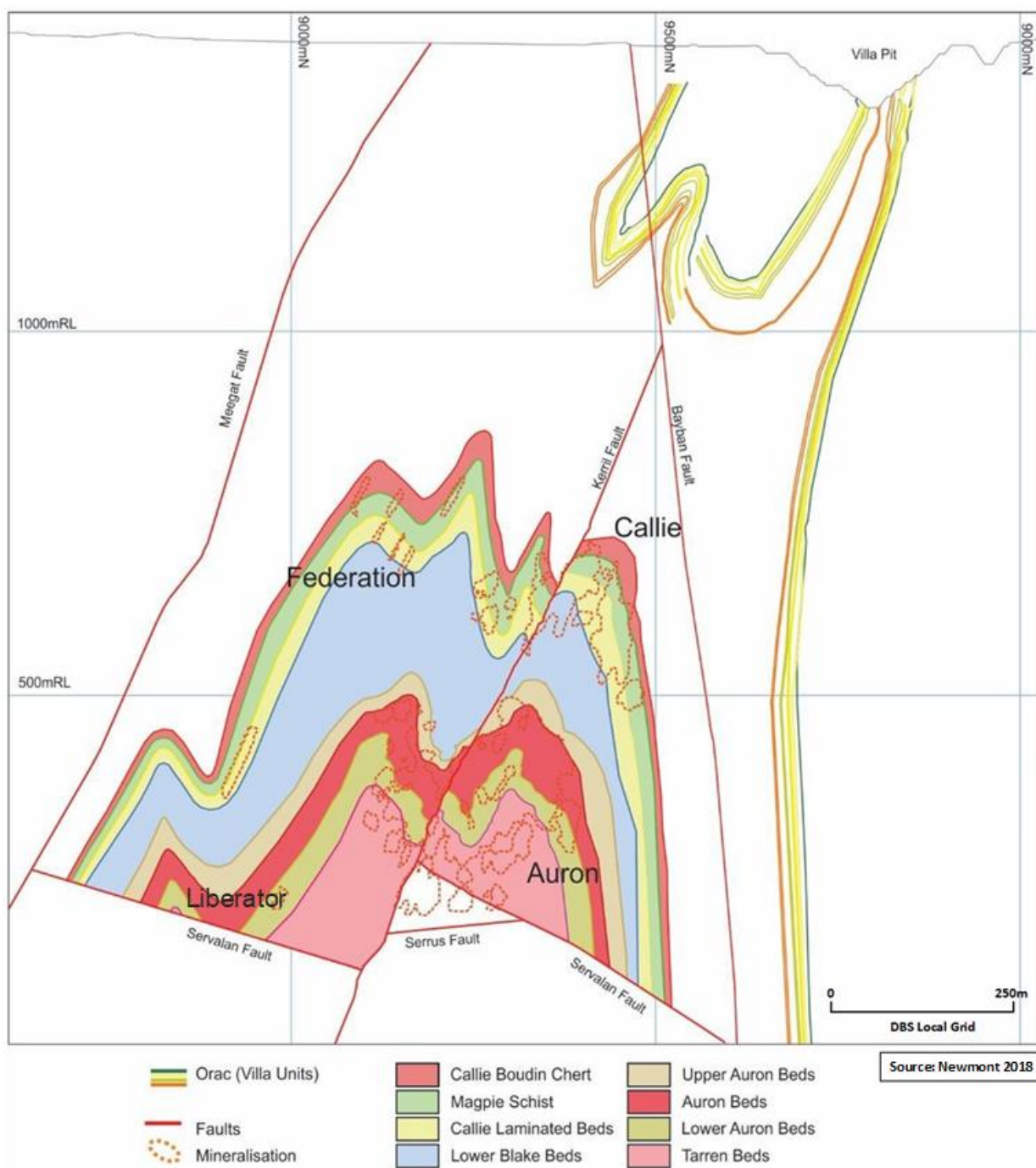
Note: Figure aligned to Mine Grid

Figure 7-4: Geological Plan of Callie, Auron, Federation and Liberator (650mRL Mine Grid)



Note: Figure aligned to Mine Grid

Figure 7-5: Geological Plan of Callie, Auron, Federation and Liberator (200mRL Mine Grid)



Note: Figure aligned to Mine Grid

Figure 7-6: Geological Cross-Section of Callie, Auron, Federation and Liberator (60200mE Mine Grid)

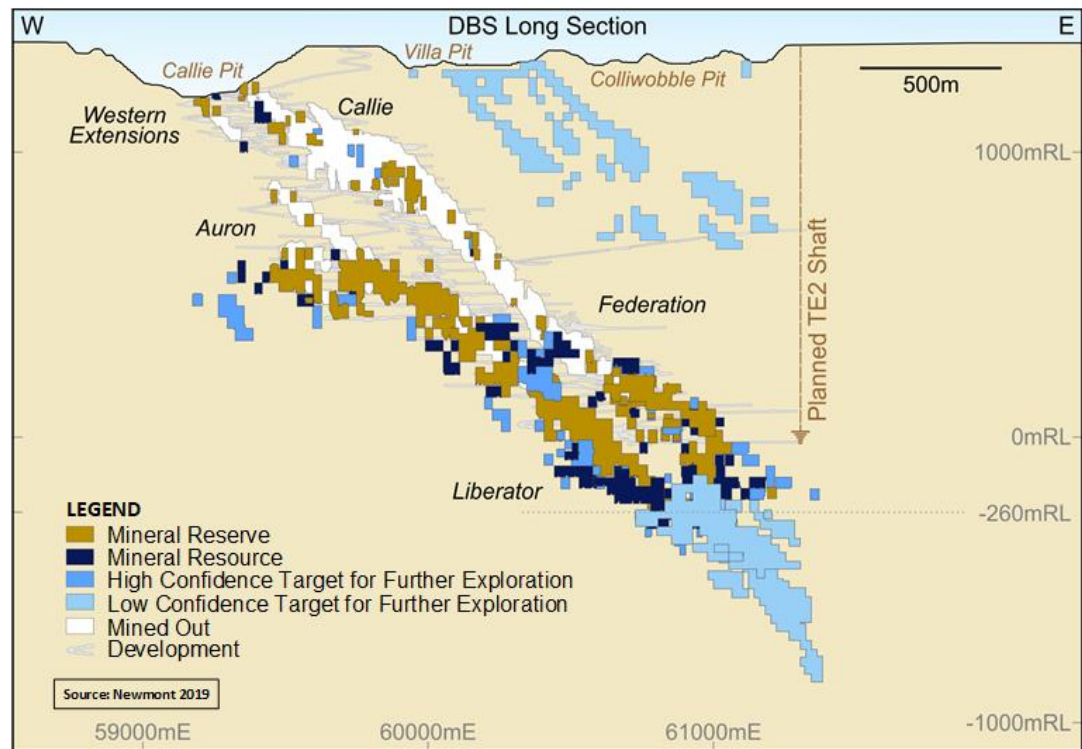


Figure 7-7: Long Section Showing DBS Underground Deposits

Gold is characteristically nuggety and occurs exclusively within the veins as grains from sub-millimeter sizes through to >20 mm. No disseminated gold is found.

The Callie mineralization was initially exploited by open pit methods and then subsequently underground by long-hole open stopeing methods to a depth of ~1.2 km. Remnant Mineral Resources and Mineral Reserves remain above this depth.

7.4.2 Federation

Federation represents an un-mined mineralization envelope hosted predominantly within the hinge and southern limb of the Federation Anticline. The Federation Anticline is parallel to the Callie and Lantin Anticlines that host the bulk of the Callie mineralization, and occurs within the same stratigraphic sequence.

The Federation mineralization has been demonstrated to have a plunge extent of at least 1 km, its extents range from 800 m to 1.5 km below the surface. The mineralization is currently open both up and down plunge. The Federation host stratigraphy is narrow, relative to Callie and is generally 20 to 30 m thick except where locally thickened in fold hinges. The mineralization intersected in drilling at Federation also show reduced widths relative to intersections at Callie.

Gold is hosted by a sheeted 70/70 vein set parallel to the one that hosts the Callie and Auron mineralization.

Figure 7-8 presents a Federation Corridor long section looking north, showing 2018 drill sections and the location of the Federation deposit.

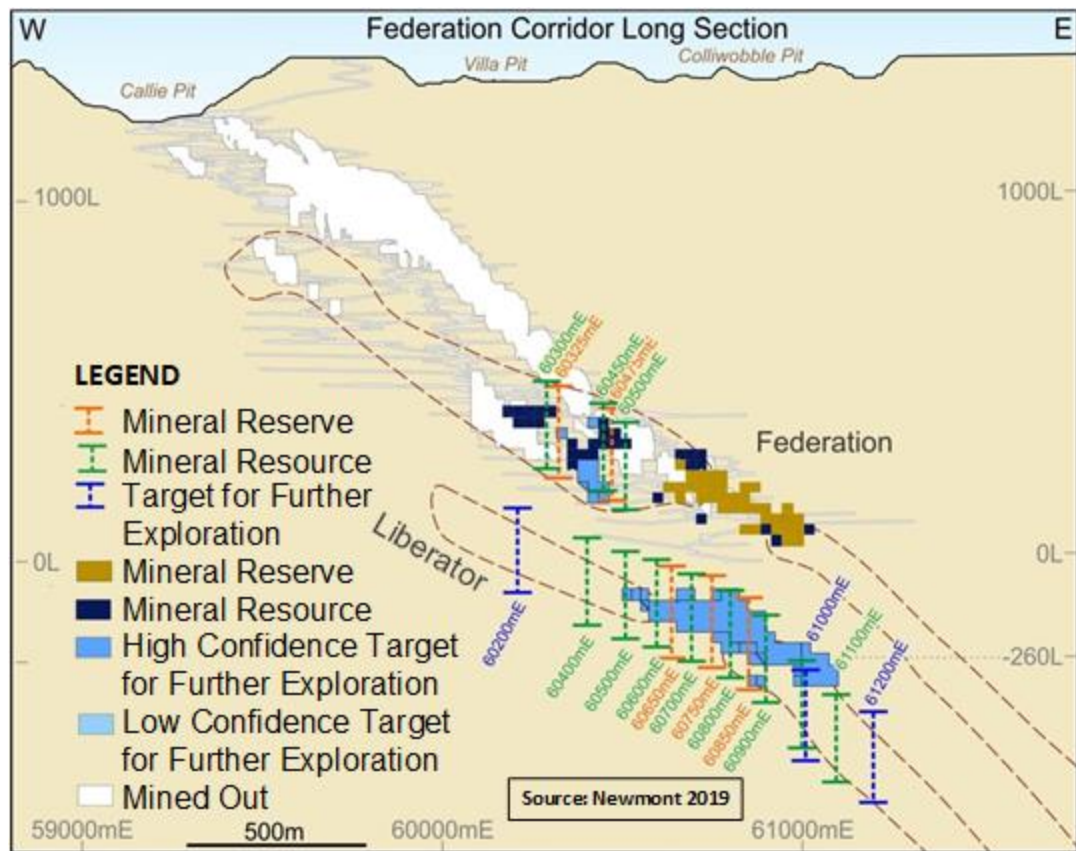


Figure 7-8: Federation Corridor Long Section Looking North and Showing 2018 Drill Sections

7.4.3 Liberator

Liberator was discovered in 2015 in the Federation Anticline. Liberator is within 250 m of existing or planned development but is located below 0mRL (1,400 m below surface). Liberator is located to the south of the Auron and Lantin orebodies, which make up Vein Corridor 1 and sits in Vein Corridor 2 with Federation but also exhibits a 45° plunge to the east.

The orebody is like Federation in its structure and geometry but is hosted in a rock mass like that of the Auron orebody, confirming prior predictions for the potential repetition in the economic mineralization in the underlying Auron stratigraphy. Due to the proximity of Liberator to both Federation and Auron, it was assumed that the orebody could use the infrastructure already in place or planned for these nearby orebodies.

Figure 7-8 presents a Federation Corridor long section looking north, showing 2018 drill sections and the location of the Liberator deposit.

7.4.4 Auron

The Auron deposit sits stratigraphically below the Callie deposit. It is hosted within the Upper Auron Beds, Auron Beds, Lower Auron Beds and Terran Beds. Lithologies have also been disrupted by the same three generations of post-mineralization faulting observed at Callie.

The majority of the economic Auron mineralization is represented by the Callie style 70-70 sheeted vein mineralization. This mineralization style is predominantly hosted

within the laminated Lower Auron Beds and to a lesser extent in the Terran Beds. Lower grade un-economic Villa style mineralization is also hosted within the Upper Auron Beds.

The mineralization hosted by the Auron Beds is similar to Villa-style disseminated mineralization, but also exhibits some characteristics of the vein-hosted Callie-style. Generally, Villa-style mineralization occurs within the sulfide rich portions of the Auron Beds. Although volumetrically extensive the Auron Beds-hosted Villa-style mineralization is lower grade than the Callie-style mineralization and is mostly un-economic. Where the core of the vein corridor intersects the Auron Beds there are localized zones where the two mineralization styles co-mingle, and elevated gold grades are observed.

Auron shows a similarly complex shape as the Callie deposit due to the intersection of the folded sequence and the vein corridors. This is further complicated by stratigraphic fault repetitions.

The Auron mineralization has been demonstrated to have a plunge extent of at least 2.5 km. The economic mineralization is truncated to the west where the vein corridor trends outside of the Lower Auron Beds at a depth of 400 m below the surface. Mineralization is open to the east where it has been intersected in deep drilling at a depth of 1.9 km below the surface. Below the main plunge of Auron, the mineralization hosted in the Terran Beds is still open in some areas.

The width of the Auron mineralization varies significantly due to the interaction of the fold, fault and quartz vein corridor geometries, at its thickest point it is ~150 m wide. It gradually pinches out up-plunge and is attenuated on the fold limbs. Underground production stoping commenced in late 2011.

Figure 7-9 presents a Wilson Corridor long section looking north, showing 2018 drill sections and the location of the Auron deposit.

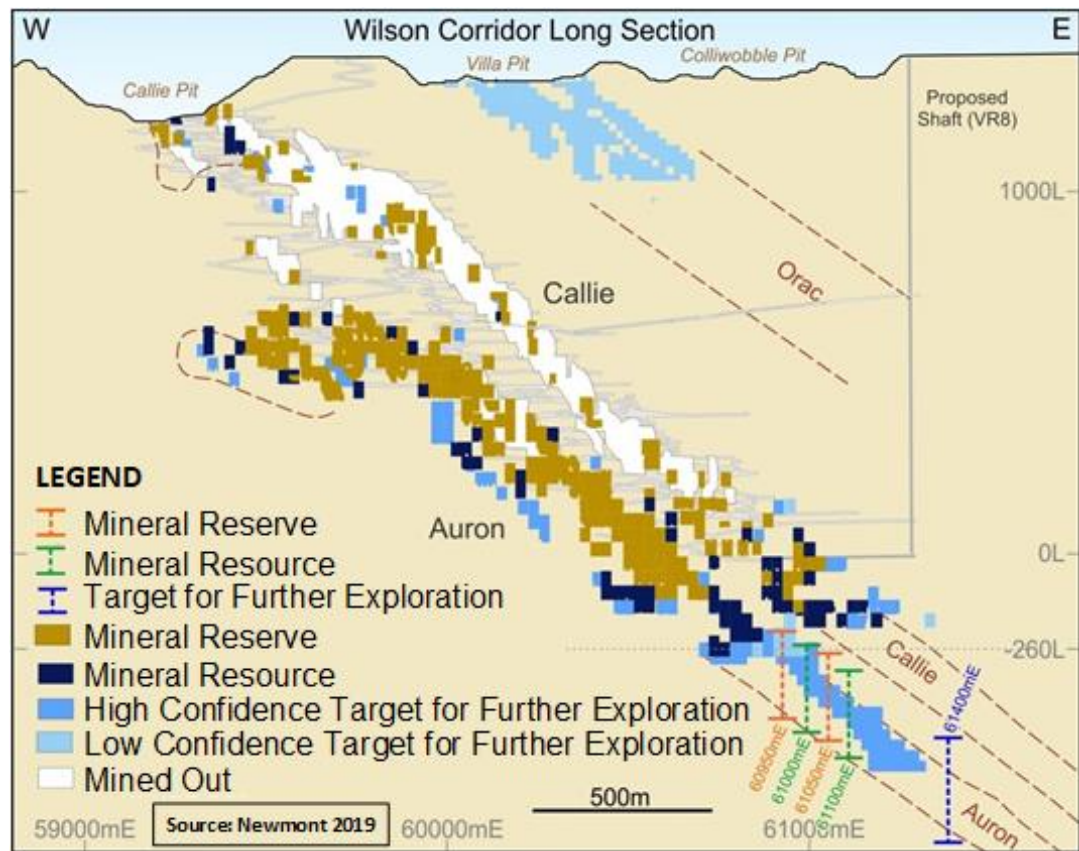


Figure 7-9: Wilson Corridor Long Section Looking North and Showing 2018 Drill Sections

7.4.5 Dead Bullock Ridge

At the Dead Bullock Ridge deposit, the Schist Hill Iron Member dips steeply southward in the overturned northern limb of the DBS anticlinorium. Boudinaged chert beds and parasitic folds resulting from multiple deformations are common features. Post-mineralization, west-northwesterly reverse faults have sinistral movement of up to 200 m (Tate, 1995c). The fault zones consist of 1 to 2 m thick anastomosing shears with sericite-carbonate-pyrite alteration.

The post-mineralization reverse faults caused a repetition of the mineralized host unit. Consequently, the Dead Bullock Ridge deposit comprised four mineralized zones with strike lengths of 100 to 150 m.

Gold occurs in amphibole chlorite rocks containing abundant pyrite $\pm$ arsenopyrite and shows a strong association with arsenopyrite.

7.4.6 Triumph Hill

The Schist Hill Iron Member forms the northern limb and the hinge zone of an overturned tight anticline. Due to the extent of post-mineralization faulting, Triumph Hill is structurally the most complex deposit in the Dead Bullock Soak Goldfield. Faulting is dominated by the northwest-trending Stag Fault and North Stag Fault. The Stag Fault dips at 60° northeast. In brittle dolerite, these fault zones contain breccia, while in ductile carbonaceous phyllite, they form anastomosing shear zones.

Gold occurs in amphibole chlorite rocks containing abundant pyrite $\pm$ arsenopyrite and shows a strong association with arsenopyrite.

7.4.7 Villa–Fumerole

The Villa and Fumerole deposits were mined in the Villa open pit. The Villa mineralization was shown by drilling to extend over 1 km of strike and to a depth of 500 m; however, within the mined pit the mineralization had a strike length of ~600 m and extended to a depth of 85 m.

The Villa host structure is an east-striking, overturned isoclinal F_2 anticline plunging 55 to 60° east, with common parasitic folds. Three generations of quartz veins have been identified. Gold values were anomalous in all the quartz veins; however, ore-grade mineralization was restricted to chert host units.

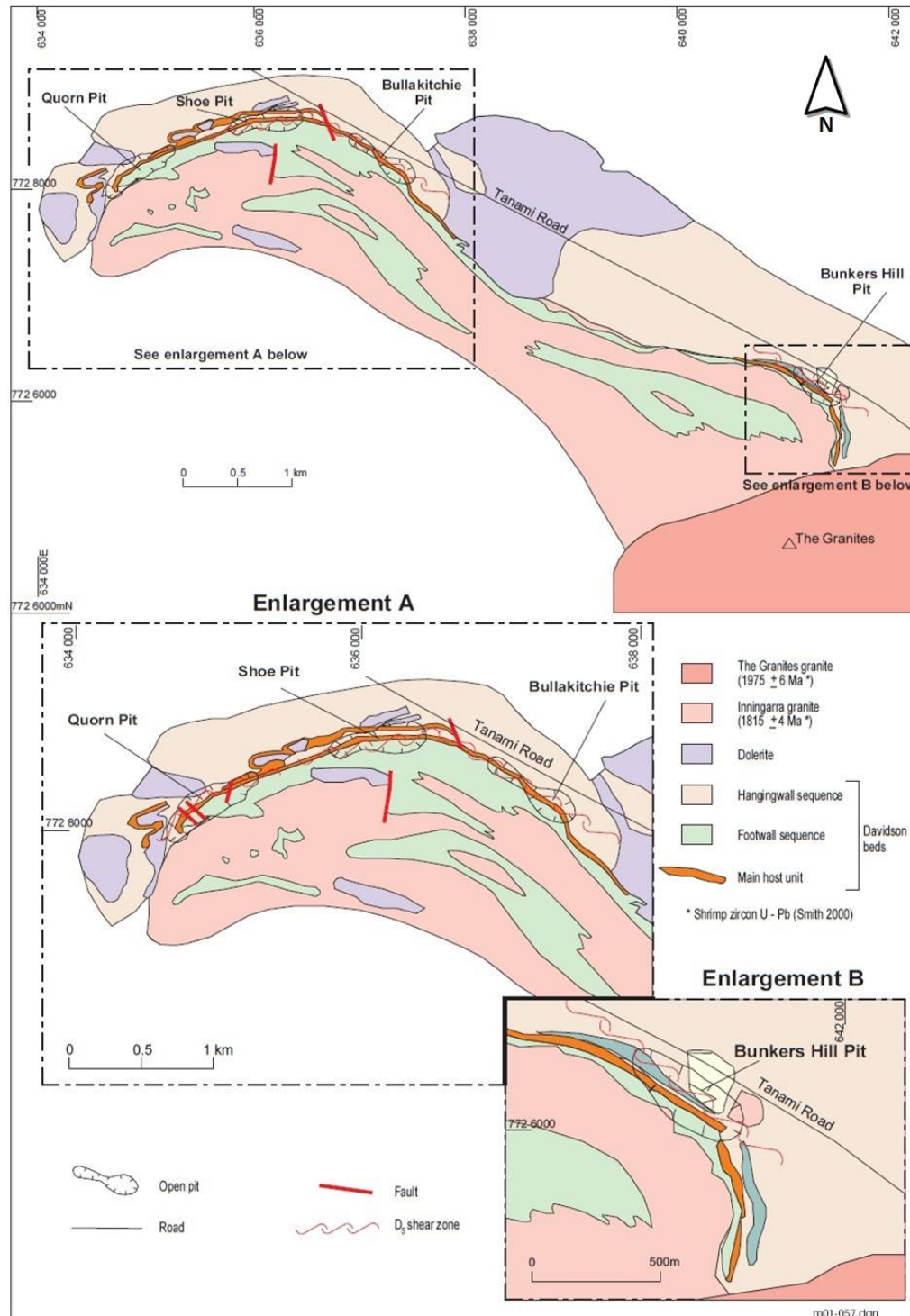
7.5 The Granites

The geology of The Granites area is characterized by the Paleoproterozoic Killi Killi Formation and Dead Bullock Formation that have been deformed under amphibolite- facies conditions. The base of the Dead Bullock Formation to the south is obscured by the intrusion of a large volume of Inningarra Suite granodiorite. Gold mineralization is restricted to the Dead Bullock Formation.

Four main deposits hosted economic mineralization at The Granites, namely Bullakitchie, Shoe, Quorn and Bunkers Hill. As the deposits have been mined out, only a summary of the mineralization setting is provided (refer to Table 7-4). A general geological plan is presented as Figure 7-10.

Table 7-4: The Granites Gold Deposits

Deposit	Mining Method	Description
East Bullakitchie	Open Pit and Underground Methods	Four lodes were mined. A fifth lode was too thin to be economically mined. Most of the gold mineralization in the 'Main Host Unit' was hosted in concordant quartz, quartz–calcite, calc-silicate and calcite veins associated with D1–D3. Calcite veins carried only part of the mineralization, as some gold was also found in narrow alteration zones surrounding veins and in host rock. The lodes had a low (3%) sulfide content compared with elsewhere in The Granites goldfield.
West Bullakitchie	Open Pit Methods	Spatially associated with East Bullakitchie, but structurally and lithologically different. Consisted of three lodes that were characterized by low (<4%) sulfide content. Most of the gold mineralization was in the 'Main Host Unit'.
Shoe	Open Pit and Underground Methods	Three lodes were mined. Sulfide concentration within the 'Main Host Unit' averages 10%, with higher concentrations in amphibole-rich rocks. Quartz veins are dominant; calc-silicate and quartz–calcite veins are less common.
Quorn	Open Pit Methods	Three lodes of the 'Main Host Unit' (A, B and C), as well as portions of the 'Footwall Host Unit', were mined. Sulfide content of the 'Main Host Unit' lodes averages 7%, but locally can be as high as 30%. Highest gold values are associated with areas of high-density arsenopyrite-bearing quartz veins.
Bunkers Hill	Open Pit Methods	Was the lowest-grade deposit. Lodes are contained within a 15 to 25 m thick 'Main Host Unit' and are abruptly truncated at shallow depth by The Granites Granite. Sulfide content reaches 5%. Vein compositions and densities are similar to those at Shoe and Quorn.



Source: Wygralak et al., 2004. Figure aligned to True North

Figure 7-10: Geological Plan of The Granites Area

7.6 Windy Hill Area (Minotaur Deposit)

The Windy Hill area is located 30 km northeast of The Granites and hosted the Minotaur deposit that was mined by open pit methods between 2003 and 2004.

In the deposit area, the Taurus Formation is interpreted to belong to the Davidson Beds and has a chemical signature equivalent to the Schist Hills Iron Member and the Orac Formation that host mineralization at DBS. The biotite schists of the Footwall Schist are considered to belong to the Blake Beds while the greywackes of the Hangingwall Schists represent the Madigan Beds.

The Minotaur deposit is located on the vertical to slightly overturned western limb of the major F1 Windy Hill Anticline with the variation in stratigraphic unit thickness a function of interfering F1, F2 and F4 folds. The Taurus Formation is host to most of the significant mineralization; however well-mineralized zones were also found within both the Footwall and Hangingwall Schists (refer to Figure 7-11).

Late-stage granitic dykes and veins occur as relatively flat-lying continuous bodies. The late-stage Hades Dolerite located between the Hangingwall Schist and the Taurus Formation has reduced the thickness of the mineralization host unit.

Gold principally occurred within fine chloritic fractures and veinlets but was also associated with sulfides, primarily arsenopyrite and pyrrhotite. Higher-grade mineralization was largely confined to the oxide portion of the deposit and it appears that extensive mobilization of gold occurred within the weathering profile.

7.7 Titania Area (Oberon Deposit)

The Titania area is located approximately 28 km to the north of the DBS mining operations. A summary of the local geology of the Oberon deposit is presented in Table 7-5 and the bedrock geology is presented in Figure 7-12. The basal beds indicated in the table are interpreted as being equivalent to the Davidson Beds at DBS.

The Oberon deposit has a complex multiple event deformation history. The deposit is hosted by a single, doubly-plunging, tight to isoclinal anticline, the Oberon Anticline. The stratigraphy is inconsistent between the limbs of the Oberon Anticline, with the most notable differences being the presence of the Hera and Echo Dacites and a repeat of the Selene/Danae Beds in the northern limb. Several faults have been identified, primarily by offsets of the folded stratigraphy.

Two porphyritic dolerite intrusions ranging in thickness from 20 m to 125 m, collectively termed the Nemesis Dolerite, cut the stratigraphy at Oberon with one dike on each limb of the fold. There are also two felsic-intermediate intrusions in the fold limbs of the Oberon anticline, the Hera and Echo Dacites.

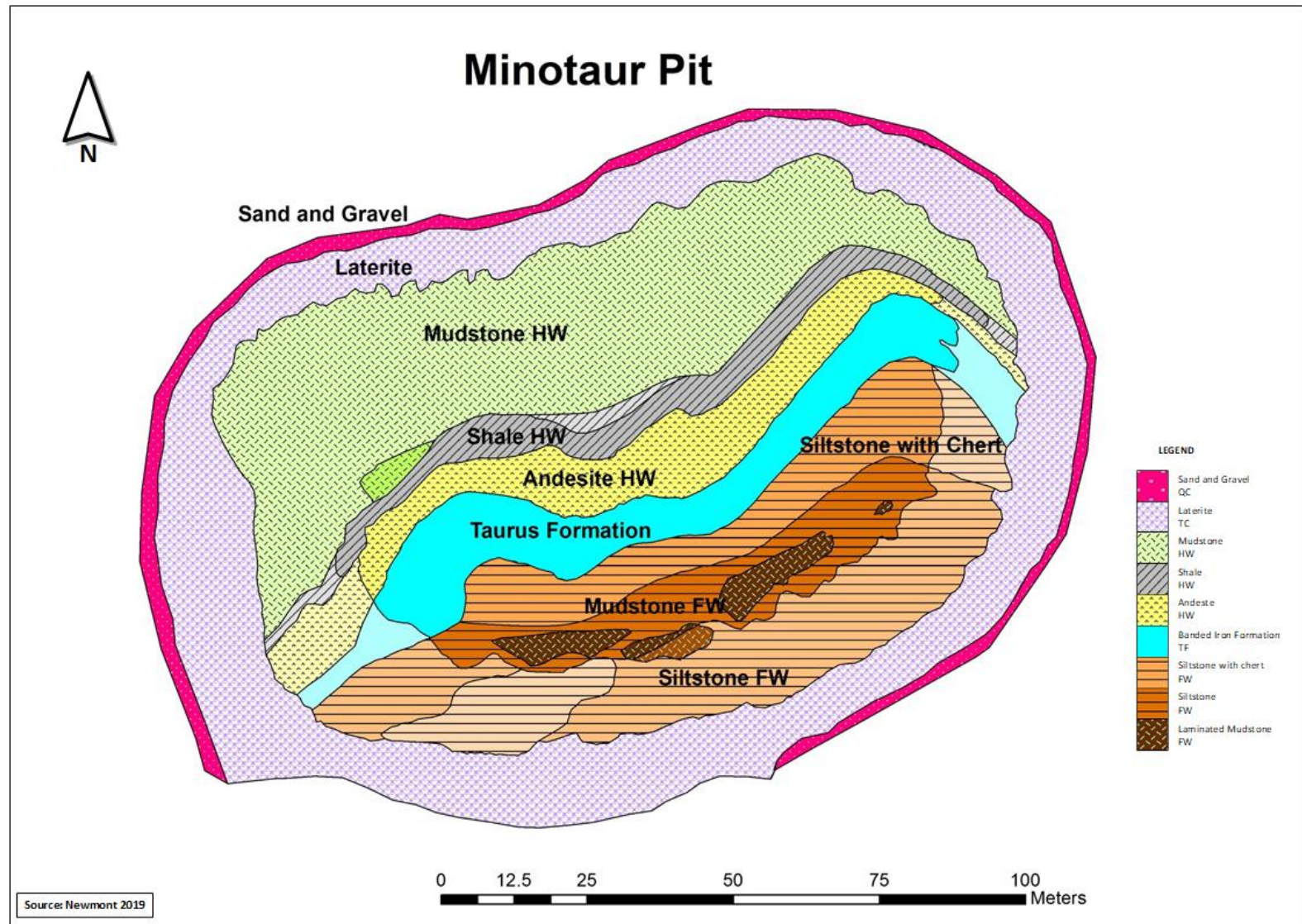
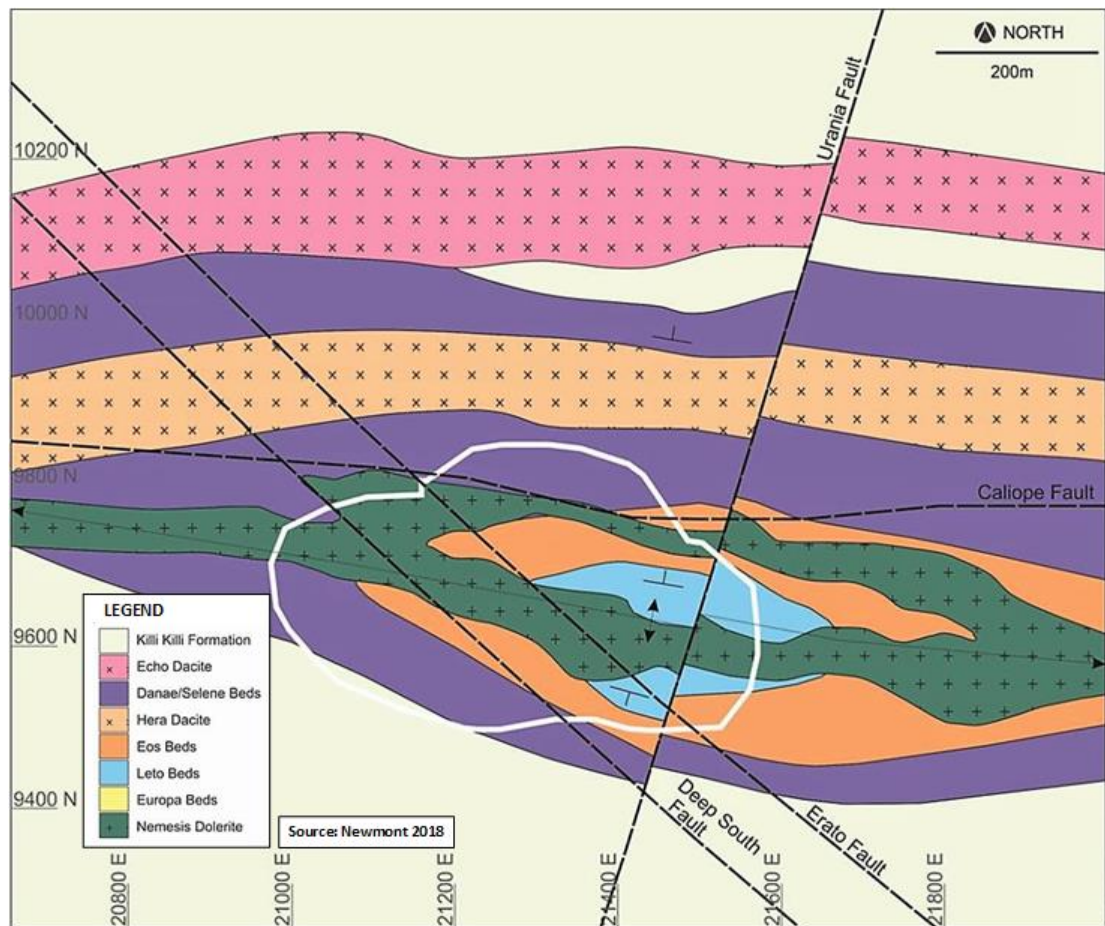


Figure 7-11: Geological Plan of the Minotaur Deposit

Table 7-5: Lithological Units of the Oberon Deposit

Unit	Description
Killi Killi Formation	Interbedded medium grained feldspar-quartz-sericite greywacke and siltstone turbidite sequences
Selene/Danae Beds	cm-scale bedded siliciclastic and carbonaceous siltstone
Eos Beds	Pyritic carbonaceous shale with amphibole-rich cherts and lesser interbedded horizons of immature sandstone.
Leto Beds	Fining upwards package of basal coarse quartz-lithic wackes, fine sandstone to siltstone, and interlayered siliciclastic and carbonaceous siltstones locally containing chert nodules
Europa Beds	Massive to bedded, fine grained, locally carbonate-bearing sandstone and siltstone



Note: White outline on plan shows the better mineralized area. Figure aligned to Mine Grid

Figure 7-12: Oberon Bedrock Geological Map

The bulk of mineralization occurs in the Europa and Leto Beds, minor mineralization also occurs in the Nemesis Dolerite and the Eos Beds. The more felsic-derived Eos Beds, Selene/Danae Beds, and Killi Killi Formation contain little to no mineralization.

Mineralization is hosted in two main mineralized structures (the North and South Lodes) and two minor structures (the Far North and Mid Lodes) and a zone of mineralization occurs within the enriched overlying colluvium. Mineralized zones typically have a strike length of 325 to 350 m. Mineralization styles include:

- Free, coarse native gold in quartz veins that do not contain arsenopyrite;
- Gold in close association with arsenopyrite as free Au in arsenopyrite-bearing veins or on cracks and grain boundaries in arsenopyrite.

Secondary weathering processes have resulted in depletion zones near surface and supergene enrichment towards the base of the oxidized zone. Within the oxide zone, mineralization is more complex due to chemical ($\pm$ physical) remobilization. The result is several locally enriched horizons (up to 30 m thick) and depletion zones (5 to 20 m thick).

7.8 Comments on Geological Setting and Mineralization

In the opinion of the QP:

- The understanding of the lithologies, and geological, structural, and alteration controls on mineralization, and the evaluation of the mineralogy is sufficient to support the estimation of Mineral Resources and Mineral Reserves;
- The geological knowledge of the area is adequate to reliably inform mine planning.

8.0 DEPOSIT TYPES

The deposits discovered to date are considered to be examples of Proterozoic orogenic gold deposits. Information on the deposit type is abstracted from Moritz, 2000 and Groves et al., 1998.

Orogenic gold deposits are a structurally-controlled gold occurrences formed during periods of tectonic stress, where gold is generally transported from depth by accompanying hydrothermal fluids. Any rock type within a greenstone or schist belt, or a metamorphosed supracrustal rock, dyke, or intrusion within, or intrusion bounding, a greenstone or schist belt can host an orogenic gold deposit.

There is strong structural control of mineralization at a variety of scales. Deposits are normally located in second- or third-order structures, most commonly near crustal- to regional-scale deformation zones, in brittle, brittle–ductile and ductile deformational environments. The structures hosting the gold deposits (shear zones, faults, extensional veins, breccias) are typically discordant with respect to the stratigraphic layering of the host rocks, but in some cases, they can be parallel to bedding planes and fold hinges or intrusive contacts.

Rocks are frequently metamorphosed to greenschist or amphibolite-facies grade. The mineralization is syn- to late-deformation and typically post-peak greenschist-facies or syn-peak amphibolite-facies metamorphism.

Deposits consist of simple to complex networks of gold-bearing, laminated quartz–carbonate fault-fill veins that are generally hosted by moderately to steeply dipping, compressional brittle-ductile shear zones and faults. Locally, shallow-dipping extensional veins and hydrothermal breccias can be associated with the faulting.

Deposits typically have a plate-like, ellipsoidal shape. Thicknesses and continuities vary; a mineralized zone can be 0.5–50 m wide, 100 m to as much as 2 km long and consist of a number of networked veins, or an en echelon vein swarm, or one large single vein. Depth extents of vein systems are variable and have been traced for over 2–3 km of vertical extent in some deposits. Typically, there is generally little change in mineralization grade, and mineralization and gangue mineralogy with depth.

Veins in the orogenic gold deposits are dominated by quartz with subsidiary carbonate and sulfide minerals, and less abundantly, albite, chlorite, white mica (fuchsite in ultramafic host rocks), tourmaline, and scheelite. Carbonate minerals consist of calcite, dolomite, and ankerite.

Gold is typically hosted in a quartz–carbonate vein network but may also be present within iron-rich sulfidised wall-rock selvages, or within silicified arsenopyrite-rich replacement zones. Gold mineralization can be in the form of native gold, as free gold grains in gangue and sulfides, and in fractures developed in gangue and sulfides.

In some deposits, gold may occur as either sub-microscopic inclusions in, or in the lattice of, pyrite or arsenopyrite. The gold to silver ratio is consistently >1 and more typically can reach 5–10.

8.1 Comment on Deposit Types

In the opinion of the QP:

- The understanding of the deposit type was appropriate in guiding initial exploration activities, is suitable for current exploration programs, and is sufficient to support estimation of Mineral Resources and Mineral Reserves.

9.0 EXPLORATION

Gold was discovered in the Tanami region in the early 20th century, where prospecting and exploration was undertaken by several companies (refer to Section 6.0). Exploration has been undertaken within the Project area by Newmont (including Normandy NFM) or by contractors (e.g. geophysical surveys).

9.1 Grids and Surveys

Data collection for each of the respective prospects and deposits within the Project area have used local grids; the conversion parameters between these local grids and UTM (AMG 66 – Zone 52) co-ordinates is presented in Table 9-1.

9.2 Geological Mapping

Geological mapping at the Project has been ongoing since the early 1980s; however, the relatively deep cover and sparse outcrop has meant mapping has been concentrated within exposures created by the mining operations.

Mapping programs completed include:

- The Project area is covered by the NT Geological Survey (NTGS) 1:100,000 scale geological map series;
- Detailed open pit mapping was completed at a range of scales for the open pits mined at The Granites, Windy Hill and DBS;
- Underground mapping has been completed at the Callie underground mine at 1:250 scale. The upper levels of the mine were comprehensively mapped; however, development completed since 2005 has only been mapped in targeted areas;
- Outcrop mapping has been undertaken at a range of scales for regional exploration projects.

9.3 Geochemical Sampling

Extensive surface geochemical samples were collected as an orientation tool in support of regional exploration programs. Sample point locations are presented in Figure 9-1. Table 9-2 presents the types of samples collected.

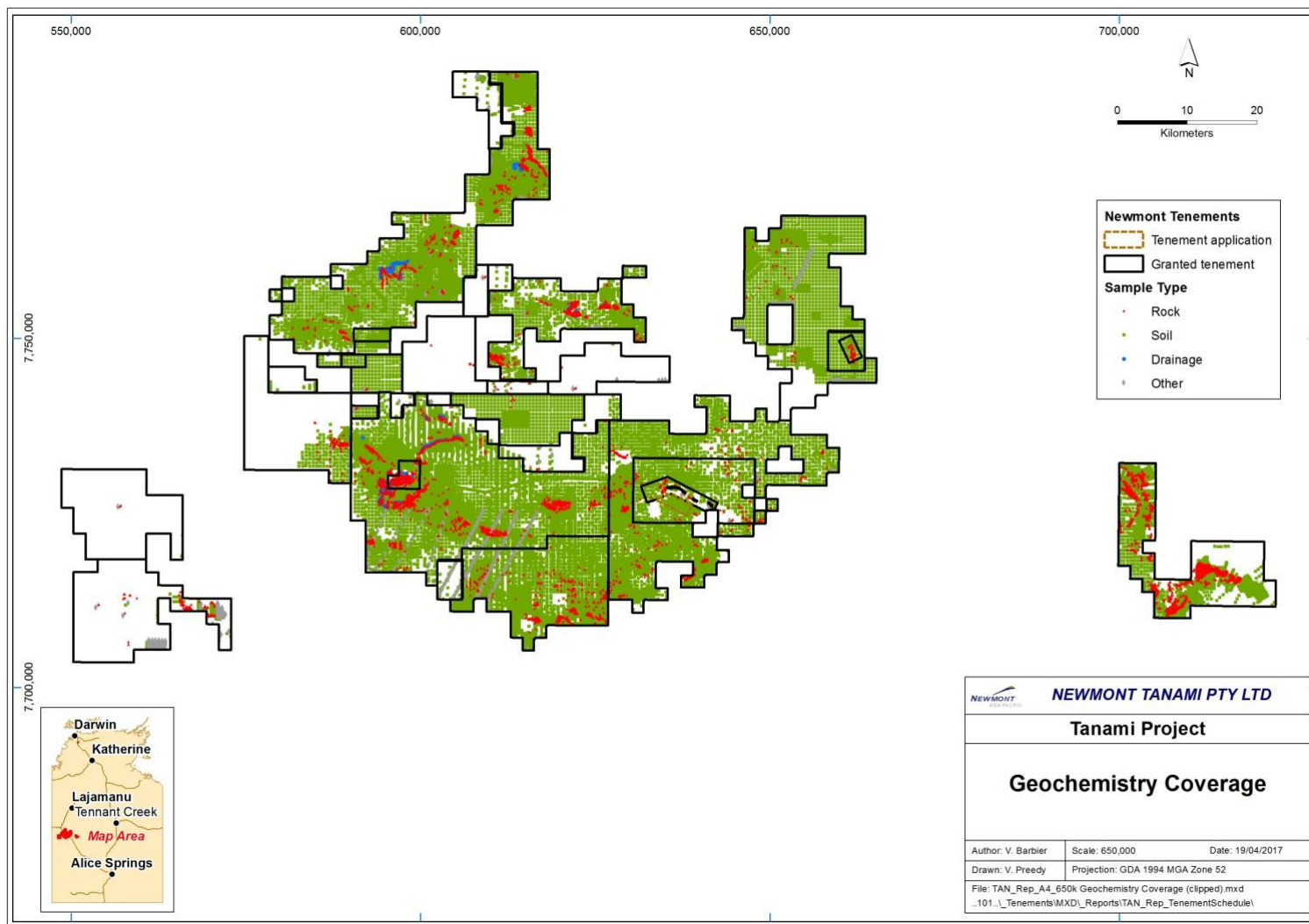
The majority of samples were recorded as soil (62,779) and rock samples (12,022) and were subject to gold and multi-element analysis. Samples were collected using a number of sieve sizes and analysis techniques that varied significantly with multiple digest, leaches, analysis techniques applied over the years.

Information generated from the sampling programs were used to vector into targets.

In 2017, following the establishment of Newmont's proprietary Deep Sensing Geochemistry (DSG) technique, an exploration campaign was aimed at achieving a broad but comprehensive geochemical re-evaluation of all areas of Newmont's tenure considered to be conceptually prospective. This evaluation was completed by way of systematic, wide spaced DSG sampling, with samples taken on offset lines on a 1 km x 1 km spacing. These samples provided a new geochemical base layer for Newmont's tenure, which has since been followed up with infill sampling (800 m x 50 m) over prospective anomalous areas, leading to subsequent drill testing.

Table 9-1: Conversion Parameters for Local Grids

Grids		Point 1						Point 2				Conversion Parameters				
From Grid	To Grid	From Y	From X	From Z	To Y	To X	To Z	From Y	From X	To Y	To X	Theta	Scale Factor	False Easting	False Northing	False RL
Anomaly 2	AMG66_52	10000	20000	1400	7726093.87	619356.19	1400					-4.1	1	598692.4	7717549.411	0
Bulla	AMG66_52	4594.153	50285.960	1400	7728027.32	636580.14	1400	3828.87	49139.09	7728175.82	635209.68	-39.89862883	0.999802546	595063.097	7756752.092	0
Bunkers	AMG66_52	4833.930	55842.098	1400	7724648.25	640995.68	1400	4594.15	50285.96	7728027.32	636580.14	-39.89675487	0.999789934	595062.681	7756750.371	0
Challenger	AMG66_52	7400	9000	1400	7775708.53	615146.45	1400					-4.65	1	605576.165	7769062.506	0
Chapmans	AMG66_52	6100	55300	1400	7725966.31	641392.70	1400					-84.9	1	630400.995	7780505.126	0
DBS	AMG66_52	7342.891	58139.736	1400	7727732.64	595279.93	400	10000	60600	7730307.45	597825.44	-1.874971995	0.999851268	536939.752	7722296.737	-1000
MGA94_52	DBS	7755651.280	601004.442	335	8600	19200	1335	7757744.55	605076.97	11000	23100	4.404392277	1.000065535	15568.9859	-7770812.02	1000
Horden	AMG66_52	10000	30000	1400	7716253.37	638061.24	1400					28.5	1	616468.314	7693150.436	0
Hyatt	AMG66_52	20000	91000	1400	7708198.11	578455.39	1400					22.5	1	502036.021	7654896.327	0
Ivy	AMG66_52	8000	36000	1400	7728499.74	635721.35	1400					25.9	1	606831.684	7705578.413	0
Ivy West	AMG66_52	12000	70000	1400	7727844.64	634372.46	1400					18	1	571506.708	7694800.772	0
Lennard	AMG66_52	10000	12200	1400	7754759.70	617995.60	1400					-23.5	1	602819.976	7750453.838	0
Madam Pele	AMG66_52	10000	18400	1400	7715282.48	594837.86	1400					13.6	1	579305.199	7701236.255	0
Magellan	AMG66_52	10000	30000	1400	7723332.90	606381.50	1400					-4.1	1	575743.302	7715503.416	0
Majestic	AMG66_52	10000	50000	1400	7732336.19	599852.91	1400					-19	1	549321.3	7739159.412	0
Margi East	AMG66_52	16000.000	76332.100	1400	7734017.04	592691.81	1400					-40.6	1	524322.649	7771543.662	0
Moon	AMG66_52	5000	50000	1400	7728522.01	636621.07	1400					-39.9	1	595055.564	7756758.666	0
Neverest	AMG66_52	6806.940	30934.810	1400	7726057.80	596527.62	1400					26	1	571707.563	7706378.835	0
Officer	AMG66_52	10000	80000	1400	7711602.09	567608.10	1400					0	1	487608.1	7701602.09	0
Ptilotus	AMG66_52	11595.450	15786.030	1400	7754782.84	621919.00	1400					-23.5	1	602818.587	7750443.781	0
Quorn	AMG66_52	7199.831	36566.134	1400	7728027.32	636580.14	1400	7932.18	35397.96	7728175.82	635209.68	25.90006567	0.99980209	606837.601	7705582.914	0
Schist	AMG66_52	6806.940	30934.810	1400	7726057.80	596527.62	1400					26	1	571707.563	7706378.835	0
Shoe	AMG66_52	4506.382	24446.829	1400	7728027.32	636580.14	1400	4688.78	23080.19	7728175.82	635209.68	1.418106481	0.999801947	612257.14	7722918.32	0
Titania	AMG66_52	9200	20000	1400	7756022.60	601714.31	400	9769.77	21498.73	7756475.54	603252.38	-4.406268522	0.999996986	581066.667	7748386.377	-1000
Titania	MGA94_52	8600	19200	1335	7755651.28	601004.44	335	11000	23100	7757744.55	605076.97	-4.404392277	0.999934469	581201.999	7748551.615	1000
Twin	AMG66_52	5250	75000	1400	7726725.36	639159.69	1400					-13	1	564900.942	7738481.246	0
Tyred	AMG66_52	10000	10100	1400	7747368.26	610568.86	1400					-4.35	1	599739.466	7738163.141	0
Windy	AMG66_52	49953	58300	1400	7749953.00	658300.00	1400					0	1	600000	7700000	0



Note: Figure aligned to True North

Figure 9-1: Surface Geochemical Samples Location Plan

Table 9-2: Geochemical Sample Summary Table

Year	Drainage	Soil	Rock	Other	Comments
Not Recorded	-	241	254	2,068	
Pre-1976	-	-	11	-	
1976 to 1985	-	-	6	-	Limited exploration while land access was negotiated
1986 to 1995	255	14,191	9,723	1	Soil/lag samples and regional BLEG drainage samples. Multi-element analysis
1996 to 2005	-	47,883	2,027	-	Soil/lag samples with multi-element analysis
2006 to 2010	-	464	1	-	Mining operations are the focus of work
2016 to 2018	-	-	-	14,027	DSG sampling as part of a comprehensive geochemical re-evaluation of all areas of Newmont's tenure considered to be conceptually prospective

Note: BLEG = bulk-leach extractable gold

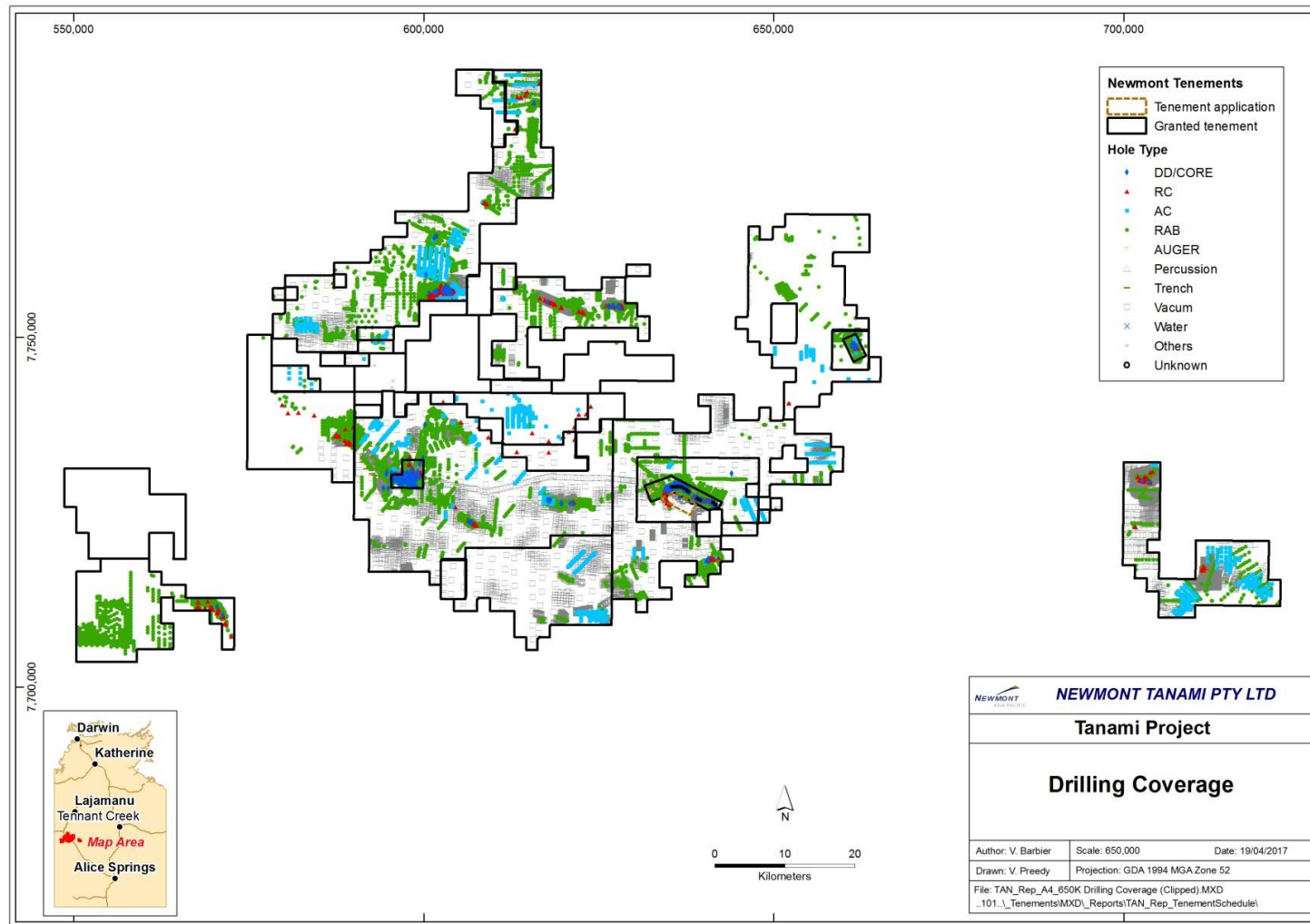
Shallow percussion, AC, RAB and vacuum drilling were used to obtain sub-surface geochemical samples for initial exploration or to follow up anomalies due to the mostly deep cover across the Project area. Collar locations are presented in Figure 9-2 and a summary of the number of non-core drill holes is presented in Table 9-3.

As with the surface geochemical samples, metadata on the sampling protocols applied to the shallow exploration drilling and analysis techniques is incomplete in the Project database and hence their reliability is uncertain. Most work was undertaken prior to 2009, with a total of 464 soil samples collected since this time. No recent shallow drilling has been undertaken on the Project.

9.4 Geophysics

Several regional and localized geophysical surveys have been carried out over the property in support of regional exploration activities. These have included; gravity, ground and airborne magnetics, seismic, radiometrics, and electromagnetics (EM). A listing of the surveys is presented in Table 9-4 and survey extents are presented in Figure 9-3.

Newmont also contributed to a regional seismic survey, which was undertaken in 2005 and covered part of the Project area.



Note: Figure aligned to True North

Figure 9-2: Sub-surface Geochemistry and Drilling

Table 9-3: Sub-Surface Drilling Summary

Year	Percussion	AC	RAB	Vacuum
Date not recorded	200	-	3,912	2,160
1976 to 1985	133	-	121	-
1986 to 1995	-	242	14,161	8,545
1996 to 2005	-	953	4,608	3,503
2006 to 2018	-	463	279	-

Note: Core and RC drilling is discussed in Section 10.0.

Table 9-4: Geophysical Survey Summary Table

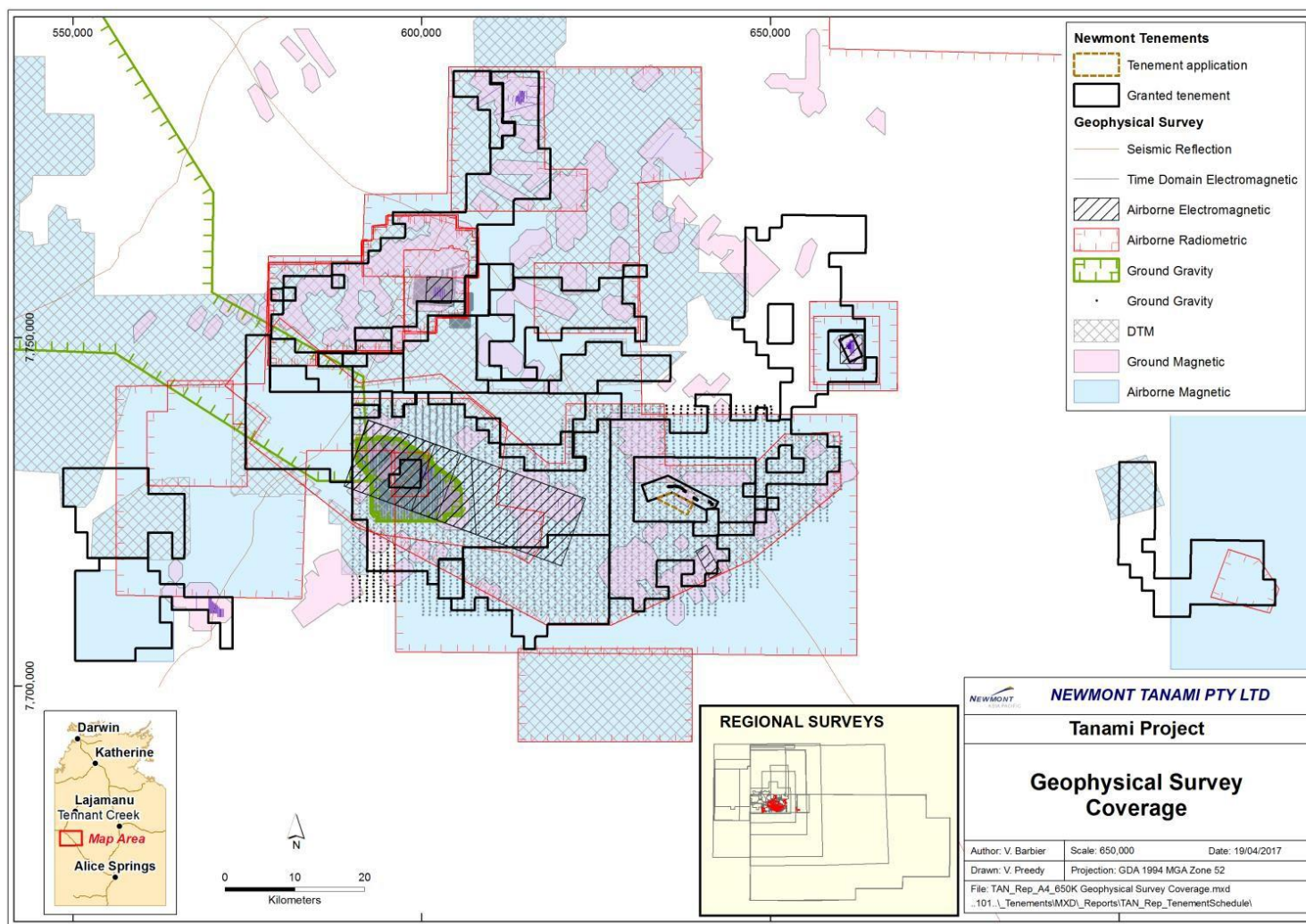
Year	Survey Type	Property	Survey Details	Contractor	Comments
2018	2D Seismic	DBS	3 lines of 2D seismic (~10 km each, 15 m shot spacing) for the purposes of assessing and designing a 3D seismic survey	HiSeis	For the purposes of assessing and designing a 3D seismic survey
2018	Airborne Gravity Gradiometry	Magellan-Pegasus	200 m line spacing, 60 m flight height	CGG	
2017	Airborne Magnetics/Radiometrics	Oberon	25 m line spacing, 25 m flight height	MagSpec Airborne Surveys	
2016	Ground gravity	DBS	400 x 100 m, with key areas identified infilled to 200 x 100 m. 4,497 stations on 64 x south to north trending lines. (GDA94) - (MGA94) Zone 52. Gravity Datum: Isogal 1984 / IGSN 71.	Haines Survey	Infill of existing Gravity dataset: from 1 x 1 km to a max of 400 x 100 m, with key areas infilled to 200 x 100 m. New gravity and existing mag data used to construct new "camp scale" geology interpretation and resultant map.
2012	Ground gravity	Titania	Projection GDA94 MGA52		In-house survey extending existing detailed data. 200 x 100 m stations
2012	Time domain EM	Titania	Dipole 100 m, static line spacing 200 m, projection GDA94 MGA52		
2005	Airborne magnetics/radiometrics	Minotaur	95 m sensor height, static line spacing 500 m		
2005	Seismic reflection	DBS	10 km line, 240 channel, shot/receiver spacing 10 m/10 m, split spread with roll on/off	ANSIR (Geoscience Australia), Fugro, HiSeis	In-house survey that aimed to image the shallow rock profile
2005	Ground gravity and magnetics	McFarlanes	Static line spacing 1,000 m	Fugro	
2004/03	Time domain EM	Titania & Symmington	MLEM Array, static line spacing 50 m		

Year	Survey Type	Property	Survey Details	Contractor	Comments
2002	Airborne magnetics/radiometrics/DEM	East Ptilotus	Static line spacing 50 m		Gridded data only (10 m cell).
2001	Airborne magnetics/radiometrics/DEM	DBS, The Granites	25 m sensor height, static line spacing 50 m	UTS Geophysics	UTS area name is "DBS Granites" (Survey name Schist_Hills_01).
2000	Airborne EM	Horden Hills, Titania, Windy Hill	75 m sensor height, static line spacing 200 m		Hoist EM
2000	Airborne magnetics	Tanami Final Stitch			Grid stitch
2000	Airborne magnetics/DEM	Tanami Area C - Area J	60 m sensor height, static line spacing 150 m	Kevron Geophysics Pty Ltd	
2000	Ground gravity	Tanami Central			Combined central data
2000	Ground magnetics	Tanami	Static line spacing 500 m–11 km		Regional data from the NTGS
1999	Airborne magnetics/radiometrics/DEM	Abbots Bore	20 m sensor height, static line spacing 200 m	UTS Geophysics	
1999	Time domain EM	Magellan and Officer Hill	Static line spacing 100 m		
1998	Airborne EM	Schist Hills	70 m sensor height, static line spacing 200 m		NewTEM,
1998	Airborne EM	Schist Hills	33 m sensor height, static line spacing 200 m		Hoistem.
1998	Airborne magnetics/radiometrics/DEM	Rabbi Flat, Smoke Hills & Redeye	40 m sensor height, static line spacing 100 m	UTS Geophysics	
1998	Airborne magnetics/radiometrics	Schist Hills	60 m sensor height, static line spacing 150 m	Aerodata	
1998	Time domain EM	Windy Hill	FLEM array, static line spacing 25 m		
1998	Time domain EM	Jumbuck	Static line spacing 100 m		
1997	Airborne magnetics	Golconda	20 m sensor height, static line spacing 20 m	UTS Geophysics	
1997	Airborne magnetics	Tanami Downs	20 m sensor height, static line spacing 150 m	UTS Geophysics	

Year	Survey Type	Property	Survey Details	Contractor	Comments
1997	Airborne magnetics/radiometrics/DEM	Legend	20 m sensor height, static line spacing 50 m	UTS Geophysics	
1996	Airborne magnetics	Callie Helimag	Static line spacing 50 m		
1996	Airborne magnetics	Muriel Range	25 m sensor height, static line spacing 100 m	UTS Geophysics	
1996	Airborne magnetics	Otter merge			Merge of many individual surveys; mostly Otter Resources surveys
1996	Airborne magnetics/radiometrics	Callie	60 m sensor height, static line spacing 100 m		
1996	Airborne magnetics/radiometrics/DEM	Central_GR OUP	Static line spacing 50 m		
1996	Airborne magnetics/DEM	Apertawong a	25 m sensor height, static line spacing 50 m	UTS Geophysics	
1996	Ground gravity	Homestead	Static line spacing 100 m		
1996	Ground gravity	Titania	Static line spacing 50 m		
1996	Ground gravity	Callie	Static line spacing 500 m		
1996	Time domain EM	Windy Hill	MLEM array, static line spacing 50 m	Solo Geophysics	
1995	Airborne magnetics	South Tanami & Tanami			Merged data: "South Tanami" area, and "Tanami"
1995	Airborne magnetics/radiometrics		60 m sensor height, static line spacing 100 m	Tesla Airborne Geoscience	
1994	Airborne magnetics/radiometrics	Windy Hill	60 m sensor height, static line spacing 100 m	Aerodata	
1994	Time domain EM	Anomaly 2	MLEM array, static line spacing 50–100 m		
1993	Airborne magnetics	Lennards	25 m sensor height, static line spacing 100 m and 800 m		
1993	Time domain EM	Challenger	MLEM array, static line spacing 50–100 m	Solo Geophysics	

Year	Survey Type	Property	Survey Details	Contractor	Comments
1986	Airborne magnetics/radiometrics	Granites A– D Grimwade	90–100 m sensor height, static line spacing 500 m	Geoterrex	Reprocessed by Pitt Research 1997–1998

Note: EM = electromagnetic; DEM = digital elevation model; FLEM = fixed loop electromagnetic; MLEM = moving loop electromagnetic



Note: Figure aligned to True North

Figure 9-3: Location Plan of Geophysical Surveys 1986 to 2018

9.5 Exploration Potential

9.5.1 Dead Bullock Soak Area

Mineralization within the Callie Deposit is open both down-plunge to the east and up-plunge to the west (although likely to a lesser extent).

Auron mineralization is open up-plunge to the west and down-plunge to the east at depth. Some exploration potential also exists immediately below the existing resource areas where mineralization is hosted within the Terran Beds. Site geologists are cautious with mineralization to date hosted within the Terran Beds as it appears discontinuous relative to the typical Auron host-units.

There is potential to further expand known mineralization in the Federation area down-plunge and potentially further down the southern limb of the anticline to the south. Recent exploration success has defined a repetition of the Federation mineralization within the Auron Stratigraphy at depth, now identified as the Liberator orebody.

Future drill targets may include known mineralization hosted within the more iron-rich units of the SHIM or Orac Members.

9.5.2 The Granites

The Granites area has not been a focus for exploration activity for several years as the area was considered mature and lacking exploration potential.

Reprocessed aeromagnetic data in 2013 was interpreted to suggest the mineralized main host unit may continue up to 2 km south of the drilled mineralization in the Quorn and Ivy areas. A shallow RAB drilling campaign in 2002 was considered to have closed off mineralization in the area after intersecting only marginal gold grades. The main host unit was not intersected during this program, however, and as such a program of 19 RC drill holes completed across five lines to a maximum depth of 100 m was devised to test it. This program was completed in 2014, and although it was interpreted that the main host unit was intersected in the drilling no further potential for economic mineralization was recognized.

At the time of reporting the only plan for further exploration at The Granites involves a compilation of the fragmented historical data into a consistent 3D environment.

9.5.3 Regional Exploration

Regional exploration activities across the extensive Project area was substantial and ongoing from the mid-1980s; however, despite considerable expenditure no significant exploration success was made after 1998.

The Tanami Regional Framework Study (TaRFS) was completed in 2006 after reviewing all available data and indicated that the Tanami Region retained excellent potential for discovery of Callie-style mineralization. Following the completion of a target ranking exercise, 20 targets were determined for further work. Testing of these targets resulted in generally disappointing outcomes.

In late 2007, following the completion of a revised prospectivity analysis several targets were selected for follow-up. Initial success in 2009 targeting Callie-style mineralization under the previously defined Oberon deposit saw it become the focus of exploration work for the following years. A single drill hole completed in the Windy Hill area failed to intersect any mineralization. The cessation of the Oberon exploration

project in 2013 marked the beginning of a hiatus in greenfields exploration at the Project.

The current regional exploration strategy being applied aims to capitalize on recent learnings and success at DBS, and the vast bulk of current expenditure is planned for the area immediately around the mine, on EL2367. Exploration planned for this area in the foreseeable future includes geological mapping, the collection of gravity data, soil sampling and diamond drilling.

Other regional prospects that are likely to see expenditure in the near term is Oberon (EL23662) and Officer Hill (EL23150). Exploration planned will likely include soil sampling, the collection of gravity data and possible diamond drilling.

9.6 Comments on Exploration

In the opinion of the QP:

- The exploration programs completed to date are appropriate to the style of the deposits and prospects within the Project;
- Although moderate exploration potential exists in greenfield locations within the Project area the significant challenges and costs posed by the deep cover has meant the probability of success is low and exploration resources are preferentially being directed to brownfields deposit extension targets;
- The Project retains significant brownfields exploration potential, and additional work is planned.

10.0 DRILLING

Approximately 19,240 core and RC drill holes are recorded as having been completed on the Project. A further 39,280 shallow percussion, AC, RAB and vacuum holes have been completed to support exploration.

A summary of the number and type of drill holes is presented in Table 10-1. The core and RC drilling subset completed on the DBS deposits is presented in Table 10-2. Table 10-3 shows the drill holes which are used to support the Mineral Resource estimates.

Figure 9-2 shows the locations of the drill hole collars of all drill types completed across the entire Project area. Figure 10-1 to Figure 10-4 to present the locations of the drill holes for The Granites, DBS, Windy Hill and Oberon respectively.

10.1 Drilling Methods

Over the Project history, several drilling companies have been used. Where these are known, they are presented in Table 10-4.

Core drilling since 1997 has primarily been completed at NQ/NQ2 (47.6 mm core diameter). PQ (85 mm core diameter) and HQ (63.5 mm core diameter) have been used in initial portions of deep surface drilling, areas of poor ground and for geotechnical drill holes. Minor conventional LTK60 (43.9 mm core diameter) core has been drilled for steep underground up-holes that exceed +30° dip.

RC drilling has not been routinely undertaken at the Project since 2005 and is not included in the estimate supporting any current Mineral Resources. Surface RC drilling was typically completed with a 140 mm (5½") face-sampling hammer; other bit sizes were used occasionally. In 2014, a short exploration program totaling 19 drill holes were completed using a 133 mm (5¼") hammer.

Percussion, AC, RAB and vacuum drilling were extensively used to obtain sub-surface geochemistry samples for initial exploration due to the deep cover across most of the Project area. These samples are not used to support Mineral Resource estimates.

Table 10-1: Summary of Drill Holes Completed

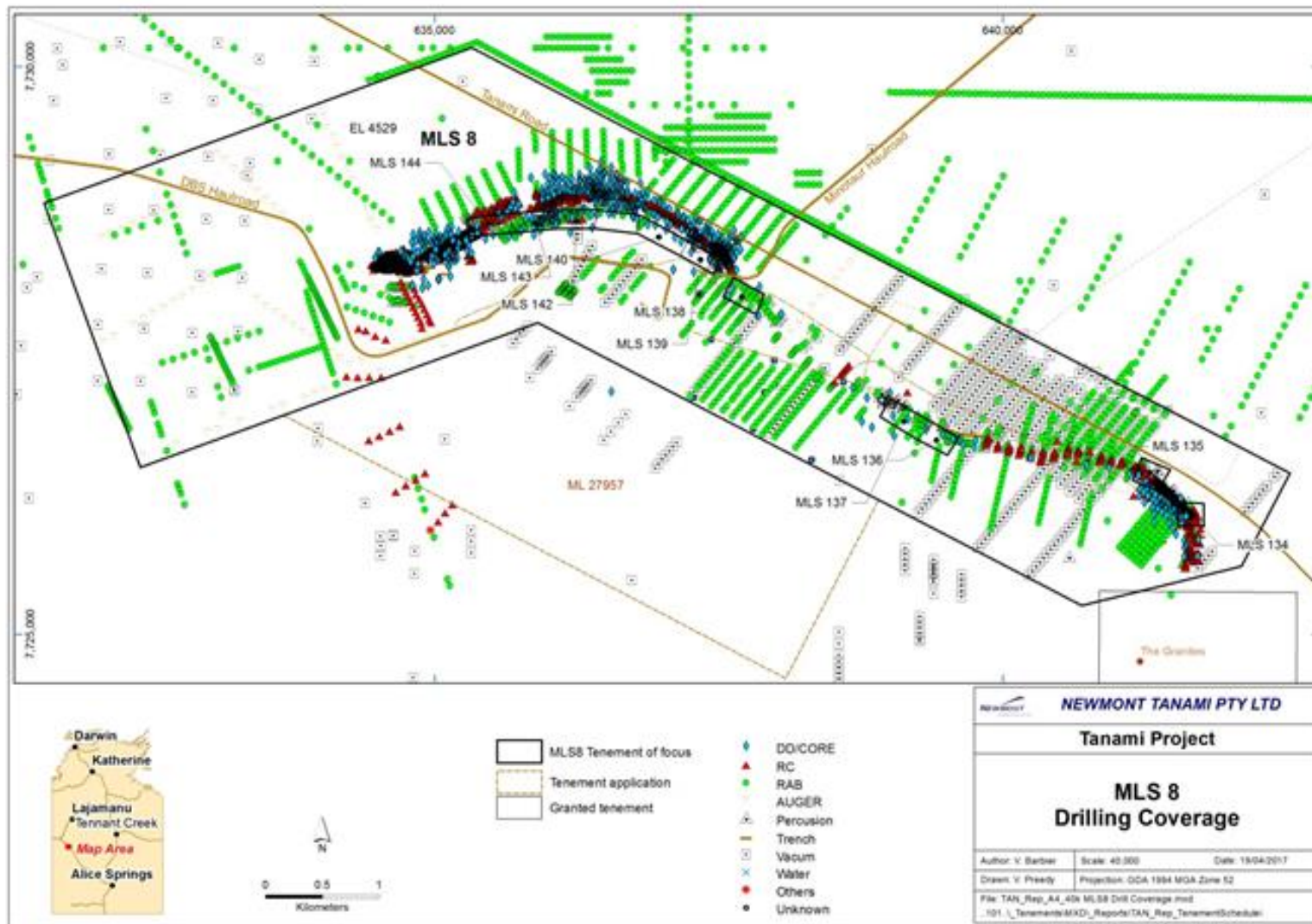
Year	Surface Core	Underground Core	RC	Water or Other
No date recorded	462	103	2,663	127
Pre-1976	6	-	-	-
1976 to 1985	67	565	194	-
1986 to 1995	719	72	911	44
1996 to 2005	285	2,429	527	26
2006 to 2018	285	10,432	35	-

Table 10-2: Summary of Core and RC Drill Programs

Company	Years	Surface RC		Surface Drilling (Core)		Underground Drilling (Core)	
		Holes	Meters	Holes	Meters	Holes	Meters
Not recorded	-	3,396	214,317	754	165,672	739	63,102
North Flinders	1989 to 1997	658	41,317	599	171,112	651	28,525
Normandy NFM	1998 to 2002	164	12,602	125	103,402	1,270	168,551
Newmont	2003 to 2018	112	12,052	346	332,869	10,941	1,954,872

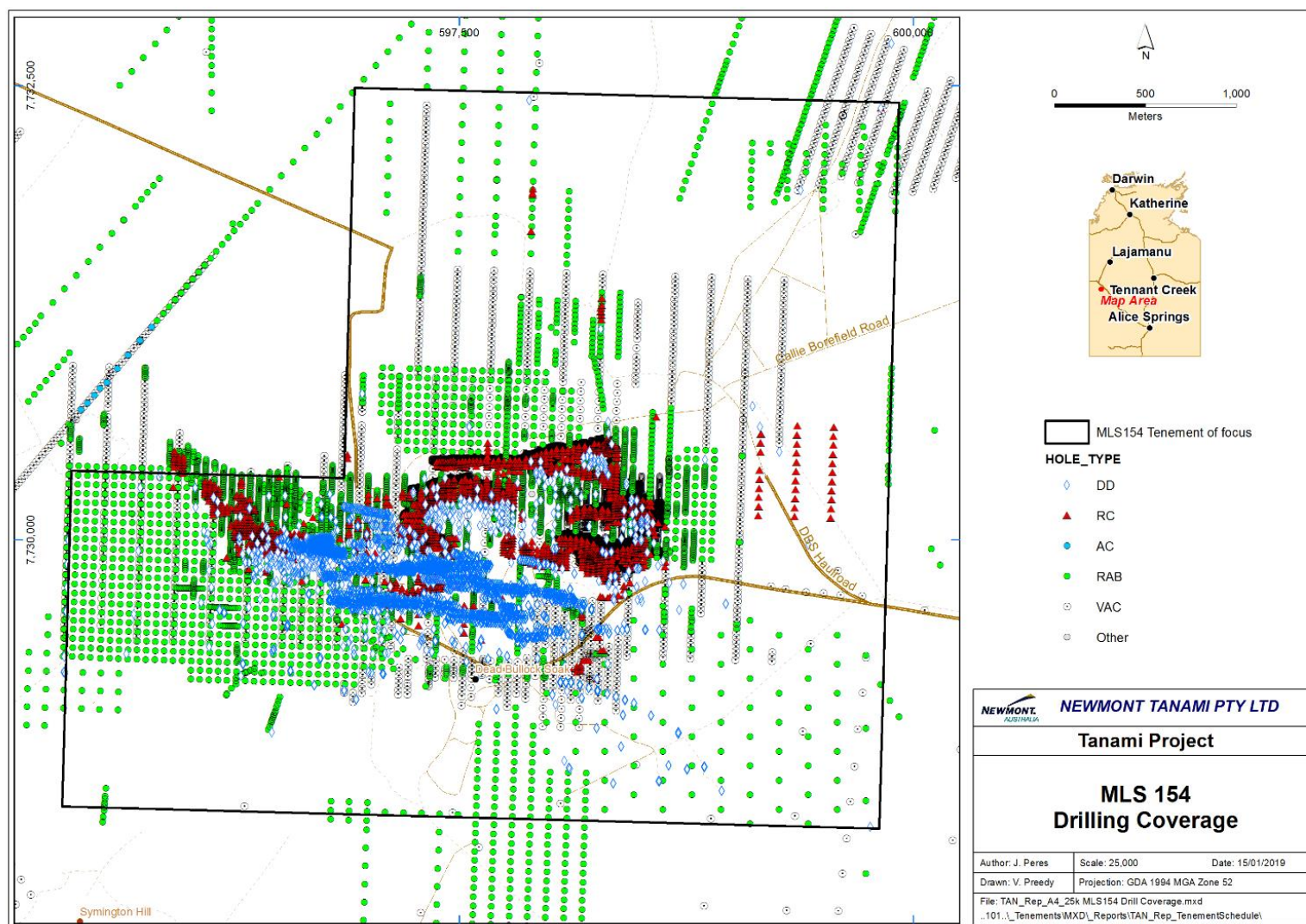
Table 10-3: Drill Data Used in Mineral Resource Estimation

Deposit / Estimate	Operator	Surface Drilling (Core)		Underground Drilling (Core)	
		Holes	Meters	Holes	Meters
Kerril South South Mineral Resource 2013 May	(Not Recorded)			54	9,040
	North Flinders Mines	43	30,112	0	0
	Normandy NFM	33	30,001	734	112,980
	Newmont			663	107,840
Callie Grade Control 2016 April (up-plunge of 60852.5mE)	(Not Recorded)			95	13,633
	North Flinders Mines	159	53,190	112	12,097
	Normandy NFM	1	576	1,780	203,746
	Newmont	1	850	3,511	556,742
Auron Grade Control 2018 March (59400.5mE - 60600.5mE)	(Not Recorded)			984	204,971
	Normandy NFM			2	550
	Newmont			2,346	394,095
Federation Mineral Resource 2016 September	(Not Recorded)			8	1,068
	North Flinders Mines	31	23,454		
	Normandy NFM	75	83,838	3	742
	Newmont	93	150,367	648	173,095
Callie Deeps Mineral Resource 2016 September (downplunge 60852.5mE)	Newmont	48	92,240	350	98,589
Auron Mineral Resource 2018 September	(Not Recorded)	31	33,333	3,328	502,807
	North Flinders Mines	20	11,379	4	561
	Normandy NFM	16	17,573	39	5,416
	Newmont	65	71,045	3,240	713,687



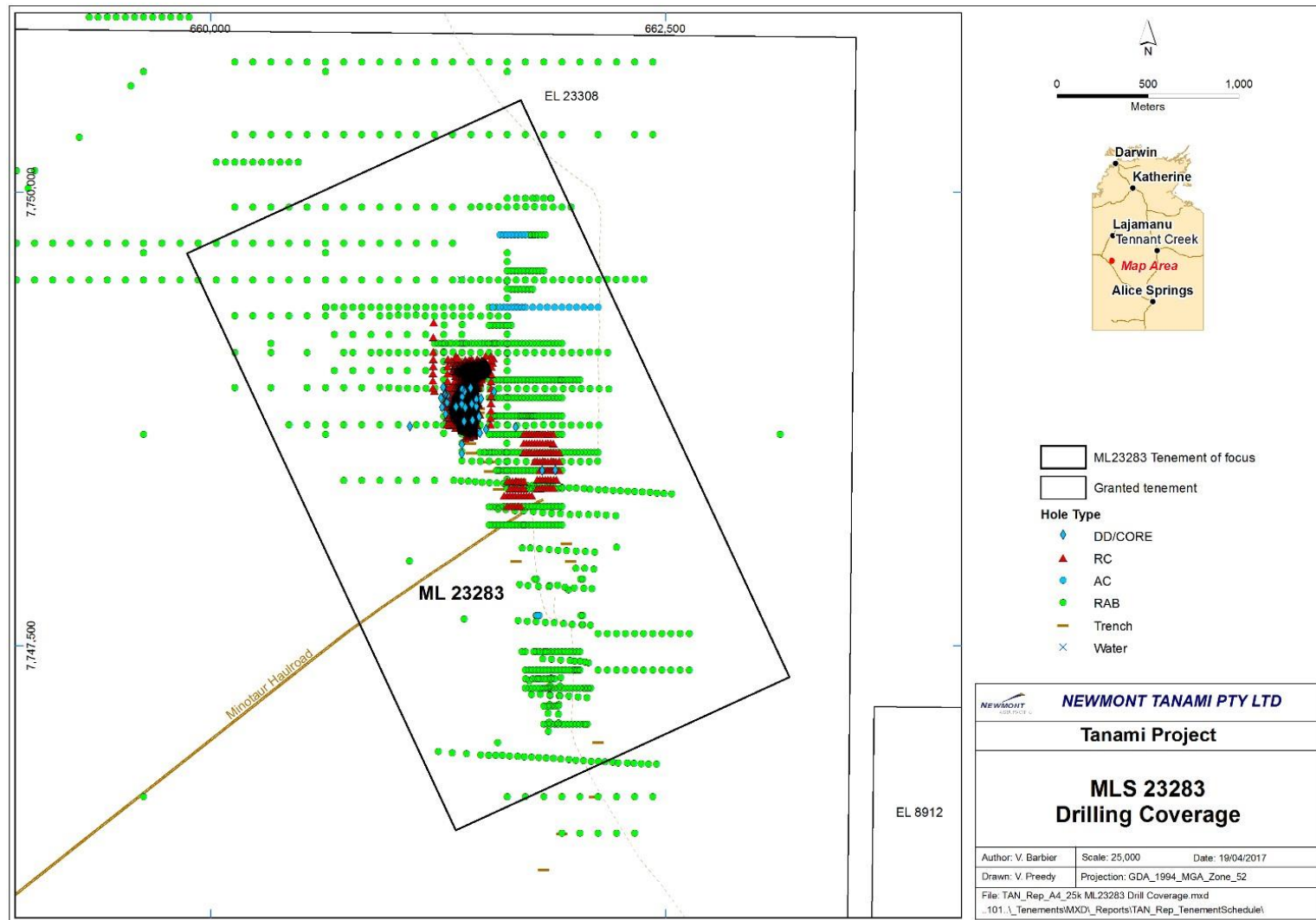
Note: Figure aligned to True North

Figure 10-1: Drill Collar Location Plan for MLS8 (The Granites)



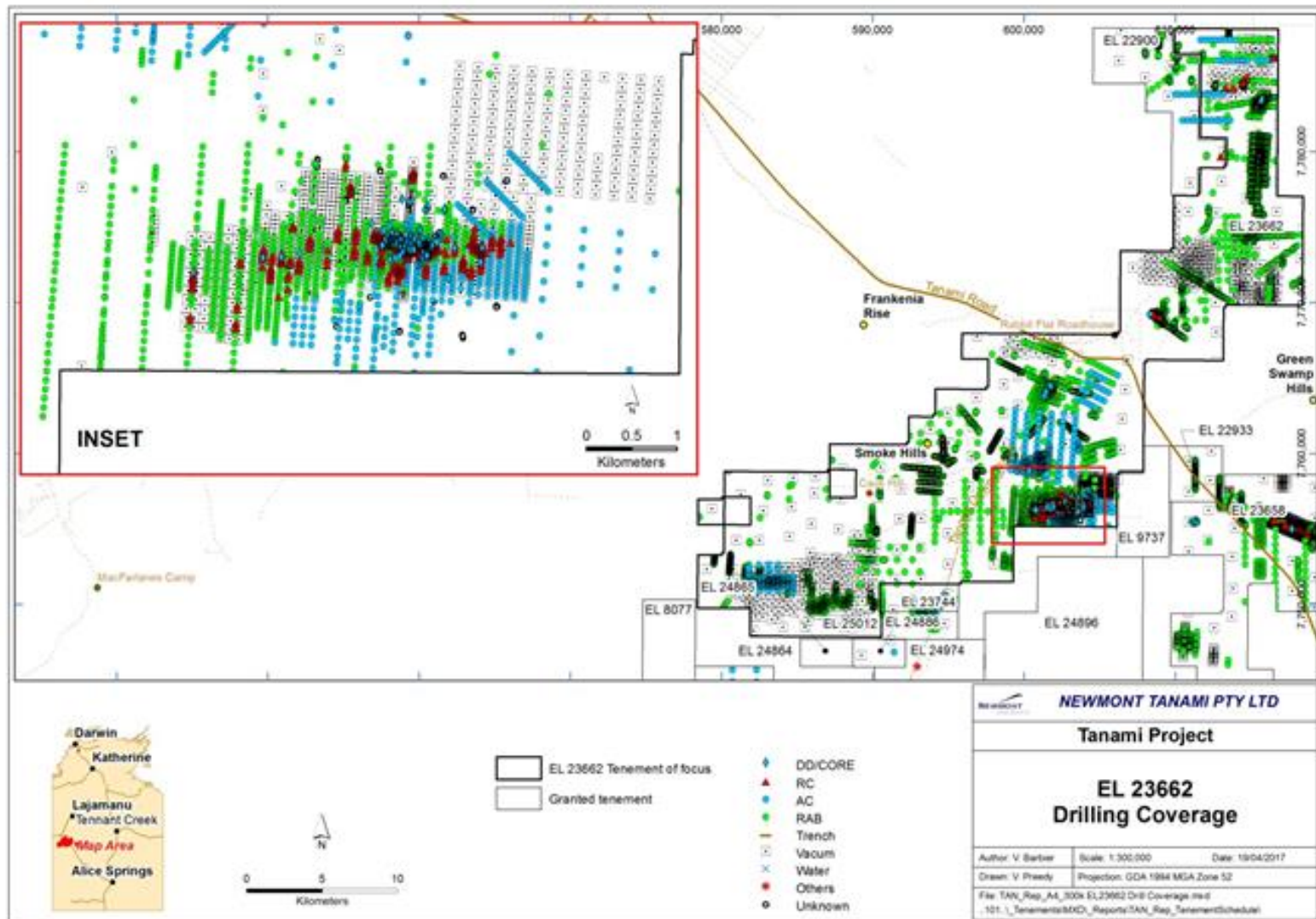
Note: Figure aligned to True North

Figure 10-2: Drill Collar Location Plan for MLS154 (DBS)



Note: Figure aligned to True North

Figure 10-3: Drill Collar Location Plan for MLS23283 (Windy Hill)



Note: Figure aligned to True North

Figure 10-4: Drill Collar Location Plan for ELS23662 (Oberon)

Table 10-4: Drilling Companies Used 1997 to 2018

Drilling Company	Period	Drill Rigs	Core Sizes	Comments
Henry Walker Eltin	1997 to 2010	LM75/LM90/Bazooka	NQ/NQ2/HQ2	Underground drilling at Callie
Centralian Drilling	1998 to 2002	Unknown	NQ/NQ2/HQ2	Surface exploration DBS
Mosslake Drilling	2001 to 2002 and 2010 to 2011	UDR1000/UDR1200	NQ/NQ2/HQ/HQ2/PQ	Surface exploration DBS/Oberon
Drill Corp	2002 to 2003	UDR1000/UDR1500	NQ/NQ2/HQ3	Surface exploration DBS
Layne Drilling	2003	Unknown	NQ/NQ2/HQ/HQ3/PQ	Surface exploration DBS
Barmenco	2005	LM75/LM90	Unknown	Underground drilling at Callie
Major Drilling	2004 to 2012	UDR1000/UDR1200/UDR1500/UDR3000	NQ/NQ2/HQ/HQ3/PQ/4.5"	Surface exploration DBS/Oberon
Swick Drilling	2007 to 2018	LM90 (H104 carrier)	NQ/NQ2/HQ2/LTK60	Underground drilling at Callie
Sanderson Drilling	2011 to 2013	UDR1000/UDR1200	NQ/NQ2/HQ/HQ3/PQ	Surface exploration Oberon/DBS
Pinnacle Drilling	2014	Sandvik DE880	RC 5.25"	Surface Exploration The Granites
DDH 1 Drilling	2018	UDR200/EDR200 /Long wear 900 Kilowatt (kW)/Longwear 120 kW	BQ/NQ/NQ2/HQ/HQ3/PQ/LTK60	Surface Exploration at DBS, Underground Exploration at Callie

10.2 Geological Logging

10.2.1 North Flinders Mines and Normandy NFM Programs

RC

For RC drilling, the main observations recorded information pertaining to the sample regolith affinity, lithological characteristics, geology, mineralogy, alteration type, weathering intensity and color was collected for each meter. A comments column allowed for further description. All logging has been recorded onto paper log sheets and manually entered into the database.

Core

Normandy NFM staff completed both geological and geotechnical logging.

Logging was undertaken on a single system with whole core logging at half meter intervals and one-meter intervals in regolith.

The main geological observations recorded on drill logs included lithology, texture and grain size, sulfide type and content, vein type and abundance, and alteration type and intensity. Basic geotechnical logging includes rock quality designation (RQD), core

recovery, bedding and non-bedding breaks, rock strength, micro-roughness, infill type and state of weathering. Depth, alpha and beta angles were recorded for valid oriented structural features, including bedding, veining, foliation, joints, contacts, shears and faults.

All logging was recorded onto paper log sheets and manually entered into the database.

Most of the Normandy NFM core was photographed onto 35 mm slides, although most of these still exist they are not catalogued or sorted and are not generally available.

10.2.2 Newmont Programs

RC

Only limited RC drilling has been undertaken by Newmont. The observations recorded used a similar process to the predecessor companies outlined in Section 10.2.1.

Core

Logging for the Newmont drill programs includes the following steps;

- Core is laid out, cleaned if required and core blocks are inspected;
- For oriented core the core is re-assembled, and an orientation line drawn on the core;
- Meter marks are drawn on the core with permanent marker and core recovery is recorded;
- Measurements of the RQD are taken;
- Measurements of the magnetic susceptibility are taken from surface exploration core;
- Core is geologically logged, and sampling intervals selected using direct capture into tablet computers;
- Measurements of the rock density are taken from selected intervals using the water displacement method;
- Core is photographed and subsequently sampled;
- The data from the core logging are uploaded to the database incorporating automated validation checks.

The drill logging scheme used for geological logging includes the following:

- Weathering – oxidation intensity, oxidation style, weathering event depth (i.e. base of oxidation);
- Lithology – regional stratigraphy, local stratigraphy, lithology, texture, grain size, sedimentary structure, younging, graphite, TiO, color and comments;
- Structures – structure type, intensity, structure name, structure orientation (alpha and beta) and comments;
- Alteration – the alteration type, style and intensity;
- Mineralization – the ore mineralization types, their percentage abundance and style;
- Veining – vein density, texture, orientation are recorded;

- Visible gold – host vein type/style, gold grain size, side of the orientation line and any other comments.

Depending on the purpose of the drilling the detail captured varies. Table 10-5 presents the logging captured for each main drill purpose. The logging system also captures the drill hole identifier, drill type and core size and the name of the logging geologist.

Prior to July 2008, data were logged into a Microsoft Excel® spreadsheet on a tablet PC. The logging spreadsheet had inbuilt code validation. Following completion of the log, further validation was undertaken on site before data were forwarded to centralized database administrators for transfer into the Structured Query Language (SQL) database.

From July 2008 to late 2013, data were logged into a Microsoft Access database program called Logchief. Logchief has inbuilt validation routines that are run prior to transmission of the data to the database team in Perth.

From late 2013, geology logs entered are directly uploaded from Newmont's in-house Visual Logger software to Newmont's standardized SQL Global Exploration Database (GED) software interface which includes validation routines.

10.2.3 Geotechnical Logging

All drilling of the 'Tanami Operations Optimisation Project' areas in 2008 included detailed oriented logs of geotechnical criteria (recovery, RQD, fracture density, number and orientation of joint sets, rock strength, fracture morphology, fracture fill, etc.). These data were captured on a separate Microsoft Excel spreadsheet by site geotechnical engineers. Prior to, and since 2008, drill holes were selected for geotechnical logging on an ad-hoc basis at the hole planning stage by geotechnical engineering staff. This equated to typically one to two drill holes on each of the resource-spaced drill sections.

For all other drilling, field technicians captured the RQD, fracture counts and core recovery as per site procedures.

Each core drill hole is photographed, and the photographs are stored on the DBS server.

10.3 Recovery

Sample recovery is recorded for all core drilled at the Project (both surface and underground). In slightly weathered to fresh rock, recovery is typically 100%, with minor loss adjacent to major fault zones. Excellent core recovery has been maintained on all exploration and underground sections. No preferential core loss has been detected.

Table 10-5: Geological Logging Requirements by Drill Purpose

Drill Type	Weathering	Lithology	Structures	Alteration	Mineralization	Veining	Visible Gold
Grade Control	Not applicable	Logged to the nearest 1 m. Includes all basic lithology properties	Structure descriptions recorded	Not logged	Logged to the nearest 1 m interval	Ore stage vein density recorded	Logged

Drill Type	Weathering	Lithology	Structures	Alteration	Mineralization	Veining	Visible Gold
Exploration	Logged on all surface drill holes	Logged to the nearest 0.1 m interval. Includes additional fields of sedimentary structures, younging and color	Structure descriptions and orientations to be measured where possible.	Logged to the nearest 0.1 m interval	Logged to the nearest 0.1 m interval	Vein density and orientation to be measured where possible. One vein per vein set per tray to be recorded where possible. Texture and mineralogy collected on off-lease holes	Logged

10.4 Collar Surveys

10.4.1 North Flinders Mines and Normandy NFM Programs

Locations of all underground and most surface drill hole collars were picked up by the mine and exploration surveyors. Instruments have included a theodolite and total station instruments.

10.4.2 Newmont Programs

Almost all Newmont drill hole collar surveys are performed by Newmont staff surveyors with collars established by total station or differential global positioning system (GPS) instruments, this includes recent surface RC holes at The Granites. The orientation of the collars of most underground drill holes are picked up by surveyors using a rod inserted in the collar. On the rare occasion that underground drill hole collars cannot be picked up, a collar position is interpolated based on the surveyed position of other holes on the same drill sections.

10.5 Downhole Surveys

10.5.1 North Flinders Mines and Normandy NFM Programs

Most of the North Flinders Mines and then Normandy NFM RC and core drill holes were inclined. Downhole surveys were taken every 30 m to 50 m on all core and RC drill holes using Eastman tools.

Downhole surveying was rarely completed to determine true dips of the mostly vertical percussion, RAB, and vacuum drill holes drilling in regional exploration.

10.5.2 Newmont Programs

All Newmont core drill holes were down-hole surveyed prior to 2006 initially at 15 m and then on 30 to 50 m intervals using Eastman camera tools. From 2006 surface drill holes were down-hole surveyed utilizing electronic Reflex EZ-Shot tools. The EZ-Shot is a solid-state electronic, magnetic single-shot instrument with stated accuracy of $\pm 0.5^\circ$ azimuth and $\pm 0.2^\circ$ dip. Underground drilling transitioned to the electronic Reflex tools in 2007–2008. While deep diamond drill programs were being undertaken between 2006 and 2009 gyro surveys were completed on an ad hoc basis on the parent holes.

In 2012 a gyro tool was made available permanently on site, due to issues encountered with magnetic ground while drilling the up-plunge portions of the Auron deposit. This tool was used whenever magnetic ground is encountered and as a periodic QA/QC check against the magnetic single shot surveys. In 2015 the gyro tool was replaced by a strain gauge-based tool known as a DeviFlex. In 2016 the DeviFlex was replaced by a North Seeking solid state gyro system tool known as an Access Champ Gyro tool.

The downhole survey data is downloaded electronically from the survey tools on a weekly basis and validated prior to entry into the database.

Surface RC drilling at The Granites in 2014 used digital survey tools with downhole survey points taken at 30 m downhole and at bottom of hole.

10.6 Metallurgical Drilling

Dedicated drilling is not completed for metallurgical testing. The site geologists, in consultation in the site metallurgical staff, select a number of planned core drill holes for testwork at the design stage. These drill holes are designated with an “M” prefix on their drill hole ID and the core is cut and only half core sampled. Composite samples are selected from these drill holes following the return of the sample assay results.

10.7 Sample Length/True Thickness

Drill holes are designed whenever possible parallel to the mine grid. Design criteria are in place with minimum and maximum allowable drilling angles for dip and strike to ensure that drill holes intersect the mineralization as perpendicular as possible; reported mineralized intercepts are typically longer than the true thickness of the mineralization.

Figure 10-5 and Figure 10-6 provide examples of the drilling directions in relation to the orientation of the lithologies and mineralization. Table 10-6 provides selected examples of drill intercepts to illustrate the range of grades and thicknesses that may be encountered.

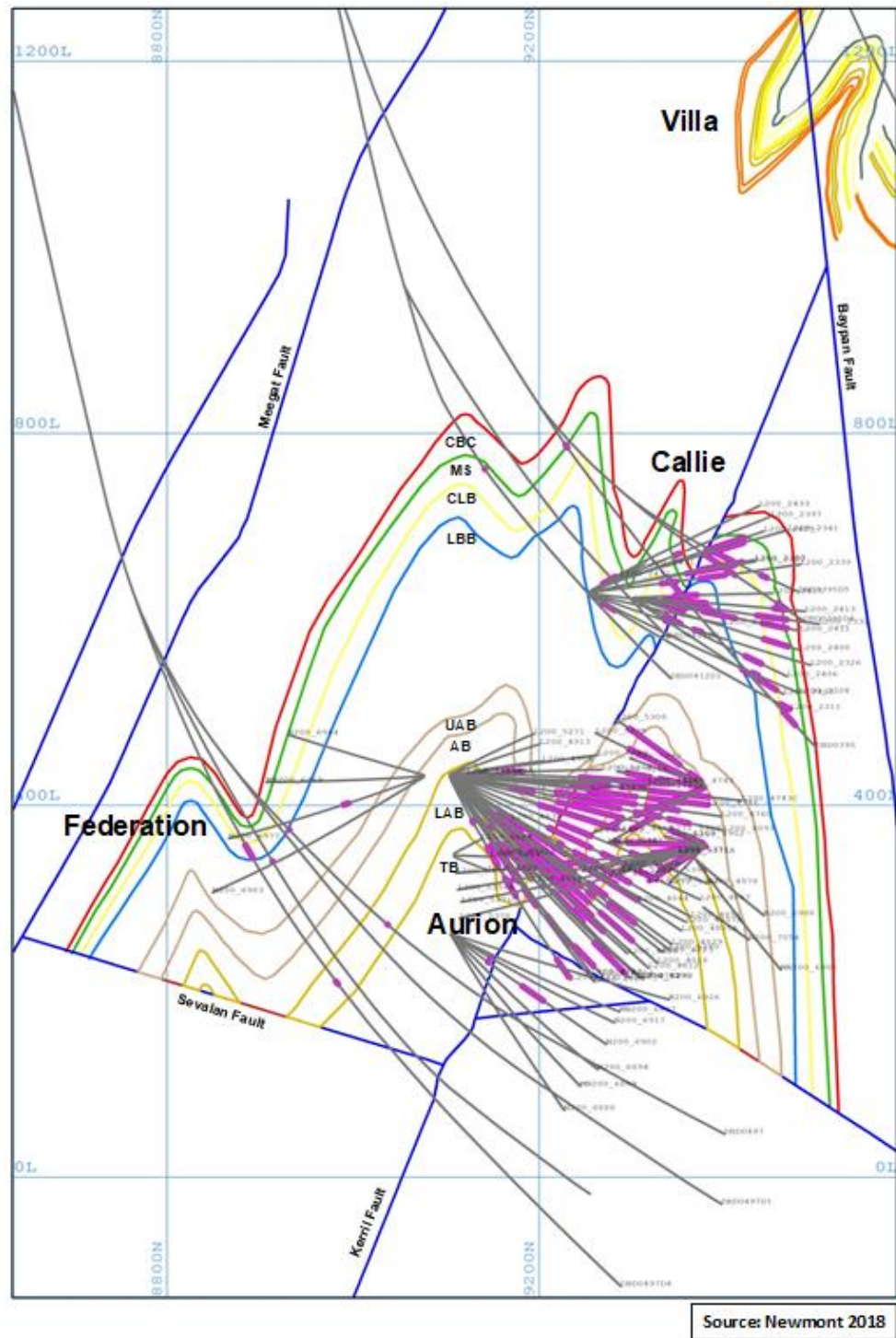


Figure 10-5: Example Drill Cross-sections from the DBS Area – 60200mE (looking west)

Note: Grey lines are drill traces, purple lines indicate the mineralized intercept encountered by the drill hole. Lithological codes are explained in Table 7-1

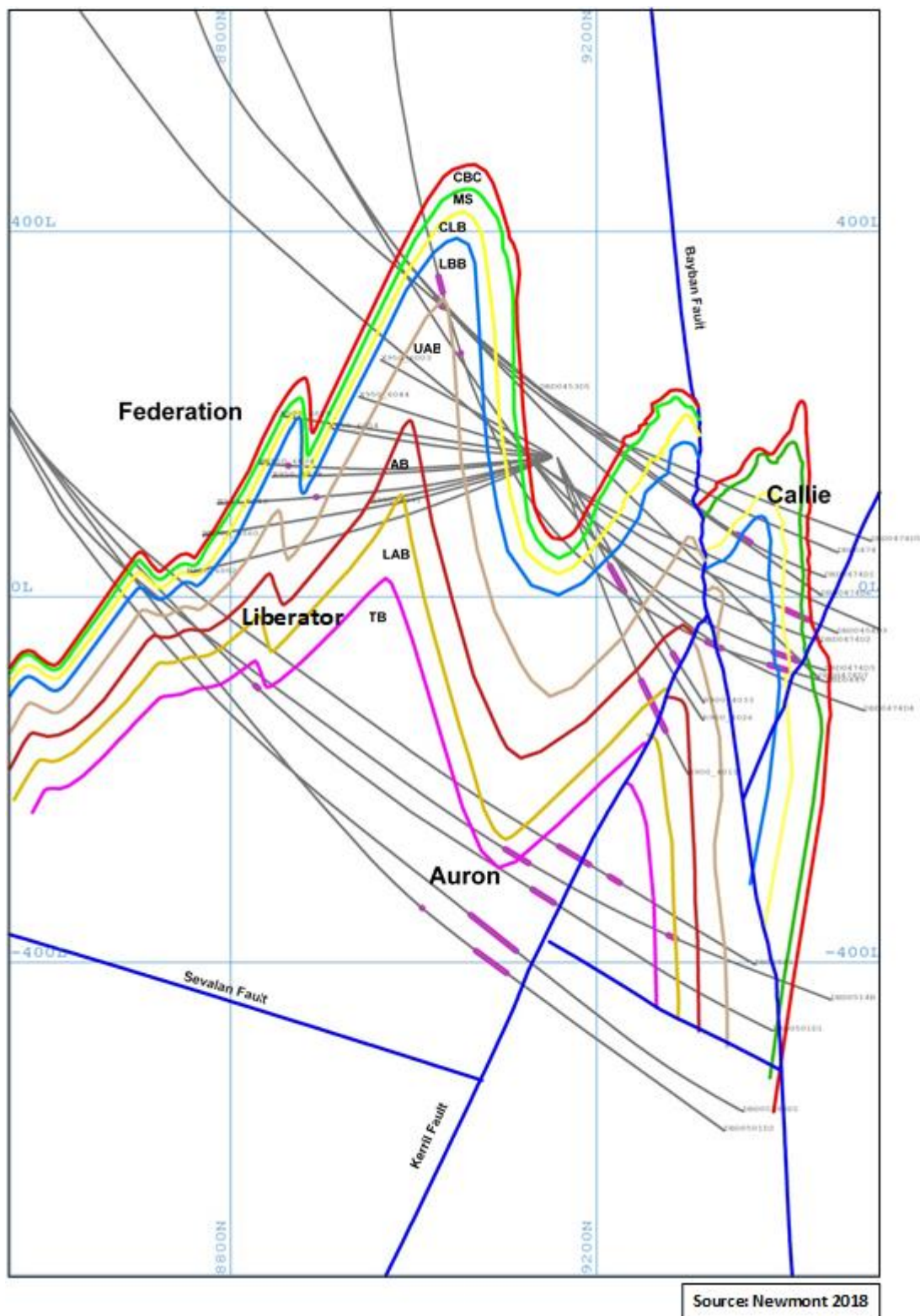


Figure 10-6: Example Drill Cross-section from the DBS Area – 60900mE (looking west)

Note: Grey lines are drill traces, purple lines indicate the mineralized intercept encountered by the drill hole. Lithological codes are explained in Table 7-1

Table 10-6: Example Drill Intercept Summary Table

Hole ID	Easting	Northing	Elevation	Azimuth	Dip	From	To	Length (m)	True Width (m)	Au (g/t)	Comment
Dead Bullock Soak - 60200mE											
DBD0395	60151.94	8941.31	1397.51	13.2	-71	678	682	4.0	3.2	27.96	Federation - Drilled from Surface
					including	678	679	1.0	0.8	110.00	Federation - Drilled from Surface
					and	862	891.3	29.3	24.1	7.04	Callie - Lantin - Drilled from Surface
					and	1049	1069	20.0	17.2	2.55	Callie - Lantin - Drilled from Surface
DBD0395D4	60151.94	8941.31	1397.51	13.2	-71	869	908	39.0	35.6	9.57	Callie - Lantin - Drilled from Surface
					including	874	876	2.0	1.8	77.60	Callie - Lantin - Drilled from Surface
					including	898	900	2.0	1.8	79.20	Callie - Lantin - Drilled from Surface
					and	962	974	12.0	11.0	20.44	Callie - Lantin - Drilled from Surface
					including	966	967	1.0	0.9	235.00	Callie - Lantin - Drilled from Surface
DBD0395D5	60151.94	8941.31	1397.51	13.2	-71	867	897	30.0	28.2	9.19	Callie - Lantin - Drilled from Surface
L200_2326	60200.57	9252.64	627.61	359.5	-20	88	95	7.0	6.6	8.72	Callie - Lantin - Drilled from UG
					and	120	129	9.0	8.5	6.41	Callie - Lantin - Drilled from UG
L200_2332	60200.58	9252.55	627.9	2.8	-6.5	77	132	55.0	50.1	10.64	Callie - Lantin - Drilled from UG
					-6.5	183	216	33.0	30.3	4.13	Callie - Lantin - Drilled from UG
L200_2339	60200.56	9252.51	628.3	3.2	7.1	80	96	16.0	13.1	4.39	Callie - Lantin - Drilled from UG
DBD0497	60395.85	8588.7	1399.63	4.5	-79	1106.1	1107	0.9	0.8	367.00	Federation - Drilled from Surface
					and	1258.2	1259	0.8	0.8	677.00	Auron - Drilled from Surface
L200_4732	60200.31	9103.69	430.64	2.6	-64.8	53	56	3.0	2.2	49.05	Auron - Drilled from UG
					including	53	54	1.0	0.7	145.00	Auron - Drilled from UG
					and	86	108	22.0	16.0	19.92	Auron - Drilled from UG
L200_4732	60200.31	9103.69	430.64	2.6	-64.8	124	147	23.0	17.2	33.48	Auron - Drilled from UG
					including	143	147	4.0	3.1	187.27	Auron - Drilled from UG
					and	228	251	23.0	17.8	4.82	Auron - Drilled from UG
					including	250	251	1.0	0.8	95.90	Auron - Drilled from UG

10.8 Comments on Drilling

In the opinion of the QP:

- Geological logging of drill core (surface and underground) and RC chips meets industry standards for gold exploration and Mineral Resource definition;
- Collar surveys have been performed using industry-standard instrumentation;
- Downhole surveys were performed using industry-standard instrumentation;
- Recovery data from core drill programs are acceptable;
- Geotechnical logging of drill core meets industry standards for established underground operations;
- Drill orientations are generally appropriate for the mineralization style, and have been drilled at orientations that are optimal for the orientation of mineralization for the bulk of the areas of the deposits (refer to Figure 10-5 and Figure 10-6);
- Drill hole intercepts as presented in Table 10-6 appropriately reflect the nature of the gold mineralization. The table demonstrates that sampling is representative of the gold grades in the deposits, reflecting areas of higher and lower grades;
- The quantity and quality of the lithological, geotechnical, collar and downhole survey data collected in the exploration and infill drill programs completed by Newmont and the preceding operating companies are sufficient to support Mineral Resource and Mineral Reserves estimation;
- No material factors were identified with the data collection from the drill programs that could materially impact the accuracy and reliability of the results or that would affect Mineral Resource estimation.

11.0 SAMPLE PREPARATION, ANALYSES AND SECURITY

Newmont and previous operators' staff throughout the duration of the Project have been involved with, or responsible for the following:

- Sample collection;
- Core splitting;
- Delivery of samples to the analytical laboratory;
- Sample storage;
- Sample security.

All analytical procedures that support Mineral Resource estimation, including sample preparation and analysis, were performed by independent analytical laboratories.

11.1 Sampling Methods

11.1.1 Geochemical Sampling

Normandy NFM Programs

Shallow percussion, AC, RAB and vacuum drilling was heavily applied to obtain sub-surface geochemical samples for initial exploration due to the deep cover and scarce outcrop. Detailed sampling protocols and sizes are generally not recorded. Most samples consisted of soil and lag.

Newmont Programs

As for the Normandy NFM programs above, Newmont geochemical sampling relied heavily on shallow drilling techniques for sample collection. Other than the drill method sampling details have not been recorded. Newmont conducted a soil sampling program at DBS. Sample collection methodology is proprietary to Newmont.

11.1.2 Underground Sampling

No information is available on face sampling procedures undertaken at The Granites by North Flinders Mines.

Assays from underground rock sampling are not used in the current Mineral Resource estimation dataset.

In the initial years of the Callie underground mine development, rock sampling from underground was trialed to increase the sample data in the Mineral Resource database. Three types of underground sampling were trialed with limited success:

- Drawpoint grab sampling;
- Quartz vein / host rock sampling;
- Face chip channel sampling.

By 1998, it was concluded that any sample taken in isolation was not a reliable indicator of grade and that drill samples were less subject to sampling bias and allowed a better approximation of lower and higher-grade ranges. Routine rock sampling from underground was discontinued.

11.1.3 RC Drilling

Sampling procedures for RC drilling have varied over the life of the Project and detailed records of the subsample size and splitting methodology applied to individual drill holes has not been consistently recorded. In 2001, it was reported that:

"RC samples are split off the rig using a two or three stage riffle splitter to produce a 1.5 to 2.0 kg sample."

The only RC drilling completed at the Project in the last 10 years was a program of 19 drill holes completed at The Granites in 2014. Sampling in this program involved the use of a Sandvik DE880 multi-purpose drill rig operated by Pinnacle Drilling. The drilling was completed using a 133 mm (5¼") face-sampling hammer. Sampling was undertaken on meter intervals, with sample weight varying from roughly 25 kg to 35 kg, depending on the degree of oxidation of the rock. Meter intervals of drill spoil were deposited directly into plastic bags from the cyclone. Sub-samples of composited 4 m intervals weighing roughly 3 kg were then produced for geochemical assay from the spearing of the 1 m intervals.

No RC drill holes are used in the current Mineral Resource estimate.

11.1.4 Core Drilling

Normandy NFM Programs

Underground drill core was sampled and logged at a nominal 1 m interval, with a 0.5 m minimum sample length and a 1.5 m maximum sample length. Prior to late 1998, the core was routinely sawn and sampled as half or three quarter-core; however, from late 1998, whole-core sampling was standard for all infill drilling.

Surface drill hole samples varied in length from 0.5 m to 2.0 m. Surface exploration diamond cored holes were sawn and mineralized intervals half-sampled with the remaining half retained as a permanent record.

The standard procedure used by Normandy NFM Exploration for core samples in areas of unlikely mineralization was to scan sample (0.1 m of half-core at 0.3 m intervals; averaged for a 3.0 m scan sample length). This procedure was replaced in the mine areas by quarter coring and sampling 2 m intervals in areas of potential mineralization. Known barren units (such as the Coora Dolerite) were not sampled routinely.

Newmont Programs

Underground drill core is sampled and logged at a nominal 1 m interval, with a 0.3 m minimum sample length and a 1.3 m maximum sample length. Whole core sampling is routine for underground drill holes although one drill hole per section is sawn and half core sampled to allow a permanent record of the targeted deposits to be retained. Furthermore, occasionally drill holes are sawn, and half core sampled when samples are required for metallurgical or other testwork.

At the exploration and Mineral Resource conversion stage sample intervals are broken across lithological boundaries while infill drill holes are nominally sampled to the meter marks.

Not all core is sampled. The aim of the underground infill drilling programs (Mineral Resource conversion and grade control) at Callie is to provide a 12.5 m by 12.5 m sample spacing. At the collar of a drill fans there is a high clustering of data. As a cost saving measure, selected infill holes are not sampled from the collar, but from a point

lower down so there is a minimum 10 m distance between adjacent drill hole samples. The steepest up-hole and downhole, and the two drill holes closest to the horizontal are sampled in their entireties. Routinely core containing visible gold is not sampled. This procedure has been in place since early 2001.

Surface exploration drilling is half core sampled.

For whole core sampling the core is broken within the core trays by field technicians using a geological hammer and the sample placed in pre-numbered calico bags. The samples are collected in polyweave bags which are subsequently placed into bulk transport cages for shipment.

For half core sampling the core is sawn in half following the orientation line where present. One half of the sample is placed into the sample bag and the other returned to the core tray and retained. The same side of the half core is consistently selected for assay.

Sample shipments are prepared and dispatched to Adelaide via Alice Springs twice a week by road and rail. Shipment information is communicated to the laboratory and the database administrators.

Recent soil sampling conducted by Newmont at DBS was analyzed for multiple elements at Newmont's laboratory in Colorado using proprietary methods.

11.1.5 Metallurgical Sampling

Metallurgical testwork is performed on selected half-core intervals of drill core typically from exploration drilling. On occasion supplementary intervals are selected from infill drilling if additional testwork is required.

11.2 Density Determinations

11.2.1 Normandy NFM Programs

The frequency of density determinations varied by area and program. Typically, every tenth meter was tested from underground core while the surface exploration core had two or three 200 mm samples tested per core tray. Determination was performed by site personnel using the water displacement method.

11.2.2 Newmont Programs

Prior to 2016, specific gravity measurements were undertaken on all core drill holes. A 20 to 40 cm interval is selected approximately every 5 m. The procedure for measuring the specific gravity is as follows.

- Select a piece of core 20 to 40 cm in length;
- Weigh the core in air on the scale;
- Weigh the core in water in the bucket beneath the scale.

The technician records the interval, weight in air and the weight in water into a computer and the specific gravity value is automatically calculated for the interval.

Since 2009, approximately 5% of density determinations are nominated for a check determination at the assay laboratory's sample preparation facility.

In 2016 a modified process was put in place where the following changes were made:

- All diamond drill core drilled on grids spaced from wide exploration holes down to 50 m apart along strike and 25 m apart on dip have one specific gravity measurement taken for each stratigraphy unit or every 10 m, whichever is shorter;
- All underground diamond drill core drilled on grids 25 m apart along strike and 25 m apart on dip have one specific gravity measurement every 10 m;
- Infill grade control drill holes do not have specific gravity measurements taken;
- Standard weights are used before measurements of each drill hole each day to identify drift in the measuring equipment;
- Data is captured electronically directly from the scales.

11.3 Analytical and Test Laboratories

Prior to 1999, all exploration RC and core samples were assayed by Analabs in Adelaide. Accreditations at the time the assays were performed are not known.

ALS has performed sample preparation and fire assay on all geology samples between 1999 and 2015. The sample preparation is primarily conducted at their Alice Springs facility. Sample pulps are forwarded to the ALS Perth facility for analysis. Occasional batches were prepared and analyzed in the ALS facilities in Adelaide and Orange due to an overload of work. The ALS laboratory in Kalgoorlie acted as the umpire or check laboratory until 2009. The ALS facilities held ISO17025 accreditation.

Bureau Veritas (BV) has performed all sample preparation and fire assay on all geology samples from 2015. The sample preparation and fire assay is conducted at their Wingfield facility in Adelaide. Solutions are couriered to the Thebarton laboratory in Adelaide for analysis.

Independent Assay Laboratories (IAL) have completed umpire check assays since 2009 in Perth; IAL does not hold an ISO accreditation.

Metallurgical testwork has been completed at several laboratories, but primarily by Newmont Metallurgical Services in Denver and the AMMTEC Laboratory, Perth.

11.4 Sample Preparation and Analysis

11.4.1 Normandy NFM Programs

Prior to 1998 a series of sample preparation and analysis techniques were applied. Most typically samples were dried, crushed to 10 mm prior to being riffle split and a subsample being pulverized to pass 75 µm.

Following a review in 1998 which revealed an unacceptably poor precision, the preparation process was changed so that the entire core sample crushed and pulverized prior to a sub-sample being split for analysis.

Surface exploration drilling was routinely assayed using an aqua regia (AQ) digest with AAS finish. Any RC results >1 g/t and <8 g/t were fire assayed, and any core results between 2 g/t and 8 g/t were fire assayed. Both RC and core samples with AQ assays greater than 8 g/t, and any samples with visible gold were screen fire assayed. Where multiple assays were completed for a sample, screen fires took precedence over fire assays, and fire assays took precedence over aqua regia assays.

Underground core samples were assayed using a fire assay with a 50 g charge and samples containing visible gold were screen fire assayed.

Fire assays were carried out with an AAS finish.

11.4.2 Newmont Programs Prior to 2010

A similar sample preparation protocol to that discussed in Section 11.4.1 was applied by Newmont. Following drying the entire sample was crushed prior to being pulverized in an LM5 to 85% passing 75 μm . Barren flush material was inserted in the pulverize at the beginning of each job and after samples specified by Newmont (typically containing visible gold). The pulp was then subsampled prior to final analysis.

Gold analysis was completed using a standard fire assay with a 50 g charge with an AAS finish. Samples with logged visible gold were analyzed with a screen metallic fire assay on a 1,000 g sample split, which was screened to 75 μm . Duplicate assays were completed on the undersize fraction and an assay on the entire oversize fraction.

Initial testwork on the Auron deposit showed S and As were of interest for metallurgical reasons. From July 2009 samples from the Auron host units were separately analyzed for S and As. Following a four-acid complete digest the resulting solution was then analyzed by inductively coupled plasma–mass spectrometry (ICPMS).

11.4.3 Newmont Programs 2010 to 2014

Post 2010, minor modifications were made to the sampling protocol. Following drying the entire sample was crushed to -12 mm prior to being pulverized in an LM5 to 85% passing 75 μm . Where a sample exceeded 2.5 kg, the pulverization was completed in two to three increments followed by a re-homogenization step.

Subsampling of the LM5 bowl was conducted with a modified scoop with one or two passes across the diagonals of the bowl to full depth. Four splits of the pulp are generated to allow efficient processing of subsequent analysis. Unused pulp is returned to site and stored for exploration drill holes. For infill drilling, pulps are retained for several months then disposed of.

Following a review of the analysis methodology completed in 2009 the analysis technique was modified so that screen fire assays for logged visible gold were replaced with six 50 g fire assays. Samples that had an initial fire assay result above 2 g/t Au would be repeated a further five times as single fire assays above 2 g/t Au did not achieve an acceptable precision.

The weighted average sample grade of these six fire assay results was calculated by the laboratory and was the grade used for Mineral Resource estimation purposes. Testwork demonstrated this would produce similar assay precision at a lower cost and reduce turnaround time. Fire assays were carried out with an AAS finish prior to May 2013 where upon it was changed to an ICP finish following an extensive period of comparative testwork.

The five additional fire assays were reduced to three in October 2014 to lower assay costs while still maintaining an acceptable level of precision. The four fire assay results were used as a weighted average for Mineral Resource estimation.

Samples from the Auron host units continued to be separately analyzed for S and As utilizing a four acid complete digest. The resulting solution is then analyzed using an ICP finish.

11.4.4 Newmont Programs Post-2014

Samples that have an initial assay greater than 2 g/t have an additional three fire assay with ICP finish with the calculated weighted average over the four results used in

estimation. In April 2015, any samples that have an initial assay greater than 20 g/t trigger four fire assays with gravimetric finish to calculate a weighted average of the additional four samples only. Testwork was conducted on Au prills by both gravimetric and acid digest/ICP to verify that the gravimetric method was acceptable prior to implementation.

Newmont's current sampling preparation and assay flow chart is presented as Figure 11-1.

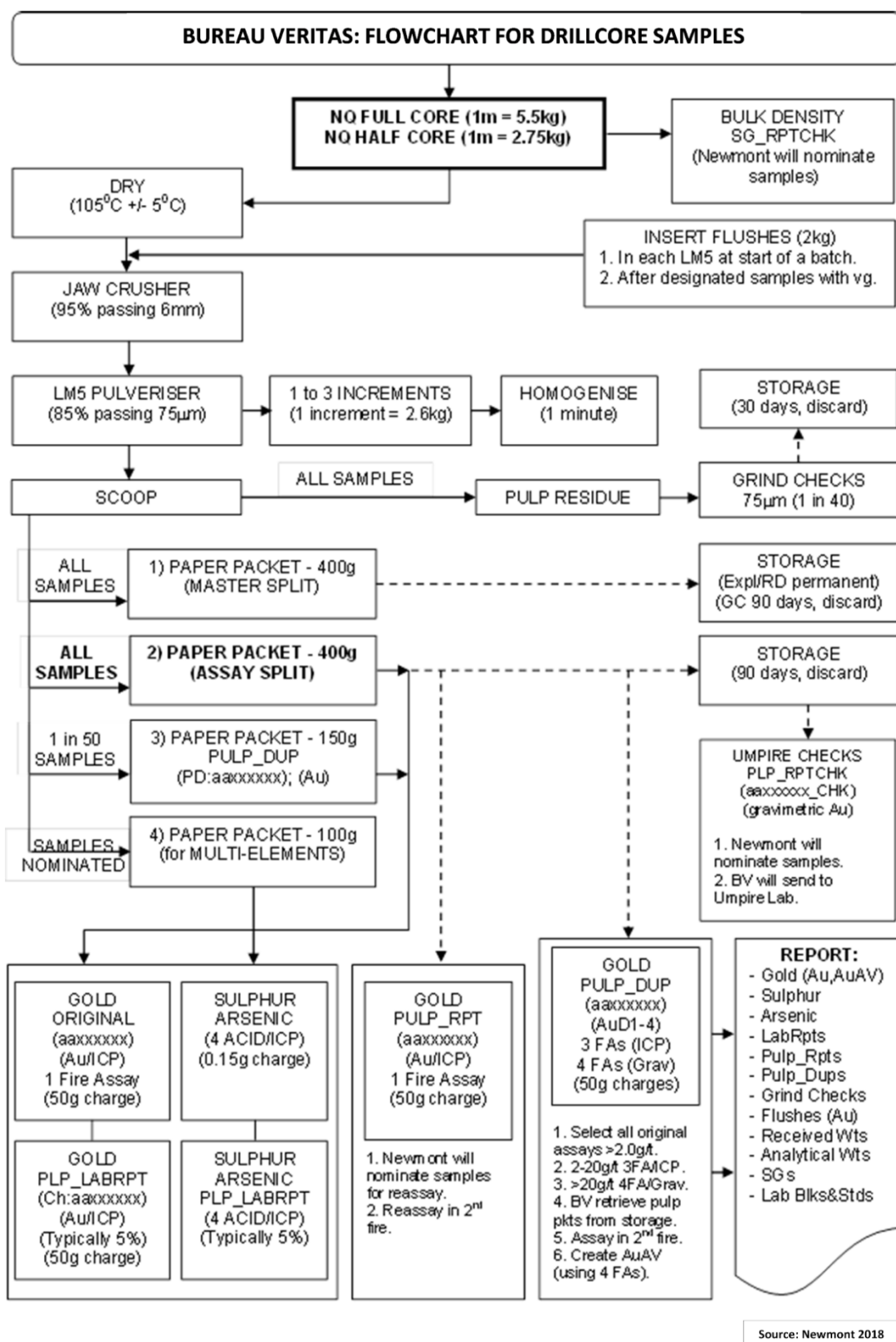


Figure 11-1: Newmont Assay Procedure – Post-2014

11.5 QA/QC

11.5.1 Programs Prior to 2004

Prior to 2004 only sporadic QA/QC data are recorded within the database. There are references in annual Mineral Resource reports to QA/QC processes being in place in various forms from 1998. Duplicate assays were reportedly completed approximately every 20th sample, with standard reference materials (SRMs) and blanks inserted every 10 to 30 samples.

Details of the standards used prior to 2004 are not available, although a QA/QC report from 2007 referred to early pre-2004 standards as being produced from Callie pit ore. Variations in results indicated that these were not mixed to a satisfactory level of precision or accuracy, and that they had been discontinued.

Since 1999, the Project has used ALS for its core assaying. ALS has an internal quality control process including a consistently applied system of SRMs, pulp duplicate samples and check assay submittals.

11.5.2 Newmont Programs Post-2004

A comprehensive QA/QC system had been in place at the project since 2004 with blank and SRMs routinely inserted into the sample stream.

Sizing checks are made in the sample preparation process. Duplicate assays are routine, as are check re-assays undertaken by an external laboratory. Regular un-announced laboratory inspections are undertaken by Project staff.

The QA/QC of batches loaded into the database is analyzed weekly and is aided by automated reports. If necessary, batches are re-analyzed (from pulp or reject material) together with the remaining standard controls. Monthly QA/QC reports are produced to allow the identification and rectification of longer-term biases and trends in results.

11.5.3 Standards

Since 2004 commercially-produced certified reference material (CRM) standards have been inserted in each assay batch. These were sourced from Ore Research and Exploration Ltd. until 2007, and then subsequently from Geostats Ltd. Due to the extremely coarse nature of the Callie gold mineralization, producing reliable standards from the operation is not feasible. Whenever possible CRMs are selected that best match the project mineralogy. Typically, four to six standards are in use at any one time with a grade range up to 50 g/t Au, these are regularly switched. Typically, standards are inserted at a rate of one per 25 samples.

With the introduction of regular assaying for S and As in 2009, a series of multi-element standards were introduced.

Standard results are reported by the laboratory, together with the primary assay results and are captured by the database during the normal importing routine.

Upon receipt, the results for the standards are compared weekly by the relevant geologist on site with the certified values using automatically generated database reports. If a standard is reported outside of three standard deviations from the certified value it is investigated and if required following a standardized workflow, the batch is re-analyzed. The standardized workflow determines if all or part of the batch requires re-assay and a subsequent log records all failed standards and the action taken to flag the associated risk for Mineral Resource estimation.

11.5.4 Blanks

Until mid-2012, two types of blank material were inserted with core samples submitted to the laboratory:

- A certified blank standard. This is a pulp packaged in the same way as the standards;
- 'Controls' which were blank core samples sourced from local Coora Dolerite. These were used to monitor contamination in the sample preparation area.

Between 2012 and 2018 only the certified blank standard has been included in batches sent to the laboratory.

In 2016, coarse blank material sourced from the barren Bunbury Basalt were introduced and replaced the certified blank standards after a period of testing.

Where any failure in the performance of blanks may have occurred, the disputed result is investigated for incorrect assignment, swaps and smearing. The dispute process follows a standardized flow of re-assay, then review. If a sample swap has occurred, then a performance review of the remainder of the batch is required, followed by swap identification. If smearing is suspected and is sustained, this indicates a calibration issue, which requires flagging with the laboratory.

11.5.5 Flushes

Two types of quartz flushes were used to clean pulverizing equipment; pre-batch flushes and mid-batch flushes. They were all assayed to monitor the level of contamination.

- Pre-batch flushes: Flushes are run through each pulverizing mill before commencing a batch of samples. The flushes are used to thoroughly clean the equipment. Prior to 2015, 6 flushes were completed, one for each LM5 at a pulverizing station. This has been reduced to 4 at Bureau Veritas due to the 4 LM5 machines at each workstation;
- Mid-batch flushes: Mid-batch flushes are run through the pulverizing mill immediately after samples with visible gold. The locations are flagged by the geologist submitting the samples. Prior to mid-2012, where visible gold is observed to be <5 mm in size, one quartz flush was requested while any visible gold >5 mm is followed by two quartz flushes. Since 2012, all visible gold occurrences are requested to be followed by two quartz flushes.

11.5.6 Duplicates

A pulp duplicate is nominated at one per 50 samples by the logging geologist to check the repeatability of the assay results. Due to the practice of whole core sampling and entire sample pulverization, no coarse duplicates are completed.

11.5.7 Grind Checks

Grind checks are performed by the laboratory to ensure that sample pulverization was achieving 85% passing 75 µm. Approximately 2.5% of samples are checked. This is compared weekly between the primary commercial laboratory and umpire laboratory to ensure grind performance and assay precision at specified ranges. This data is monitored for deviations between laboratories and in the event of trending deviations, flagged to the corresponding laboratory for potential re-adjustment and calibration.

11.5.8 Umpire Laboratory Check Assays

Umpire laboratory pulp repeat checks are undertaken at the frequency of 2% i.e. one in every 50 samples. Samples are selected weekly for umpire checks with samples from various predetermined grade ranges.

In 2013, umpire laboratory grind checks were introduced to verify the grind check results from the primary laboratory.

The umpire laboratory is QA control on the main commercial laboratory. Comparisons for accuracy, precision and grind size are compared between both laboratories monthly and quarterly to examine trends in commercial laboratory performance. Where material deviations occur, this is communicated to the commercial laboratory and examined at audit stage for process control. Equipment calibration is reviewed and adjusted until deviations are controlled or within acceptable limits.

11.6 Databases

Prior to 2001, the Project maintained a series of Microsoft Access® based databases, with separate databases for each of the underground, exploration and open pit operations.

From the beginning of 2001 the drill hole data on site was consolidated and captured in an SQL database that used DataShed® software. These data include drill hole collars, down hole surveys, assays and geological descriptions. Standard database management techniques were used that limited access and user rights to ensure data integrity. The drill hole and assay data was managed from a central database group in the Newmont Perth office.

In December 2013, the Project migrated drilling data from the DataShed® system to the Newmont-proprietary GED SQL database system. GED is a highly normalized custom-built system designed specifically for Newmont's needs.

Primary original documents, logs, down-hole surveys, core photographs, and assay certificates are cataloged and stored onsite. Digital copies are stored on networks drives that are routinely backed-up.

Prior to the implementation of GED, sampling details were captured in a Microsoft Access® based data entry form. Each file contained all the samples for one drill hole and includes the capture of sample ID interval, sample type, analytical suite, assay priority and sampler name. The files were forwarded to Perth office for entry into the database. Sample intervals are currently generated per assay batch through the Newmont-proprietary eSample web browser and automatically saved to the GED database once marked for dispatch.

Assays are received from the laboratory in text format via e-mail. On loading, the file structure, sample identifiers and analytical standards are validated. Assay certificates are received electronically.

Collar survey data are received from the surveyors in electronic form and is validated by the site geologists prior to being auto-loaded to the database through a stored SQL procedure. Validation of drill hole identifiers and survey readings are completed within this stored procedure and files are rejected if the validations fail.

Downhole survey data are downloaded electronically directly from the survey tools and after validation on site, are auto-loaded to the database through a stored procedure. Validation of drill hole identifiers, downhole survey data and survey priority are completed within this stored procedure and geologists are notified of files that have been rejected due to validation errors. An automated validation procedure flags potentially excessive deviation which is automatically reported to the site geologists for checking.

Geotechnical data are stored in a Microsoft Excel® file that is auto-loaded to the database via stored procedure. Files that have failed validation are rejected and geologists notified.

A final validation and signoff process is completed by Project geology staff for each drill fan on its completion. This involves a validation of all collar, downhole survey and logging data.

11.7 Sample Security

No information is available on the sample security protocols practiced by Normandy NFM; however, it is expected that it was substantially similar to current processes.

Newmont maintains formal procedures for sample shipping, which include:

- Samples are delivered directly to the core shed by the drilling contractors at the completion of each drill shift. The extremely remote nature of the site and its access restrictions prevent unauthorized access to the core and subsequent samples;
- Samples are individually bagged at the core shed, then placed in larger bags and finally bulk shipping cages;
- Samples from each drill hole are identified as separate shipments, so that all values on an assay certificate relate to the same drill hole;
- Shipments leave the site for Alice Springs usually no more than one or two days after sampling is completed. Sample dispatch documentation is sent directly to the lab via electronic mail;
- Formal consignment notes are generated for each sample dispatch. The samples can be tracked from site to a secured yard belonging to the transport operator, Northline, in Alice Springs;
- Northline load the bulk crate containers into containers under video surveillance and the containers are then locked and sealed with tamper-proof seals;
- Containers are transported to the Alice Springs rail yards and loaded directly onto rail cars for transport to Adelaide;
- In Adelaide the containers are offloaded from the rail cars and transported to the secured Northline yard where container seals and locks are inspected;
- The crates are then unloaded under video surveillance and loaded onto the delivery truck to the Bureau Veritas assay laboratory at Wingfield;
- Bureau Veritas receipt samples and reconcile against the sample dispatch documentation. Any discrepancies are reported to Newmont;
- Bureau Veritas facilities are secured and monitored with only authorized personnel having access to samples which are typically stored for only short periods prior to being processed;
- All parties including Bureau Veritas, the Newmont delivery provider (Toll) and Northline maintain detailed chain of possession logs which are available to the project staff upon request;
- Bureau Veritas maintains detailed records of all sample preparation and assay steps that can be viewed by Project personnel;
- Final analysis results are transmitted directly to the Newmont central database team in electronic format and are auto-loaded following validation of the files. If any discrepancies

are identified the assay laboratory investigates and re-issues assay certificates electronically.

11.8 Sample Storage

Core from the various drilling programs is stored in the open in core “farms” adjacent the main mining areas. Due to the whole core sampling practiced at DBS, core storage is relatively limited.

Currently, samples are shipped off site to preparation laboratories in Adelaide. Rejects are discarded, and pulps are stored at the laboratory for three months before either being returned to site for storage (for exploration drilling) or discarded.

11.9 Comments on Sample Preparation, Analyses, and Security

In the opinion of the QP:

- Sample collection, preparation, analysis and security for drill programs provide data that are sufficiently bias and error free to support Mineral Resource and Mineral Reserves estimation;
- Sample preparation for core samples has followed similar procedures since 1998; all currently reported resources are based on data collected since this time;
- Sample preparation and analysis for core samples has used industry-standard methods for the time, and has been performed by independent laboratories;
- The Project has assay measures in place to deal with the coarse nuggety gold that is a feature of the DBS mineralization;
- Drill programs included insertion of blank, duplicate and standard reference material samples;
- QA/QC program results do not indicate any material assay biases or problems with the analytical programs;
- Data are subject to validation, which includes checks on surveys, collar co-ordinates, lithology data, and assay data. The checks are appropriate, and consistent with industry standards;
- Data audits have been conducted, and indicate that the sample collection and database entry procedures are acceptable;
- Chain-of-custody and sample security procedures consist of filling out sample submittal forms that are sent to the laboratory with sample shipments to make certain that all samples are received by the laboratory. The remote nature and access control procedure on site prevent unauthorized access to the samples.

12.0 DATA VERIFICATION

12.1 Site Visits

Mr. Doe, the QP, visited the Project between 13 and 17 April 2018.

During his site visit to the Project, Mr. Doe inspected the underground operations, and viewed the process plant and associated general site infrastructure, including the current tailings storage facility (TSF) operations. While on site, he discussed aspects of the operation with site-based staff and assessed the knowledge and abilities of the site staff to carry out their duties as required. These site discussions included the overall approach to the mine plan, anticipated mining conditions, selection of the production target and potential options for improvement. Other areas of discussion included plant operation and recovery forecasts, capital and operating forecasts and results.

Mr. Doe receives and reviews monthly reconciliation reports from the mine. These reports include the industry standard reconciliation factors for tonnage, grade and metal; F1 (Mineral Reserve model compared to ore control model), F2 (mine delivered compared to mill received) and F3 ($F1 \times F2$) along with other measures such as compliance of actual production to mine plan and polygon mining accuracy. The reconciliation factors are recorded monthly and reported in a quarterly control document. Through the review of these reconciliation factors, the QP is able to ascertain the quality and accuracy of the data and its suitability for use in the assumptions underlying the Mineral Resource and Mineral Reserves estimates.

Mr. Doe also reviews Newmont's processes and internal controls at the mine site with operational staff on the work flow for determining Mineral Resource and Mineral Reserves estimates, mineral process performance, mining costs, and waste management.

12.2 Legacy (pre-Newmont) Programs

No information is available on any data verification programs undertaken by project operators prior to Newmont.

12.3 Newmont Programs

12.3.1 Data Verification on Database Upload

Newmont has the following protocols in place during data upload to the Project database.

The digital logging is undertaken in the Newmont Proprietary Visual Logger software. Logging is loaded directly to the database using the logging interface and with the loading procedure running inbuilt validation routines. These checks test for overlapping or inconsistent intervals, validity of logging codes and missing data. Errors are reported upon loading and resubmitted once amended.

Assay results are received from assay laboratories as digital text files via email directly to the database administration team. Assay certificates are also received electronically in pdf format and are archived on the file servers. Assay result text files are imported directly to the database via an automated process in order to minimize the possibility for error. An automated daily email, alerts users to what files have been loaded each day and how the QA/QC samples are performing. Approximately 5% of all assay

certificates received during a year are manually checked against the assays in the database once a year. No errors have been identified.

Downhole survey data are downloaded electronically directly from the survey tools and after validation on site are submitted to the Newmont Perth Office for automatic entry into the database. All data is processed via an SQL Server Integration Services package that loads surveys to a temporary table and checks the validity of the data prior to loading to the database. Checks include valid ranges for dip and azimuth, correct instrument, existence of a collar record in the database, and checks against the end of hole depth. Problematic data is rejected, and email alerts are sent to site geologists to review. An additional automated validation procedure flags potentially excessive deviation once the surveys are entered which is automatically reported to the site geologists for checking and further correction if necessary.

The GED database has numerous in-built constraints to prevent incorrect data being entered.

12.3.2 Data Validation Following Upload

Following the completion of each drill fan a final validation and signoff process is completed by project geology staff. This involves checks made to the collar, downhole survey, logging and geotechnical data both in tables and visually in 3D when extracted to the Vulcan® modeling software. Checks are made by both the drilling geologist and supervising project geologist.

Additional validation of data extractions is carried out at the Mineral Resource estimation stage or as required. The procedure involves downloading the data into Vulcan® and includes checks for mismatches between collar, survey and sample tables, and gaps or overlaps in the sample intervals, missing assay data, completion of assay QA/QC, plus additional downhole survey and azimuth checks.

12.3.3 2009 Data Review

An external review of the Project database was conducted by Valerie W. Laidlaw in Denver on 6 April 2009, using the Newmont's Stones® drill hole consistency check. The review comprised:

- Missing downhole intervals;
- Overlapping intervals;
- Negative or zero intervals;
- Data below the collar total depth;
- Blank collar coordinates;
- Surveys greater than the total depth;
- No surveys for a drill hole;
- No downhole data;
- Surveys without collars;
- Suspicious surveys;
- White-space before or after drill hole names;
- Miscellaneous errors.

The review flagged few unexplainable errors and an action plan was put in place to rectify the issues identified. Missing metadata were flagged in the review; however, in most cases this information was not recorded originally and could not be corrected.

12.3.4 November 2010 Internal Review of the Project Database

An internal review of the Project database was conducted by W. Hermawan, Project Geologist, in November 2010. Approximately 5% of the data were checked against original assay reports. There were no errors found.

12.3.5 November 2011 Auron Deposit Review

Golder Associates Pty Ltd (Golder) completed a "Review of Auron Underground Gold Resource" in November 2011; of its conclusions the following statements were relevant to data verification:

"The sample preparations and assaying procedures are well documented and thoroughly audited."

"Golder has reviewed the QAQC data and the conclusions and recommendations seem reasonable. The monitoring and follow-up of issues is timely and well documented."

"The density checks by ALS should be performed on the same core sample as a direct comparison of the data and check on instrument calibration."

"The drill hole data used for the 2010 resource model seems reasonable."

12.3.6 2013 Downhole Survey Re-validation

In mid-2013, issues were identified with downhole surveys during a validation check conducted during a Mineral Resource estimation. Most involved magnetically influenced surveys that had not been made inactive and collar pickups which had been transposed. These issues were rectified in the database and a program put in place to recheck the priorities for all surveys completed in the previous four years. In all, 323 issues were identified and amended in the database.

12.3.7 2013 Database Migration Preparation

In December 2013, the DataShed® SQL database system was being migrated to a new system the Newmont GED. In preparation for the migration a detailed data audit was undertaken, and issues were identified. The bulk of these were addressed prior to the migration.

The GED database has much tighter constraints on meta data and stronger validation rules. Many of the issues related to missing metadata and values missing from library tables.

The migration preparation involved mapping the existing data in DataShed® to the relevant tables in GED and identifying all shortcomings with the existing data. These were then fixed as much as possible. Once the data was validated the various library and metadata values were migrated to the relevant tables in GED. SQL scripts were then written to do bulk inserts into the GED tables from the DataShed® database. Given the work completed previously on the data this process went smoothly on 14 December 2013.

Post migration the data was compared by site geologists with the data in DataShed®. Minor issues with rounding of interval depths were experienced due to the change

from DataShed's extensive use of floats to the more rigorous use of numeric datatypes in GED.

12.3.8 External Reviews

External reviews were undertaken as follows:

Golder, 2017. Review of the 2016 Mineral Resource estimate and 2016 Mineral Reserves estimate and associated mine planning, geotechnical and metallurgical aspects of the Project.

12.4 Comments on Data Verification

In the opinion of the QP, the data verification programs undertaken on the data collected from the Project adequately support the geological interpretations, the analytical and database quality, and therefore support the use of the data in Mineral Resource and Mineral Reserves estimation, and in mine planning, based on the following:

- Sample data collected adequately reflect deposit dimensions, true widths of mineralization, and the style of the deposits;
- Database audits confirm the data are acceptable for use in Mineral Resource and Mineral Reserves estimation with no significant database errors identified;
- No bias corrections were considered warranted on drill and analytical data.

13.0 MINERAL PROCESSING AND METALLURGICAL TESTING

13.1 Metallurgical Testwork History

A significant number of metallurgical investigations have been undertaken over the life of the Project. The majority of the early testwork is no longer relevant due to the deposits being mined out.

The most relevant of remaining tests that relate to the underground ore processing from the DBS area that is currently under active mining are presented in Table 13-1.

The processing plant was originally set up in the 1980s to treat oxide ores from open pit mines. In subsequent years, blends of 80%:20% underground:oxide ores were treated. The Granites processing plant now treats 100% underground ore from the Callie deposit area. The underground ore is much harder than open pit ores as throughput rates decreased due to the 100% proportion of underground mill feed material, the grind size also coarsened prior to the process plant upgrades completed in 2017 for the TEP. The metallurgical testwork programs have been updated to reflect the post-TEP flowsheet.

Metallurgical testing programs continue to be conducted as required to evaluate different ore lithologies from new mining areas that may require changes in processing to improve throughput and/or recoveries. Metallurgical testwork programs are active in the Auron, Callie, Callie Deeps (extension of the Callie deposit below ORL) Federation and Liberator areas, and continue to evaluate the impact of gravity gold, and grind size P_{80} on overall gold recovery.

13.2 Grindability Tests

Samples from Auron were tested at three different grind sizes in 2011 at an average P_{80} s of 180, 150 and 106 μm . Evaluation of laboratory testwork results conducted at that time suggested that grinding finer from a P_{80} of 150 μm to a P_{80} of 106 μm was likely to result in a 2% increase in recovery for mineralization in both the Auron Beds and Lower Auron Beds. However, prior to the commissioning of the TEP upgrades, the throughput rate of the plant would have needed to be reduced significantly to achieve this target size reduction. Hence overall gold production would have been lowered as a result, leading to an increase in the overall cost/oz of production. Given the sensitivity to grind size, this relationship influenced the design basis for the TEP plant upgrades.

Similar testwork was conducted on Federation samples in 2014, with a P_{80} s of 212, 150, 125, 106 and 75 μm evaluated. Head grades for this study ranged from 0.25 g/t Au to a maximum of 118 g/t Au. The former samples results were varied with no consistent improvement as grind size P_{80} reduced. The latter returned a 1.6% recovery improvement, with consistent improvements for each finer P_{80} .

Following the TEP, the future ores metallurgical testwork program was adjusted so the gravity and leach testwork of each sample, rather than a composite of the samples, was conducted at 150 μm , 106 μm and 75 μm . For the 21 Auron Deeps samples tested in 2018, the average increase in recovery from a P_{80} of 150 μm to 75 μm was 3.1% from head grades which ranged from 1.18 g/t Au to 22.67 g/t Au.

13.3 Mineralogical Studies

Mineralogical studies conducted on the Callie and Auron ores indicate that the predominant host minerals are quartz, chlorite and albite, with minor biotite. Sulfide minerals are more

abundant in the Auron Beds mineralization, and are predominantly pyrrhotite and minor arsenopyrite; however, the sulfide content still only comprises 2% to 7% at most. Gold mineralization is relatively coarse as evidenced by the high gravity recoverable gold content in most samples tested, which is 60 to 70% on average but can range from 30% to 90%. Gold is predominantly present as native gold and electrum.

Table 13-1: Recent Metallurgical Testwork Studies

Title / Year	Facility	Description
Stage 3-4 report, Section 4.2 – Metallurgical Assessment, Jan 2011	AMMTEC Laboratory, Perth	Mineralogy and flowsheet development; comminution testing; gravity and cyanide leaching; gold recovery and grind size sensitivity testing and ore variability analysis. Callie and Auron deposits.
Metallurgical study results from seven Auron samples, Nov 2011	Newmont Mining Technical Services	Mineralogy, gravity gold recovery and cyanide leach gold recovery.
AMIRA P420D – gold processing technology. Newmont Tanami Gold Mine gravity survey Nov-12	Curtin University, Gold Technology Group	Plant survey incorporating classification/comminution circuit and gravity gold recovery circuit. Survey conducted March 2012.
Tanami cyclone gravity recoverable gold testwork. Report No. 20130612-1. May 2013	Newmont Mining Technical Services	Laboratory gravity recoverable gold testing on 11 samples of cyclone overflow samples collected during the March surveys.
Metallurgical study results from several samples collected from the Auron deposit Tanami. 2011, 2012, 2014 and 2015	AMMTEC Laboratory, Perth	Comminution, gravity and leach testwork. Grind size versus gold recovery sensitivity testwork.
Metallurgical study results from several samples collected from the Federation deposit. 2014, 2015 and 2016	ALS Metallurgy Laboratory, Perth	Comminution, gravity and leach testwork. Grind size versus gold recovery sensitivity testwork.
Metallurgical Study Results from several samples collected from the Auron deposit Tanami (2011, 2012, 2014 and 2015)	ALS Metallurgy Laboratory, Perth	Comminution, gravity and leach testwork. Grind size versus gold recovery sensitivity testwork.

13.4 Current and Proposed Testwork

Callie Underground

To determine metallurgical characteristics of the Callie underground mill feed, metallurgical samples were selected on available core at approximately 200 m spaced sections to provide an indication of both comminution parameters and gold recovery at varying depth. A summary of key results is presented in Table 13-2.

Table 13-2: Callie Physical Testwork Summary from 2010

Area	Section	Density (t/m ³)	Ball Mill Work Index (kWh/t)	Abrasion Index
Callie Deeps Anticline 1	60,910 E	2.86	15.8	0.1422
Callie Deeps Anticline 2	60,910 E	2.84	16.1	0.1331
Callie Deeps Anticline 3	61,315E	2.83	16.0	0.1435
Callie Deeps Anticline 4	61,315E	2.89	15.8	0.1403
Callie Deeps Anticline	60,030E	2.87	17.8	0.2292
Callie Deeps Anticline	60,490E	2.90	17.0	0.1579
Callie Deeps Anticline	60,950E	2.90	16.4	0.1571
Average for Callie		2.87	16.4	0.1576
Callie Anticline Waste	Various	2.88	16.3	0.1147

Gravity/leach tests were also conducted on samples taken from the Callie Deeps area, with the summarized results presented in Table 13-3. The dominant ore types/lithologies from these sections are from the Lower Blake Beds and Callie Laminated Beds. Calculated head grades were like current ore feed grades to the mill, and metallurgical performance was in line with the current model for Callie ores.

Table 13-3: Callie Gold Recovery Testwork Summary

Cross-section	Drill Hole	Composite No	Calc Head Grade Au (g/t)	Gold Recovery P80 150 µm	Callie Model Estimate
60,400E	DBD 396D4	1	3.95	91.0%	94.2%
60,400E	DBD 396D4	2	5.23	94.1%	95.2%
60,700E	DBD 398D4	3	3.22	89.8%	93.5%
60,800E	L780_3647	5	2.54	95.3%	92.7%
61,000E	DBD 468D4	4	7.27	97.2%	96.4%

Based on the plant surveys and comminution testwork programs performed in 2007 and 2009, the range of the SMC and Bond ball mill work index (BBWi) comminution test results demonstrated that for the Callie ore source, an average throughput rate of 280 tph or 2.3 Mtpa should be achievable for LOMP purposes. It was recommended that when the ore sources show high SMC and BBWi the grind size could be relaxed to 180 µm, and at the low end of these comminution parameters the grind size target could be lowered. In 2016, throughput rates of up to 300 tph were achieved in the plant on a blend of Callie and Auron ores, with P₈₀ up to 190 µm being produced. In 2017 and 2018, following the plant expansion (and addition of a secondary ball mill), throughput rates of up to 370 tph were achieved with P₈₀s typically between 60 and 80 µm.

No metallurgical testwork has been conducted on Callie since 2010. Samples for testwork have been requested for when drilling of the deeper parts of the Callie deposit is scheduled.

In summary, and in the view of the QP, there is a high degree of confidence that the Callie ore source should continue to perform to historical levels of gold recovery and throughput rates. Further work on the lower levels of Callie/Lantin and the Kerril South South mineralization will need to be undertaken in future testwork programs to support metallurgical understanding of these areas.

Auron Underground

Comprehensive metallurgical testwork programs were performed on the Auron mineralization in 2010, followed by further testing each year from 2012 to 2015. A total of 21 samples from deeper in the Auron deposit (Auron Deeps) were tested in 2018. Key outcomes are presented in Table 13-4.

Table 13-4: Summary of 2018 Auron Deeps Comminution Testwork Results

Parameter	Units	Auron Beds	Lower Auron Beds
Specific Gravity	-	2.98	2.84
Unconfined Compressive Strength	MPa	75 (40-134)	78 (41-160)
Bond Ball Mill Work Index	kWh/t	19.8	21.1
Abrasion Index	-	0.097	0.085
Mia (range)	kWh/t	20.5-28.6	20.3-30.6
Mih (range)	kWh/t	15.7-23.5	11.6-26.9
Mic (range)	kWh/t	8.1-12.2	6.0-13.9
Ta (range)	-	0.22-0.32	0.20-0.48

A total of 26 Auron Bed and 55 Lower Auron Bed comminution tests have been conducted to the end of 2018. It should be noted that the range provided for SMC test results (Mia, Mih and Mic) are those of individual samples, representing the softest and hardest material, and are not mixed from within the sample set.

The two major ore lithologies appear to have very similar comminution parameters, with the Auron Beds slightly softer than the Lower Auron Beds, evidenced by the 1.3 kilowatt hour per tonne (kWh/t) lower BBWI. The higher specific gravity (2.98 g/cm³ versus 2.84 g/cm³), more than likely reflects the higher sulfide content of the Auron Beds, and may explain the difference in hardness, with breakage being more favorable along the sulfide boundaries.

Comparing Auron ore with the current Callie and Callie Deeps ores it is notable that the Auron ore is harder, with an average work index of ~20 kWh/t, compared with Callie at ~17 kWh/t. The abrasion index for both ores was similar at 0.10.

Throughput rates have been modelled both internally and by external consultants. These investigations concluded that for the Auron ore, maximum throughput of the pre-TEP process plant was approximately 280 tph. 2016 operating performance exceeded this level, with monthly ranges from 287 to 299 tph. The throughput rate of Auron ore was expected to be slightly lower than that of Callie ore given the historical comminution testwork properties. In 2017 and 2018, after the addition of the secondary ball mill as part of the TEP, the plant regularly achieved throughput rates of up to 370 tph, with grind size P₈₀s ranging from 60 to 80 µm.

The metallurgical testwork program also entails gravity separation followed by cyanide leaching. A summary of the 53 Auron Beds and 71 Lower Auron Beds tests, conducted at a grind size P₈₀ of 150 µm and completed to the end 2015 is presented in Table 13-5.

Table 13-5: Summary of Auron Gravity/Leach Testwork Results

Parameter	Units	Auron Beds	Lower Auron Beds
Gold head grade	g/t	7.11	18.44
As head grade	ppm	784	202
C(org) head grade	%	0.04	0.07
S head grade	%	1.63	0.25
Grind size P ₈₀	mic	150	150
Gravity recovery	%	41.0	66.6
Au recovery (24 h)	%	90.0	94.6
Au recovery (48 h)	%	91.6	95.5
NaCN consumption	kg/t	0.28	0.21
Lime consumption	kg/t	0.42	0.26

The average sulfur head grades for the two lithologies are clearly different, with the Auron Beds containing more sulfur (present as arsenopyrite). It is possible that some of this gold is associated with the sulfides (evidenced by diagnostic leaching studies of leached tailings samples conducted during these programs). The gold appears to be finer and less gravity recoverable, with Auron Beds achieving a 41% gravity gold recovery on average compared to Lower Auron Beds at 67% gravity gold recovery. The Lower Auron Beds appear to be more closely aligned with Callie ores, with a higher coarse / free gold component.

The overall gold recovery for Auron Beds is lower than Lower Auron Beds at 91.6% versus 95.5% based on 48-hour leach test results, but both are lower than Callie ore, which typically achieves 96.5% recovery. During 2018 metallurgical testwork on 21 Auron samples from deeper in the mine (Auron Deeps) was conducted at grind size P₈₀s of 150 µm, 106 µm and 75 µm, as actual post-TEP grind sizes were typically in the 50 µm to 70 µm range. The gravity and 48-hour leach recoveries at 75 µm grind size for Auron Beds and Lower Auron Beds respectively were 40.9%/94.4% and 54.6%/96.8%. Future testwork will continue to complete the leach tests at a range of grind sizes for each sample, instead of on a master a composite.

While cyanide consumptions are similar for the two ore types (the Auron Beds is slightly higher), the lime consumption was much higher for the Auron Beds, probably due to the elevated sulfide content. Actual plant cyanide and lime consumptions were significantly higher. Compared to Lower Auron Beds they were over 80% for lime and 130% for cyanide.

Federation Underground

Testwork on Federation ore samples was conducted in 2014, 2015 and 2016. Summaries of the comminution and gravity/leach testwork results are presented in Table 13-6 and Table 13-7 respectively.

Table 13-6: Summary of Federation Comminution Testwork Results

Parameter	Units	Auron Beds
Specific Gravity	g/cm ³	2.88
Unconfined Compressive Strength	MPa	70 (48–86)
Bond Ball Mill Work Index	kWh/t	22.3
Abrasion Index	lb/kWh	0.09
Mia (range)	kWh/t	23.0–31.5
Mih (range)	kWh/t	17.9-26.6
Mic (range)	kWh/t	9.3-13.8
Ta (range)	-	0.20-0.30

Table 13-7: Summary of Federation Gravity/Leach Testwork Results

Parameter	Units	Auron Beds
Gold head grade	g/t	22.64
As head grade	ppm	320
C(org) head grade	%	0.19
S head grade	%	0.32
Grind size P ₈₀	mic	150
Gravity recovery	%	69.6
Au recovery (24h)	%	91.7
Au recovery (48h)	%	94.2
NaCN consumption	kg/t	0.34
Lime consumption	kg/t	0.29

Federation ore is harder than Auron, which is harder than Callie. Throughput modelling has suggested a feed rate of 324 tph at the post-Tanami Expansion Project (TEP) target grind size of 125 µm. In the proportions of mill feed being proposed for Federation and the currently expected feed rates, there is not expected to be any constraints to throughput from Federation ore.

During the metallurgy gravity and leaching testwork Federation returned similar results to those of Lower Auron Beds ore samples for gravity and 48 h leach recoveries and lime consumption. Cyanide consumption was closer to Auron Beds. It is expected actual plant consumption of cyanide and lime to reflect the significantly higher rates experienced when treating Auron Beds and Lower Auron Beds.

Liberator Underground

In 2018, two samples from the Liberator deposit were tested, with further samples to be submitted as they are made available. There was insufficient sample to complete all the comminution component of the metallurgical testwork program. Gravity and 48-hour leach results averaged 75.4% and 98.2% respectively.

13.5 Recovery Estimates

A review of recovery models was conducted in 2015 and new models were developed for Callie, Auron Beds and Lower Auron Beds domains.

The Auron Beds recovery model was developed from results from 40 sets of tests conducted from 2011 to 2015. The model calculates a tails grade from the head grade. It replaced the flat 90.2% recovery model used prior to the 2016 Mineral Reserves estimate.

The Lower Auron Beds recovery model was developed from results from 49 sets of tests conducted from 2011 to 2015. The model calculates a tails grade from the Head grade. It replaced the flat 94.0% recovery model used prior to the 2016 Mineral Reserve estimate.

The Callie ore recovery model was developed from actual reconciled monthly performance data from January 2012 to December 2012, when only Callie sourced ore was treated through the processing plant. The model calculated a recovery from the head grade. A recovery model for Federation was developed in 2015 and was based on the results of twenty metallurgical gravity and leach testwork results. Additional metallurgical testwork in 2016 on five samples returned similar results and, as such, the recovery model was not updated.

Liberator was declared a Mineral Resource during 2018 and metallurgical testwork had been conducted on only two samples chosen from a single drill hole. Liberator comprises the same stratigraphy as Lower Auron Beds, therefore, the Lower Auron Beds recovery model was applied with additional 0.05 g/t Au increase in tails as a conservative approach.

Gold recovery functions are summarized below:

- **Callie Gold Recovery** = $0.6166 \times (\text{Au g/t}) + 94.08\% + 1.55$ – Capped at 98%;
- **Auron Beds Tails Gold Grade** = $((\text{Au g/t}) - (0.0656 \times (\text{Au g/t}) + 0.0771)) / (\text{Au g/t}) \times 100 + 1.55$;
- **Lower Auron Beds Tails Gold Grade** = $((\text{Au g/t}) - (0.0064 \times (\text{Au g/t}) + 0.1978)) / (\text{Au g/t}) \times 100 + 1.55$;
- **Federation Tails Gold Grade** = $((\text{Au g/t}) - (0.0074 \times (\text{Au g/t}) + 0.1861)) / (\text{Au g/t}) \times 100 + 1.55$;
- **Liberator Tails Gold Grade** = $((\text{Au g/t}) - (0.0064 \times (\text{Au g/t}) + 0.1978 + 0.05)) / (\text{Au g/t}) \times 100 + 1.55$.

Recovery testwork on the Oberon deposit was completed by Newmont Metallurgical Services in 2005 as part of the Oberon project FS. Results indicated that expected recovery is approximately 95.2% (oxide / primary ore types) at a P₈₀ grind size of 150 µm, with an average gravity gold recovery of 65.2% achieved across all five samples

The Lower Auron Beds material continued to show improved recoveries compared with Auron Beds. The Auron Beds have a higher proportion of sulfides, which may have impacted the metallurgical performance in these recent testwork programs. It should be noted that the Auron Beds will comprise up to 20% of ore delivery to the processing plant. Federation ore is expected to comprise up to 30% of mill feed when it is processed.

Callie gold recovery is capped at a maximum of 98% in the Mineral Reserves economic evaluations, as based on historical performance, this is likely to be the upper limit of the process plants ability to recover the gold. This prevents the model operating at close to 100% recovery in the event that higher-grades are encountered in the block model and adds some conservatism to the recovered gold content.

New recovery models will be developed for all lithologies during 2019 and will be based on the modified metallurgical testwork flowsheet which reflects finer, post-TEP grind sizes.

Actual plant performance for 2018 yielded a gold recovery of 97.7% at a feed rate of 335 tph.

13.6 Metallurgical Variability

Variability testing was carried out on samples drawn from the Auron Beds and Lower Auron Beds. The results are presented in Table 13-5 and show, in general that the Auron Beds contain a lower proportion of gravity recoverable gold (average 41% for AB compared to 67% for the Lower Auron Beds), and slightly lower overall recoveries 91.6% at 48 h compared to 95.5% for the Lower Auron Beds at 48 h leach time. Federation ore, presented in Table 13-7, returned similar results to Lower Auron Beds

Several of the Auron Beds samples failed to meet a 90% recovery level (12 in total, ranging from 83.6% to 89.5%), and it is intended in future test programs to determine whether the gold association with sulfide minerals is strong in these samples and whether recovery can be improved by flotation/regrind and leaching to enhance overall gold recovery. As the proportion of Auron Beds material is relatively small, there may be little economic benefit in targeting an alternate flowsheet, however the issue should be identified and characterized.

Ore treated through the mill is made up of a weekly mining parcel, i.e. all mined ore from a week is processed as a discrete parcel without any blending of ores from outside that week. This maintains a reasonably consistent mill feed over the period of a week the parcel is being processed.

13.7 Deleterious Elements

There are no significant levels of deleterious elements identified in either the Callie or Auron mill feed material to date. Nor have there been any flagged in the Federation assay results to date. As processing will be via gravity gold recovery, leaching, carbon adsorption, elution electrowinning and cathode sludge smelting to produce doré on site, it is not expected deleterious elements will be an issue for the Tanami Operations.

13.8 Comments on Mineral Processing and Metallurgical Testing

In the opinion of the QP:

- Metallurgical testwork and associated analytical procedures were performed by recognized testing facilities, and the tests performed were appropriate to the mineralization type;
- Samples selected for testing were representative of the various types and styles of mineralization identified to date. Samples were selected from a range of depths within the deposit. Sufficient samples were taken so that tests were performed on adequate sample mass;
- Mineralogical studies have shown that the gold is predominantly free and on average 60% recoverable by gravity concentration. Other key mineral associations are with silicates and minor sulfides in the Auron Beds stratigraphy. Silver occurs in association with metallic gold;
- The Auron Beds and Lower Auron Beds mineralization tends to exhibit higher ball mill grinding energy input compared to Callie and Federation is higher again, but all demonstrate average compressive strength, and low abrasion indices. Actual plant throughput and recoveries have realized the suspected upside opportunity in the throughput and recovery modelling.

- Callie, Auron and Federation ores are amenable to gravity concentration and depending on sample head grade and lithology, gravity recoverable gold varied between 30% and 90% at a P₈₀ grind size of approximately 150 µm;
- Recovery estimates for ores to be treated in 2019 at the budget head grade of 5.89 g/t Au were 99.0% for Callie, 93.7% for Auron Beds, 97.7% for Lower Auron Beds, 97.7% for Federation and 96.7% for Liberator;
- The current processing facility is considered appropriate to treat up to 3 Mtpa of Callie, Auron and Federation mill feed material identified above, with minimal modifications required and the existing testwork flowsheet supports the existing facility layout and operation, with both Callie and Auron ores processed to date matching or bettering the testwork predictions.

14.0 MINERAL RESOURCE ESTIMATES

14.1 Introduction

Four underground deposits in the DBS area and the Oberon open pit deposit have current Mineral Resource estimates; these encompass seven separate Mineral Resource model estimates.

Figure 14-1 presents a location plan.

Close-out dates for the databases used in Mineral Resource estimation are as indicated in Table 14-1.

Drill data used to support the Mineral Resource estimates comprises drill core data only at DBS and a combination of surface drill core and RC at Oberon.

The co-ordinate system used for Mineral Resource modelling is the local DBS mine grid, except for Oberon, which uses the Titania grid. The co-ordinate systems used at the Project, and transform properties are discussed in Section 9.1.

14.2 Geological and Mineralization Models

Geology models were constructed by the Project resource development geologists. Surface wireframes were constructed representing interpreted structure (faults) and stratigraphic units. The geological modelling was primarily based on stratigraphy, however; geological mapping, structural logging and Au and S assay data is also used to guide the interpretation.

Faults were modeled based on the logged or mapped hanging wall contact of each fault. Fault interpretations are reviewed in 3D, plan and section views, and against drill hole data.

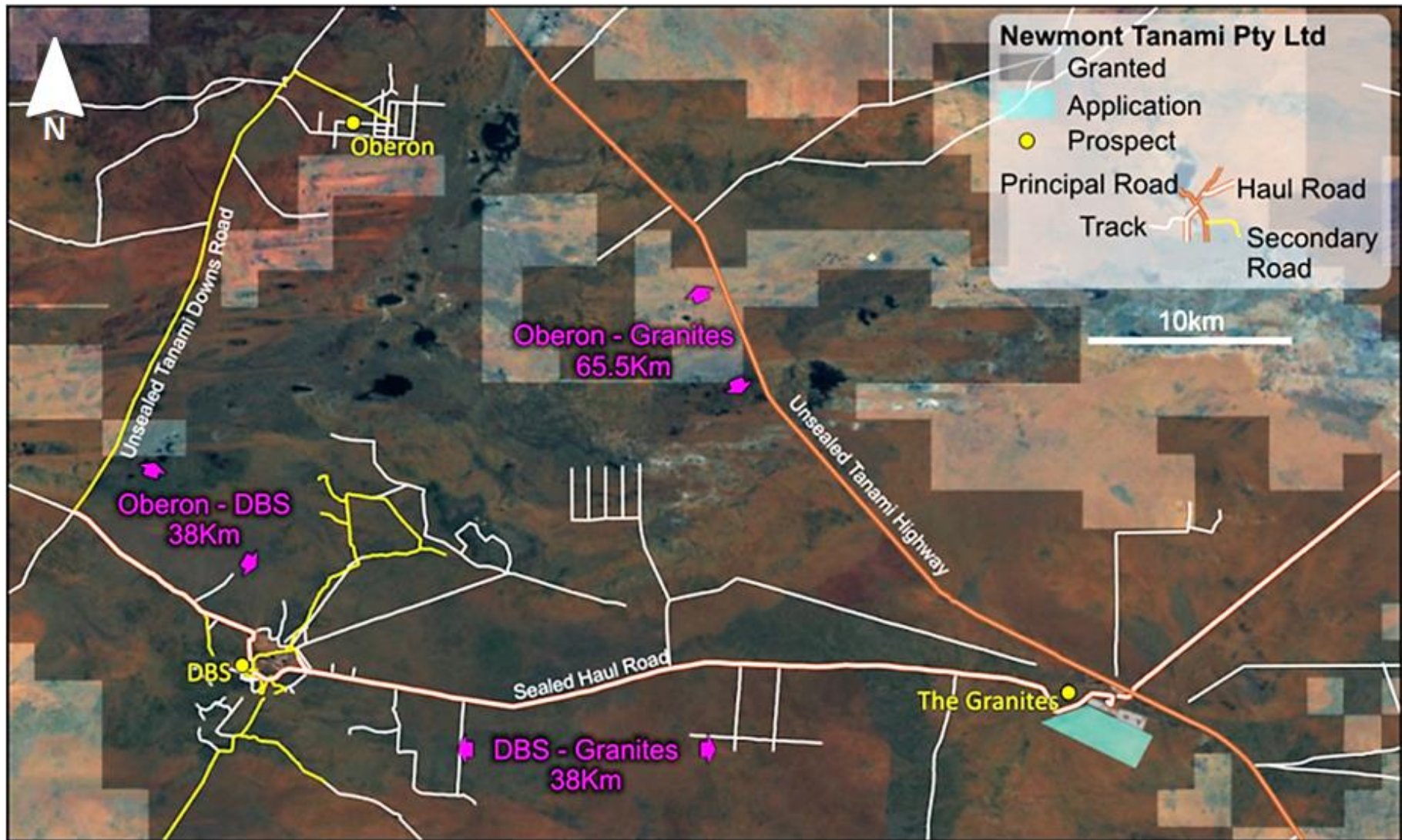
Stratigraphy contacts were digitized on section (south–north DBS grid), separately for domains defined by the major faults.

Wireframe surfaces were generated from stratigraphy sectional interpretation strings.

Table 14-1: Drill Database Close-out Table

Model	Last Drill Hole Used in Model	Drill Data Close-out Date	Block Model Release Date
Kerril South South Mineral Resource 2013 May	L025_5936	April 14, 2013	June 3, 2013
Oberon Mineral Resource Model 2013 April	TID0088	October 16, 2012	April, 2013
Callie Grade Control 2016 April (up-plunge of 60852.5mE)	R650_8110D1	November 20, 2014	March 7, 2016
Auron Grade Control 2018 March (59412.5mE - 60587.5mE)	L537_11367	January 2, 2018	February 2, 2018
Federation Mineral Resource 2016 September	MN700_9557	May 24, 2016	July 4, 2016
Callie Deeps Mineral Resource 2016 September (downplunge 60852.5mE)	N100_9675	July 20, 2016	September 7, 2016
Auron Mineral Resource 2018 September	L962_12214	September 29, 2018	October 24, 2018

For all Mineral Resources (open pit or underground) mineralized zone grade shell solids were constructed to enclose high grade portions of the sheeted quartz vein corridor, based on a cut-off grade of 0.5 g/t Au for close-spaced drilled regions and 1 g/t Au within continuous zones for wider spaced drilled regions.



Note: Source = Newmont 2019

Figure 14-1: Location Plan for DBS, Oberon and The Granites

The high nugget nature of the DBS deposits means that these mineralized zones commonly contain a substantial proportion of samples below the nominal cut-off grades while isolated high-grade samples are occasionally not captured inside the solids. The mineralized zones typically trend along the intersection of the sheeted 70-70 quartz vein corridor and along the more prospective host stratigraphic units.

Stratigraphy and mineralized zone wireframes were reviewed for consistency in 3D, plan and section views and validated for consistency.

14.2.1 Callie Underground

Three models cover the Callie mineralization; these models collectively cover the areas known as Lantin, Lantin North Limb, and Kerril South.

The upper contacts of the Callie stratigraphic units are interpreted and modeled based on 10 m spaced south–north sections west of 60600mE, 12.5 m spaced sections between 60600mE to 60850mE and then 25 m sections down plunge of 60850mE.

The multiple-folded Callie stratigraphy is offset by the Kerril, Bayban and Servalan Faults which form major domain boundaries.

The Callie grade control model mineralized zone solids are generated from a sectional interpretation that encloses blocks estimated in a preliminary median indicator kriged (IK) “domain definition” estimation as being greater than 0.5 g/t Au. Five mineralized zones were defined, one for each of the major fault-bound domains. These high-grade mineralized zones formed the primary estimation domains with the surrounding regions assigned to four low grade halo estimation domains.

The Callie Deeps model contains four separate high-grade mineralized zones enclosing samples greater than 1 g/t Au in continuous zones. Interpretation strings are snapped to drilling to ensure correct sample coding and solid wireframes were truncated against fault surface wireframes. The mineralized zones predominantly cover the intersection of the sheeted vein corridor and the main mineralized stratigraphic units.

The Kerril South model contains two high-grade mineralized zones that were digitized similarly to that described for the Callie Deeps Mineral Resource estimate.

14.2.2 Auron Underground

The Auron deposit is covered by two Mineral Resource estimates, namely:

- The Auron Mineral Resource model (calAUR201809_rsv), which covers the known extents of the Auron mineralization from 58787.5mE to 61,412.5mE;
- The Auron (GC201803_AUR) grade control model, which covers a closely spaced drilled area in the central part of the deposit between 59,400.5mE to 60,600mE.

The upper contacts of the Auron stratigraphic units are interpreted and wireframed on 12.5 m spaced south–north sections through the grade control model areas while to the east and west 25 m spaced sections are used. The folding of the Auron stratigraphy is informed by structural logging data and repetitions of stratigraphy. The form of the folds is modelled on the fold shapes of the generally more extensively drilled overlying Callie stratigraphic units. As with the Callie models, the Auron stratigraphy is offset by several major faults. In addition to the Kerril, Bayban and Servalan Faults the Serrus, Soolin and Xenon Faults also which form major domain boundaries within Auron.

The Auron grade control model mineralized zone solids are generated from a sectional interpretation that encloses blocks estimated in a preliminary median indicator kriged “domain definition” grade estimation as greater than 0.5 g/t Au. This domain definition estimate is confined to the major fault-bound domains. High-grade lodes were defined which traverse the contact of the Lower Auron, underlying Terran and overlying Auron Beds. These combinations of stratigraphy and lodes formed high-grade estimation domains with surrounding low-grade halo estimation domains.

The Auron Mineral Resource model contains 32 separate mineralized zones (including two domains for the Liberator deposit) comprising samples greater than 1 g/t Au in continuous zones. The central core of the Auron mineralization is represented by two large solids; MZ35 and MZ45, which, north and south of the Kerril Fault respectively, enclose over two-thirds of the currently known Auron deposit extent. The remaining mineralized zones consist of narrower parallel subsidiary zones that extend through the Auron stratigraphy to the north and south. As for the Callie stratigraphy, these zones are primarily controlled by the intersection of the sheeted vein corridor and the host stratigraphy. A further broad low-moderate grade mineralized zone (MZ945) is defined only within the Auron Beds in the Federation Anticline.

14.2.3 Federation Underground

The Federation mineralization is hosted primarily by the Callie stratigraphy within the Federation Anticline to the south of the main Callie mineralized zones. The Federation model was estimated within nine sub-parallel mineralized zone envelopes manually digitized to enclose samples greater than 1 g/t Au. These mineralized zones were mostly hosted in the hinge zone and southern limb of the Federation Anticline within the main mineralized Callie stratigraphy.

14.2.4 Liberator Underground

In a similar relationship to that of Callie and Federation (hosted by the Callie Laminated Beds), the Liberator deposit is south of the Auron deposit (in the Federation anticline) and hosted by the same Lower Auron Beds.

Drilling completed in 2017 continued to demonstrate good continuity between drill sections, with significant visible gold regularly intersected. Liberator is the southern extension of Auron and the geological model and Mineral Resource estimate is included in the Auron model.

Figure 14-2 to Figure 14-6 present estimated gold grades and Mineral Resource categories in plan and cross-section views for the DBS deposits.

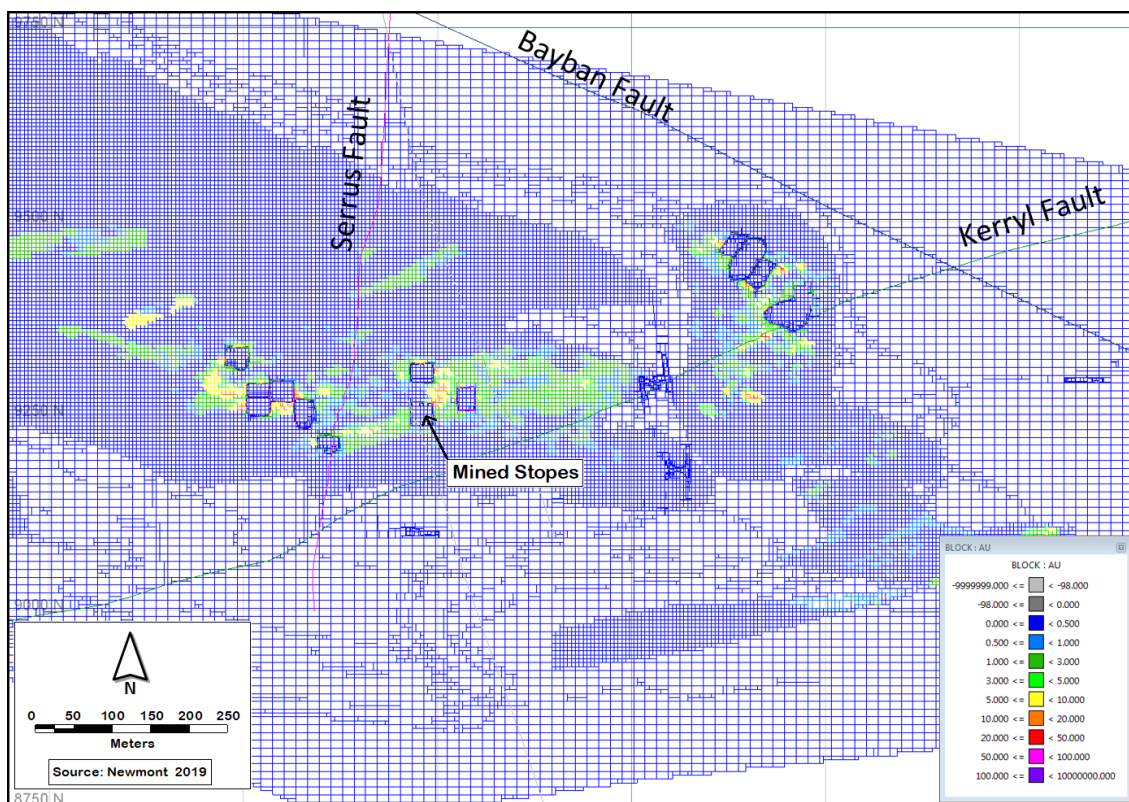


Figure 14-2: Gold Grade Plan at 650mRL Mine Grid

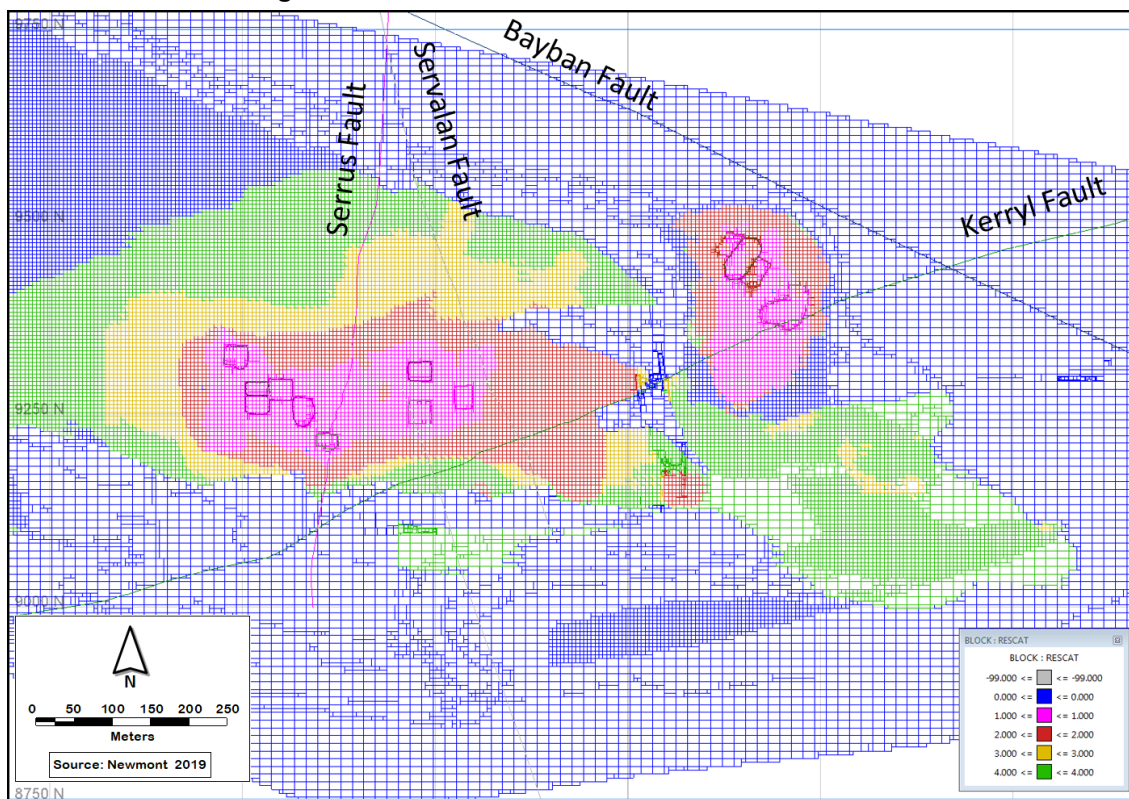


Figure 14-3: Mineral Resource Category Plan at 650mRL Mine Grid

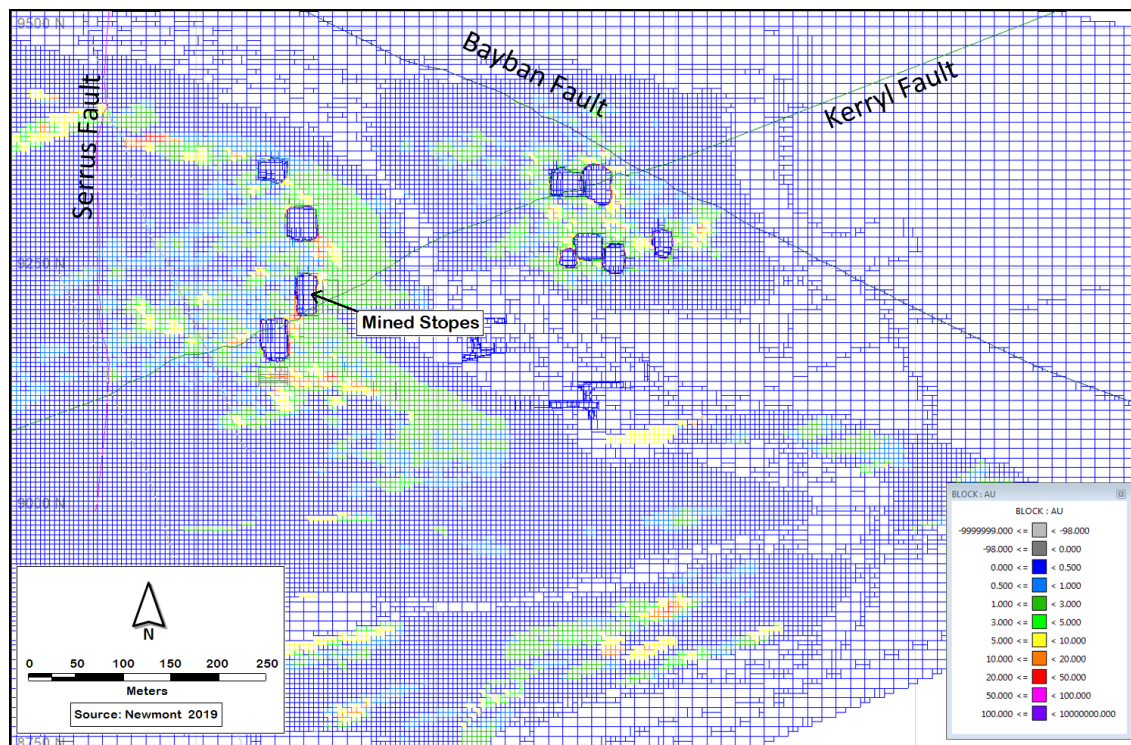


Figure 14-4: Gold Grade Plan at 200mRL Mine Grid

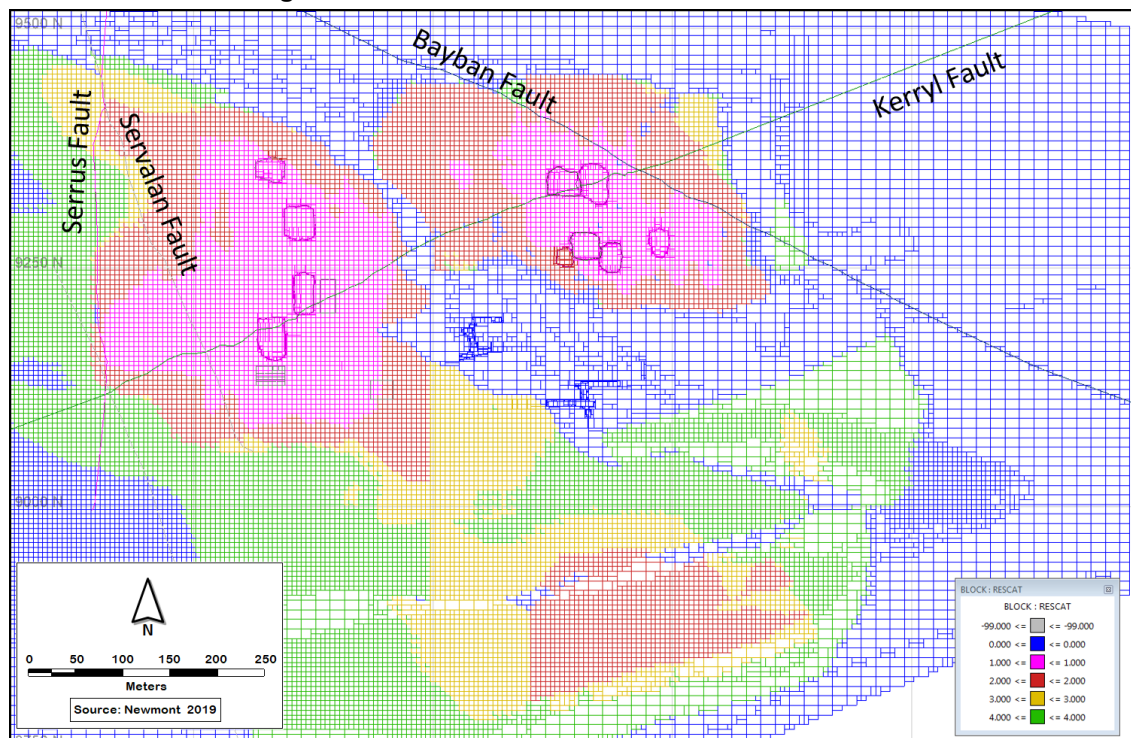


Figure 14-5: Mineral Resource Category Plan at 200mRL Mine Grid

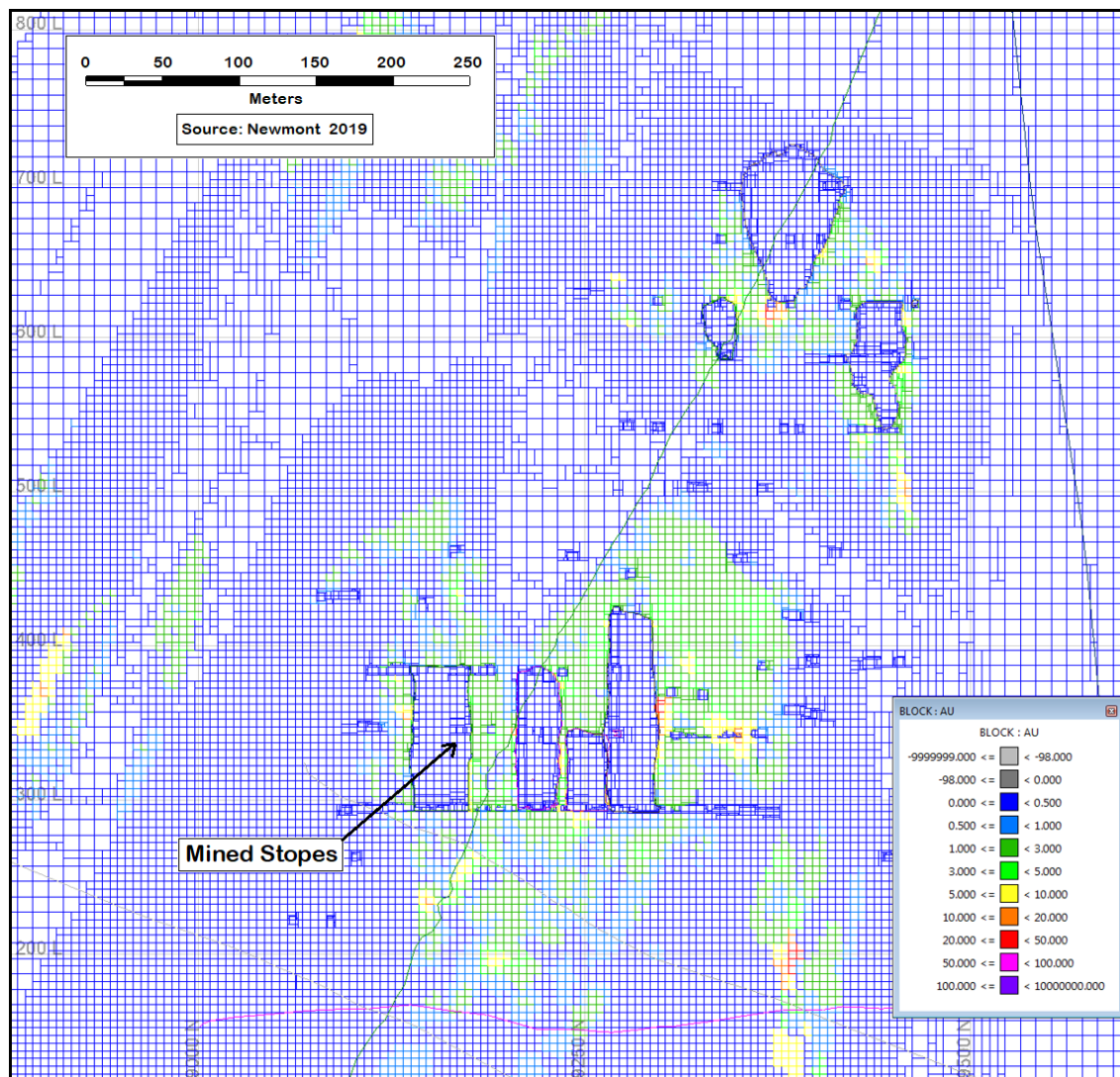


Figure 14-6: Gold Grade Cross-section at 60200mE (looking north [Mine Grid])

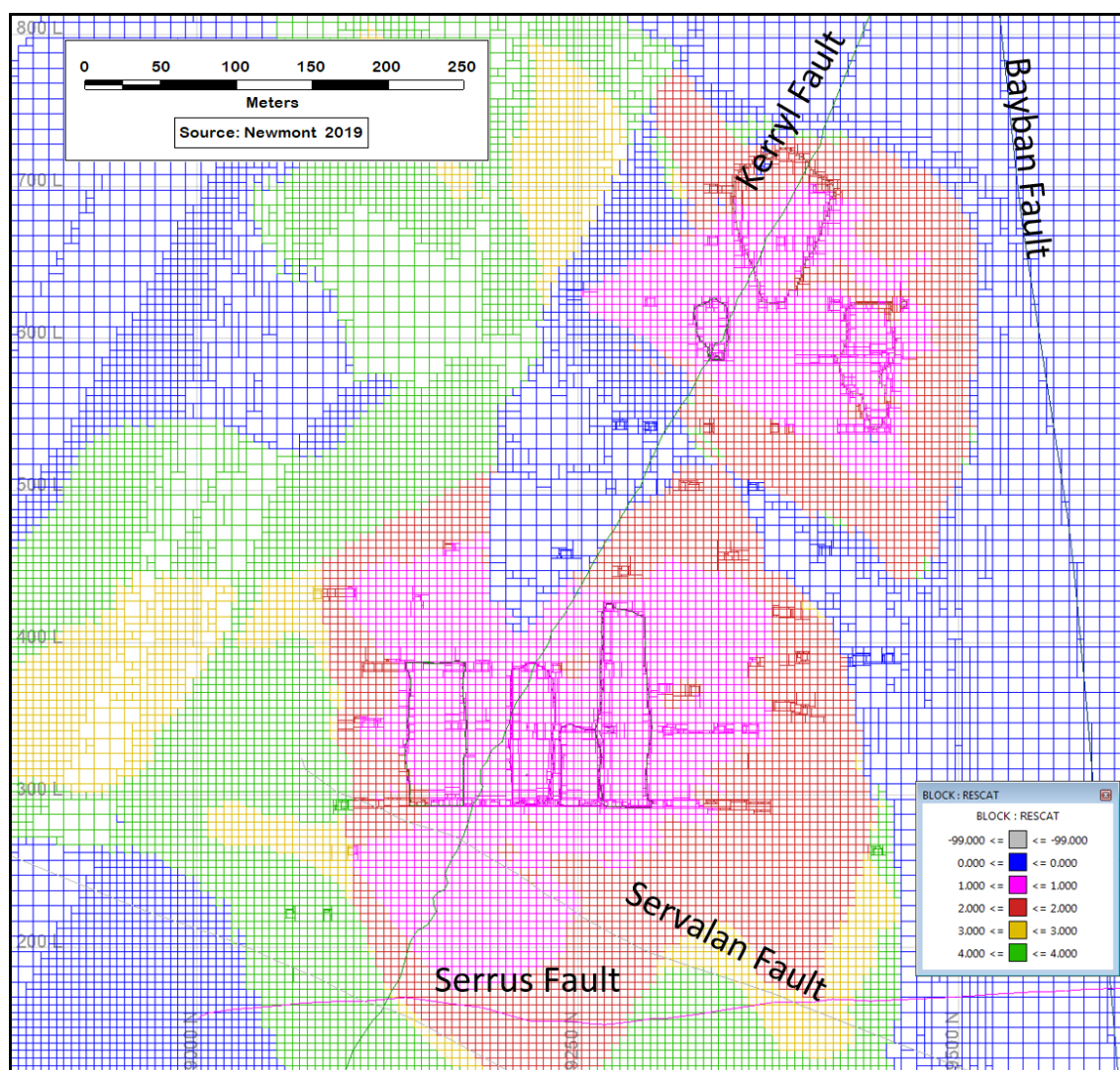


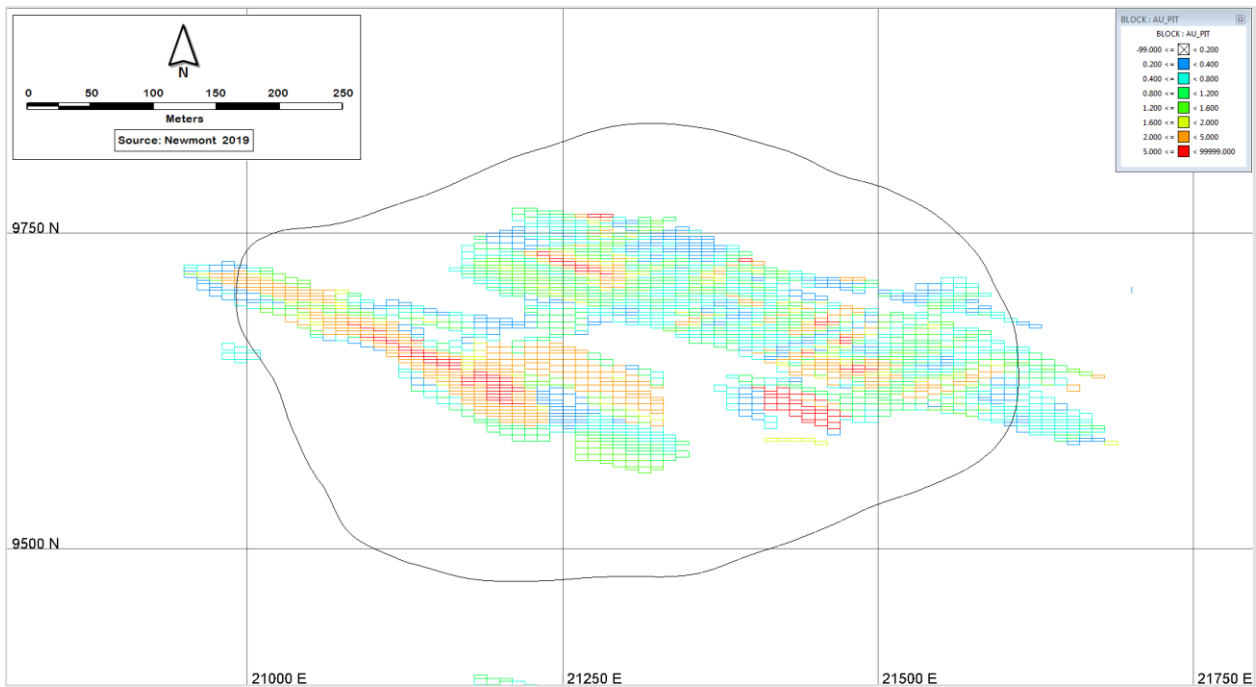
Figure 14-7: Mineral Resource Category Cross-section at 60200E (looking north [Mine Grid])

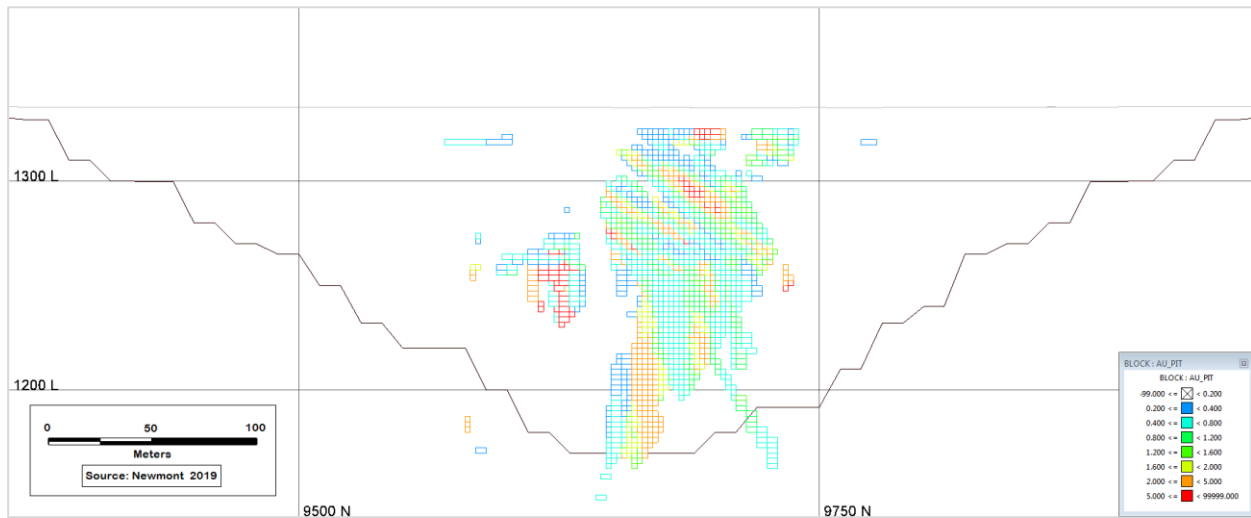
14.2.5 Oberon Open Pit

The Oberon Mineral Resource model, last updated in 2013, used a combination of surface core and RC drilling to create the geological interpretation. The infill drilling completed during 2018 was not used for the geology model or Mineral Resource estimate as assay results had not been received by the time modelling took place. The geology model defined the main lithological units and structural controls. Four major faults (Caliope, Erato, Deep South, and Urania) were interpreted based upon drilling and/or aeromagnetic imagery.

Mineralization at Oberon was modelled in two main mineralized lodes (the North and South Lodes) and two minor lodes (the Far North and Mid Lodes). Oxidation surfaces were modelled to define the base of cover, base of oxidation and top of fresh rock.

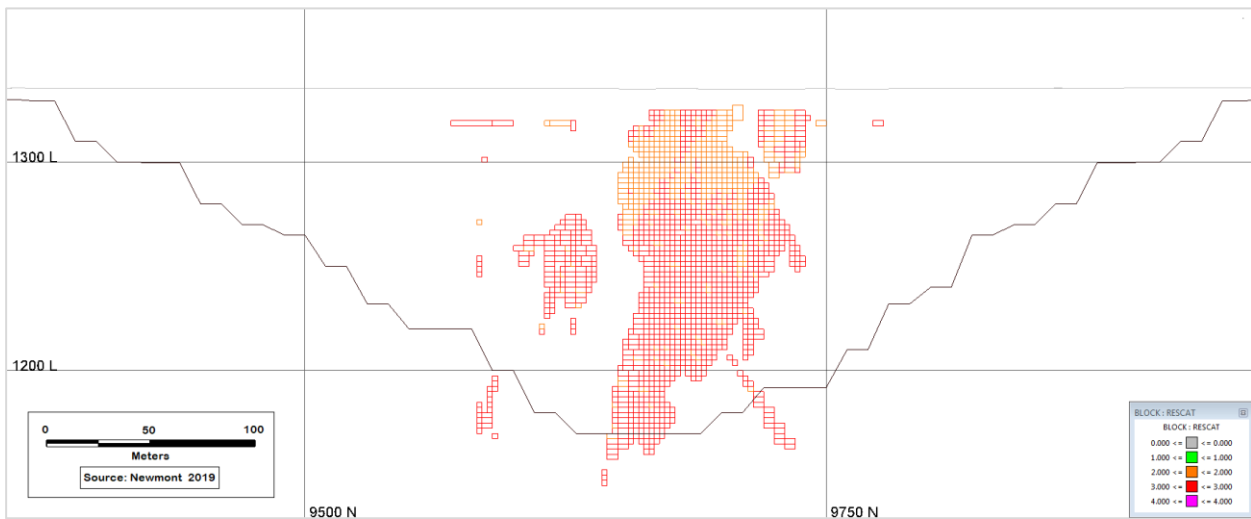
Figure 14-8 to Figure 14-11 present estimated gold grades and Mineral Resource categories in both plan and cross-section views for the Oberon deposit.





Note: gray line = Mineral Resource pit design based on a gold price of US\$1,400/oz, equivalent to AU\$1,750/oz

Figure 14-10: Gold Grade Cross-section at 21400mE (looking west [Mine Grid])



Note: gray line = Mineral Resource pit design based on a gold price of US\$1,400/oz, equivalent to AU\$1,750/oz

Figure 14-11: Mineral Resource Category Cross-section at 21400mE (looking west [Mine Grid])

14.3 Composites

A nominal composite length of 2 m for Au was used for both the Callie and Auron models and 1 m for Federation, with a tolerance of 0.5 m. The composite length is a multiple of the nominal 1 m sampling interval. As and S were composited along with Au grades.

The composites were flagged based on fault, stratigraphy and mineralized zone wireframes and assigned an estimation zone code and visually validated against the interpretation wireframes.

14.4 Statistical and Exploratory Data Analysis

Drill holes drilled from the surface (832 drill holes) were excluded from the grade control estimates where close spaced underground diamond drilling was available, due to the lower

confidence in the spatial position of the surface drill holes at deep intercept depths. A small number of underground drill holes (47 drill holes in total) were excluded from both EDA and grade estimates due to their non-representative nature if drilled parallel to the 70-70 quartz vein corridor.

The data were flagged by fault bound domains and stratigraphy and analyzed statistically to determine domain selection for Mineral Resource estimation purposes. EDA comprising statistics, log histograms and log probability plots was performed on the raw drilling data and the composited data.

Due to the fan drilling completed underground at DBS, there is clustering of data in collar area of drill fans. Cell or NN declustering weights were calculated for both raw drilling data and composite data and used in statistical analysis. Samples and composites were also length weighted.

Many drill holes contain areas with non-assayed intervals. These are commonly the collar portion of drill holes which are deliberately not sampled to reduce the impact of data clustering. No inference can be made that un-sampled intervals are barren, hence unsampled intervals are excluded from the statistical analysis.

The EDA reveals that the gold distribution for all domains at DBS have an extremely skewed, log-normal like distribution.

14.5 Variography

Downhole and directional variography was undertaken using Snowden Supervisor software to quantify the spatial relationships of composited data within the selected geological/mineralized domains.

Due to the extremely skewed and highly variable gold grade distribution, experimental variograms were generated using a normal score transform and subsequent variogram models transformed back to grade space.

For grade control estimates, variography was undertaken on indicator transformed data. For preliminary domain definition estimates and low-grade domains, the median indicator only is modelled. For high-grade domains, several thresholds were modelled to represent the spatial relationships across the grade distribution.

Most variograms are modelled with two to three spherical models. The nugget variance was modelled from down-hole variogram, ranging from 49 to 79% of the total sill.

Most domains displayed directional anisotropies within the plane of the 70–70 sheeted vein corridor. Most commonly the direction of maximum continuity follows the plunge of the intersection of the vein corridor and the plunge of the stratigraphy. The minimum continuity is typically perpendicular to the vein corridor.

The variograms yield short range structures (less than 25 m). The overall sill typically reached between 30 m and 90 m, except for the Auron Beds domains which yielded continuity ranges up to 350 m.

14.6 Density Assignment

Density values were defined for each stratigraphic unit based on statistical analysis of available density data. Density values through Callie stratigraphy are very consistent showing low variability. From 2016, Mineral Resource estimates hosted by the Callie stratigraphy

(Callie Deeps and Federation), were assigned average densities based on stratigraphic coding, ranging from 2.84 g/cm³ to 2.89 g/cm³.

The Auron units show significantly greater variability in density both between and within units, particularly within the Auron Beds due to the elevated sulfide abundance.

Average density values are assigned to the Auron estimates based on stratigraphic coding with values ranging from 1.5 g/cm³ to 4.98 g/cm³ within the Auron Beds.

For Oberon, average densities were assigned by lithological unit and weathering based on 964 density samples. The assigned density for oxide was 2.45 g/cm³. Densities assigned to fresh rock varied from 2.39 g/cm³ to 2.8 g/cm³ depending on the lithology.

Table 14-2 provides the average density values assigned to the Mineral Resource models.

Table 14-2: Mineral Resource Model Assigned Density

Model	Stratigraphy/Weathering	Density(g/cm ³)	No Samples
Kerril South South Resource 2013 May	All	2.84	4,379
Oberon Mineral Resource Model 2013	Oxide	2.45	964
	Fresh (by lithology)	2.39 - 2.80	
Callie Grade Control 2016 March (up-plunge of 60852.5mE)	Upper Blake Beds	2.88	730
	Callie Boudin Chert	2.89	746
	Magpie Schist	2.86	2,281
	Callie Laminated Beds	2.86	2,612
	Lower Blake Beds	2.88	7,395
Federation Resource 2016 September	Upper Blake Beds	2.85	923
	Callie Boudin Chert - Magpie Schist - Callie Laminated Beds	2.86	2,597
	Lower Blake Beds	2.89	3,733
	Upper Auron Beds	2.88	2,347
	Auron Beds	2.91	922
	Lower Auron Beds - Terran Beds	2.82	478
Callie Deeps Resource 2016 September (down-plunge 60852.5mE)	Upper Blake Beds	2.84	1,721
	Callie Boudin Chert, Magpie Schist, Callie Laminated Beds	2.86	1,170
	Lower Blake Beds	2.86	749
Auron Grade Control 2018 March (59412.5mE - 60587.5mE)	Lower Blake Beds	2.88	12,313
	Upper Auron Beds	2.87	16,077
	Auron Beds	2.90	35,359
	Lower Auron Beds	2.82	24,062
	Terran Beds	2.83	43,499
Auron Resource 2018 September	Upper Auron Beds	2.87	19,879
	Auron Beds	2.90	37,949
	Lower Auron Beds	2.82	25,310
	Terran Beds	2.83	46,643

14.7 Grade Capping/Outlier Restrictions

Grade capping was used to limit the spatial extrapolation of occasional isolated high grades in the Mineral Resource model estimates. The capping analyses included the use of log histograms, probability plots, ranked composites, and outlier analysis. This involved removing the highest-grade values sequentially and analyzing the effect on the coefficient of variation (CV) of the remaining data. Capping was applied to the composites at the time of grade estimation.

Due to the significant variability in drill data density within each domain, different cap grades were evaluated and applied to each estimation pass. For Auron, the cap grade was reduced with subsequent passes to reflect the lower support for high grade data and greater influence of high-grades in areas of wide-spaced drilling.

The Callie and Auron grade control estimates are interpolated using a MIK estimate for which the grade of the highest indicator threshold was capped for selected domains to reduce the impact of high-grade outliers.

High-grade caps range from 3 g/t Au in the Callie low-grade domains to 275 g/t Au in Auron (2 m composites) and 730 g/t Au in Federation (1 m composites).

Both high and low caps were also applied to density data, to address potential poor-quality determinations, suspected data entry errors and spurious values. Overall, the impact of removing the outliers on density measurements is considered minor.

14.8 Estimation/Interpolation Methods

A block model was constructed using sub-blocking to define narrow zones and to maintain volume integrity with the geology wireframes.

Estimation block sizes range from 15 x 10 x 10 m, to 5 x 5 x 5 m depending on drill density with sub-blocks of 5 x 5 x 5 m.

All the block models were non-rotated and aligned north–south with the primary DBS mine grid.

Kerril South South, Auron, Federation and Oberon Mineral Resource models were estimated by OK. Callie Deeps model was estimated with an ID method. The Callie and Auron grade control models were estimated using MIK for high grade domains and median IK for low-grade areas. The indicator methods were applied to address the mixed grade populations and highly skewed grade distribution.

14.9 Block Model Validation

Estimation validation involved the following:

- Visual inspection of results on plane and section compared to the sample data;
- Comparison of the estimate against alternative estimation techniques and the NN distributions;
- Analysis of grade profiles by easting, northing and elevation using swath plots;
- Checks for global bias between block grades and sample and composite grades;
- Back reconciliation of previously milled portions of the estimate to mill production.

The checks showed the models were acceptable for use in Mineral Resource and Mineral Reserves estimation.

14.10 Classification of Mineral Resources

The existing Mineral Resources at the project (which are all located at the Callie Mine) are classified using criteria based primarily on drill spacing and a minimum number of drill holes informing each estimated block.

Measured Mineral Resources require a drill spacing of 12.5 m x 12.5 m and at least three drill holes per estimate. The Measured Resource classification is applied where there is an MIK interpolated grade control estimate. Portions of the deposit drilled to the nominal Measured Resource spacing are classified as an Indicated Resource where there is no grade control estimate.

Indicated Mineral Resources at Callie and Auron require a maximum spacing between sectional drill fans of 50 m and a drill hole spacing of 25 m within fans and at least three drill holes per estimate. For the Federation deposit, Indicated Mineral Resources require 25 m x 25 m drill spacing and at least three drill holes per estimate.

Inferred Mineral Resources at Callie and Auron require a maximum spacing between sectional drill fans of 100 m and a drill hole spacing of 25 m within fans and at least three drill holes per estimate. The classification of Inferred Resources at Federation requires 50 m x 50 m drill spacing.

Risk assessments completed by Newmont have identified that the drill spacing is the dominant risk factor. Drill spacing studies completed in 2014 on Federation demonstrated that the narrower Federation mineralization required a closer spacing between drill sections for the definition of Indicated Resource (25 m) and Inferred Resource (50 m) compared to the established 50 m and 100 m respectively, used elsewhere in Callie and Auron.

14.11 Post-processing

Estimated and classified block models were coded with several reporting variables and combined into a single mine block model. Metallurgical domains were coded using a combination of the stratigraphy and model codes identifying Callie, the Auron Beds, below Auron Beds and Federation.

A depleted version of the combined model is completed with blocks within previously mined areas depleted. Where areas are back-filled, these blocks were coded by fill type and their density reset to reflect the fill type.

14.12 Reconciliation

There are no active open pit operations, hence; this section relates to the underground operations, for which reconciliation data is available.

Direct analysis of the Mineral Resource estimates performance is difficult as the reported reconciliation factors incorporate mining modifying factors and often reflect the performance (or otherwise) of mining. Ore control estimates are more straightforward to reconcile. For all stopes completed since 2013, the estimates have been comprehensively reconciled against the Cavity Monitoring System (CMS) solids and compared to the allocated mill and stockpiled reconciled production.

New and updated grade control estimates are evaluated against the CMS and their high-grade cap grades and other estimation parameters calibrated accordingly.

Reconciliations at site attempt to measure the performance of the Mineral Resource estimates through comparing the performance of the Mineral Reserves stopes, through the mining cycle, to the final tonnage and metal physicals achieved following processing of the ore as an approximate measure of the Mineral Resource performance. This measure is strongly influenced by both mining engineering factors (e.g. stope optimizations after Mineral Reserves with changed planned dilutions and recoveries) and physical mining effects (e.g. actual

dilutions and underbreak) that can obscure the real performance of the Mineral Resource model.

This process uses three measuring points or factors, as follows:

- F1 – Performance of the Mineral Reserves Stope Design to the Optimized Final Mine Design of the equivalent stopes;
- F2 – Performance of the tonnes, and metal “Delivered to the Mill” calculated based on the final design of stopes and development, to the physical tonnes and metal measured following processing of the ore (Received at Mill component);
- F3 – Performance of the physical tonnes and metal measured following processing of the ore (termed Received at Mill) against the tonnages and grades predicted by the equivalent Mineral Reserves Stope Designs.

2018 reconciliation performance was as follows:

- F1 – Gold reconciled to 98.3%;
- F2 – Gold reconciled to 107.5%;
- F3 – Gold reconciled to 105.7%.

14.13 Reasonable Prospects of Eventual Economic Extraction

Over 18 years of underground production demonstrates that the orebodies at DBS are amenable to the long-hole open stoping method.

Mineral Resources for the Callie, Auron, Federation, Liberator and Oberon deposits have been evaluated and reported utilizing essentially the same methodology as applied to the evaluation of Mineral Reserves. The key differences are that the evaluation of Mineral Resources also includes material classified as Inferred Mineral Resources and use an elevated gold price of US\$1,400/ oz or AU\$1,750/ oz.

Underground Mineral Resources are reported within designed stope and development solids with modifying factors applied to allow for additional dilution and mining recovery losses based on recent mining performance. As for Mineral Reserves, the prospects of eventual economic extraction are evaluated separately for each area, mine level and stope. As factors such as trucking distance, access development, ventilation, metallurgical recoveries, backfill type and drilling requirements vary between stopes, a single cut-off grade is not applied when reporting Mineral Resources.

Mineral Resources for Oberon are reported within designed open pits with modifying and economic factors applied based on estimates completed during Q4 2018 to a pre-feasibility level. There is no Mineral Reserves estimate for Oberon.

The Mineral Resource estimate for Oberon was prepared by Newmont following a process of review of the input parameters, pit shell generation and selection, mine design, mine scheduling, cost estimation, and financial modelling. Pit shells and stage selection were undertaken using parameters current for the 2018 Mineral Reserve and Mineral Resource pricing provided by the Group Executive Reserves. The pricing is based on a US\$1,200/oz or AU\$1,600/oz gold price for the initial phase and a US\$1,400/oz or AU\$1,750/oz gold price for the final pit. The financial analysis is based on a US\$1,200/oz or AU\$1,600/oz gold price.

The open pit mining study assumes that the sub-blocks which contain mineralisation are wholly and discretely mineable but can be mined as part of a contiguous mineralised zone.

The mining study has relied on the interpretation in that there is sufficient geological continuity of economic mineralisation at a mining scale of 10 m by 2.5 m by 2.5 m.

The base open pit mining costs were based on a non-binding bill of quantities submitted by a mining contractor in September 2018. Rates were provided on an ore/waste load and haul and ore/waste drill and blast basis. Indicative quantities for mobilisation, infrastructure development and topsoil stripping were also provided.

Processing rates for each open pit ore type were applied in a uniform manner, assuming oxide/cover, transitional and fresh ore would have similar throughput ratios. Unit costs were developed from BP19 Tanami variable rates applied directly to Oberon ore. Fixed costs from BP19 were apportioned based on initial ore (in the case of direct processing fixed costs) and total tonnes (in the case of indirect processing costs) between Oberon and Tanami.

Calculated costs included into the processing costs were incremental ore haulage/drill and blast ore costs, estimated surface haulage from Oberon to The Granites mill (including road maintenance), aquifer re-injection costs, TSF incremental raise, and a CRF on sustaining capital.

Table 14-3 presents the economic parameters used in the 2018 underground Mineral Resource stope evaluation.

Table 14-3: Economic Parameters used in Underground Mineral Resource Stope Evaluation

Description	Unit	Rate	\$/ore tonne
Variable Mining Costs			
Lateral Ore Development	per metre advanced	\$3,320	\$42.36
Slot Rise Development	per metre	\$1,066	\$0.55
Diamond Drilling	per drill metre	\$146	\$6.89
Production Drilling	per stope drilling m	\$29.0	\$1.94
Stope Blasting	per stope tonne	\$2.01	\$2.01
Loading	per tonne	\$3.78	\$3.78
Haulage	per tkm	\$1.29	\$3.71
Hoisting	per tonne	\$4.87	\$4.87
Surface Haulage	per ore tonne	\$4.96	\$4.96
Paste Filling	per fill tonne	\$21.35	\$14.21
Utilities	per tonne	\$0.76	\$0.76
UG Mine Support	per tonne	\$0.43	\$0.43
Other Variable Costs			
Process	per ore tonne	\$12.60	\$12.60
Support (Site & Region)	per ore tonne	\$1.36	\$1.63
General Sustaining Capital	per ore tonne	\$7.12	\$11.02
Fixed Costs			
Mining	per year	\$88,000,000	\$33.85
Sustaining Capital	per year	\$11,400,000	\$5.69
Milling	per year	\$26,600,000	\$10.23
Support (Site & Region)	per year	\$38,600,000	\$14.85

The sustaining capital includes an allowance for the CRF. The cut-off grade varies with depth. All the costs listed in Table 14-3 were used in a cashflow evaluation for each stope and each level.

Table 14-4 presents the economic parameters used in Oberon open pit Mineral Resource evaluation.

Table 14-4: Economic Parameters used in Oberon Mineral Resource Evaluation

Description	Unit	Cover	Oxide	Transition	Fresh
Processing – Incremental Ore Haulage/D&B Costs	AU\$/tonne	0.03	0.12	0.08	0.17
Processing – Surface Ore Haulage to Granites	AU\$/tonne	9.50	9.50	9.50	9.50
Processing – Variable	AU\$/tonne	21.1	21.1	21.1	21.1
Processing – Fixed	AU\$/tonne	5.4	5.4	5.4	5.4
Dewatering/Aquifer Reinjection operating costs	AU\$/tonne	1.2	1.2	1.2	1.2
Closure Cost	AU\$/tonne	0.0	0.0	0.0	0.0
Tailings Capacity – Sustaining Capital	AU\$/tonne	1.2	1.2	1.2	1.2
CRF on Sustaining Capital (TSF)	AU\$/tonne	0.2	0.2	0.2	0.2
Total Process Costs	AU\$/tonne	38.56	38.66	38.61	38.71
Processing Sales – WGC	AU\$/oz	0.08	0.08	0.08	0.08
Processing – Recovery	%	95.2	95.2	95.2	95.2
Price	AU\$/oz	1,600	1,600	1,600	1,600
Cut-Off Grade	g/t Au	0.788	0.789	0.789	0.790

14.14 Mineral Resource Statement

Mineral Resources consider geological, mining, processing and economic constraints, and have been confined within appropriate parameters for underground mining scenarios, and therefore are classified in accordance with the CIM Definition Standards for Mineral Resources and Mineral Reserves.

The Mineral Resource estimates were prepared by Mr. Shaun Schneider FAIG, Geology Manager and Newmont employee under the supervision of the QP. Mineral Resources are reported exclusive of Mineral Reserves.

Mineral Resources are reported exclusive of those Mineral Resources that were converted to Mineral Reserves. Mineral Resources that are not Mineral Reserves do not have demonstrated economic viability.

Mineral Resources are reported in Table 14-5 at gold price of US\$1,400/oz or AU\$1,750/oz and have an effective date of 31 December 2018.

Table 14-5: Mineral Resource Statement – Gold at the Effective Date of 31 December 2018

	Measured Resource			Indicated Resource			Measured + Indicated Resource			Inferred Resource		
	Tonnage (kt)	Au Grade (g/t)	Au Metal (koz)	Tonnage (kt)	Au Grade (g/t)	Au Metal (koz)	Tonnage (kt)	Au Grade (g/t)	Au Metal (koz)	Tonnage (kt)	Au Grade (g/t)	Au Metal (koz)
Tanami UG	400	3.15	40	3,800	4.73	570	4,200	4.57	610	6,300	5.67	1,140
Tanami OP	-	-	-	4,200	2.21	300	4,200	2.21	300	3,900	1.88	230
Total UG + OP	400	3.15	40	8,000	3.40	870	8,400	3.39	910	10,200	4.22	1,370

Notes to accompany the Mineral Resource table:

- UG = underground and OP = open pit;
- UG Mineral Resources presented are aggregations for the Auron, Callie, Federation and Liberator UG mine areas and comprise weighted average values;
- OP Mineral Resources presented represent Mineral Resources for the Oberon deposit;
- Mineral Resources have an effective date of 31 December 2018;
- Mineral Resources are reported exclusive of Mineral Reserves; and are reported on a 100% basis;
- Mineral Resources are reported using a gold price of US\$1,400/oz, equivalent to AU\$1,750/oz at an exchange rate of US\$0.80 = AU\$1.00;
- UG Mineral Resources (Auron, Callie, Federation and Liberator) are evaluated and reported within conceptual stope designs;
- Mining, processing, site and regional G&A costs, metallurgical recoveries, and royalties are considered in the cut-off determination and in individual stope economic evaluations. The lowest cut-off grade applied was 1.6 g/t Au (ore development). Table 14-3 presents the economic parameters used in the 2018 underground Mineral Resource stope evaluation;
- For UG operations, the Mineral Resource was evaluated following a similar process to that used for the Mineral Reserves; tonnages are based on technically and economically viable minable shapes applied to the Mineral Resource model and include modifying factors for stope dilution (7 to 20% dilution) and mining recovery losses (92.5% recovery);
- OP Mineral Resources (Oberon deposit) are reported within a Mineral Resource pit design based on a gold price of US\$1,400/oz, equivalent to AU\$1,750/oz. Table 14-4 presents the economic parameters used in Oberon open pit Mineral Resource evaluation;
- Tonnages are rounded to the nearest 100,000 tonnes;
- Ounces are estimates of gold metal contained in the Mineral Resource and do not include allowances for processing losses. Ounces are rounded to the nearest 10,000 ounces;
- Rounding as required by reporting guidelines may result in apparent summation differences between tonnes, grade and contained metal content;
- Tonnage and grade measurements are in metric units. Gold ounces are reported as troy ounces.

14.15 Factors That May Affect the Mineral Resource Estimate

Factors which may affect the geological models and the preliminary stope designs used to constrain the Mineral Resources, and therefore the Mineral Resource estimates include:

- Commodity price assumptions and overall operating and capital costs;
- Mine design and operating assumptions, including haul distances;
- Dilution assumptions;
- Changes to geotechnical parameters, mining and metallurgical recovery assumptions;
- Changes to royalties levied;
- Changes in interpretations of mineralization geometry and continuity of mineralization zones;
- Changes to assumptions made as to the continued ability to access the site, retain mineral and surface rights titles, maintain the operation within environmental and other regulatory permits, and retain the social license to operate.

14.16 Comments on Mineral Resource Estimate

In the opinion of the QP, the Mineral Resource estimate for the Project conforms to industry standard practices and satisfies the requirements of the CIM Definition Standards.

15.0 MINERAL RESERVES ESTIMATES

15.1 Basis of Mineral Reserves

Mineral Reserves were estimated from the Mineral Resource model.

The current Mineral Reserves comprise Lantin, Kerril, Auron, Lantin North and Federation.

NTO used the following criteria to convert Mineral Resources to Mineral Reserves:

- Only Measured and Indicated Mineral Resources are considered;
- Dilution is included in the Mineral Reserves estimate based on economically viable mining shapes applied to the Mineral Resource model that incorporate dilution;
- A mining recovery of 92.5% was applied;
- A 3.5 g/t Au cut-off grade was used to generate the stope shapes. Final stopes were designed manually. Each stope and each level were run through a cashflow evaluation;
- Mineral Reserves are supported by an economic mine plan;
- For Oberon, the modifying factors are not estimated to a level adequate to declare a Mineral Reserve.

15.2 Stope and Mine Design

Stope and development design was completed via an interactive scheduling process to generate the mining activities and associated physicals used in the economic evaluation.

LOMP and tabulation of Mineral Reserves was completed, accounting for all development, production and backfilling activities in each zone of the mine.

Mineable shape optimizer (MSO) software was used to obtain an indication of the relative tonnes, ounces and corresponding head grade attained at various stope cut-off grades.

The stopes were then adjusted manually to create mineable shapes for addition into the mine plan, with due consideration of the location of the stopes and their economic viability.

Stope designs were based on the following criteria:

- Ore body geometry;
- Geotechnical and backfilling constraints;
- Known safety constraints;
- Limitations of the equipment fleet;
- Reconciliation of mined stopes.

Stopes were designed using a transverse, long-hole open stoping mining method with paste fill, generally using a primary/secondary mining sequence. Some tertiary stopes were designed where the hydraulic radius exceeded the geotechnical recommendation. Stopes can be mined over multiple levels.

Most stopes were designed to be 30 m wide along the strike. Width by length by height (W x L x H) dimensions of 30 m x 30 m x 40 m have proved to be stable. Where possible, stopes

were designed to the full height of the ore body, up to 120 m. The transverse length of stopes is determined by orebody thickness. Stopes are designed at 40 m sublevels for Callie and Auron, and at 30 m sublevels for Liberator and Federation, due to the narrower nature of the mineralization. It is desirable for geotechnical reasons that secondary stopes do not exceed 30 m in width.

Stope shapes are made as regular as possible in the interests of stability and dilution control.

A 92.5% mine recovery factor was used after a review of the latest mined stope reconciliation results. A dilution factor between 7% and 20% was assumed as per geotechnical recommendations. The dilution grade is generally between 1.6 g/t or 2 g/t Au based on geological recommendations. Cable bolts are used to support internal brows. This has historically resulted in lower dilution despite sometimes irregular stope shapes.

LOMP and tabulation of Mineral Reserves were completed using mainly the Deswik® suite of planning software. This software allowed for a comprehensive three-dimensional design and interface between all development, production and backfilling activities in each zone of the mine.

Standard mine design basis and design procedures were followed in accordance with the established site design criteria and approved business plan strategy.

15.3 Cut-off Grade

The BP19 costs were used to define the breakeven cut-off grade, in conjunction with the forecast rates of ore production, both by year and over the whole life-of-mine.

A cut-off grade of 3.5 g/t Au was used for the Mineral Reserves, which yielded the best Net Present Value (NPV) over the current cost and revenue profile defined by BP19. The cut-off grade incorporates a CRF. The CRF reflects an allowance for a pre-tax return on the investment capital and is added to the expenditure to ensure an adequate return.

Analysis has shown that including marginal ore into the mine plan is necessary at NTO to maximize annual underground production considering operational constraints relating to mining area availability, core drilling, Mineral Resource conversion and lateral development rates.

Marginal stopes with overall grades between 3.0 and 3.5 g/t Au were also included in the Mineral Reserves, based on the individual economic evaluation of each stope. Typically, these stopes were adjacent to higher grade stopes and thus require minimal waste development and infrastructure. Economic analysis was conducted for every stope and level with verification that the Mineral Reserves, if mined alone, will be NPV positive.

Table 15-2 presents the economic parameters used in the 2018 underground Mineral Reserves stope evaluation.

Table 15-1: Economic Parameters used in Underground Mineral Reserves Stope Evaluation

Description	Unit	Rate	\$/ore tonne
Variable Mining Costs			
Lateral Ore Development	per metre advanced	\$3,320	\$42.36
Slot Rise Development	per metre	\$1,066	\$0.55
Diamond Drilling	per drill metre	\$146	\$6.89
Production Drilling	per stope drilling m	\$29.0	\$1.94

Description	Unit	Rate	\$/ore tonne
Stope Blasting	per stope tonne	\$2.01	\$2.01
Loading	per tonne	\$3.78	\$3.78
Haulage	per tkm	\$1.29	\$3.71
Hoisting	per tonne	\$4.87	\$4.87
Surface Haulage	per ore tonne	\$4.96	\$4.96
Paste Filling	per fill tonne	\$21.35	\$14.21
Utilities	per tonne	\$0.76	\$0.76
UG Mine Support	per tonne	\$0.43	\$0.43
Other Variable Costs			
Process	per ore tonne	\$12.60	\$12.60
Support (Site & Region)	per ore tonne	\$1.36	\$1.63
General Sustaining Capital	per ore tonne	\$7.12	\$11.02
Fixed Costs			
Mining	per year	\$88,000,000	\$33.85
Sustaining Capital	per year	\$11,400,000	\$5.69
Milling	per year	\$26,600,000	\$10.23
Support (Site & Region)	per year	\$38,600,000	\$14.85

The sustaining capital includes an allowance for the CRF. The cut-off grade varies with depth. All the costs listed in Table 15-2 were used in a cashflow evaluation for each stope and each level.

15.4 Mineral Reserves Statement

Mineral Reserves for the Project are presented in Table 15-2 and are reported using the CIM Definition Standards.

Mineral Reserves have an effective date of 31 December 2018 and are reported using a gold price of US\$1,200/oz or AU\$1,600/oz.

The Mineral Reserves estimates were prepared by Mr. Johannes Grobler, MAusIMM CP (Min), Technical Services Superintendent and Newmont employee, under the supervision of the QP.

Table 15-2: Mineral Reserves – Gold at the Effective Date of 31 December 2018

	Proven Reserves			Probable Reserves			Proven + Probable Reserves		
	Tonnage (kt)	Au Grade (g/t)	Au Metal (koz)	Tonnage (kt)	Au Grade (g/t)	Au Metal (koz)	Tonnage (kt)	Au Grade (g/t)	Au Metal (koz)
Tanami UG	10,200	5.46	1,780	16,400	5.54	2,910	26,600	5.51	4,690
Total UG	10,200	5.46	1,780	16,400	5.54	2,910	26,600	5.51	4,690

Notes to accompany the Mineral Reserves table:

- UG = underground;
- UG Mineral Reserves presented are aggregations for UG operations and comprise weighted average values for several distinct underground mine areas;
- Mineral Reserves are reported on a 100% basis;
- Mineral Reserves are reported to a gold price of US\$1,200/oz, equivalent to AU\$1,600/oz at an exchange rate of US\$0.75 = AU\$1.00;
- UG Mineral Reserves are evaluated and reported within stope and development designs;
- Mining, processing, site and regional G&A costs, metallurgical recoveries, and royalties are considered in the cut-off calculations and in individual stope economic evaluations. The lowest cut-off grade applied was 1.6 g/t Au (ore development). Table 15-2 presents the economic parameters used in the 2018 underground Mineral Reserves stope evaluation;
- For UG operations, the Mineral Reserves tonnages are based on technically and economically viable minable shapes applied to the Mineral Resource model and include modifying factors for stope dilution (7 to 20% dilution) and mining recovery losses (92.5% recovery);
- Tonnages are rounded to the nearest 100,000 tonnes;
- Ounces are estimates of metal contained in the Mineral Reserves and do not include allowances for processing losses. Ounces are rounded to the nearest 10,000 ounces;
- Rounding as required by reporting guidelines may result in apparent summation differences between tonnes, grade and contained metal content;
- Tonnage and grade measurements are in metric units. Gold ounces are reported as troy ounces.

15.5 Caution Regarding Forward-Looking Information

The Mineral Reserves are forward-looking and actual results may vary. The risks regarding Mineral Reserves are summarized in the Report (refer to Section 15.6 and Section 25.0). The assumptions used in the Mineral Reserves estimates are summarized in the footnotes of the Mineral Reserves table, and in Section 14.0.

15.6 Factors That May Affect the Mineral Reserves Estimate

Factors that may affect the Mineral Reserves estimates include:

- Metal price assumptions;
- Assumptions relating to geotechnical parameters used in mine design;
- Assumptions that go into defining the unit costs used to evaluate Mineral Reserves;
- Dilution estimates based on past performance and geotechnical input including pastefill / backfill integrity of adjacent stopes;
- Mining and metallurgical recovery assumptions;
- Variations to the expected revenue from short-term marketing and sales contracts;
- Royalty or taxation assumptions;
- Permitting, operating or social license regime assumptions.

15.7 Comments on Mineral Reserves Estimates

In the opinion of the QP, the Mineral Resources and Mineral Reserves have been performed using industry-accepted practices and conform to the requirements of the CIM Definition Standards. The Mineral Reserves are adequate to support mine planning.

16.0 MINING METHODS

16.1 Geotechnical Considerations

Ground conditions at Callie are generally good and the stress issues at present depths are manageable. Stress modelling indicates that stress related issues will increase with stoping depth, but that these changes are manageable and less severe than experienced at most other deep mines in Australia.

Stress measurements completed during 2018 indicate that the stress gradient gradually increases with depth. Typical rock mass uniaxial compressive strength (UCS) values range from 60 megapascals (MPa) to 150 MPa in the deeper working areas (below 700 m below surface) where the majority of the stoping currently takes place.

Ground support is achieved using a combination of split sets, cable bolts, and mesh. Pattern bolting and mesh is used in development. Permanent development consists of mesh and 2.4 m long split sets with rows spaced a maximum of 1.6 m apart. Drive intersections, stope draw points and areas encountering adverse geological conditions are cable bolted with 6 m twin strand bulbed cables. Fibrecrete is available on-site for use as surface support as required, particularly in lower levels due to deterioration of ground conditions.

Stope backs are assessed using a Modified Stability Number and several other empirical charts using factors such as 'Q', hydraulic radius, stress ratio, cable bolt density and cable bolt lengths to estimate ground support required. Numerical modelling is done to help achieve the most favorable extraction sequences from a stress perspective.

The Callie underground geotechnical parameters and ground support program are documented in the Callie Ground Control Management Plan (GCMP).

The main geological structures and lithology units identified at Callie are modelled as wireframes. Their geometry and characteristics are considered during geological block modeling and the stope and development design processes.

The mechanical properties of the different lithologies are similar. Outside weathered and faulted zones, a single geotechnical domain is used. Geological structures that could potentially affect stoping are identified and characterized in advance from geotechnical logging results and geotechnical mapping.

The geotechnical department has compiled a LOM Planning Guideline which outlines the maximum hydraulic radius to design to by mine area.

16.2 Hydrogeological Considerations

In the Tanami region, surface drainage is generally undefined with water movement predominantly by sheet flow. There are no significant permanent surface water bodies within or near The Granites or DBS, however small seepages exist around The Granites outcrop east of The Granites lease.

At DBS, ephemeral streams drain to the north after significant rainfall events. The DBS occurs to the north of the lease boundary and is a fenced exclusion zone under Newmont's agreement with the CLC.

During the wet season, salt lakes form in the vicinity of the palaeochannel and depending on the extent of rain, these may persist well into the dry season.

Groundwater at DBS is limited with the best flows achieved in fault sets within the lease. Very little by the way of water flow is achieved outside fault zones. Mining activity has modified the water table lowering it by some tens of meters (Robertson GeoConsultants, 2007). Groundwater quality is brackish and inflows into the underground mine are variable and estimated at between 4 to 10 L/sec. In general, the Callie underground mine is considered a 'very dry' mine (Robertson GeoConsultants, 2009).

Groundwater monitoring bores were installed on the DBS mineral lease and surrounds in 2005 and 2007. Groundwater results indicate very little, if any, impact on water quality from past or ongoing mining activities with a circum-neutral pH and low concentrations of dissolved metals (Robertson GeoConsultants, 2008).

However, the bores do show significant variations in major ion composition and salinity which varied from fresh (total dissolved solids [TDS] 320 mg/L at DBS10) to saline (TDS 9,100 mg/L at DBS9). Fresher groundwater is typically observed in the western portion of the DBS lease with more saline groundwater to the north and south. This range in salinity was also observed in the historic groundwater quality data set. In general, the groundwater is brackish, and the major ions include sodium, magnesium, chloride and sulfate (Robertson GeoConsultants, 2008).

Overall, the Project mining operations are considered to be dry, with annual pumping from operations being in the range of 20 L/m.

16.3 Mining Method

Currently the Project comprises no active open pit operations. All mining production is from the Callie underground mine.

16.3.1 Method Selection

The levels are named based on the surface Reduced Level of 1,400 m, so 400mRL is located 1,000 m below surface.

Access to the mine is via the Callie Decline located in the footwall of the Callie mine area. At the end of December 2018, the decline had reached the elevation of M20mRL, where "M" represents levels located below 0 mRL ("Minus"). Nominal level spacing is 40 m, with additional sub-levels in the upper part of the mine. For many years, mine production has focused on the Callie mine area and peripheral extensions. The Auron orebody was identified relatively recently, with production activities commencing in 2013. Development of an independent decline system for Auron commenced in 2013 and production activities started late in the same year.

The general geometry of the mine can be seen in the long projection presented in Figure 16-1. This long projection shows the development and stopes already mined.

The mining method used at the Project is long-hole open stoping with backfill, using a 92.5% mine recovery factor and a stope dilution factor incorporated in the reserve mine plan (as applied to the Mineral Resource model prior to consideration as Mineral Reserves) between 7% and 20% as per geotechnical recommendations (based on the stope location).

Prior to the commencement of the Callie underground mine the plan was to use sub-level caving (SLC). However, shortly after the commencement of underground development it became apparent the ore body differed from the early interpretations and was found to be smaller with discrete high-grade zones. Geological models using hard lithological boundaries favored a more selective mining method; hence sub-level open stoping (SLOS) was adopted.

SLOS enables a higher recovery and lower dilution to be achieved than SLC and is not as development-intensive as SLC. However, SLOS incurs higher cost primarily because of the need to backfill stopes. In addition to higher recovery, other factors that influenced the decision not to use SLC were the elimination of air blast potential and the plunge of the ore body, approximately 45° and could potentially cause problems with the behavior of the cave.

To date SLOS has proven to be a successful mining method for the various mining areas within Callie.

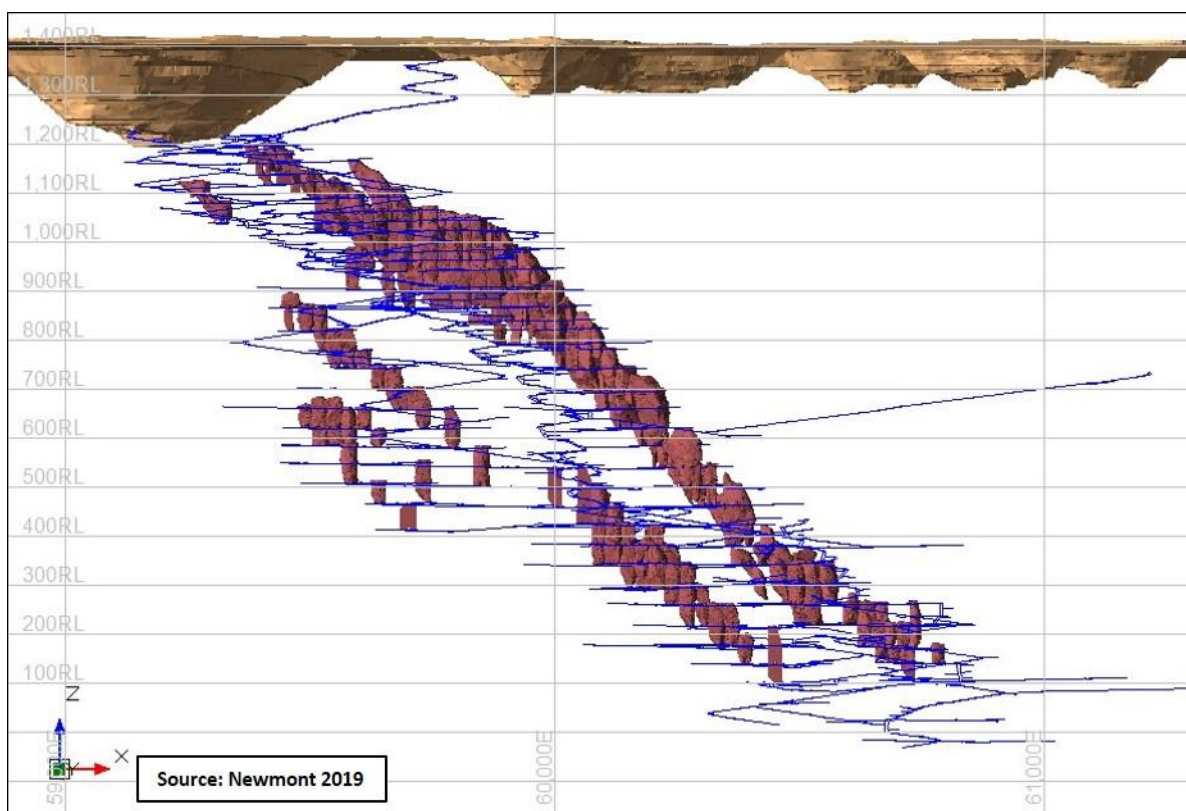


Figure 16-1: Long Section of Callie Mine Showing Existing Development and Stopes

16.3.2 Mine Development

Mine development is drilled using twin boom electric hydraulic jumbos. The jumbos are also used to install ground support.

The mine is accessed using a 5.5 m wide x 5.8 m high fully arched 1:8 to 1:7 decline. The decline spirals down following the plunge of the ore body on the footwall side. Levels are generally accessed every 40 m with the occasional intermediate sub-level in the upper levels. Level development typically consists of an access drive 5.0 m wide x 5.5 m high running along the footwall of the ore body.

Drill drives 5.0 m wide x 5.6 m high are developed off the access drive into the ore body. Stopes typically have one drill drive per level and two draw points drives (5.5 m wide x 5 m high) on their extraction level.

In addition to the decline and associated development for the primary ventilation system, other capital development designed includes stockpiles, sumps, diamond drill cuddies, escape ways and ventilation rises.

16.3.3 Auron Decline

Haulage efficiency is a key factor in the design basis and overall mining strategy and truck haulage was previously identified as one of the main mining constraints at the Project. The Auron Decline link from 1200RL to 425RL was completed in 2016 to create a one-way traffic haulage loop to support mining down to -260RL, which decongested the access network for the mine.

The stopes targeted for mining in the early years are from material in the upper levels of the mine, above ORL because areas are diamond-drilled to a higher drill density and confidence level. Balancing and sequencing tonne-kilometers (tkms) between upper levels and lower levels is an important strategy so that the mine continues to be economic and well exploited. This strategy is mainly based on the availability of work areas for the stoping sequence.

Mining of stopes below ORL will commence in 2020, once development is completed. The benefit of this mining option is that it produces a sustained ounce profile for a prolonged period and assists in maximizing the extraction of the resource.

16.3.4 Tanami Expansion 2 (TE2)

A current project developed during 2018 is TE2. TE2 aims to maximize value from the Project by determining the optimum method of extracting the next block of ore beyond the current business plan, thereby debottlenecking mine production, reducing operating costs and extending mine life. This project is currently at feasibility study level.

TE2 is focussed primarily on a 1,460 m production shaft and associated infrastructure. A shaft is considered to present best long-term value for the Project to reduce operating costs, and to access ore at depth. The shaft option requires fewer trucks underground, thus reducing labor cost, and reducing energy requirements both for fuel and refrigeration.

The upfront investment for TE2 is US\$635M.

TE2 will facilitate future growth opportunities through optimized production, lower operating costs and improved access to exploration platforms.

16.4 Backfill

To achieve higher recoveries, primary stopes need to be filled using engineered fill. This allows extraction of adjacent ore in secondary stopes. Secondary and tertiary stopes are filled with rock fill.

Until 2008, Cemented Aggregate Fill (CAF) was used to fill primary stopes using a dedicated surface borehole for each stope. CAF filling was suspended in October 2008 after proving unsuccessful at depth. The boreholes were closing, and the fill rates could not sustain production. This caused disruption to the schedule and affected recovery and dilution

performance until a paste fill plant was commissioned in June 2011 and engineered filling activities were able to resume.

Currently paste fill is the used as the primary method, with waste rock fill used as a secondary stope fill method.

Historically the main constituent of the paste was material recovered from old tailings dams. This material typically contained a significant proportion of clay which provided the rheological characteristic required for reticulation. The paste plant is in the process of switching to a tail provided directly from the processing plant, rather than recovering tails from the tailings dam. A desliming and filtering plant is being commissioned to supply a consistent tails product to the paste plant.

The underground reticulation system consists of two surface boreholes, 368 m in vertical length, which intercept the 1020L of the mine. The pipeline then splits to support filling both the Callie and Auron mining areas.

Rock fill comprises loose rock from underground development waste, where waste is defined as material with a grade below 1.6 g/t Au. Rock fill is a significantly less expensive fill material and provides a cost saving relative to hauling waste rock to surface. Concurrent scheduling of stoping and development activities is undertaken to ensure waste rock use underground is maximised.

16.5 Ventilation

The current ventilation system involves three exhausts and five major intakes two of which provide chilled air. Two of the intakes also serve as major access declines to the mine. Currently the ventilation draw is 1,100 cubic meters per second from three exhaust fans with a total for a total of 7 MW of power draw. There are two refrigeration plants providing a total of 24 MWR of cooling power (6MW of installed power). The current Mineral Reserve and Mineral Resource mine plan requires an additional exhaust ventilation rise (VR8) and upgrade of the refrigeration system for VR3. This is required to mine Federation and the extensions to Callie and Auron orebodies at depth (refer to Figure 16-2).

Primary ventilation control is achieved using drop board regulators at the breakthroughs to rises. For the control of airflow throughout the primary ventilation system, ventilation walls combined with other control devices are used. Secondary ventilation involves the use of underground development fans blowing air into headings through flexible ventilation ducting. Figure 16-2 and Figure 16-3 provide schematics of the ventilation system proposed under the current Mineral Reserves mine plan.

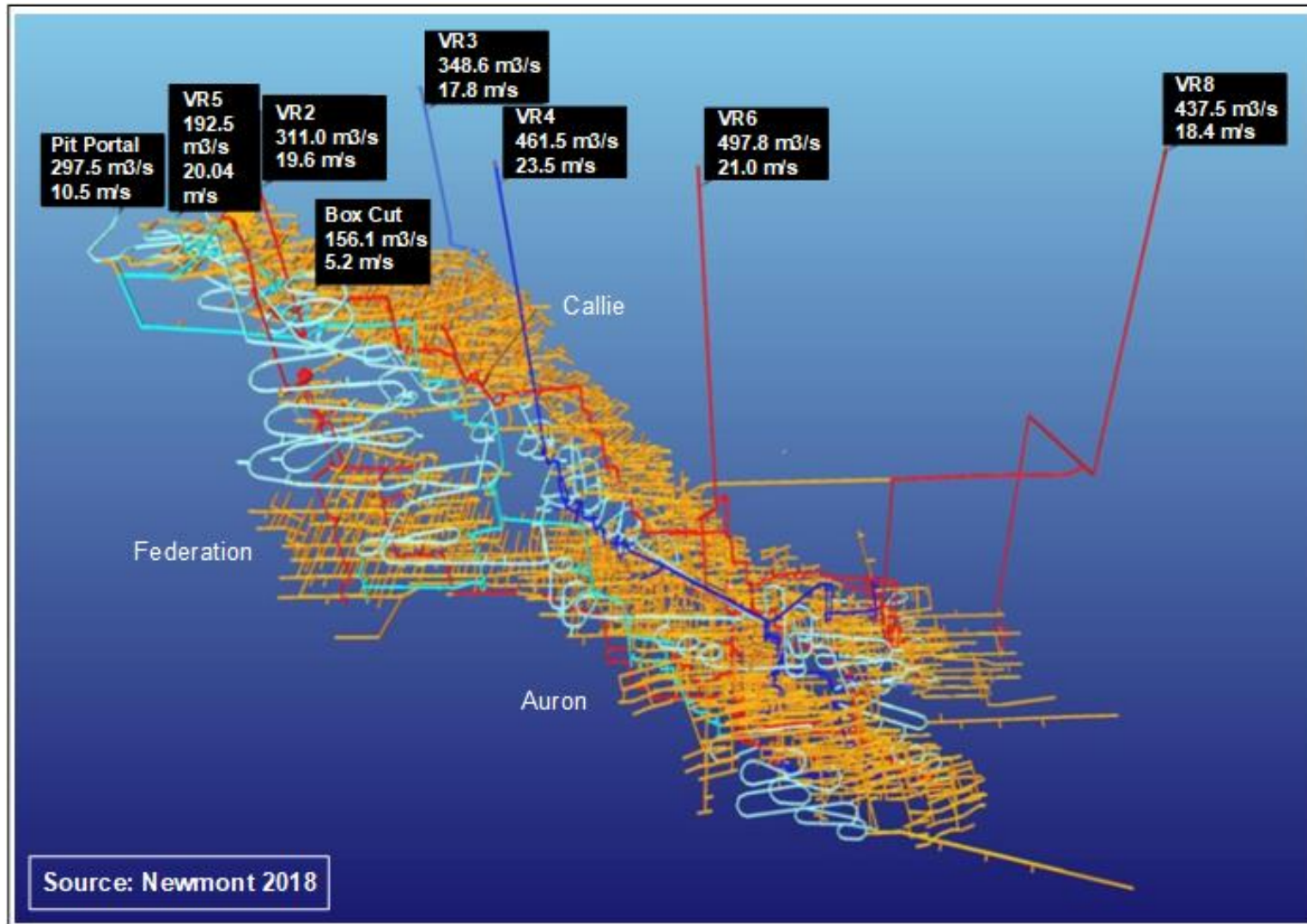


Figure 16-2: Schematic Showing Proposed Ventilation Network Required for the Current Mineral Reserves Mine Plan

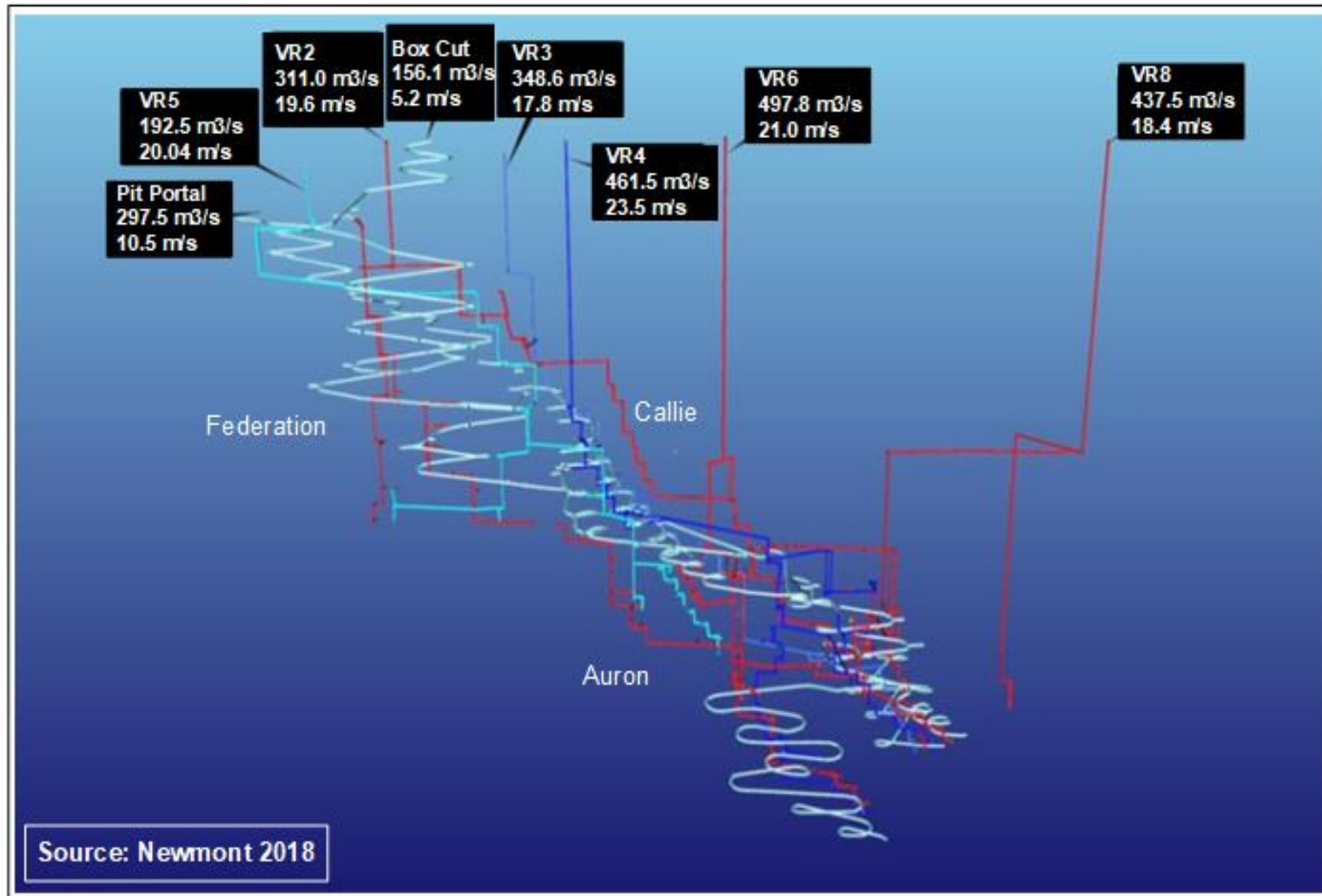


Figure 16-3: Schematic Showing Proposed Ventilation Network Required for the Current Mineral Reserves Mine Plan

16.6 Underground Infrastructure Facilities

Key underground infrastructure that supports the Project is presented in Table 16-1.

Table 16-1: Summary of Key Underground Infrastructure

Key Underground Infrastructure	Location (Mine Level)
Major Pump Stations	060L, 260L, 460L, 810L
Workshops	460L, 790L
Magazine	980L
Fresh Air Rise	
<i>Vent Rise 4</i>	Surface to 100L
<i>Vent Rise 5</i>	Surface to 260L
Return Air Rise	
<i>Vent Rise 1</i>	Surface to 820L
<i>Vent Rise 2</i>	Surface to M020L
<i>Vent Rise 3</i>	Surface to M020L
Auron Decline	1190L to 425RL
Callie Decline	Surface to M020L
Wilson Decline	425L to M020L
UG Water Storage Facilities	1010L, 750L, 700L, 540L and 220L
UG Compressors	380L (Planned)
UG Power	Auron, Callie Ring Mains
Paste Reticulation	Surface to 180L

16.7 Production Schedule

The base case assumption for the Mineral Reserves Plan is that the TE2 project will deliver the shaft by 2023. The TE2 project allows for debottlenecking of the mine production, improves mine haulage, and provides for more efficient usage of the current mill capacity.

Underground ore production is presently scheduled at 2.7 Mtpa. The waste produced accounts for up to an additional 800 kt, which results in total tonnes mined of around 3.5 Mtpa. Annual jumbo development has been scheduled at around 10 km per annum for the next five years, before decreasing as requirements drop off.

The business plan is developed on a month by month basis for 2019, 2020 and 2021, and then reported on an annual basis from 2022 through the end of mine life.

Mining operations are currently scheduled to 2028 in the Mineral Reserves Plan.

16.8 Blasting and Explosives

Production drilling incorporates up holes, down holes, full 360° rings or partial fans. Production drilling primarily utilizes 102 mm diameter down holes. In some instances, 89 mm and 64 mm diameter holes are used. The standard long-hole burden for 102 mm blast holes at Callie is 3.0 to 3.5 m. The toe spacing for the 102 mm blast holes is 3.0 m to 3.8 m. The production drilling rate averages 12,000 m per month. The current preference is for drilling down holes.

Blasting is carried out with the use of mobile explosives vehicles primarily utilizing Emulsion explosives. Non-electric detonators are used in mine development and electronic Ikon

detonators are used in production stoping. Blasting typically takes place at the end of shift with only the blasters remaining in the mine; however, localized blasting on demand can be permitted in designated “safe” zones when applicable.

16.9 Mining Equipment

Conventional underground mining equipment is used to support the underground mining activities. This equipment is standard to the industry and has been proven on site. The underground equipment fleet is in good working condition and a large percentage has recently been replaced or overhauled as part of the natural equipment rebuild/replacement schedule.

Equipment assumptions are based on BP19. The equipment list is as follows:

- 5 x Elphinstone 2900 LHD (6th on site) underground loaders;
- 17 x Atlas Copco MT6020 trucks;
- 4 x Tamrock twin boom jumbo;
- 2 x Tamrock solo production drills;
- 2 x charge-up vehicles;
- 6 x integrated tool carriers;
- 1 x Robbins 41R raisedrill;
- 1 x Robbins 73R raisedrill;
- 12 x LM75 diamond drill rigs;
- 1 x Cable bolt Rig – Sandvik DS421-C.

Comprehensive maintenance tracking and reporting systems, in addition to preventative maintenance (PM) programs are well established.

Site maintenance facilities are considered adequate to support the fleet and exist in centralized facilities both underground and on surface.

PMs are routinely performed as per the manufacturer’s recommendations, which is generally at 250-hour intervals.

Frame-up rebuilds are performed based on engine hours, as recommended by the equipment supplier, and/or based on component wear factors. Major overhauls and rebuilds are often done offsite at a contracted facility. The LOM equipment replacement/rebuild schedule is adequate to support the mine plan.

16.10 Mine Plan Considerations

While the effective date of the mine plan is 31 December 2018, Newmont will undertake routine reviews of the mine plan. Alternative scenarios and reviews are based on ongoing or future mining considerations, including but not limited to the application of continually updated modifying factors.

16.11 Comments on Mining Methods

In the opinion of the QP:

- The mining methods used are appropriate to the deposit style and employ conventional mining tools and mechanization;
- The life of mine underground mine plan has been appropriately developed to maximize mining efficiencies, based on the current knowledge of geotechnical, hydrological, mining and processing information on the Project;
- The equipment and infrastructure requirements for LOM operations are well understood. Conventional underground mining equipment is used to support the underground mining activities. This equipment is standard to the industry and has been proven on site. The underground equipment fleet is in good working condition and a large percentage has recently been replaced or overhauled as part of the natural equipment rebuild/replacement schedule. Appropriate allocation has been made for overhaul and rebuild of underground equipment, as required. The LOM fleet requirements are appropriate to the planned production rate and methods;
- The predicted mine life to 2028 is achievable based on the projected annual production rate and the Mineral Reserves estimated.

17.0 RECOVERY METHODS

17.1 Process Flow Sheet

The current plant circuit, which includes gravity concentration, leach, CIP and gold recovery circuits, is presented in Figure 17-1.

17.2 Plant Design

The initial process plant was constructed in June 1986 with an initial throughput of 0.36 Mtpa. It consisted of a crushing circuit and one ball mill feeding a CIP plant. This plant was upgraded in September 1989 via the addition of a low-aspect semi-autogenous grind (SAG) mill to maintain gold production as the mill feed head grades declined.

In the upgrade, a SAG primary mill was installed ahead of the ball mill and an additional 2,400 m³ of leach capacity was installed together with the replacement of cyclones, pumps and screens to enable the higher flowrates to be achieved. A mill scat handling and crushing system was installed in late 1990 which further increased the mill capacity to around 1 Mtpa. A third mill was installed in 1998, and the processing plant ran in this configuration (SAG mill and two ball mills) until November 2003 when all three mills were decommissioned and a single 5.5 m x 9.3 m, 4,700 kW Outokumpu ball mill was installed. This mill is currently operating today, fed by a 3-stage contract crushing facility.

Previous modelling of plant throughput rates indicated a plant capacity of up to 2.3 Mtpa (230 tph) prior to the TEP. Achieved rates during 2015 to 2017 were sustainably higher, with weekly averages feed rates exceeding 300 tph when ore availability permitted.

To process additional ore that arose from increased mining rates as a result of completing a second decline link in the underground mine, various improvements to the plant were designed, constructed and then commissioned in Q3 2017 as part of the TEP. Specifically, these included:

- A second ball mill to allow the annual average throughput to be increased from 2.3 Mtpa up to 2.6-2.8 Mtpa;
- Classification and efficiency improvements due to the installation of a pre-leach thickener;
- Wet plant upgrades, such as sampling stations, a tailings deslime circuit for the paste filter plant feed preparation and a tailings filter plant installation;
- An upgraded gravity concentration circuit to maximize gravity gold recovery.

Post-TEP, and to the end of 2018, significantly higher feed rates than the 333 tph design were achieved and sustained, in the order of 380 tph to 390 tph. Operational issues prevented further increases and very few bottleneck conditions were experienced.

Current mining forecasts provide ore feed rates well within the above ranges, between 300 to 350 tph. The post-TEP operating experience demonstrates a high level of confidence that the planned throughput rates will be achieved.



17.3 Crushing and Ore Handling

Underground ore is transported 44 km by trucks to The Granites processing plant site and stockpiled on a ROM pad adjacent to the crushing plant. The ROM ore is crushed in a 3-stage crushing circuit to an approximately P_{80} of 12 mm. The equipment utilised to crush the ore is as follows:

- Primary Jaw Crusher (Vicker Reiwooldt 42" x 30" double toggle);
- Secondary Cone Crusher (Symonds 7' SXHD);
- Tertiary Cone Crusher (Symonds 7' SXHD).

The primary jaw crusher is fed ROM ore that is smaller than 800 mm.

The secondary crusher is fed with the product from the primary jaw crusher and continues to reduce the ore size. Ore from the secondary crusher then passes to the tertiary crushers.

The tertiary crushers operate in a closed circuit with double and triple deck vibrating screens to produce a coarse ore (less than 30 mm) and fine ore (less than 14 mm) product. The tertiary crusher product is fed onto the screens in order to separate the smaller particles (undersize) from the larger particles (oversize). The undersize is the final product from the crushing circuit and is conveyed into a large ore holding bin (with a 3,500 t capacity) for storage. The oversize is conveyed back to the tertiary crusher for further size reduction.

17.4 Grinding

Crushed ore is recovered from the ore bin via two apron feeders and then ground in a Primary overflow ball mill (4,700 kW installed power) operated in closed circuit with a hydrocyclone distributor consisting of ten 400CVXFB Weir Warman Cavex Flat Bottom cyclones. Quicklime is added to the mill feed conveyor via a screw feeder located beneath a lime silo. Trommel oversize is discharged into a scats bay and periodically recycled back to the ball mill via an emergency feeder located adjacent to the ore bin.

The primary ball mill circuit cyclone overflow is fed to a secondary grate discharge ball mill (2700 kW installed power). This mill is in closed circuit with ten 400CVX10 Weir Warman cyclones. The targeted liberation grind size from the grinding circuit is within a P_{80} range of 50 μm to 80 μm , when both the primary and secondary ball mills are operating. The cyclone overflow is sampled via an automated, cross-stream sampler and reports to a pre-leach thickener ahead of the leach/CIP circuit.

17.5 Gravity Separation

Gravity recoverable gold is recovered from the grinding circuit in four distinct circuits. The primary milling gravity circuit comprises Dutch State Mines (DSM) screening of two primary cyclone underflow streams, with the DSM screen undersize reporting to two distinct gravity separation circuits. One circuit consists of rougher and cleaner spirals and Wilfley tables, and the other circuit uses a 30" Knelson concentrator. Each circuit is fed from the underflow stream of a dedicated 400CVXFB cyclone located within the primary grinding circuit hydrocyclone distributor. A second primary milling gravity circuit was installed as part of the TEP project, however; pumping issues to and from the circuit has meant that this system is currently not operating (as of December 2018). The circuit is expected to be fully commissioned and

operating in 2019. The secondary grinding gravity circuit comprises a split of the secondary cyclone underflow, which is fed to a single 48" Knelson concentrator

Each of the gravity cyclone underflow streams passes over a dedicated DSM screen. Screen underflow from the spiral's DSM is pumped to a parallel bank of ten triple feed rougher spirals followed by three double feed cleaner spirals. The concentrate from the cleaner spiral is fed to two Wilfley tables in series. Tailings and middlings from both spirals are pumped back to the feed end of the ball mill. The second Wilfley table concentrate is pumped to a hopper in the gold room, which also collects concentrate from the 30" Knelson and is periodically transferred to the hopper above the TEP installed CS4000 Acacia reactor.

Underflow from the Knelson DSM screen flows under gravity to a 30" Knelson concentrator. Concentrate from the Knelson concentrate flows to a hopper in the gold room, which also collects the concentrate from the spirals Wilfley tables circuit and is periodically transferred to the hopper above the TEP install CS4000 Acacia reactor.

Concentrate from the Secondary milling circuit 48" Knelson concentrator also reports to the Acacia reactor feed hopper. The Acacia reactor is operated on a cycle of one (up to two) batches per day. Electrowinning of the Acacia pregnant solution is carried out in a dedicated 18-cathode electrowinning cell.

Overall, the proportion of feed to the gravity section has increased from about 9% of circulating load to about 25% and will increase further up to 40% once the new primary gravity circuit is fully commissioned.

17.6 Leach and Adsorption

Cyclone overflow from the grinding circuit gravitates over a vibrating trash screen for the removal of wood pulp and plastic material and then reports to the Pre-leach thickener ahead of the leach circuit. The pre-leach thickener has allowed the milling circuit to operate at more traditional operating densities with cyclone overflows of 35% solids increased to 50% solids for feed to the leach tanks.

The leach circuit consists of three 1,500 m³ tanks and one 600 m³ tank. Cyanide is added to the head of the circuit with a top up to the third leach tank. A cyanide level of 140 to 150 g/m³ is maintained within the first and third tanks of the leach circuit, with tailings exiting the plant at less than 45 g/m³. Oxygen can be injected into all four leach tanks via the shaft of the tank agitators.

Oxygen is produced on site from two pressure swing adsorption (PSA) plants – via two tpd and three tpd units. Destruction of the cyanide on the CIP tail is achieved by injecting Caro's Acid into the tailings line. The total leach and adsorption residence time at a throughput rate of 295 tpd and slurry density of 50% solids is about 22 hours.

The CIP circuit consists of seven 600 m³ adsorption tanks fitted with air agitated vertical wedge wire inter-tank screens. Carbon movement is via Weir Warman recessed-impeller pumps. Carbon concentrations are maintained at 8-10 g/L in each tank, depending if there are six or seven tanks on line due to maintenance inspection requirements.

17.7 Deslime and Filtration

As part of the TEP, a deslime section and filter plant were installed. The deslime section consists of a cluster of 14 x 250 CV20XV Weir Warman Cavex cyclones, which are fed at a

constant rate, with the balance of tailings are directed to the tailings thickener, thus bypassing the deslime cyclones. Overflow from the deslime cyclones also reports to the tailings thickener, and the combined thickener underflow is pumped to the TSF. Deslime cyclone underflow is pumped to the filter plant.

The filter plant consists of a 158 m² Jord vacuum filter belt. Filtercake from the filter belt is discharged onto a conveyor belt which feeds a radial stacker. This discharges onto a 10,800 t capacity stockpile.

Both the deslime section and filter plant generated material as per the design criteria of 20% passing 20 µm at 130 tph. However, ongoing investigations are underway to optimise the specification of material delivered by these plants and it is expected future modifications to the plant and operational practices will be required to meet these specifications. While this has been underway, material provided to the mine for paste backfill has consisted of a blend of filtered material and mined tailings.

17.8 Gold Recovery

Final loaded carbon is removed from the head of the adsorption circuit at a normal rate of two 3.4 t batches per day. Elution is carried out via an Anglo American Research Laboratories (AARL) process. The carbon is initially acid washed (cold) in a separate rubber lined vessel and then transferred to the stainless-steel alloy elution column.

Elution is carried out at 135 °C and a pressure of 330 kPa. To conserve water during the elution cycle, a split elution is practiced. Electrowinning is carried out in an 18-cathode electrowinning cell. Loaded stainless steel wool is high-pressure cleaned, and the gold sludge concentrate is vacuum filtered, dried, and smelted in a conventional tilting furnace.

Eluted carbon is regenerated in an Ansac horizontal rotary kiln operating continuously.

There were no changes to the acid washing, elution and thermal regeneration section as part of the upgrade. In the gold room, a new 18 cathode electrowinning cell was installed. The upgraded flowsheet allows the separate recovery and smelting of gravity and carbon recovered gold. This has assisted the monitoring of gold production from the two circuits and facilitates improved met accounting variance investigations and analysis.

17.9 Process Control and Metallurgical Accounting

The Tanami processing plant has a Supervisory Control and Data Acquisition (SCADA) control system using Citect software. Equipment protection via interlocks/alarms and control loops of the major streams/processing parameters within each process circuit are actively in use within the process plant.

The mill feed rate is controlled by a weightometer and variable speed apron feeders at the coarse ore bin. The tailings thickener is controlled via rake torque and bed pressure to maintain a clear overflow and a high-density tailings stream before Caro's Acid addition and deposition into the TSF.

Metallurgical accounting is performed using a site built, Microsoft Access database-based program. For the relatively straightforward process flowsheet this has been considered acceptable.

17.10 Gravity Circuit Optimization

A study into the Gravity Recoverable Gold (GRG) (La Plante, 2001; 2002) found that the feed to the processing plant had a high GRG content of 80% and that the 45% recovery by the gravity circuit being achieved in 2002 could be improved significantly. The study also showed that the gravity spirals that were operating at the time were probably better suited to the circuit than the mechanical centrifugal units currently common in the gold industry. The major losses to gravity recovery were from the Wilfley tables.

Circuit improvements and optimization increased the recovery of gold by the gravity circuit up to 66%. In 2007, further gravity circuit improvements were undertaken to increase gravity recovery to 70%. A 30" Knelson concentrator was installed parallel to the spirals and a CS1000 Acacia intensive leach reactor was installed as the primary intensive leach reactor.

In March 2012 an extensive gravity and milling circuit plant survey was undertaken by Curtin University Gold Technology Group as part of the AMIRA P420D project. At the time of the survey the processing plant was treating 100% ore from the underground mine, sourced from the Callie orebody. The objective of the survey was to clarify current circuit performance, identify circuit optimization opportunities as well as develop a model of the gravity circuit. Feed samples were taken of the ore treated during the survey to undertake laboratory ore characterization and subsequent GRG testing.

The GRG performance of the spirals was found to be excellent with 95% of rougher spiral feed GRG reporting to cleaner concentrate. The recovery of fine gold on the spirals ($<53\text{ }\mu\text{m}$) was lower than in the coarser fractions at 50%. The Knelson concentrator GRG recovery was unexpectedly low at 22%, with highest recovery in the finer size classes. Overall, however, the two gravity recovery devices worked synergistically, leading to high overall recovery for the circuit. The Wilfley table was set up well for the survey with continuous operation yielding 97.8% recovery to concentrate (similar to intensive cyanidation). Classification was an area requiring attention as the cut of the hydrocyclones was not particularly efficient and could be leading to mis-representation of GRG in the CIP circuit.

17.11 Diagnostic Leach Testing

The effectiveness of the process plant gold recovery and extraction is determined using diagnostic testing of the final tails. This process provides a quantitative measurement of:

- Free milling gold available for cyanide leaching;
- Gold locked up in various mineralogical species.

The sample is a composite made up of shift tail samples over a calendar month. The testing is conducted by the on-site laboratory. The results of these tests help to determine the focus of plant improvements and provide quantitative justifications for capital expenditures. Since the TEP, the grind size achieved has been significantly lower than the P_{80} of $125\text{ }\mu\text{m}$ expected during design, with P_{80} s of 50 to $70\text{ }\mu\text{m}$ over extended periods not uncommon. As expected with finer grind sizes, the quantity of gold locked in carbonates, sulfides and silicates has reduced. Silicate locked and extended leachable (cyanide soluble) gold are similar at 34% and 32% of gold in tailings solids respectively, with gold locked in carbonates and sulfides also similar at 16% and 18% respectively. All four have a range between their maximum and minimum from 21% to 43%, which reflects the variation month to month in the distribution of the locked gold.

17.12 Gold Recovery Historical Performance

A review of monthly reconciled data from 2008 to 2018:

- Feed gold grade has increased over the period from mid-2010 to the end of 2013/start of 2014, after which it has varied, but overall not increased further;
- Since the TEP, recovery has increased by approximately 1.7%. This is in line with the estimated improvement of 1.55% used in the recovery models;
- Gravity recovery has improved with the TEP expansion of the gravity circuit. Even though feed rates have increased, the quantity of gold being fed to the leach section is similar to pre-TEP;
- For October 2018, the reconciled feed grade for the month was 9.90 g/t Au. This was 1.56 g/t Au and 1.99 g/t Au higher than the next two best months in the last five years, May 2016 and April 2018 respectively. Recovery for the month was 97.9%. This confirmed the plant can recover higher quantities of gold than are normally fed to the plant and gives confidence in the ability of the plant to perform at higher throughputs in the future;
- There is an inverse relationship between gravity and leach recoveries. As gravity recovery increases, there is a decrease in leach recovery and vice versa.

17.13 Energy, Water and Process Materials Requirements

The process plant consumes approximately 30 kWh/t of underground ore processed.

Major consumables include grinding balls (0.441 kilograms/tonne [kg/t] in 2018), gold recovery reagents (sodium cyanide (0.298 kg/t), lime (0.637 kg/t), sodium hydroxide (0.209 kg/t), cyanide destruction reagents (sulfuric acid (0.632 kg/t), hydrogen peroxide (0.208 kg/t) and liquid petroleum gas (0.157 kg/t).

Raw water is pumped to the plant site at a maximum rate of 60 L/s from two bores at the Jumbuck and two bores at the Billabong borefields. This is an increase from prior to TEP, with the additional water being used in the filter plant for rinsing the filtercake to reduce residual cyanide.

The TDS for Billabong averages less than 2,400 g/m³ and at Jumbuck less than 7,000 g/m³. Two reverse osmosis water treatment plants supply potable and elution circuit water for the Project.

17.14 Comments on Recovery Methods

In the opinion of the QP:

- The mill throughput, process and associated recovery factors are considered appropriate to support Mineral Resource and Mineral Reserves estimation, and mine planning requirements for the current and future expanded operations.

18.0 PROJECT INFRASTRUCTURE

Due to the remote geographic location of the Project a completely stand-alone infrastructure has been established to cater for situations of isolation and absence of local services. The site infrastructure is predominately split between The Granites (refer to Figure 18-1) and DBS (refer to Figure 18-2) operational areas.

18.1 Road and Logistics

Project personnel primarily access the site via air. Regular charter flights to The Granites airstrip are operated by Alliance Airlines from Darwin, Alice Springs, Brisbane and Perth. In addition to providing passenger transport, the charter flights provide a limited air-freight capacity. The Granites airstrip is sealed with bitumen and flights can operate in most weather conditions.

Dirt roads extend around and throughout both The Granites and DBS providing access to waste rock dumps, tailings storage facilities and to the surface explosive and ammonium nitrate storage areas as well as the core laydown storage area. There are also bitumen roads from the accommodation village to the DBS haul road and the accommodation village to the processing plant.

The Schist Hills bore field is connected to DBS via an unsealed track that extends from the north-east of DBS mineral lease. The 22 km long bore field pipeline runs adjacent to this track. The Granites is connected to the Schist Hills bore field via an unsealed track that extends for 20 km from the western end of the mineral lease which is followed by the pipeline. The Granites is also connected to the Billabong borefield via a 30 km long pipeline along an unsealed road from the northern side of the lease.

DBS is linked to The Granites by a sealed 44 km haul road. Ore is hauled by MLG Oz haulage contractors using 220 t capacity road trains. The road trains travel from DBS RIM pad to The Granites ROM pad via the haul road. The road trains consist of a triple configuration (prime mover and three trailers). The haul road is also used for personnel transport and freight.

18.2 Waste Storage Facilities

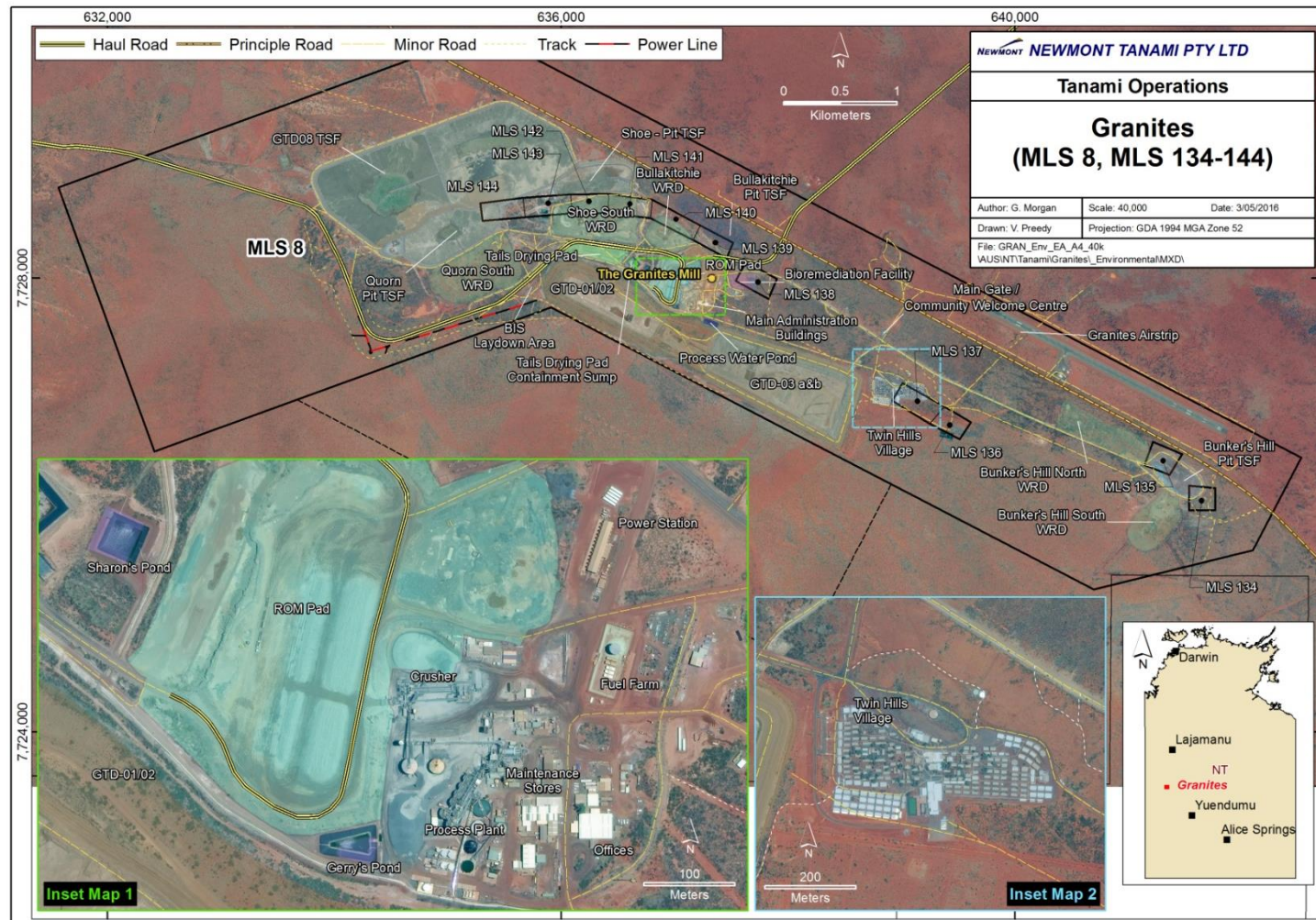
Wherever possible, waste rock is deposited underground as stope backfill and waste rock is rarely brought to surface. Table 18-1 details the waste rock facilities (WRFs) currently in existence for The Granites and DBS mineral leases and indicates the status of each facility. Of the nine waste rock dump facilities that exist on the project, only Callie North, located at DBS, is active.

Due to its inert nature, waste rock is periodically excavated from Callie North WRF and crushed to provide road base and aggregate rock for use in construction and as road base.

Table 18-1: Summary of WRF

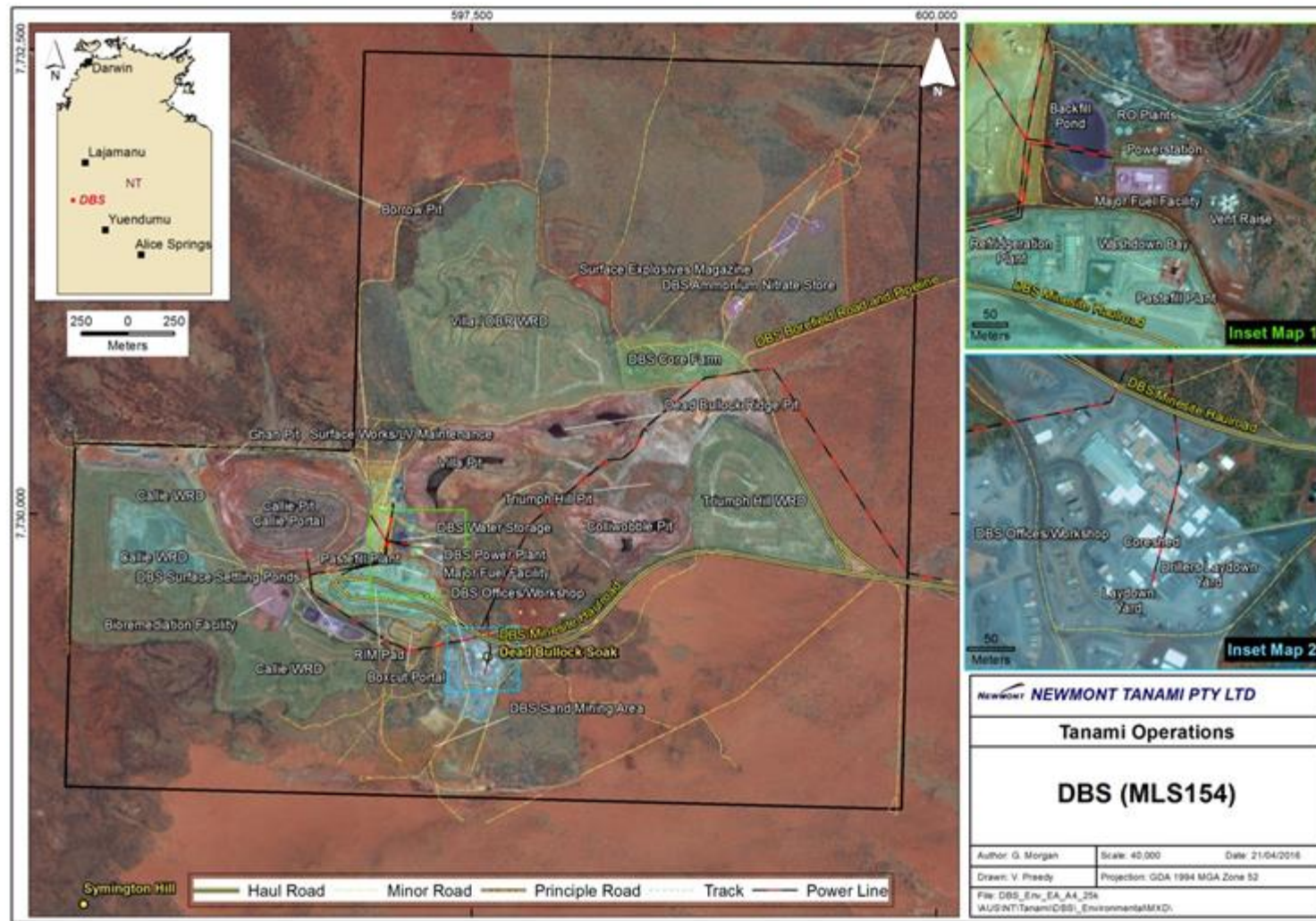
Lease	WRF	Status
The Granites, MLS8	Quorn South	Rehabilitated. Incorporated into biennial rehabilitation monitoring program.
	Shoe South	Partial use of material for TSF construction.
	Bullakitchie	Partial stockpile remaining. Some of the area is being used as the operational scrap steel laydown area.

Lease	WRF	Status
	Bunkers Hill	Partial use for raising of Bunkers wall to closure height and 2016 rehabilitation of the Bunkers Hill In-pit TSF. Remaining landform incorporated in biennial rehabilitation monitoring.
	New Bunkers Hill	Rehabilitated. Incorporated into biennial rehabilitation monitoring program.
DBS, MLS154	Callie North	Partially rehabilitated, open section utilized as alternate waste rock tip if there is no underground alternative. Incorporated into biennial rehabilitation monitoring program.
	Callie South	Rehabilitated. Incorporated into biennial rehabilitation monitoring program.
	Villa/DBR*	Rehabilitated. Incorporated into biennial rehabilitation monitoring program.
	Triumph Hill	Rehabilitated. Incorporated into biennial rehabilitation monitoring program.



Note: Figure aligned to True North

Figure 18-1: The Granites – Site infrastructure Layout



Note: Figure aligned to True North

Figure 18-2: DBS – Site Infrastructure Layout

18.3 Tailings Storage Facilities

The tailings produced at The Granites processing plant have been discharged into the following TSFs:

Table 18-2: Summary of TSF

TSF	Area (Ha)	Tailings Discharge History	Tailings Deposited (t solids)	LOM Capacity & Status
GTD01/02	75	Deposited 1986-1996	9,850,000 (estimate)	Complete – no further tailings to this facility, currently being utilized for tails harvesting for the paste backfill.
GTD03	50	2001 to 2012 2016 to 2017	12,994,685	Final perimeter lift completed in 2010 to allow for increased tailings capacity. A total of 2,001,135 t of tailings were harvested from GTD03 between 2014 and 2016. 1,807,131 t deposited 2016. 597,497 t deposited 2017.
Bullakitchie	10	Oct 1996 to Dec 1998 Topped up during 1998 to 2003	4,839,100	Complete and rehabilitated.
Quorn	26	Sept 2003 to present	11,488,735	This facility is close to completion and is being retained as an optional discharge location.
Shoe	14	Aug 2000 to Sept 2003 Topped up during 2003 to present. July 2010 to present.	7<703,309	It is estimated that ~4 months of storage remains in this facility.
Bunkers	1.3	June 2007 to present.	2,584,526	Complete and rehabilitated.
GTD08 (New Quorn North West Facility)	165 (Cell 1&2), Cell 3 - 75	Commenced July 2013	7,518,426	Discharge commenced into Cell 1 with 1,391,676 t deposited between July 2013 and March 2014, filling Cell 1 to capacity. Discharge to Cell 2 subsequently commenced in April 2014 with 1,544,723 t deposited at December 2014. 2,207,366 t deposited 2015 and 544,545 t deposited 2016. Cell 1 and Cell 2 first upstream lift completed late 2016. 1,830,115t deposited 2017.

In 2013, the Quorn North West TSF (GTD08) was constructed, located to the northwest of the plant abutting the northern embankment of the original Quorn TSF (GTD05). This paddock-type storage facility was designed to be operated and raised on a cyclic basis (stage construction and operation) to ensure that the design densities are achieved. The TSF was designed to accommodate approximately 33.2 Mt tailings within three cells, each with a starter embankment and two raises.

The combined storage capacity of Cell 1 and Cell 2 provides enough capacity to contain a total tailings production of 18.1 Mt, and the capacity of the three cells is enough for storage of the full design basis of 33.2 Mt. This is sufficient to meet the BP19 LOM storage requirements, assuming the projected pastefill requirements for approximately 30% of the tailings volume is realized.

The first upstream lift embankments for Cell 1 and Cell 2 have been completed and have a storage capacity of about 4.6 Mt. The Quorn North West TSF consists of two cells with the following components:

- Tailings transport and distribution system;
- Perimeter main embankments, dividing embankment, decant causeway embankments;
- HDPE liner system at the centre of each cell;
- Under-drainage system installed over the HDPE liner (internal drains and edge of the liner drains);
- Embankment upstream toe drains;
- Seepage collection ponds equipped with LCRS and inspection pits;
- Decant tower and return water system;
- Embankment downstream toe drain; and
- Evaporation/sedimentation pond.

Tailings disposal entails pumping wet thickened tailings (with an average solids content of 62%) out to the TSF, discharging tailings around the edge of each cell, forming a decant pond in the center of the cells, allowing the tailings to settle, consolidate and desiccate and then raising the containment when required and practical.

The tailings are reacted with Caro's Acid prior to deposition to ensure weakly acid-dissociable cyanide complexes have been destroyed prior to exposure on the dam.

To minimize the seepage from the TSF, the Quorn North West TSF is equipped with a series of underdrainage/seepage collection drains.

The Seepage Collection Ponds are double lined and have a detailed seepage monitoring and leak detection system to monitor possible leakage through the primary liner. The potential leakage from the primary liner is collected into a collection pit located at the toe of the seepage collection ponds.

At the end of the construction of Quorn North West TSF, the primary liner installed on the seepage collection ponds were tested for leakage and zero leakage was observed in the inspection pits. It is expected that this no-leak condition will remain the same during the operation of the TSF.

Tailings filtration as well as tailings reclamation is currently being undertaken as a part of requirement to backfill the required areas of the underground mine at DBS with pastefill. The paste fill cycle currently involves the harvesting of tailings from TSFs at The Granites, as well as filtering partly deslimed tailings from the process plant ahead of tailings deposition. The tailings filter plant was installed in 2017 as part of the TEP. While the plant achieved its original design criteria, the ability to discharge tailings in the underground reticulation system is hampered by the current system design and coarseness of product specification. As a result, blending of harvested oxide tailings with filtered tailings is undertaken as an interim solution. A longer-term project will evaluate the best method of tails production from the plant without the requirement for harvesting.

From 2014 to 2016, 2 Mt of tailings from GDT03 were mined. Between March 2016 and January 2017, 1.85 Mt of tailings were deposited in GTD03 while GTD08 was undergoing consolidation and upstream embankment lift construction. Following the return of tailings

discharge to GTD03, tailings harvest from GTD01/02 recommenced with 359,111 t and 391,573 t excavated in 2016 and 2017 respectively for the purposes of paste backfill.

18.4 Water Management

18.4.1 Surface Water Management

A system of water management infrastructure including draining channels, bunding, and pipework is used to:

- Ensure that any contaminated water from operational facilities is treated or recycled and remains on-lease;
- Reduce water loss and wastage due to evaporation;
- Direct storm water from operational areas via drainage channels into Storm Water Ponds or open pits or redirected natural drainage channels and non-operational catchments to evaporate or infiltrate away from operational areas.

A series of raw water settling ponds at DBS are used for the dewatering of the underground operations. Water from the settling ponds is recirculated back underground for reuse once the solids have been separated.

18.4.2 Water Treatment

Fresh/raw water is collected in various facilities including ponds and tanks to supply the operations at both The Granites process plant and DBS mine. Where potable water is required at the operation, it is supplied using on-site reverse osmosis plants. The water is used for processing operations, the mine operations including the refrigeration plant and general water use across the operation.

Grey water and sewage waste water produced on site is treated either using the sewage treatment facility located southeast of the accommodations village, Abco wastewater treatment systems or septic systems.

There is zero discharge of water off site, hence there are no other water treatment facilities installed at The Granites site.

18.5 Surface Infrastructure

The following infrastructure is located at The Granites and DBS.

- The Granites:
 - Entry gate and social responsibility office;
 - Administration office area and clinic;
 - Airstrip and associated infrastructure (Jet A1 fuel farm, passenger waiting area);
 - Emergency response training yard;
 - Diesel fuel farm;
 - LPG gas bullet tanks;
 - MLG haulage contractor offices and workshop;
 - Crushing facility (owned and operated by CSI);
 - Diesel power station (owned and operated by KPS/Pacific Energy);

- Landfill facility (burn and non-burn pit);
- Scrap metal laydown yard;
- Land farm for remediation of hydrocarbon contaminated soil;
- Light vehicle and boilermaker workshops;
- Processing plant and associated infrastructure;
- Raw water ponds (used primarily for dust suppression);
- ROM pad;
- Tailings filter plan and storage pad;
- Tailings storage facilities;
- Rubber liners workshop;
- Twin Hills Village (rooms, laundry facilities, wet and dry mess, pool and gym);
- Twin Hills Village sewage system;
- Vehicle wash-down bay;
- Contractors' workshops.
- DBS:
 - Administration office areas and health, safety and loss prevention (HSLP) clinic;
 - Core logging and sampling shed;
 - Core laydown and storage area;
 - Drillers laydown yards;
 - Heavy and light vehicle wash-down bay – workshop;
 - Heavy vehicle workshop and laydown area;
 - Paste backfill plant (including tailings stockpile area, conveyors, mixing plant, silos and other associated infrastructure);
 - Surface explosive magazines and ammonium nitrate storage area;
 - Surface water settlement ponds (for silt removal from underground dewatering water);
 - Diesel power station (housed in two separate buildings, both owned and operated by KPS/Pacific Energy);
 - Light vehicle workshop and laydown area;
 - Refrigeration plant and ventilation infrastructure;
 - Backfill surface water pond (storage of raw water);
 - CAF backfilling plant, crusher and associated workshop and crib room;
 - Land farm for remediation of hydrocarbon contaminated soil;
 - Landfill facility (including a burn and non-burn pit);
 - Diesel fuel farm;
 - Supply storage shed and laydown areas.

Personnel for the current operations are housed in the Twin Hills accommodation village located at The Granites, which is maintained and serviced by Sodexo contractors. Presently the village consists of 1,118 accommodation rooms, recreation, dining and kitchen, wet mess and laundry facilities.

18.6 Power and Electrical

Diesel-powered power stations are located at both The Granites and DBS. The power stations and generators are owned and operated by KPS/Pacific Energy Ltd. The Project purchases power under contract with KPS/Pacific Energy Ltd.

The Granites power station currently consists of a single engine hall housing 20 generators providing power for the processing plant, village and ancillary areas. Of the 20 units 17 can be run at any one time leaving three as standby sets. The maximum energy output for this facility is 14.5 megawatts (MW).

At DBS, the existing power generation equipment comprises of two connected engine halls, each with a capacity to house 10 generators, supplying underground, surface offices, workshops, ventilation fans, paste plant and refrigeration plant. Of the 20 units, a maximum of 18 units can be run at any one time leaving two generators as standby sets. There are 15 x KTA50 1.25 Megavolt amperes (MVA) generators and 5 x QSK78 2.4MVA generators, giving it a maximum generation capacity of 19.9 MW.

There is a redundant power line from The Granites to DBS which, if refurbished could potentially be used to transmit power either to or draw power from The Granites. This line has a maximum capacity of 3 MW.

In January 2019, a new gas pipeline and dual fired (diesel / gas) generators is being commissioned as part of a power supply infrastructure upgrade. A 450 km gas pipeline (buried) is being installed as a spur from the Darwin to Alice pipeline out to The Granites and DBS operations. Once commissioned, the gas turbines will reduce the cost of electricity generation by 20% and improve power supply during the wet season, when diesel cannot be trucked to site.

18.7 Fuel

Diesel fuel is delivered to the site via triple trailer road trains on the Tanami Track from Alice Springs.

Fuel facilities vary in size with the primary diesel storage facilities being the DBS and The Granites fuel farms. The maximum storage volumes for these facilities have been designed such that sufficient capacity can be held to allow for periods when the site is inaccessible by road. Optimal inventory for the wet season period is >80% capacity at all times allowing for a period of >30 days of road isolation for diesel and >6 weeks for Jet A-1.

18.8 Communications

The primary communications to the project site and between The Granites and DBS is via single mode fiber optic with a backup satellite system installed.

Vehicle safety and emergency reporting are accomplished through the use of a site and mine-wide radio system with dedicated channels for mill operations, mine operations, and road operations. The radio system extends throughout the mine by use of a leaky feeder system. The Granites site is using UHF frequency band and at DBS, including for underground operations we are using VHF frequency band

A hard-wired analog mine phone system is also installed throughout the mine and single mode fiber optic is also used for the underground mine operations.

18.9 Comments on Infrastructure

In the opinion of the QP, the existing infrastructure is appropriate to support the current LOMP.

19.0 MARKET STUDIES AND CONTRACTS

19.1 Market Studies and Contracts

Newmont has an agreement with The Perth Mint for refining of doré produced from the Project. Newmont's bullion is sold on the spot market, by marketing experts retained in-house by Newmont.

The terms contained within the sales contracts are typical of and consistent with standard industry practice and contracts for the supply of doré elsewhere in the world.

The Project has many supply contracts in place for goods and services required to operate an underground mine and processing facility at a remote site.

Large services contracts include:

- Processing plant crusher – Crushing Services International (CSI);
- Charter air services – Alliance Airlines;
- Camp catering and housekeeping – Sodexo;
- Underground drilling – Swick Mining Services;
- Power station supply and operation – Pacific Energy Ltd;
- Surface ore haulage – MLG;
- Explosives storage, mixing and transport – Orica;
- Assay and analytical services – Bureau Veritas.

Many supply contracts are in place with suppliers for the purchase of various goods, the largest contracts including purchase of fuel, reagents, ground support and leases of mining equipment. The terms, rates and charges established by these contracts are within industry norms.

19.2 Commodity Price Projections

Metal price assumptions are provided by Newmont management and are based on three-year trailing average prices applicable at the time the Mineral Reserves are estimated.

Metal pricing and exchange rate assumptions used for the 2018 Mineral Reserves estimates are as follows:

- **Gold:** US\$1,200 or AU\$1,600 /oz;
- **Exchange Rate:** US\$0.75 = AU\$1.00.

A higher metal price (US\$1,400/oz or AU\$1,750/oz) is used for the Mineral Resource estimates to ensure the Mineral Reserves are a sub-set of, and not constrained by, the Mineral Resources, in accordance with industry accepted practice.

19.3 Comments on Market Studies and Contracts

In the opinion of the QP:

- The terms contained within the sales contracts are typical of and consistent with standard industry practice and contracts for the supply of doré elsewhere in the world;

- Metal prices are set by Newmont management and are appropriate to the commodity and mine life projections and consistent with peers in the industry.

20.0 ENVIRONMENTAL STUDIES, PERMITTING AND SOCIAL OR COMMUNITY IMPACT

20.1 Baseline Studies

An environmental assessment and management report for the Tanami Granites Gold Mine (a former name of the Project) was completed in 1985. Possible environmental impacts were studied, and their management considered. Aspects covered included:

- Waste management;
- Groundwater;
- Biological environment;
- Social;
- Dust and gaseous emissions;
- Noise;
- Dangerous materials;
- Infrastructure;
- Fire.

No significant impacts were identified as likely to result from the Project as then configured.

In 1990 an Environmental Report was completed for the DBS area, outlining pre-mining site conditions. Areas covered by the study included:

- Flora and fauna survey;
- Letter from the Conservation Commission of the NT, identifying possible presence of the Rufous Hare-Wallaby to the west of The Granites site, later confirmed to be Speckled Hare-Wallaby;
- Water quality analysis;
- Dust assessment;
- Noise assessment; and
- Waste rock characteristics.

Several additional baseline studies were performed as part of the early mining operations, and in support of the current underground operations. These included:

- Flora Studies:
 - Three habitat types/landforms supporting various vegetation associations identified during botanical surveys of The Granites mineral lease and associated exploration areas. A total of 125 species were identified. No plants considered rare were recorded during the survey. Two introduced species were noted;
 - Two habitats supporting a relatively narrow range of plant communities which were generally widespread and common throughout the Tanami region were identified at DBS. A total of 198 species were recorded. No rare flora was noted. Two introduced species were identified.

- Though not listed under territory or federal law there are several species in the area that are considered significant to the Indigenous Warlpiri people. Trees of significance in the area greater than 2 m tall should not be removed unless approved by the Aboriginal Traditional Owners;
- Fauna Studies:
 - Three locally abundant but regionally and nationally rare mammals (Spectacled Hare-wallaby, Mulgara; and Bilby) were found to be widespread;
 - Evidence of populations of Greater Bilby (*Macrotis lagotis*) has been recorded along the DBS haul road and in the Billabong and Jumbuck borefields. Bilby populations were also present along the Windy Hill (Minotaur) haul road prior to and during operation;
 - No stygofauna are known to be present in groundwater of The Granites and DBS leases and the Billabong and Jumbuck borefields.
- Bird Surveys:
 - These surveys observed 82 species at DBS. For the wider Granites region, 162 bird species have been observed.
- ARD and Characterization Studies:
 - All Callie and Auron ore samples were determined to be slightly basic to basic. The Callie waste samples were determined to be slightly basic to highly basic. This preliminary assessment showed that the rocks do not have acid generating potential;
 - At DBS, the main potentially acid generating (PAG) units were identified in the Dead Bullock member of Triumph Hill pit. This material appears in the transitional to fresh zone of the pit in which the material was mined. The waste was encapsulated within Dead Bullock Ridge WRF and the location recorded;
 - Test work identified PAG material to be present within the hanging wall component of the Quorn pit at The Granites. A selective sampling and handling program was initiated to manage this material within Quorn west pit. The pit has since been used to store tailings which covered this material;
 - Sulfide bearing lithotypes are present in trace-abundance in Shoe South WRF and are largely confined to a sub unit of sulfide-graphite-schist. Due to the decades of weathering the pyrite-oxidation of this material is now well advanced. The material identified as PAG was placed on the inner walls of the Shoe in-pit TSF embankment raise, subsequently capping this material within the TSF.

Since 2005, Newmont has overseen the completion, in collaboration with the CLC, of fauna and flora surveys within a 200 km radius of the existing Mineral Leases. These surveys were initially intended to assess the impacts of the operations on biodiversity of the Tanami but have also been a source of employment for Indigenous Rangers from Yuendumu and Lajamanu and have provided substantial information on the biodiversity of the region. To date, a total of eight surveys have been completed.

The effects of season, distance to mine impact sites, latitude (north/south), fire and disturbance, land unit, vegetation species richness and percentage of ground cover have been assessed. This was achieved by using regression modelling analysis from the database to quantify and assess the significant effects of a series of explanatory variables and co-variants on a selection of flora and fauna groups (Newsome et al, 2009).

- Fauna data was highly variable between sites and surveys which suggest that local habitat and seasonal conditions are important determinants of abundance, richness and

probability of occurrence of selected fauna groups. Distance to mine site impacts did not appear to affect the majority of fauna groups although the abundance of all fauna, particularly members of the Muridae family, appeared higher in proximity to mine sites (Newsome et al., 2009).

20.2 Environmental Considerations

20.2.1 Fire Management

The fire regime in the area is managed by the CLC in consultation with Bushfires Council NT and the Traditional Owners. The area surrounding the mineral leases is a part of the Central Desert Aboriginal Land Trust and is not managed by Newmont. Risk to operations presented by fire is managed in accordance with the Newmont Bushfire Management Plan which includes the provisions for fuel reduction burning and fire break management. This management plan was developed in consultation with Bushfires NT and has been provided to and reviewed by Bushfires NT with their recommendations adopted.

20.2.2 Weed Management

Weeds are managed as per the Newmont Weed Management Plan. Primary focus is spraying of areas around the lease that have been rehabilitated or not disturbed or areas of high occurrence. Weed monitoring is undertaken along the borefield access tracks, DBS–Granites haul road and select locations at DBS and The Granites subject to occurrence.

20.2.3 Fauna

Newmont has a long-term dingo management plan which was developed in 2006. The aim of the plan is to minimize the level of dependence of the dingo on mining activities (i.e. food and water resources) and to reduce potential human-dingo interaction on the leases to ensure the protection of the dingo and the safety of site personnel.

There are infrequent sightings of rabbits along the haul road and mineral leases. Occasional reports are also received for sightings of other feral species in or around the lease areas including cats, foxes and camels.

20.2.4 Monitoring Activities

Newmont applies a consistent approach to monitoring and measurement programs and assesses the effectiveness of controls. Programs focus on those activities which are either legislative requirements, voluntary commitments or relate to significant (high or extreme) risks. Aspects incorporated in the environmental monitoring program include:

- Water, including groundwater, surface water and process water;
- Hydrocarbon, including bioremediation facility;
- Weather;
- Soil erosion;
- Weeds;
- Cyanide;
- Tailings;
- Flora and fauna;
- Waste rock;

- Reclamation and closure.

Some routine monitoring is conducted by external consultants and contractors, including:

- Biennial rehabilitation monitoring;
- Annual bore field status report;
- Annual groundwater report; and
- Regional biodiversity monitoring program.

20.3 Closure Plan

The project maintains the Newmont Closure and Reclamation Plan, which in addition to meeting regulatory requirements of the annual MMP, provides information used in decision making and project management, including planning and methods of work, and provide records of closure and reclamation activities.

The prime closure objectives included within the plan were to:

- Ensure that land is physically safe for people to access and does not pose a human health risk;
- Ensure that land is left in a stable condition that minimizes long-term environmental impacts;
- Rehabilitate disturbed land such that it is ecologically sustainable and generally reflects the surrounding ecology, geography and hydrology, or, where appropriate, the ecosystem relevant to an appropriate land use;
- Ensure that water quality does not pose an ecological risk or health risk to humans or livestock;
- Establish final land use objectives to meet land capacity and the surrounding social context and relevant agreements;
- Engage in Community / Stakeholder consultation to a degree that there is general acknowledgement that the process has been satisfactory and was delivered in a way that enabled understanding and opportunity for stakeholders to exert reasonable influence on relevant processes.

Following rehabilitation and lease relinquishment, the area will be returned to the site's Traditional Owners, the Warlpiri People, through the CLC. There are no active land uses suggested at this time; specifics of this will be discussed at length with Traditional Owners and the CLC prior to mine closure. The primary focus of post closure management will be guided by the closure objectives and targets.

As a part of Newmont's closure strategy, rehabilitation monitoring has been undertaken on an annual basis from 2006 to 2012. Following the completion of the 2012 program, monitoring activities were changed to a biennial basis with annual assessments no longer considered justified based on the maturity of the rehabilitated areas.

The Project has undertaken a third-party appraisal of the rehabilitation monitoring program to date and associated methodologies employed. This revision has assessed the suitability of alternate industry leading methodologies to the existing program that may provide more qualitative and quantitative results than the existing Ecosystem Function Analysis (EFA) program that some consider a subjective methodology. This revision has resulted in the

development of remote survey techniques using drone technology for high resolution aerial imagery collection and in field ground truthing. Data interrogations via image segmentation algorithms classify the imagery to guild and species level via unique spectral frequencies specific to the contained vegetation communities.

Monitoring information allows ecosystems development to be quantified, providing a direct measure of performance against the closure criteria and further guides continual improvement to the Closure and Reclamation Plan.

In 2018, Newmont submitted a closure estimate of AU\$41.8M for authorization 0086-02 as part of the 2018 MMP submission. In July 2018, the DPIR accepted the 2018 MMP and associated liability estimate. A bank guarantee in the amount of AU\$38,686,910 is currently held by DPIR to cover closure costs for the Project, being approximately aligned with the 10% reduced security bonding requirements of the NT Government.

20.4 Permitting

20.4.1 Current Permits

The Project currently has an authorization (0086-02) registered under an active approved MMP. This authorization allows the Project to conduct mining activities and has monitoring commitments attached. The Project must ensure it meets these commitments or risk penalties from the DPIR

The rights of the Project to extract groundwater are also managed through the MMP, under the term of the current authorization. The MMP includes details on the mine closure plan and sets out the security bond that is required to be lodged with the NT Government to cover the anticipated cost of the rehabilitation commitments associated with the Project.

An updated MMP is required to be completed and approved by the DPIR every four years, with an Operational Performance Report (OPR) to be submitted in the intervening years, outlining the production and environmental performance of the operations. The 2018 MMP was submitted on 10 May 2018 and approved by the DPIR on 3 July 2018. The 2018 MMP included the entire scope for the TE2 project and forms the required permit and approvals for the TE2 project should it successfully progress to full funding and construction.

The Project currently holds all required permits and approvals to allow for ongoing mining and processing operations as well as the growth associated with the TE2 project. Some permits must be re-applied for annually.

20.4.2 Notices of Intent

Notice of Intent (NOI) applications are presented to DPIR for assessment for mining activities presenting a significant change to operations. The level of assessment varies depending on the sensitivity of the local environment, the scale of the proposal and its potential impact on the environment. Where deemed necessary, any further assessment is presented as either a Public Environmental Review (PER) or an Environmental Impact Statement (EIS).

To date, Newmont has submitted four NOI applications to DPIR. Of the four applications submitted, all have been approved without requiring further assessment having been referred for approval under the MMP submission process.

A new mineral lease application was submitted in February 2010 to the DPIR to be located south of MLS 8 (The Granites) for operating a seepage interception system. A Sacred Sites Clearance Certificate (SSCC) for the mineral lease application (MLA27957) was received in December 2010. Newmont is currently undergoing consultation with the CLC to finalize the Mineral Lease Agreement which has had several extensions, the most recent of which has extended the assessment period to 6 March 2019.

20.5 Considerations of Social and Community Impacts

20.5.1 Sociocultural Aspects

Every five years, NTO conducts extensive research on the impact of operations through Social Impact Assessments (SIA). These assessments include the collection of insights and feedback from residents and other stakeholders about NTO and the future needs of the community across the extended life of the Project.

SIAs have been conducted in 2009, 2013 and most recently in 2018 by the Centre for Social Responsibility in Mining Pty Ltd (CSRm). The primary focus of the 2018 SIA was to provide information on understanding and quantifying priority areas, impacts on some of the more qualitative social components such as community cohesion and environmental values of the traditional owners of The Granites/DBS and the aboriginal residents of Yuendumu, Lajamanu, Willowra and Nyirripi. A final report from CSRm is yet to be received, however; findings from the 2018 SIA will be developed into an action plan for delivery during 2019.

20.5.2 Stakeholders

NTO has set a clear strategic direction for stakeholder engagement that achieves appropriate, regular and consistent engagement with key stakeholders on the performance, achievements and the future of the Project. The key stakeholders for the Project include:

- TO of the Warlpiri language and cultural group;
- CLC;
- DPIR;
- NT Department of Environment and Natural Resources (DENR);
- NT WorkSafe;
- NTO and Newmont Australia;
- NTO contractors and suppliers;
- NTO workforce;
- Newmont shareholders;
- Owners of Rabbit Flat;
- Pastoralists;
- Local indigenous communities in the Tanami Region;
- Community of Alice Springs.

NTO has developed a SRMP to set a clear strategic direction for stakeholder engagement that achieves appropriate, regular and consistent engagement with key stakeholders on the performance, achievements and future of the Project.

20.5.3 Community Affairs

The Project lies within the Central Desert Shire, which consists of nine widely-spread communities. Through the external relations department, Newmont actively engages with stakeholders in their concerns, aspirations and values regarding expansion, operational and closure aspects. The success of engagement is monitored through direct feedback from the community members, the CLC and from undertaking social impact assessments.

20.6 Comments on Environmental Studies, Permitting and Social or Community Impact

In the opinion of the QP:

- Newmont has sufficiently addressed the environmental impact of the operation, and subsequent closure and remediation requirements so that Mineral Resources and Mineral Reserves can be declared, and that the mine plan is appropriate and achievable. Closure provisions are appropriately considered. Monitoring programs are in place;
- The mine currently holds the appropriate social license to operate;
- Newmont has developed a Communities' Relations Plan to identify and ensure an understanding of the needs of the surrounding communities and to determine appropriate programs for filling those needs. The company appropriately monitors socio-economic trends, community perceptions and mining impacts;
- Permits held by Newmont for the Project are sufficient to ensure that mining activities are conducted within the regulatory framework required by NT and Federal Government regulations.

21.0 CAPITAL AND OPERATING COSTS

21.1 Capital Cost Estimates

Future capital costs are estimated based on expected sustaining capital requirements of the mine. Development costs are estimated based on previous experience and are adjusted for any future anticipated changes in factors which would affect cost and amount of development. The timing of equipment replacement and rebuilds are based on replacement and rebuild schedules, and the anticipated cost is based on actual experience. In the later years of the LOM, costs are estimated for the required equipment to sustain production.

Mine capital costs consist predominantly of replacement of equipment and major equipment rebuilds to sustain production through LOM, as well as anticipated underground development and capitalized drilling and infrastructure needs.

Capital development costs have been estimated based on the expected amount development in each year and the anticipated costs of development. This is derived from previous experience with updates to the cost based on projected changes in items that would affect costs. Based on BP19, it is estimated that the Project will develop approximately 43 km of capital development for a cost of US\$235M (US\$5,500/m) over the LOM, which includes the development required to execute TE2.

Expenditure related to the underground mobile fleet is estimated at US\$190M over the LOM. The timing of equipment replacement and rebuilds are based on replacement and rebuild schedules, and the anticipated cost is based on actual experience.

Projected process plant and tailings sustaining capital costs for the LOM are US\$115M. This includes sustaining capital, provisions for tailings dam wall lifts.

A CRF has been applied to sustaining capital as required in the Newmont cut-off grade guidelines (130%).

For TE2, US\$635M of project development capital for a hoisting shaft is included in the 2019BP, which includes a US\$25M upgrade to the recently completed gas power plant.

21.2 Operating Cost Estimates

Operating cost estimates were derived from the 2019 budget and LOMP. The budget is built using various cost inputs including operating experience, quotes from various service providers, anticipated personnel changes, and changes in production. The operating costs used in the economic evaluation of the Mineral Reserves are based on an average of future costs over the next seven years. A validation step ensured that these average costs were adequately representing LOM costs.

Mining costs are expected to average US\$48.44/t mined over the LOM. Process costs are estimated at US\$19.82/t mined. G&A costs total US\$12.14/t mined.

21.3 Comments on Capital and Operating Costs

The QP has reviewed the capital and operating cost provisions for the LOMP that supports Mineral Reserves and considers that the basis for the estimates that include mine budget data, vendor quotes, and operating experience, is appropriate to the known mineralization, mining and production schedules, marketing plans, and equipment replacement and maintenance requirements.

Appropriate provision has been made in the estimates for the expected mine operating usages including labor, fuel and power and for closure and environmental considerations.

Capital cost estimates include appropriate sustaining estimates and project expenditure for TE2.

22.0 ECONOMIC ANALYSIS

This section is not required as the Project is currently in production, Newmont is a producing issuer, and this report does not include a material expansion of current production.

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23.0 ADJACENT PROPERTIES

This section is not relevant to this Report.

24.0 OTHER RELEVANT DATA AND INFORMATION

This section is not relevant to this Report.

25.0 INTERPRETATION AND CONCLUSIONS

In the opinion of the QP:

- Information provided by Newmont's legal and tenure experts on the mining tenure held by Newmont in the Project area supports that the company has valid title that is sufficient to support declaration of Mineral Resources and Mineral Reserves;
- Information provided by Newmont's legal and tenure experts supports that the Operations hold sufficient surface rights to enable mining operations, and the declaration of Mineral Resources and Mineral Reserves. Appropriate steps, where required, have been taken to lodge either extensions or renewals of tenements as such fall due;
- Knowledge of and understanding of the deposit settings, lithologies, mineralization style, and structural and alteration controls on mineralization is sufficient to support Mineral Resource and Mineral Reserves estimation;
- The exploration programs completed to date are appropriate to the style of the deposits and prospects within the Project. The Project retains significant brownfields exploration potential, and additional work may be undertaken;
- The quantity and quality of the lithological, geotechnical, collar and downhole survey data collected in the exploration and infill drill programs completed by Newmont and the predecessor operating companies are sufficient to support Mineral Resource estimation and conversion to Mineral Reserves;
- The process of data verification for the Project has been performed by Newmont personnel, staff from Newmont's predecessors and external consultancies contracted by Newmont. The data verification programs undertaken on the data collected from the Project adequately support the geological interpretations, the analytical and database quality, and therefore support the use of the data in Mineral Resource and Mineral Reserves estimation, and in mine planning;
- Metallurgical testwork and associated analytical procedures were appropriate to the mineralization type, confirmed the optimal processing routes, and were performed using samples that are typical of the mineralization styles found within the Project;
- Samples selected for testing were representative of the various types and styles of mineralization. They were selected from a range of depths within the deposit and sufficient intersection selected so that tests were performed on a sufficient sample mass. As mining progresses deeper and/or new mining zones are identified, additional variability tests are planned to be undertaken as required;
- Testwork results have primarily been confirmed by production data in mining areas;
- Mill process recovery factors are based on production data for Callie and testwork results for Auron ore, as Auron has never been treated separately through the mill. Federation recovery is also based on testwork, but no Federation ore has yet been processed through the mill;
- Ore hardness, reagent consumptions and process conditions are based on both testwork and production data;
- Recovery factors vary on a day to day basis depending on grade and mineralization type being processed. These variations are expected to trend to the forecast average recovery value for monthly or longer reporting periods;

- The Mineral Resources and Mineral Reserves estimation for the Project have been performed to industry best practices and conform to the requirements of the CIM Definition Standards. There is an upside for Project if Inferred Mineral Resources can be upgraded to higher confidence Mineral Resource categories to support additional Mineral Reserves;
- The assumptions used in developing the mine plan are consistent with previous plant operating experience. Previous production throughputs and recoveries and the Project background history provide supporting data for the proposed production profile;
- The current process facilities are appropriate for the mineralization types provided from the mine. The post-TEP flowsheet and equipment upgrade and existing infrastructure adequately support the current mine plan;
- Newmont is readily able to market the doré produced from the Project;
- Existing mining activities are supported under existing Project permits held by Newmont; however, additional permits would be required for Project development for any other areas other than the areas currently permitted for mining activities at DBS and processing activities at The Granites; permits may require annual approvals;
- Newmont has sufficiently addressed the environmental impact of the operation, and subsequent closure and remediation requirements that Mineral Resources and Mineral Reserves can be declared, and that the mine plan is appropriate and achievable. Closure provisions are appropriately considered. Monitoring programs are in place. The mine currently has the appropriate social license to operate;
- The capital and operating cost provisions for the mine plan that supports Mineral Reserves have been reviewed. The basis for the estimates that include mine budget data, vendor quotes, and operating experience, is appropriate to the known mineralization, mining and production schedules, marketing plans, and equipment replacement and maintenance requirements. Appropriate provision has been made in the estimates for the expected mine operating usages including labor, fuel and power and for closure and environmental considerations. Capital cost estimates include appropriate sustaining estimates;
- Under the assumptions in this Report, the Project has positive economics until the end of mine life, which supports the Mineral Reserves estimate.

26.0 RECOMMENDATIONS

To facilitate future growth of the Mineral Resources and Mineral Reserves, the QP recommends the following:

- Continue to develop and implement plans to explore in-mine and more distant exploration targets on the Project tenements. The approximate expenditure over the next five years is expected to be AU\$44.8M for surface exploration drilling totalling approximately 170,000 m and AU\$72.58M for underground exploration drilling totalling approximately 407,000 m;
- Continue to infill drill areas of the existing Mineral Resources to ensure confidence is maintained at an appropriate level for ongoing planning, scheduling and ore-control. The approximate costs of underground ore control drilling over the next five years is expected to be AU\$61M for approximately 605,000 m;
- Continue with metallurgical testing programs to validate input factors and assumptions used for generating Mineral Reserves as new material is added to the Mineral Resource inventory. The approximate costs of these ongoing sampling and testing programs are expected to amount to AU\$850k over the next five years;
- Continue to review the potential for expanding mine production, particularly at depths below the current operations, and determine the requirements to convert Mineral Resource in these areas to Mineral Reserves.

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27.2 Glossary of Abbreviations, Symbols and Units

Symbol	Definition
'	seconds (geographic)
'	foot/feet
	minutes (geographic)
"	inches
#	number
%	percent
/	per
<	less than
>	greater than
µm	micrometer (micron)
a	annum/ year
Å	angstroms
asl	above sea level
BQ	36.5 mm size core
c.	circa
d	day
d/wk	days per week
dmt	dry metric tonne
fineness	parts per thousand of gold in an alloy
ft	feet
g	gram
g/cm3	Grams per cubic centimeter
g/m3	Grams per cubic meter
g/dmt	grams per dry metric tonne
g/t	Grams per tonne
Ga	billion years ago
ha	hectares
HP	horsepower
HQ	63.5 mm size core
kg	kilogram
kg/m3	kilograms per cubic meter
km	kilometer
km2	square kilometers
koz	thousand ounces
kt	thousand tonnes
kV	kilovolt
kVA	kilovolt-ampere
kW	kilowatt
kWh	kilowatt hour
kWh/t	kilowatt hour per tonne
lb	pound
M	million
m	meter/meters
m3	cubic meter
m3/hr	cubic meters per hour
Ma	million years ago
mesh	size based on the number of openings in one inch of screen
masl	meters above sea level
mi	mile/miles

Symbol	Definition
Mlb	million pounds
Mm	million meters
mm	millimeter/millimeters
Moz	million ounces
MPa	megapascals
Mt	million tonnes
Mtpa	million tonnes per annum
MVA	megavolt amperes
MW	megawatts
MWR	megawatts of refrigeration
NQ/NQ2	47.6 mm size core
°	degrees
°C	degrees Celsius
oz	ounce/ounces (troy ounce)
oz/t	ounces per tonne
p	passing
pH	measure of the acidity or alkalinity of a solution
pop	population
ppb	parts per billion
ppm	parts per million
PQ	85 mm size core
t	metric tonne
tkm	tonne-kilometer/kilometers
tpa	tonnes per annum (tonnes per year)
tpd	tonnes per day
tph	tonnes per hour
t/m3	tonnes per cubic meter
TDS	total dissolved solids
TO	Traditional Owner
TSS	total suspended solids
µm	micrometer
wt%	weight percent

Abbreviation	Definition
®	registered name
AAS	atomic absorption spectroscopy
AB	Auron Beds
AC	aircore
ALRA	Aboriginal Land Rights Act
ALS	ALS Minerals
ANC	acid-neutralizing capacity
ANP	acid-neutralizing potential
AQ	aqua regia
ARD	acid-rock drainage
ASX	Australian Stock Exchange
AuAA	cyanide-soluble gold
AU\$	Australian Dollar
AuEq	gold equivalent

Abbreviation	Definition
AuFA	fire assay
AuPR	preg-rob gold
AuSF	screen fire assay
AusIMM	Australasian Institute of Mining and Metallurgy
BBWi	Bond ball mill work index (BBWi)
BP19	Business Plan 2019
BFA	bench face angle
BIF	Banded Iron Formation
BLEG	bulk leach extractable gold
BLM	US Bureau of Land Management
BV	Bureau Veritas
C.P.G.	Certified Professional Geologist
CAF	cement aggregated fill
Capex	capital expenditure
CDALT	Central Desert Aboriginal Lands Trust
CGM	Chapman Gold Mines NL
CLC	Central Land Council
CIL	carbon-in-leach
CIM	Canadian Institute of Mining, Metallurgy and Petroleum
CIP	Carbon-In-Pulp
CMA	Consolidated Mining Agreement
CMS	Cavity Monitoring System
CN _{wad}	acid-dissociable cyanide
CP	Chartered Professional
CRF	capital recovery factor
CRM	certified reference material
CRN	Crown
CSI	Crushing Services International
CSRM	Centre for Social Responsibility in Mining Pty Ltd
CST	cleaner scavenger tailings
CTOT	carbon total
Cu Eq	copper equivalent
CuCN	cyanide-soluble copper
CV	coefficient of variation
DBS	Dead Bullock Soak
DENR	Department of Environment and Natural Resources
DGPS	differential global positioning system
DPIR	Department of Primary Industry and Resources
DSM	Dutch State Mines
E	east
EDA	exploratory data analysis
EFA	Ecosystem Function Analysis
EIS	Environmental Impact Statement
EM	electromagnetics
EOM	end of month
EOY	end of year

Abbreviation	Definition
FAusIMM	Fellow of the Australasian Institute of Mining and Metallurgy
FAIG	Fellow of the Australian Institute of Geoscientists
GCMP	Ground Control Management Plan
GED	Global Exploration Database
Golder	Golder Associates Pty Ltd
GPS	global positioning system
GRG	Gravity Recoverable Gold
GSM	Groupe Spécial Mobile
H	horizontal
Harlock	Harlock Pty Ltd
HPGR	high pressure grinding rolls
IAL	Independent Assay Laboratories
ICP	inductively-couple plasma
ICPMS	inductively-coupled plasma mass spectrometry
ICP-OES	inductively-coupled plasma optical emission spectrometry
ID	inverse distance
JCR	joint condition rating
JORC	The Joint Ore Reserves Committee of The Australasian Institute of Mining and Metallurgy, Australian Institute of Geoscientists and Minerals Council of Australia
JV	joint venture
KRS	Kerril South
KSS	Kerril South South
KV	kriging variance
LAA	Land Access Agreements
LAB	Lower Auron Beds
LAN	Lantin
L-G	Lerchs–Grossman
LNH	Lantin North Limb
LOA	length overall
LOM	life-of-mine
LSK	large-scale kinetic
MAIG	Member of Australian Institute of Geoscientists
MAN	Major Activity Notice
MAusIMM	Member of the Australasian Institute of Mining and Metallurgy
MIK	multiple-indicator kriging
MMP	Mining Management Plan
MSO	mineable shape optimizer
MTO	Minerals Title Office
MWMS	mine water management system
MWMT	meteoric water mobility testing
MWR	megawatts of refrigeration
N	north
NAG	net acid generation/net acid generating

Abbreviation	Definition
NAPP	net acid-producing potential
Newmont	Newmont Mining Corporation
NFM	North Flinders Mines
NNFM	Normandy North Flinders Mines
NI 43-101	Canadian National Instrument 43-101 "Standards of Disclosure for Mineral Companies"
NOI	Notice of Intent
Nova	Nova Minerals Pty Ltd
NN	nearest-neighbor
NNP	net neutralizing potential
NPV	Net Present Value
NSR	net smelter return
NT	Northern Territory
NTGS	Northern Territory Geological Survey
NW	northwest
OHFJVA	Officer Hill Farm-in and Joint Venture Agreement
OK	ordinary kriging
OP	open pit
Opex	operating expenditure
Pegasus	Pegasus Gold Ltd
P.Eng.	Professional Engineer
P.Geol	Professional Geologist
PAG	potentially acid-generating
PER	Public Environmental Review
PLI	point load index
PM	preventative maintenance
PoO	Plan of Operations
PSA	pressure swing adsorption
PSI	yield strength
QA/QC	quality assurance and quality control
QLT	quick leach test
QP	Qualified Person
QVC	Quartz Vein Corridors
RAB	rotary air blast
RC	reverse circulation
RIM	run-in-mine
RM-SME	Registered Member, Society for Mining, Metallurgy and Exploration
RMR	rock mass rating
ROM	run-of-mine
RPL	Environmental Monitoring Plan
RQD	rock quality designation
S	south
SAG	semi-autogenous grind
SCADA	Supervisory Control and Data Acquisition
SE	southeast
SEC	Securities and Exchange Commission
SEDAR	System for Electronic Document Analysis and Retrieval

Abbreviation	Definition
SEIS	Supplemental Environmental Impact Statement
SG	specific gravity
SHIM	Schist Hills Iron Member
SIA	Social Impact Assessments
SLC	sub-level caving
SLOS	sub-level open stoping
SME	The Society for Mining, Metallurgy & Exploration
SMU	selective mining unit
SRM	standard reference material
SRMP	Stakeholder Relationship Management Plan
SS	sulfide sulfur
SSCC	Sacred Sites Clearance Certificate
ST	scavenger tailings
STOT	sulfur total
SX-EW	solvent extraction–electrowin
TaRFS	Tanami Regional Framework Study
TE2	Tanami Expansion 2
TEA	Tanami Exploration Agreement
TEP	Tanami Expansion Project
TF	tonnage factor
TGGM	The Granites Gold Mine
TJV	Tanami Joint Venture
TO	Traditional Owners
Topo	topography
TSF	tailings storage facility
UC	uniform conditioning
UCS	uniaxial compressive strength
UG	underground
UHF	ultra-high frequency
US	United States
US\$	United States Dollar
USGS	United States Geologic Survey
V	vertical
VHF	very high frequency
VR	vent rise
WA	Western Australia
W	west
WHA	Windy Hill Agreement
WRF	waste rock facility
WSF	waste storage facility
XRD	X-ray diffraction
XRF	X-ray fluorescence
Zapopan	Zapopan NL

Symbol	Element
Ag	silver
Al	aluminum

Symbol	Element
As	arsenic
Au	gold
B	boron
Ba	barium
Be	beryllium
Bi	bismuth
C	carbon
Ca	calcium
CaCO ₃	calcium carbonate
CaO	calcium oxide
CaSO ₄ •2H ₂ O	calcium sulfide dehydrate
Cd	cadmium
Ce	cerium
Cl	chlorine
CN	cyanide
CO	carbon monoxide
Co	cobalt
Cr	chromium
Cs	cesium
Cu	copper
Fe	iron
FeOx	iron oxides
Ga	gallium
Ge	germanium
H	hydrogen
Hf	hafnium
Hg	mercury
In	indium
K	potassium
La	lanthium
Li	lithium
Mg	magnesium
Mn	manganese
Mn(OH) ₂	manganous hydroxide
MnO ₂	manganese dioxide
Mo	molybdenum
N	nitrogen
Na	sodium
Nb	niobium
NH ₃	ammonia
Ni	nickel
NOx	nitrogen oxide compounds
O ₂	oxygen
P	phosphorus
Pb	lead
Pd	palladium
Pt	platinum
Rb	rubidium
Re	rhenium
S	sulfur
Sb	antimony

Symbol	Element
Sc	scandium
Se	selenium
Sn	tin
SO ₂	sulfur dioxide
Sr	strontium
Ta	tantalum
Te	tellurium
Th	thorium
Ti	titanium
Tl	thallium
U	uranium
V	vanadium
W	tungsten
Y	yttrium
Zn	zinc
Zr	zirconium